OTA Operation Manual

Operation Manual for OTA Platform

1. Master Data Management

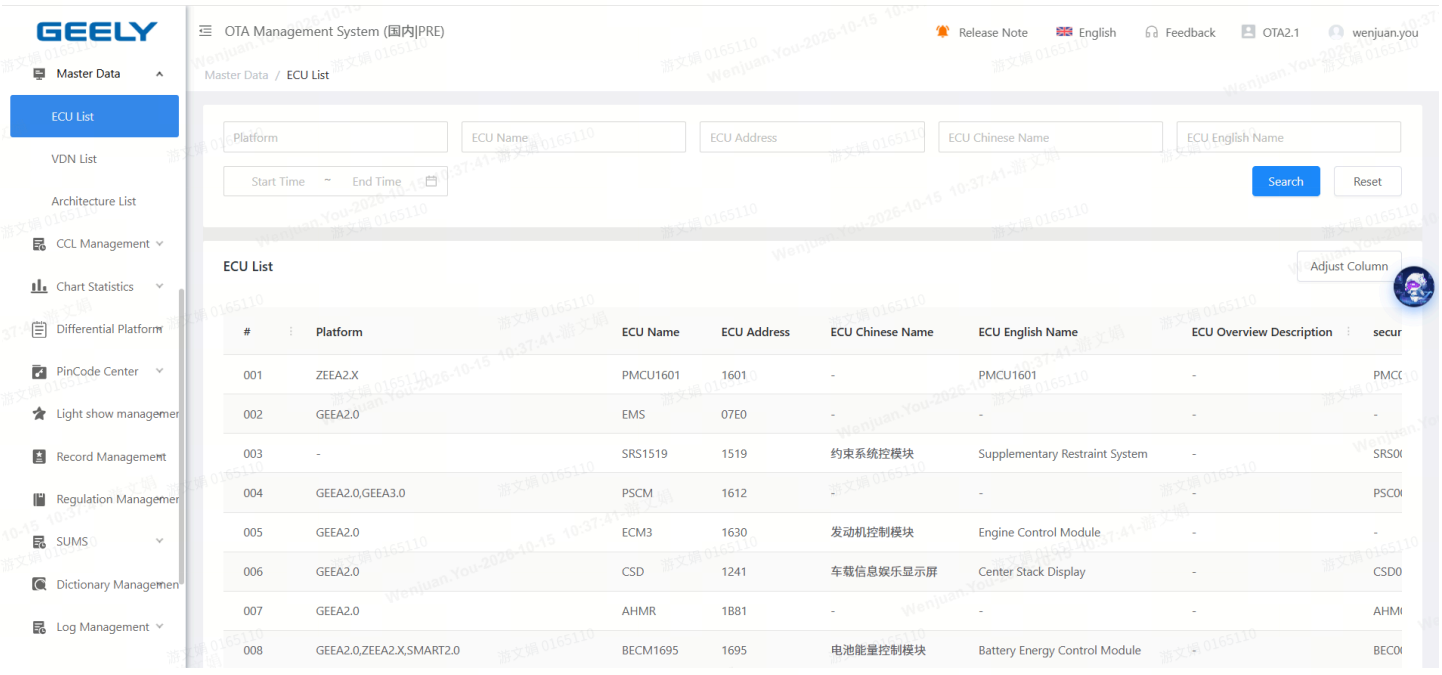
1.1 ECU List

1.1.1 Function Entry: Master Data Management >> ECU List

1.1.2 Precondition: The data comes from GDDC and is view-only, not modifiable

The fields on the ECU list page include: architecture, ECU name, ECU address, Chinese name of ECU, brief description, securityKey, changer, and change time.

If there is any change to the ECU in the GDDC system, it will be synchronized to all OTA environments.



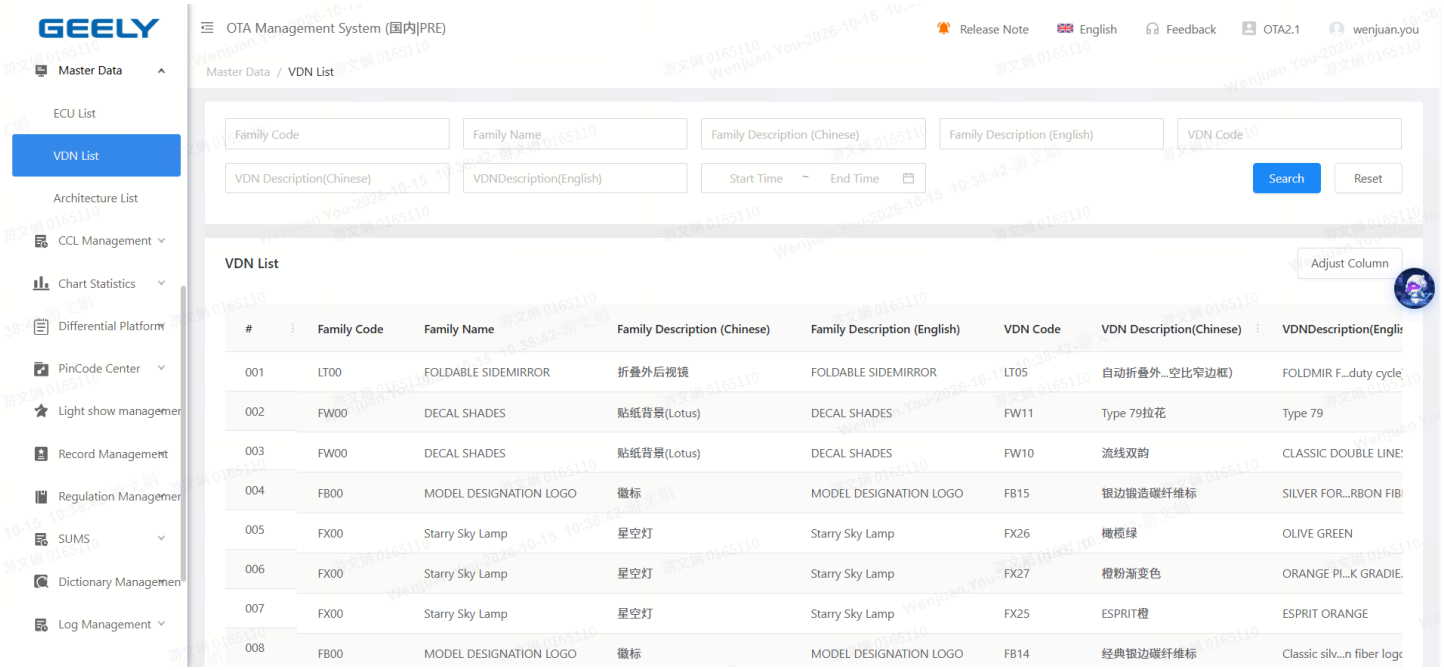
1.2 VDN List

1.2.1 Function Entry: Master Data Management >> VDN List

1.2.2 Precondition: The data comes from GDDC and is view-only, not modifiable

The fields on the VDN list page include: family code, family name, family description (Chinese), VDN code, VDN description (Chinese), VDN description (English), modifier, and modification time.

When there is a change to the VDN in the GDDC system, it will be synchronized to all OTA environments.



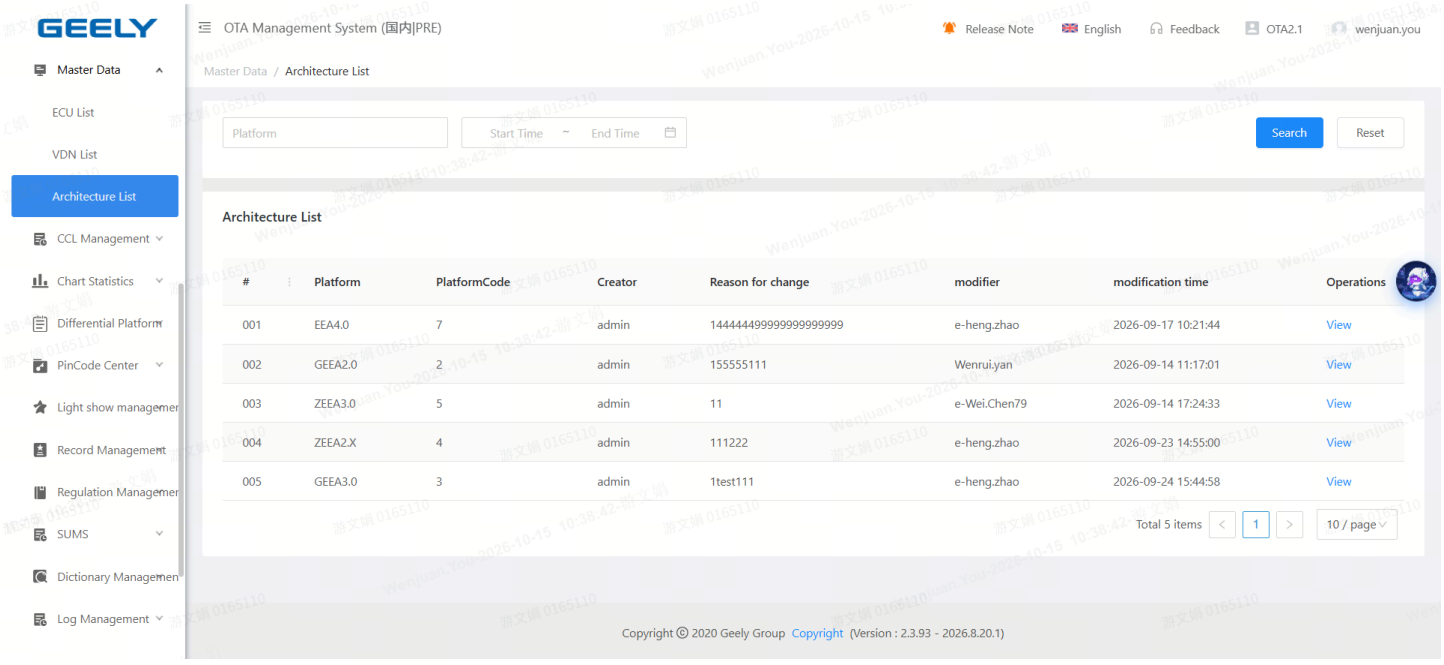
1.3 Architecture List

1.3.1 Function Entry: Master Data Management >> Architecture List

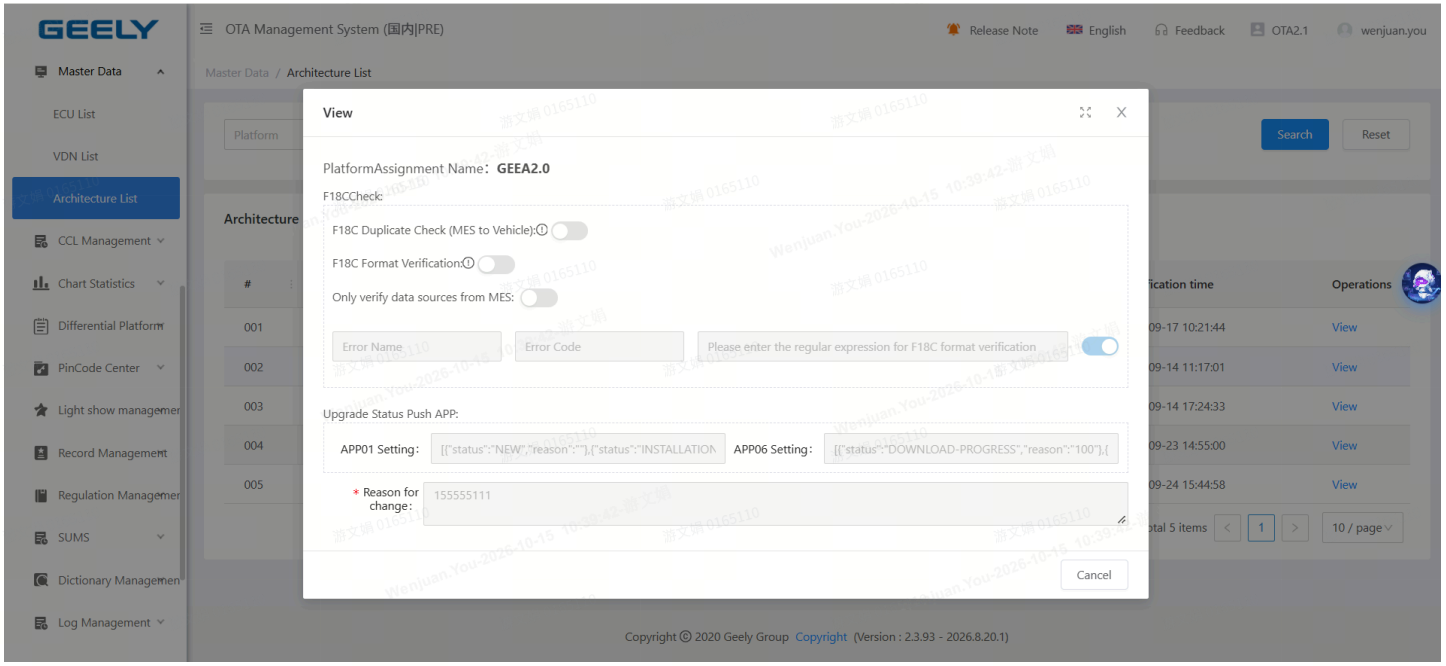
1.3.2 Precondition: The data comes from GDDC and is view-only, not modifiable

The fields on the architecture list page include: Architecture, Architecture Code, Creator, Change Reason, Changed By, Change Time, and Action (View).

If there is any change to the architecture in the GDDC system, it will be synchronized to all OTA environments.



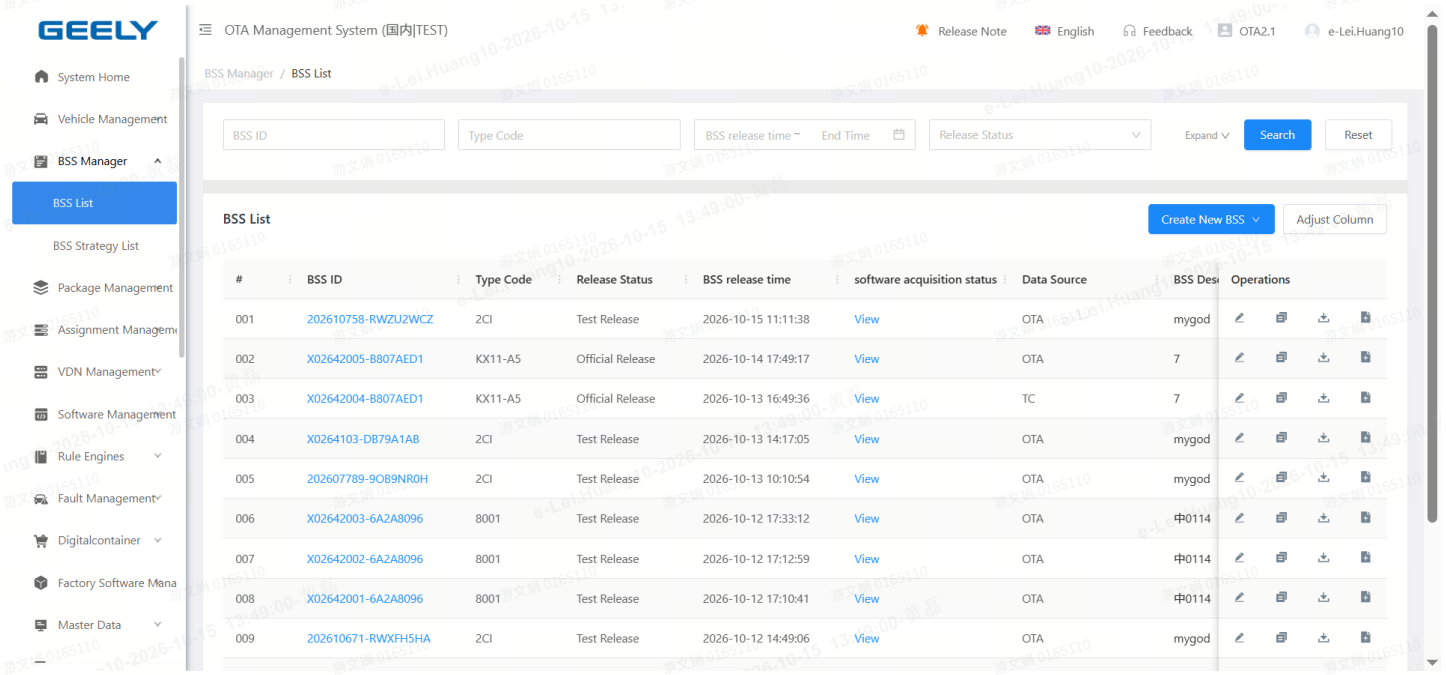
Click to view the F18C settings and APP configuration for the architecture demonstration



2. BSS Management

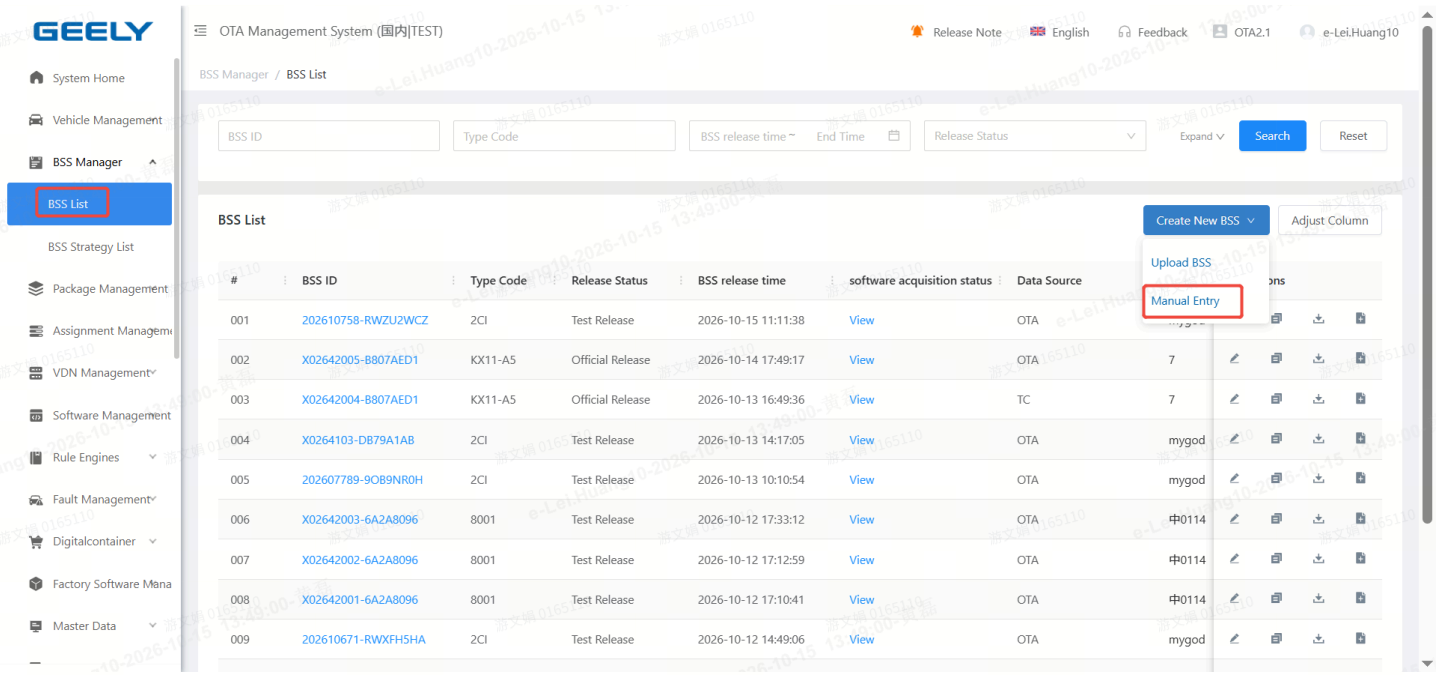
2.1 BSS List

- Note: BSS is divided into TC synchronization and manual creation. This operation manual only describes the steps for manual creation. Manual creation of BSS is not recommended during OTA activities in the production environment.
- Function Entry: BSS Management >> BSS List
- precondition: None

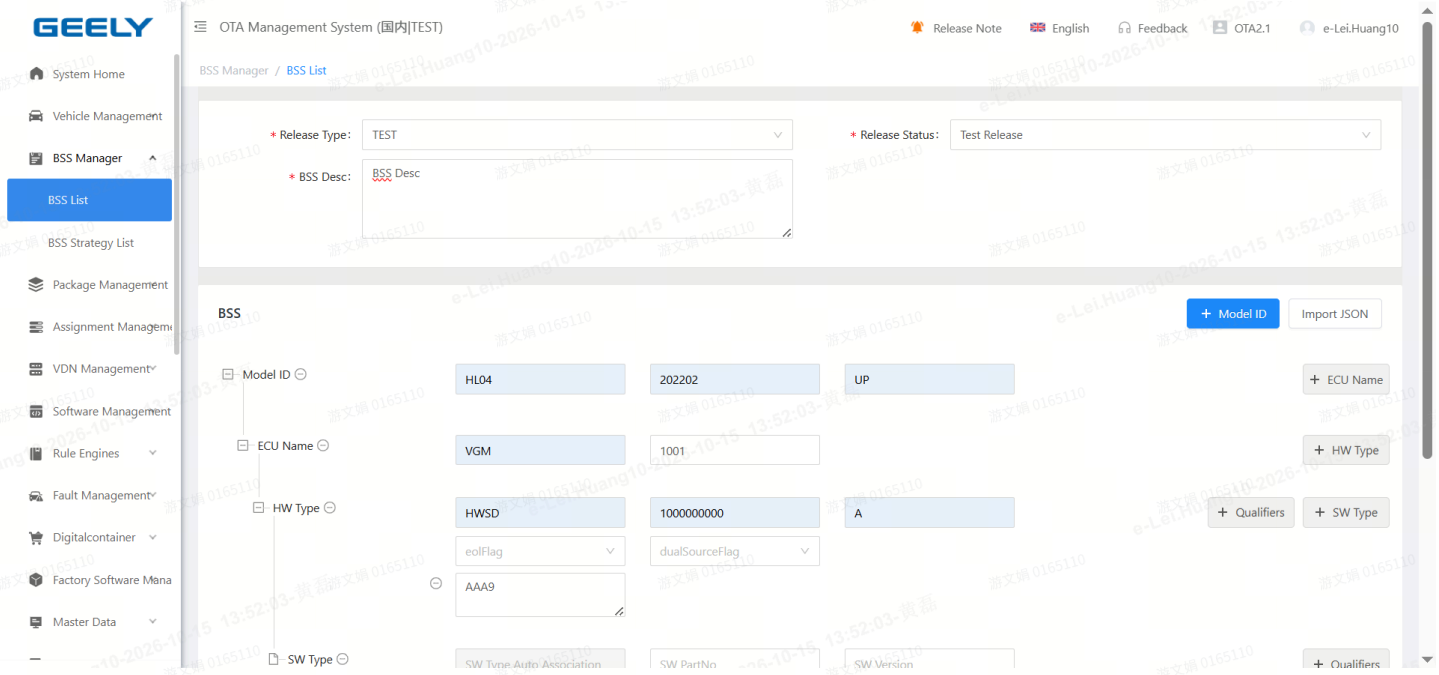


2.1.1 BSS List - Manual Addition:

- Function Entry: Menu >> BSS Management >> BSS List >> Add New BSS >> Addition Method: Manual Addition



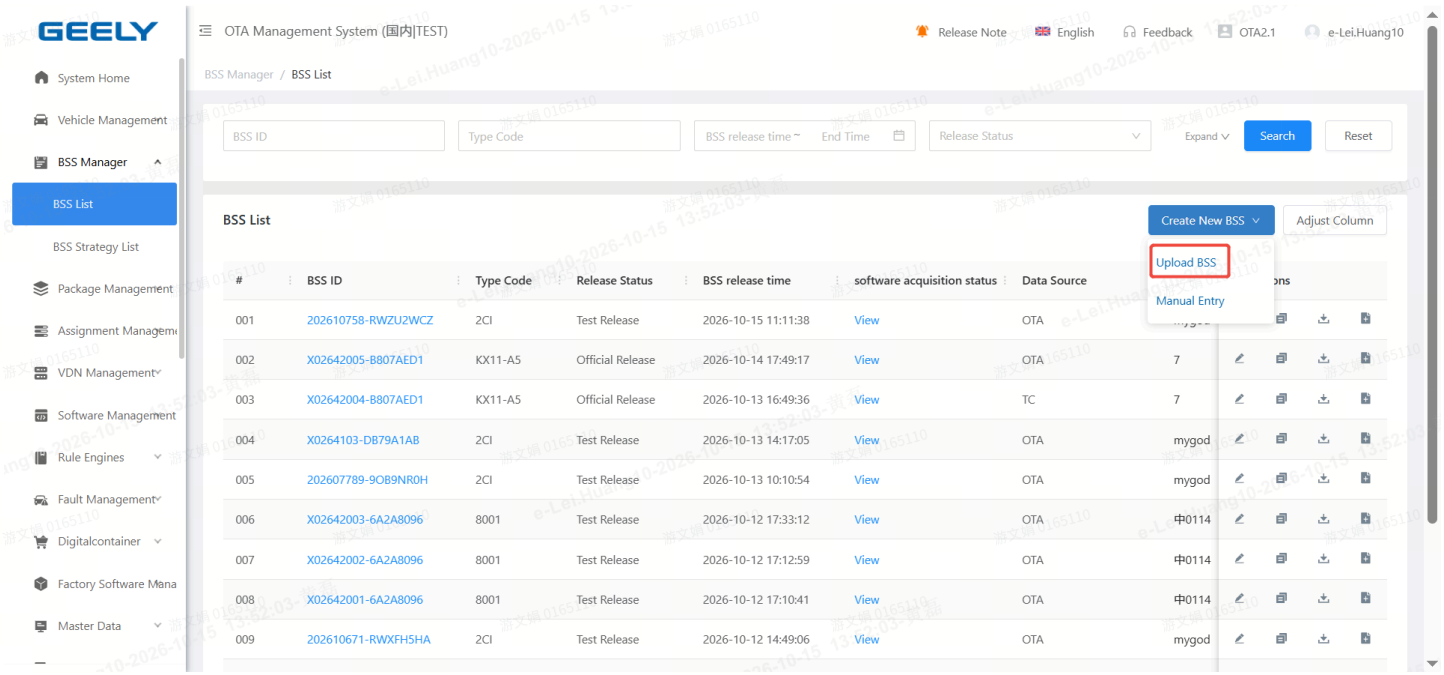
- Manually create the BSS page:



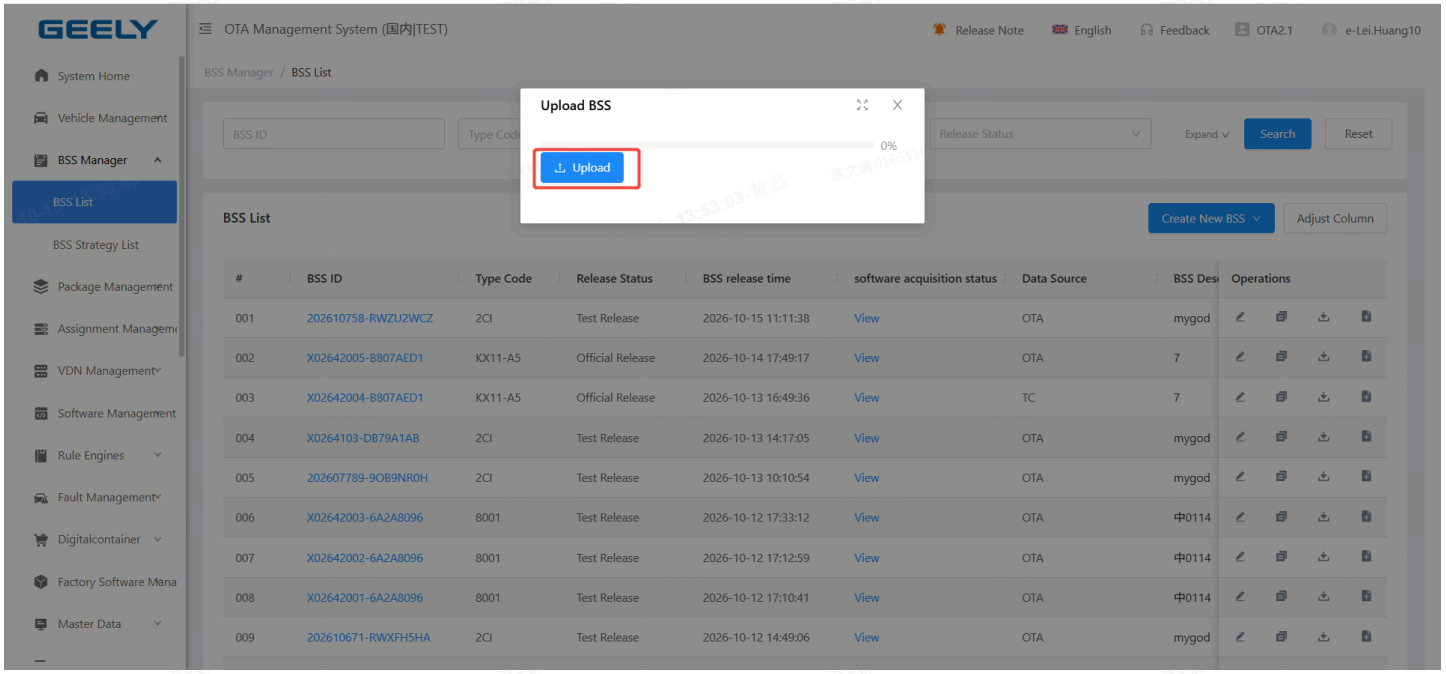
Just fill in the data on this page with reference to relevant information

2.1.2 BSS List - Upload BSS:

- Function Entry: Menu >> BSS Management >> BSS List >> Add New BSS >> Addition Method: Upload BSS



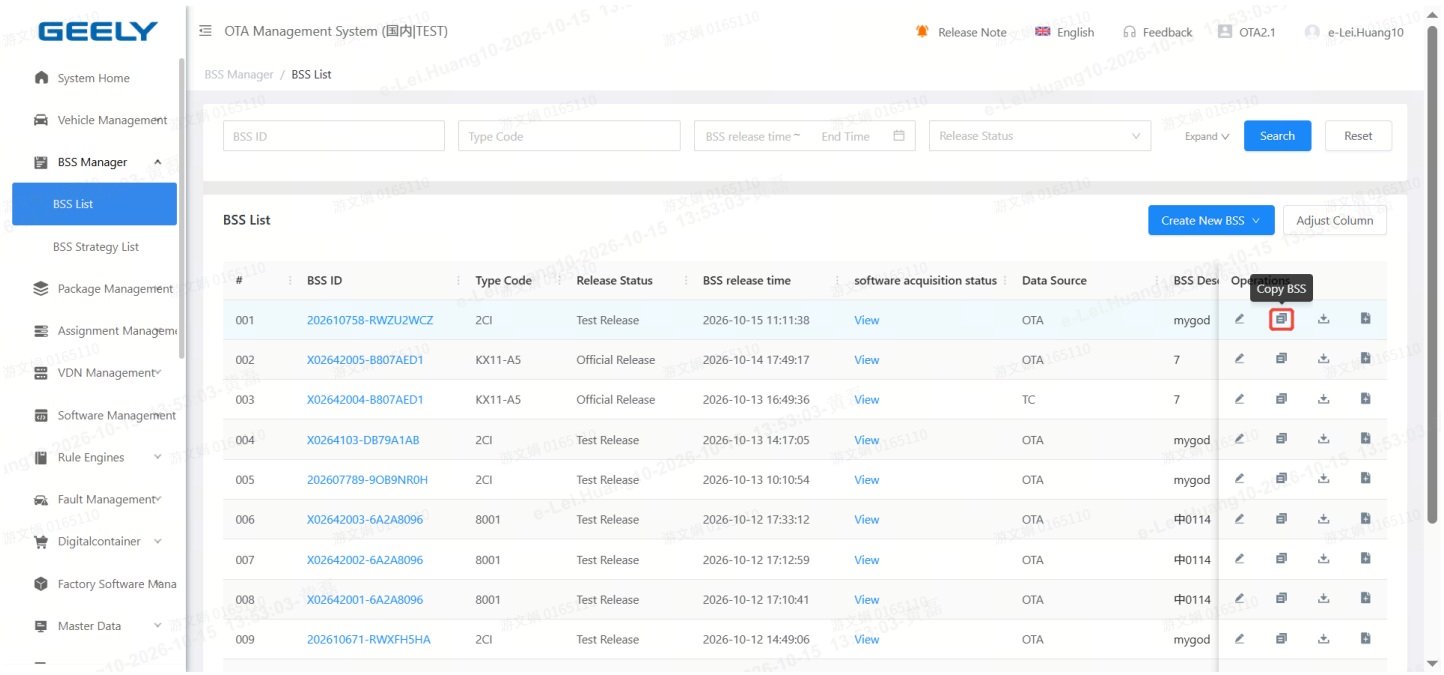
- Upload BSS file:



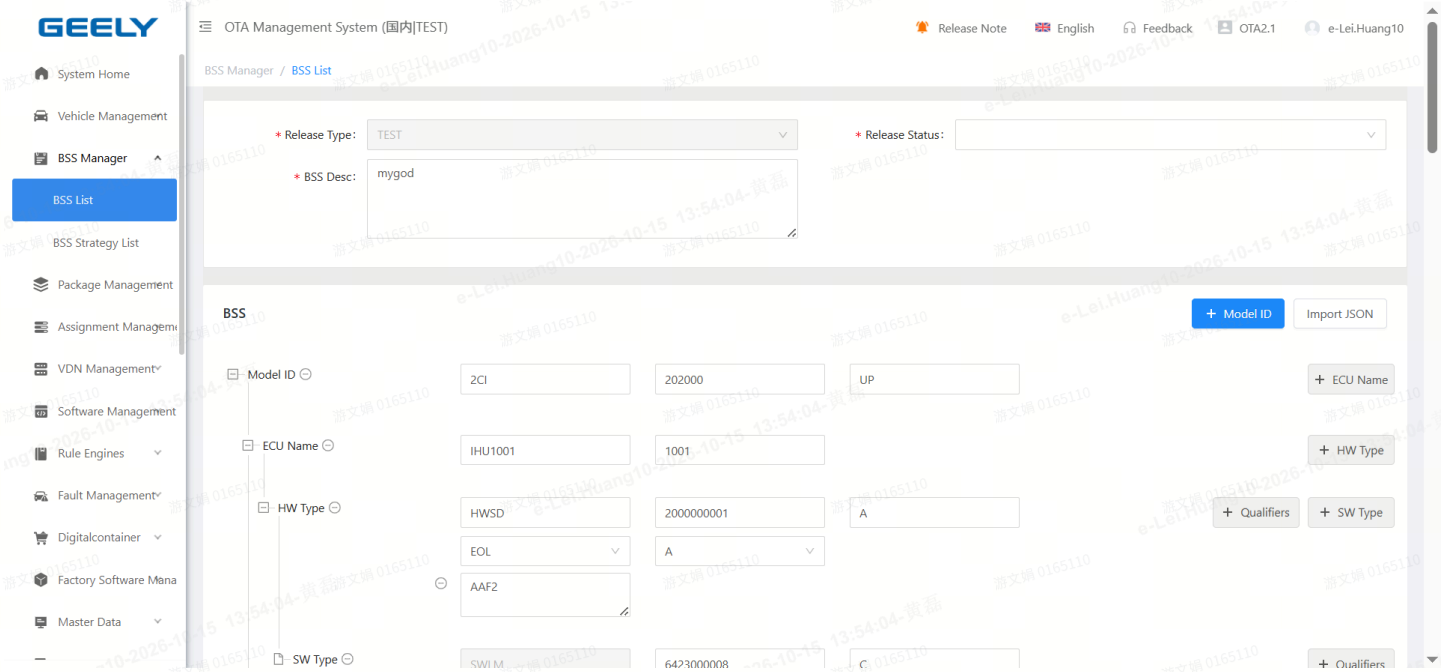
2.1.3 BSS List - Copy BSS:

Function Overview:

- Function entry: Menu >> BSS Management >> BSS List >> Operation Column >> Copy BSS

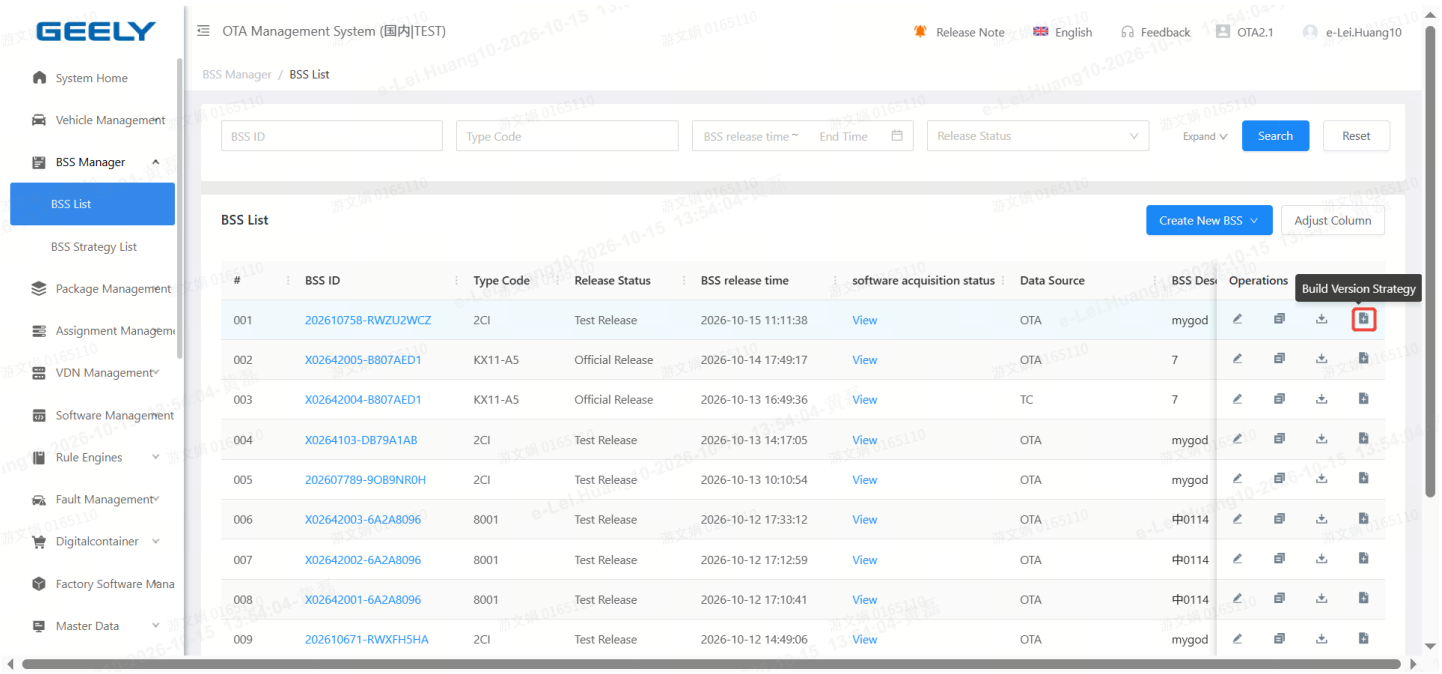


- Copy the BSS edit page:



2.1.4 BSS List - Build Version Strategy:

- Function Entry: Menu >> BSS Management >> BSS List >> Operation Column >> Build Version Policy



- Build Version Policy Edit Page:
- Basic Information:



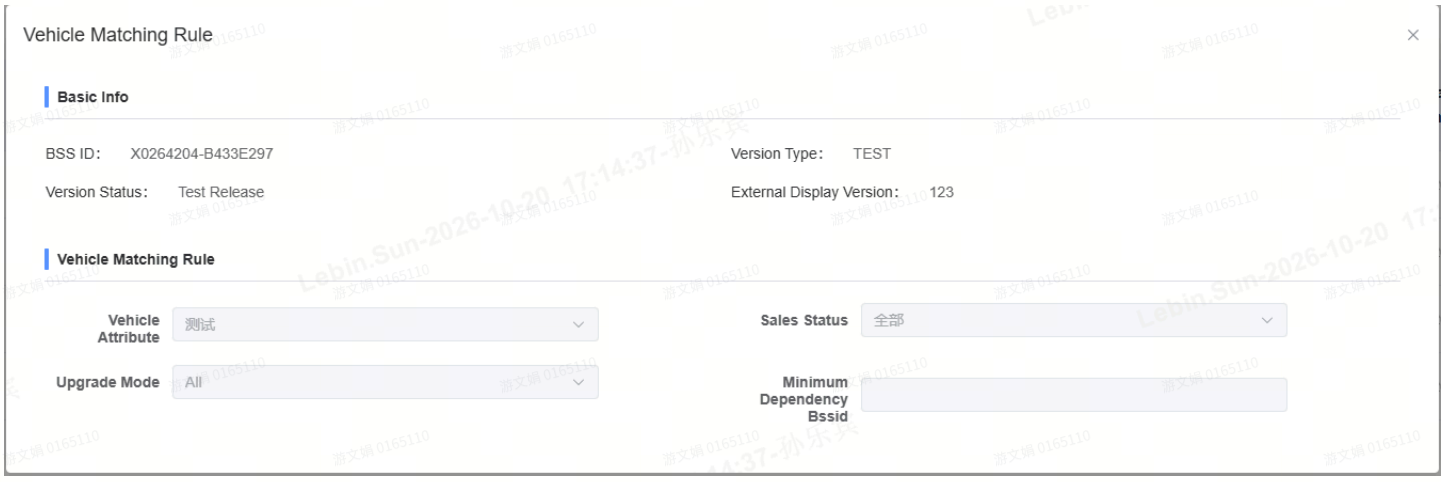
BSSID: The BSSID corresponding to this BSS, which is grayed out and cannot be modified

Version Type: Data source is BSS, grayed out and unmodifiable

Version Status: Data source BSS, grayed out and uneditable

Public Display Version: Text Box, Required

- Vehicle Adaptation Rules:

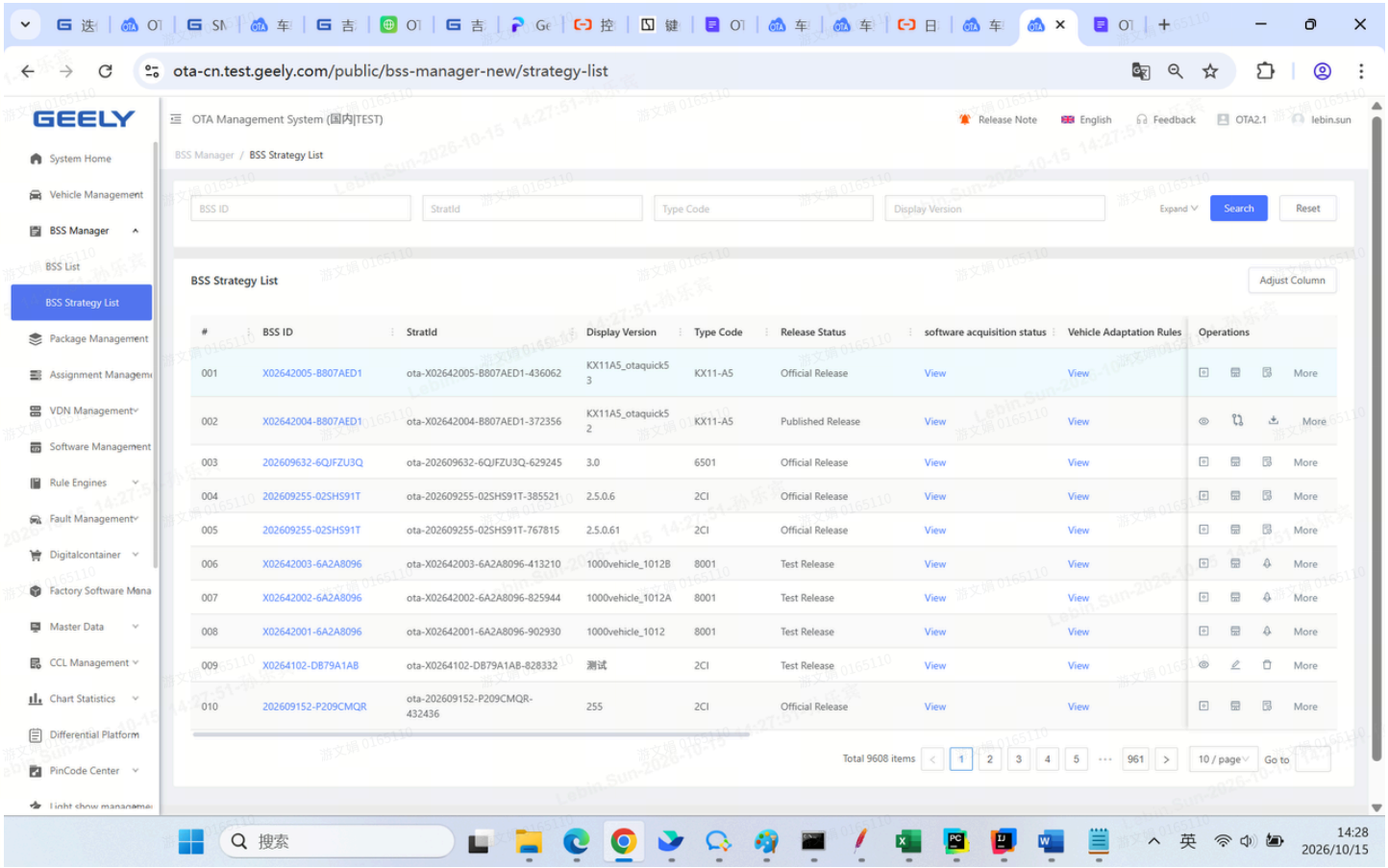


2.2 BSS Policy List

2.2.1 Function Overview

Performing new BSS splitting based on TC-based BSS can generate multiple new BSS instances (with the content of the source BSS remaining unchanged). A new ID (StratId) will be automatically generated for each split BSS as its unique identifier. Subsequent planning and business content maintenance, including displayversion, applicable vehicle rules, release notes, etc. , will be carried out on the split BSS. The purpose is to add operation-related attribute information to the split BSS, so as to facilitate cloud management and the subsequent use of the OTA platform;The BSS policy function can replace the currently used functions such as BSS pre-verification, XX family splitting, multi-pool splitting, and small market processing.

Navigation Path: Menu >> BSS Management >> BSS Policy List



The BSS policy list is shown in the figure above. All fields in this list are either synchronized from GDDC or created based on the BSS list, and their specific meanings are as follows:

BssId: The original value of TC synchronization, and the vehicle OTA upgrade and installation baseline is distinguished based on this

StratId: Policy ID. The policy ID in historical data is consistent with the BSS ID, while a new policy ID is generally in the format of Ota + BSS ID + random number

External display version: namely the configured external display version (display version), where multiple external display versions correspond to one BssId, and one external display version corresponds to one StratId;

typecode: corresponds to the typecode in the vehicle model list

Version Status: Branch Test Release, MRD Release, Official Release, ENG Release, Published Release, Suspending Official Release (the policy-based feature disablement function is not yet implemented)

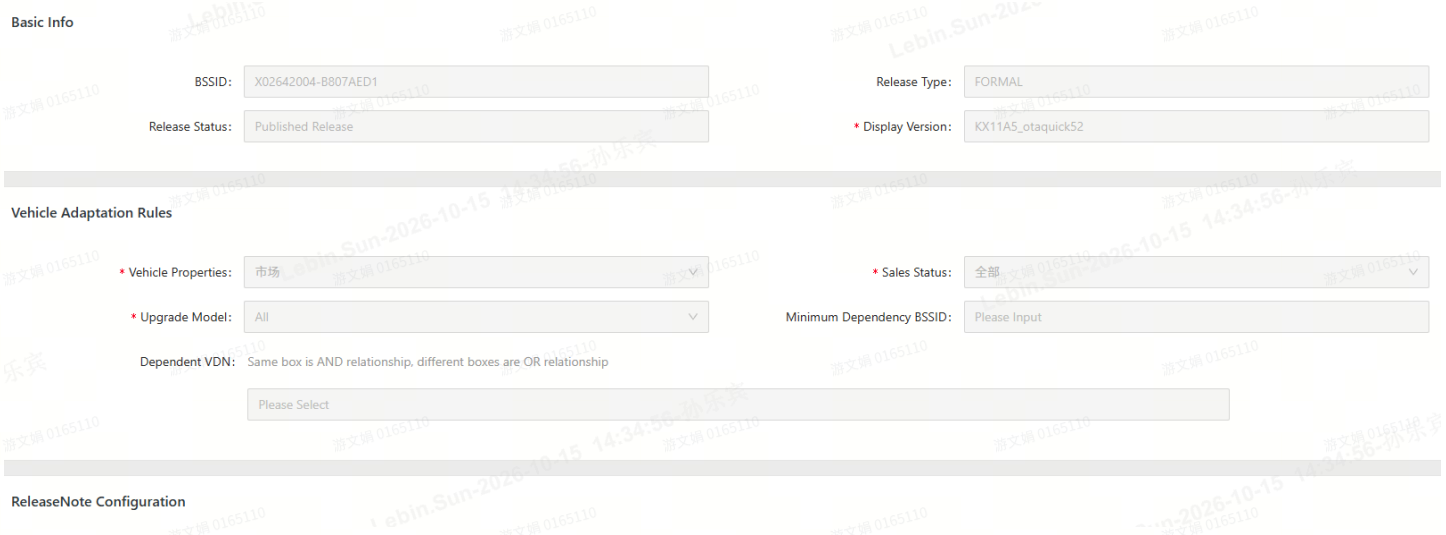
Software acquisition status: Tap to view the synchronization status of all VBF software under the baseline at the OTA site

Adaptation rules for vehicles: You can click the pop-up box to view, and the content to be viewed includes the baseline basic information configured during OS version construction, vehicle attributes, sales status, minimum dependent BssId, and dependent VDN

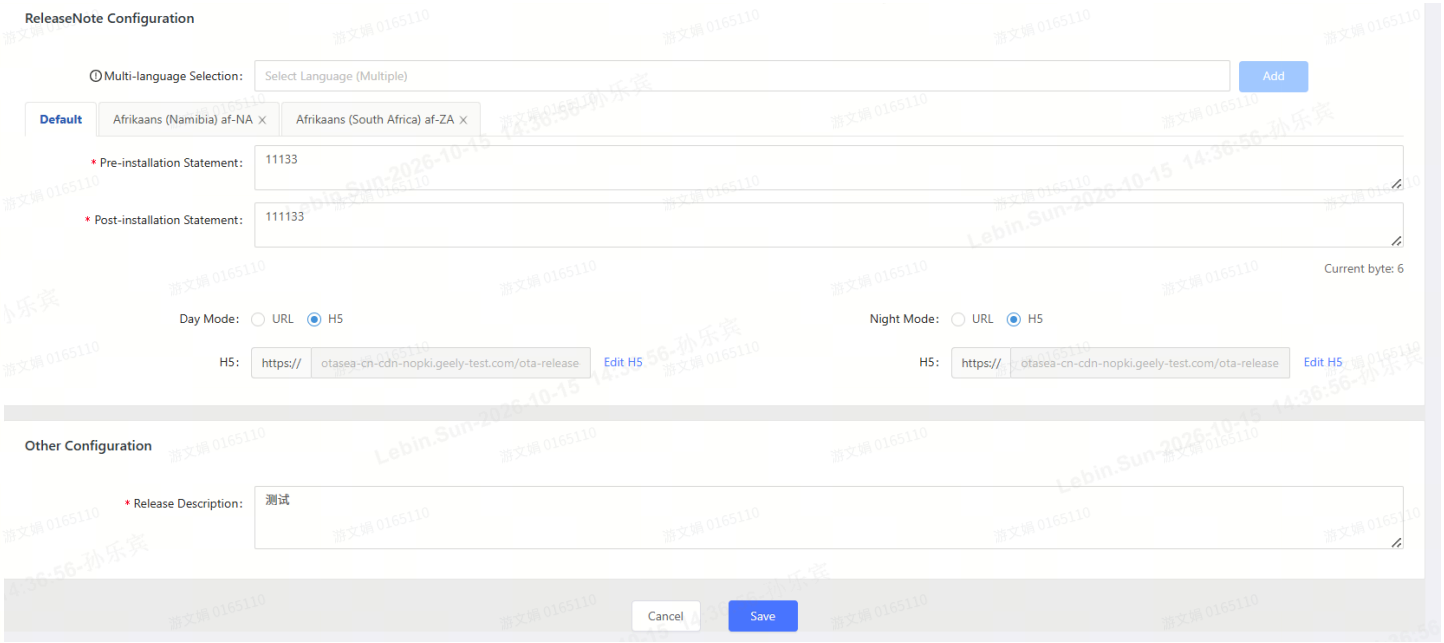
ReleaseNote: You can click the pop-up to view it, and the content displayed is the Release Note information configured when building the OS version

Status: categorized into unpublished and published

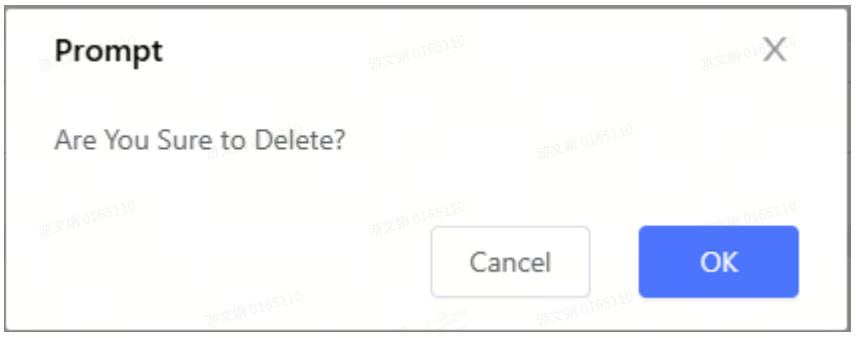
2.2.2 Available functions:



View: As shown in the figure above, to view the basic information of a strategy, this function is only supported for Published Release strategies that are either unpublished or already published.

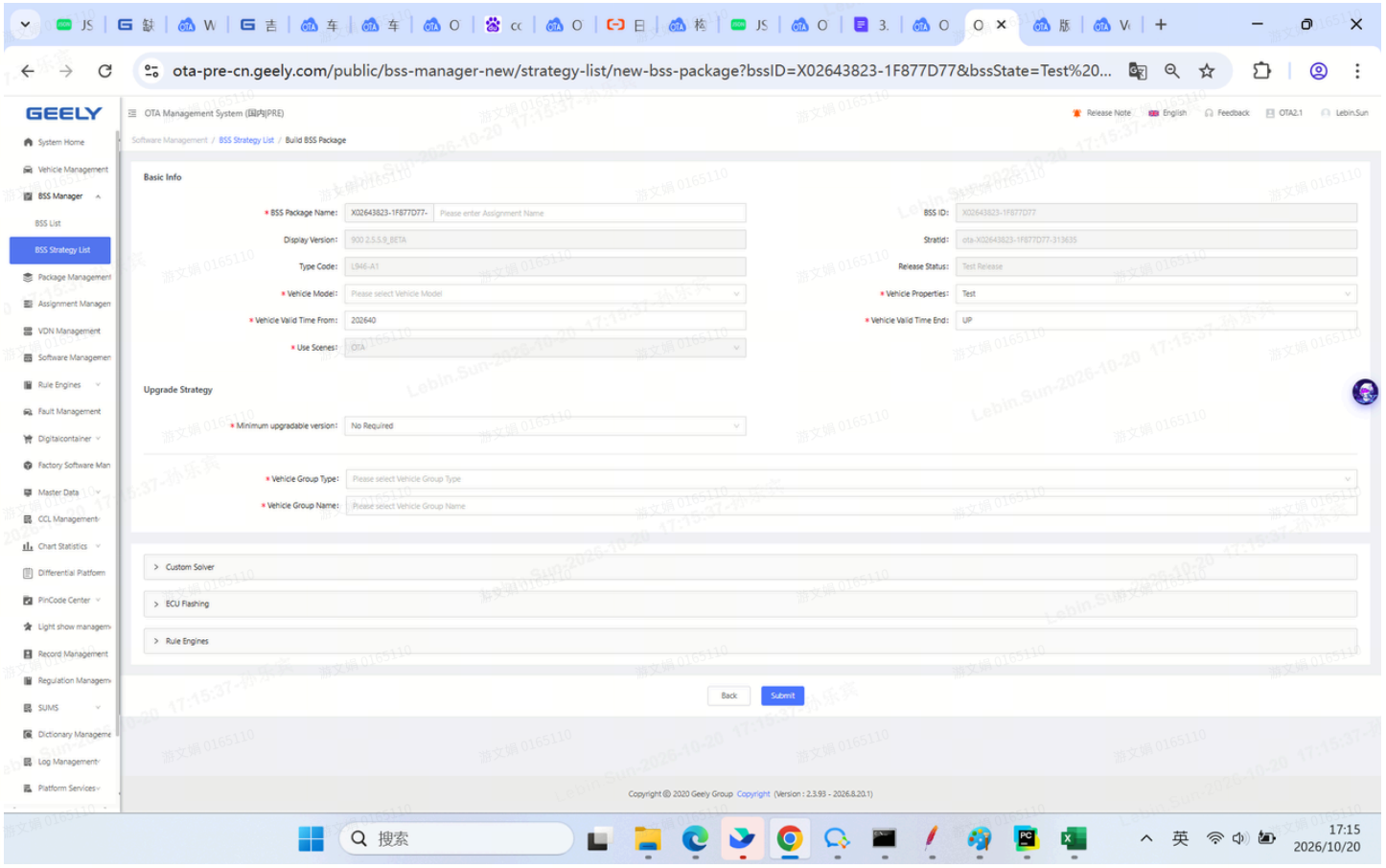


Edit: As shown in the diagram above, you can edit the basic information of the strategy and save the changes. This feature is only available for unpublished strategies.



Delete: For the deletion function of a policy, secondary confirmation is required, and this function is only supported for unpublished policies.

Publish: Change the status of a policy from unpublished to published; once published, the policy cannot be edited or deleted, and a secondary confirmation is required. This function is only applicable to unpublished policies.



Create a regular package, create a shortcut package, create an offline package, create a pre-sales package: For those in the released status, BSS of the Test Release, MRD Release, and Official Release types can be created. For parameters when creating a PKG, please refer to PACKAGE参数

Compare BSS: Published Test Release, MRD Release, Official Release Type BSS (with the possible exception of the Suspending Official Release type) all support this function; compare the difference between the baseline corresponding to this policy and other baselines of this vehicle model.

Download BSS: All types of released BSS (with the possible exception of the Suspending Official Release type) support this function. When downloading the baseline corresponding to this policy, you can choose between JSON format and EXCEL format.

Obtain VBF hash values: All types of published BSS (with the possible exception of the Suspending Official Release type) support this function, and users can download the hash values of all software under the baseline corresponding to this policy.

3. Package Management

3.1 OTA Package

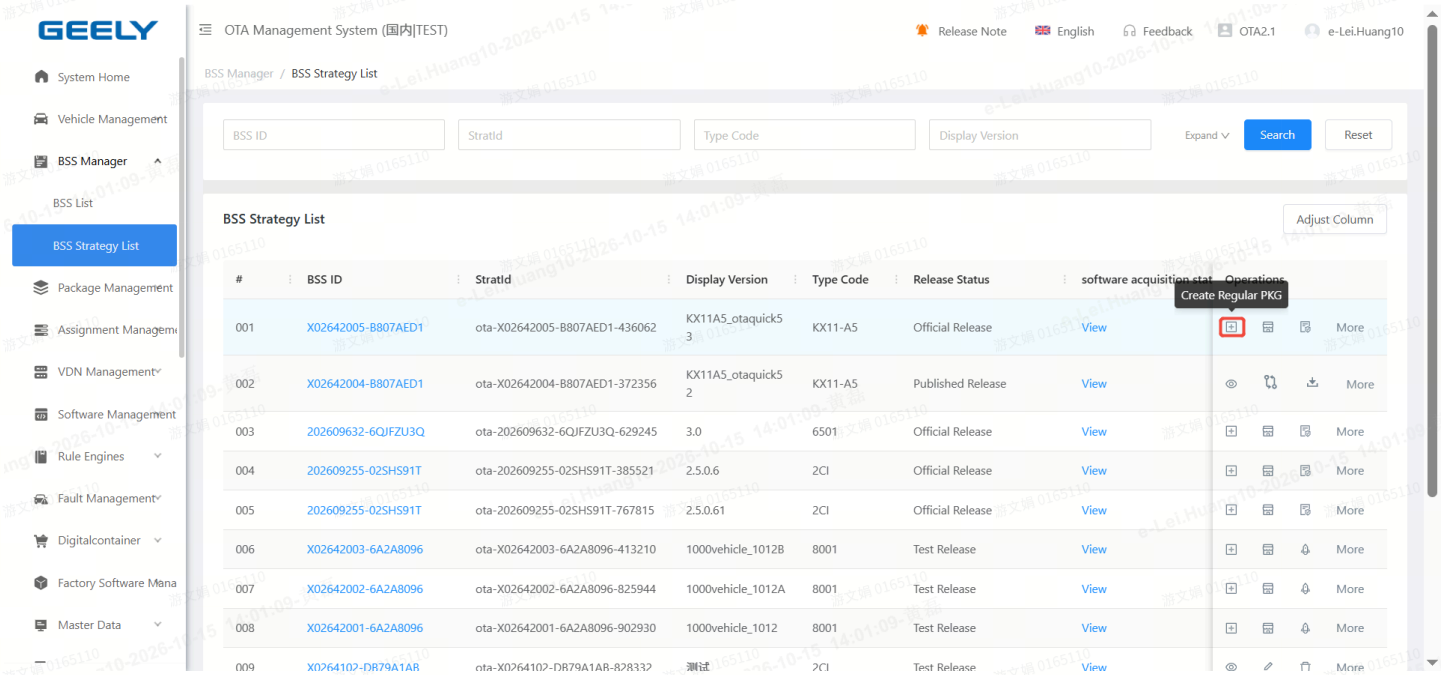
3.1.1 Function Overview

Manually enter BSS Package information on the OTA management platform; create a new BSS Package; this needs to be created manually, not automatically by the system.

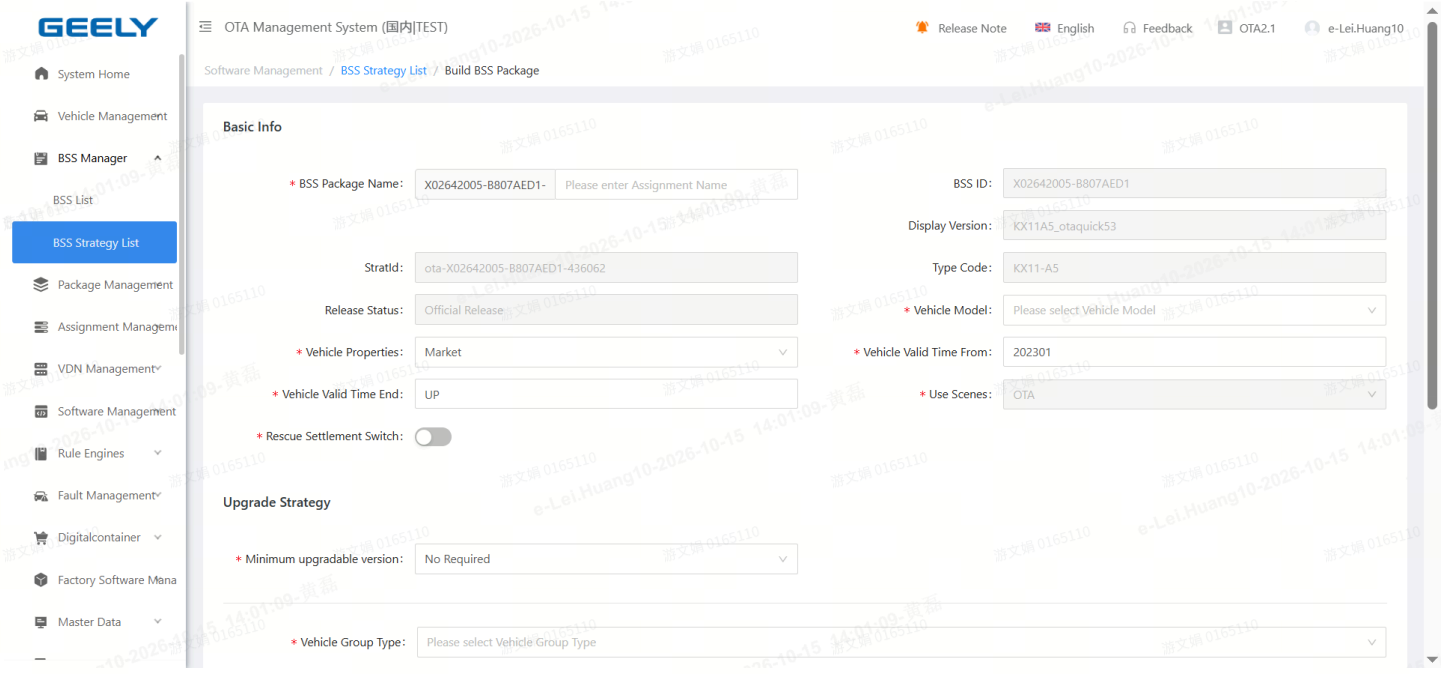
3.1.2 BSS Online (Regular) Package Creation Entry

Entry Path: Menu >> BSS Management >> BSS Policy List >> Operation Column

New Method: Create a Regular PKG from Scratch



3.1.3 BSS Online (Regular) Package Creation Form



Basic Information:

BSS Package Name: Text Box, BSSID - Fillable BSS Package Name: Limited to 50 alphanumeric characters or Chinese characters, Required

BSS ID: The ID corresponding to the selected BSS policy, which is grayed out and cannot be modified

StartID: The policy ID corresponding to the selected BSS policy, which is grayed out and cannot be modified

External display version: The external display version corresponding to the selected BSS strategy is grayed out and cannot be modified

Model: Dropdown list, all models matching the type code corresponding to this BSS strategy, required field

Type code: The Type code corresponding to the selected BSS policy, which is grayed out and cannot be modified

Version Status: The version status corresponding to the selected BSS strategy is grayed out and cannot be modified.

Vehicle Attribute: The vehicle attribute corresponding to the selected BSS strategy, presented as a drop-down menu, required field

Vehicle batch start time: text box, default value is BSS structure week, required field

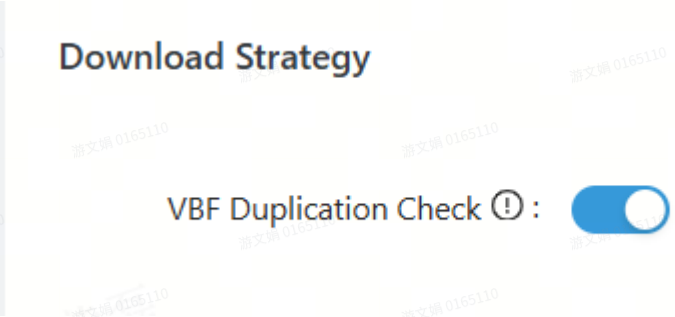
Vehicle batch end time: text box, default value is BSS structure week, required field

Usage scenario: The value is imported from BSS and the field is grayed out and uneditable

Software verify check: Whether it is necessary to verify that the software status is verify when the Package contains a vehicle pool with market attributes (this field is displayed when the version status is Official Release, and it is currently applicable to GEEA2.0, GEEA3.0, and EEA4.0)

Download Strategy:

VBF Repeatability Check**:** Enabled by default, required for EEA4.0



Upgrade Strategy:



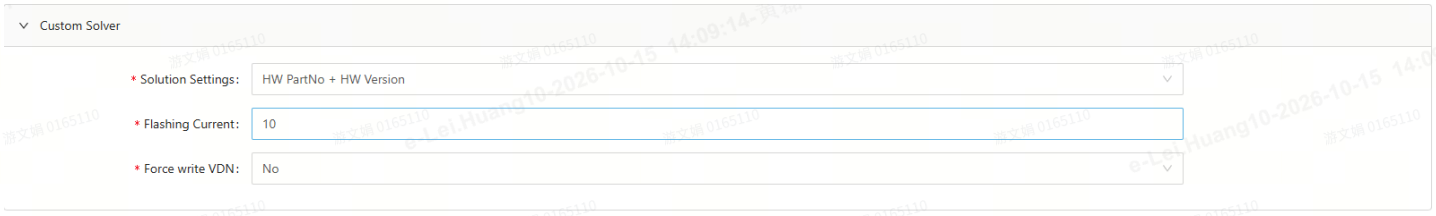
Minimum upgradable version: Dropdown box, the minimum upgradable dependent BSS ID corresponding to this BSS policy

Vehicle Pool Type: Dropdown menu, covering all vehicle pool types under the user's permissions

Vehicle Pool: Dropdown menu, showing the vehicle pools created by the user or accessible to the user

Custom Solver:

Data is synchronized with the vehicle model, the calculation settings are required, hardware part number + hardware version number and hardware part number are optional; flashing voltage is required; forced flashing VDN is required, and the drop-down box can select No or Yes



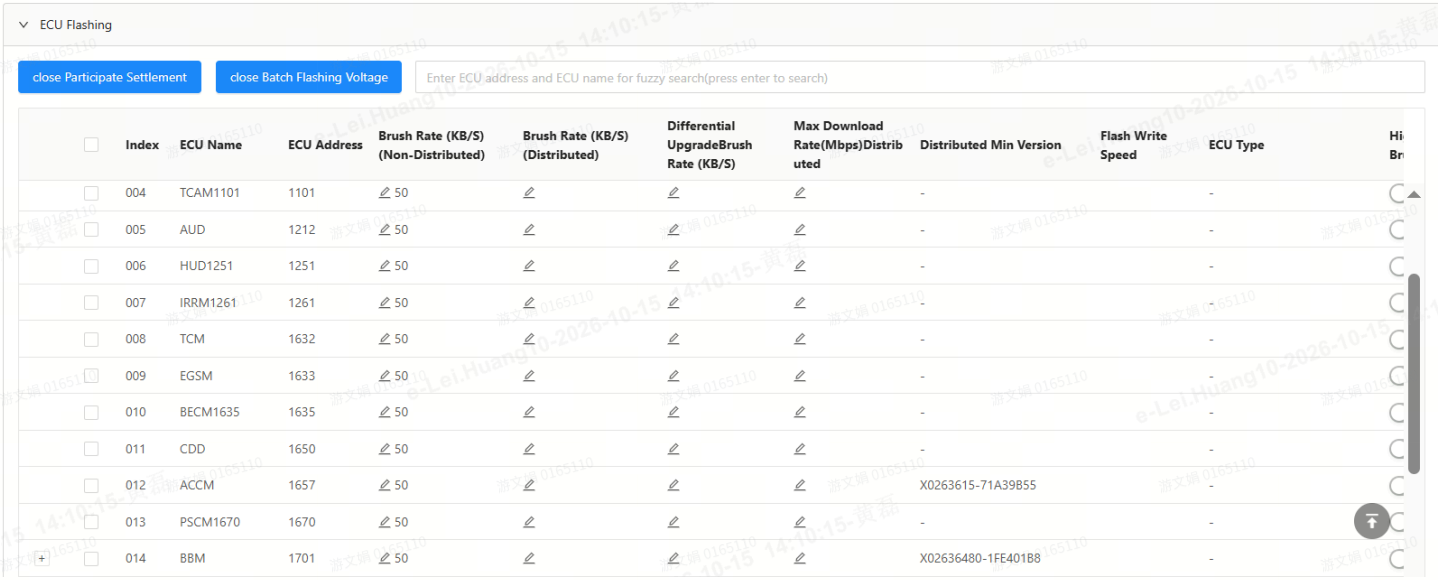
Vehicle Condition Check Configuration:

Data is synchronized with the vehicle model, and data can be deleted and added.



ECU Flashing:

The data is synchronized with the vehicle model, and can be modified for the current package only.



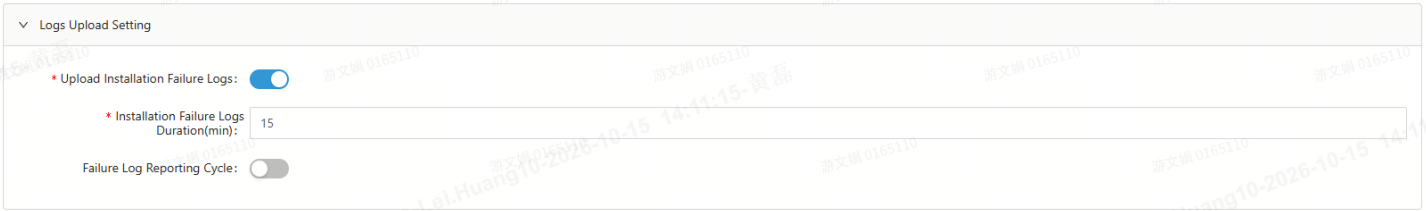
Rule Engine:

Data Source, Rule Engine



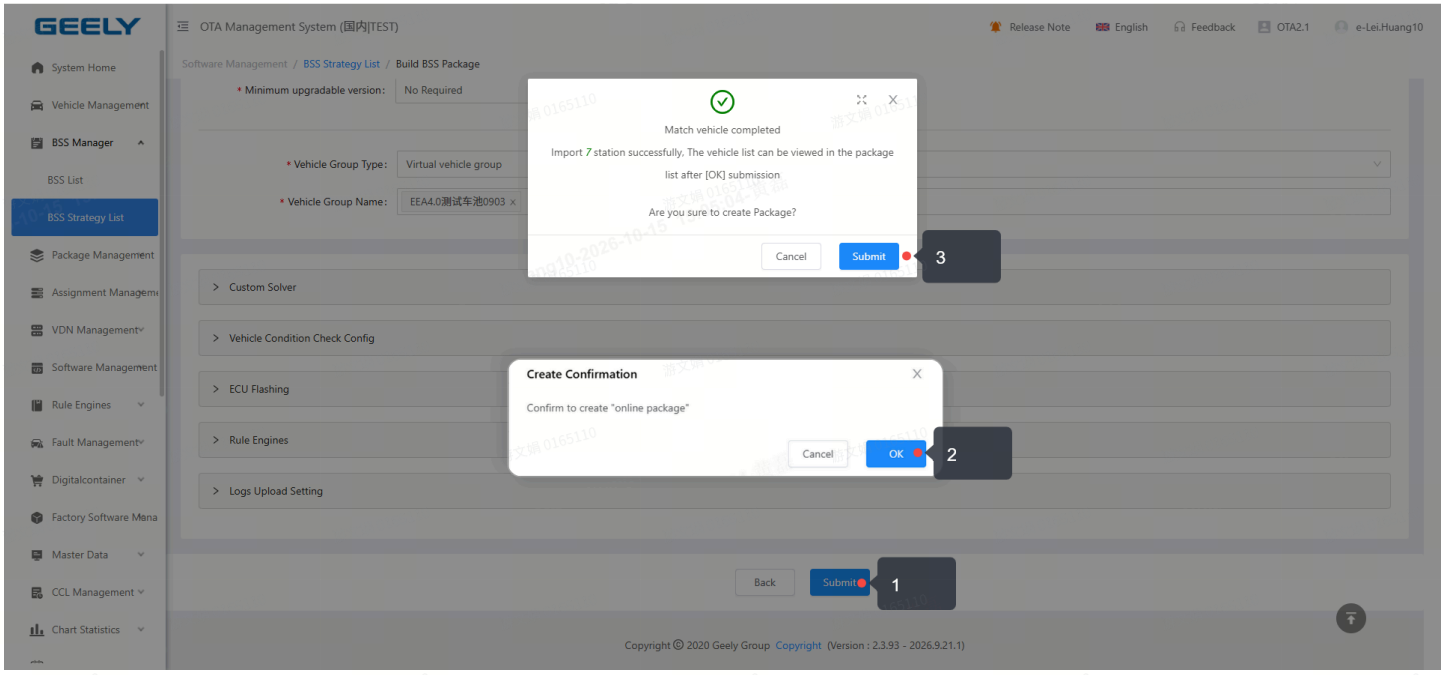
Log Upload Settings:

Configure according to the actual scenario



3.1.4 BSS Package Creation and Submission Process

Form Submission > Submission Confirmation Pop-up > Vehicle Matching Result Submission

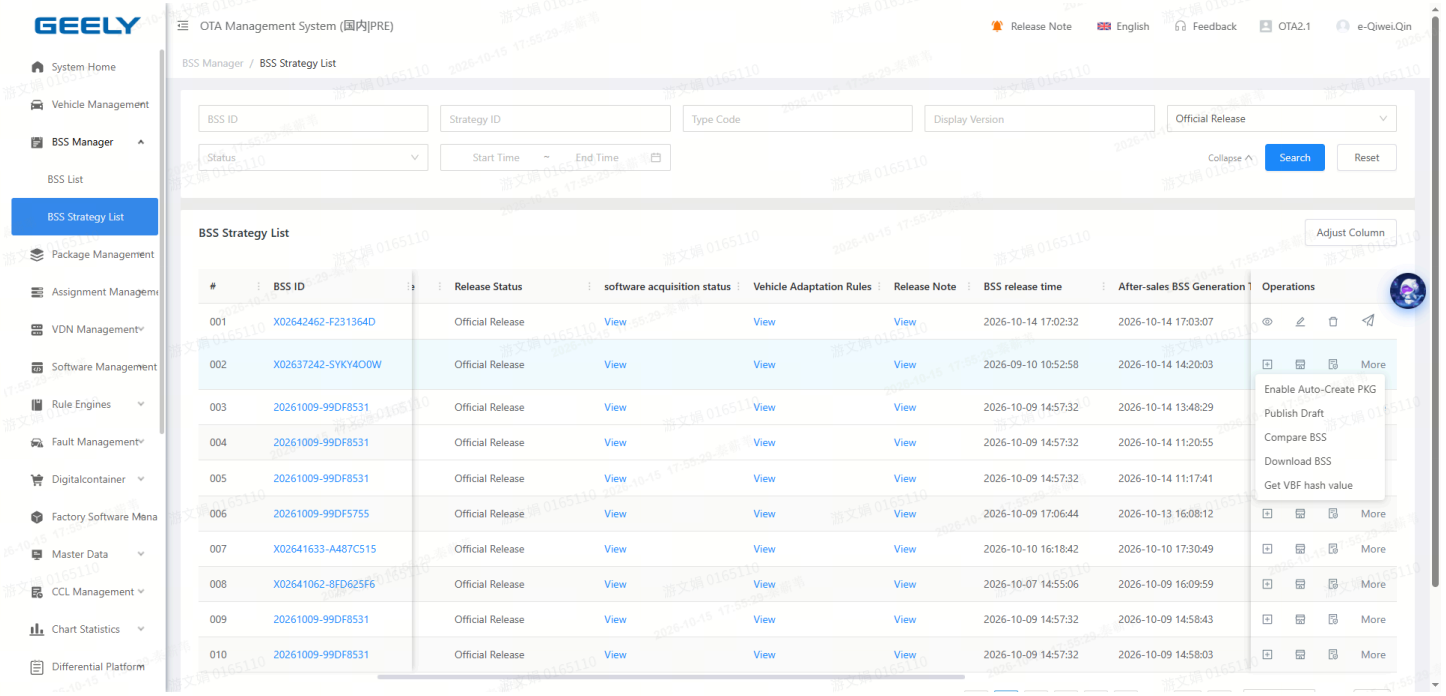


3.2 Quick Package

3.2.1 Function Overview

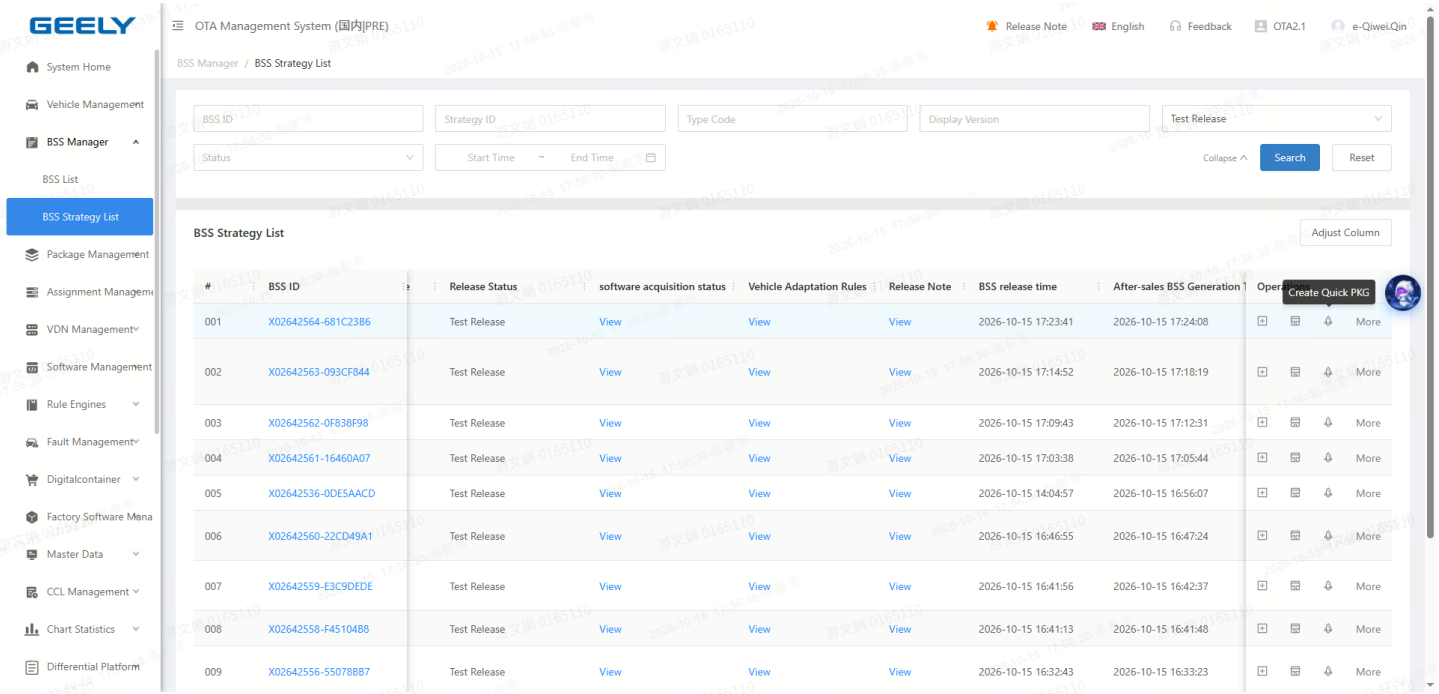
3.2.2 Automatically Create Quick Package

1. Function Entry: Navigation Bar > BSS Management > BSS Policy List
2. Precondition: The data source of the BSS policy is **TC** and the version status is **Official Release,** and the BSS policy status is released
3. BSS Policy List Action Bar > Select Enable Auto Package Creation (after the auto quick package is created, the system will automatically generate a quick OTA task)

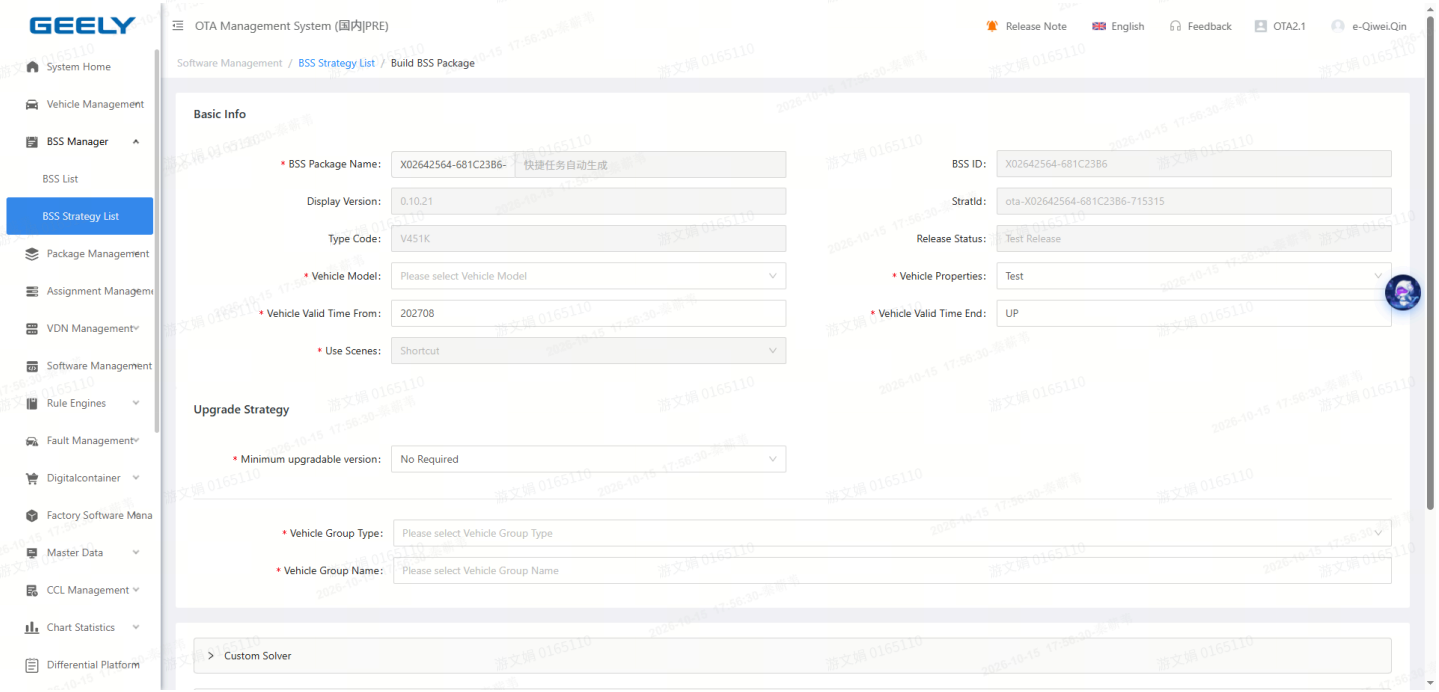


3.2.3 Manually create a shortcut package

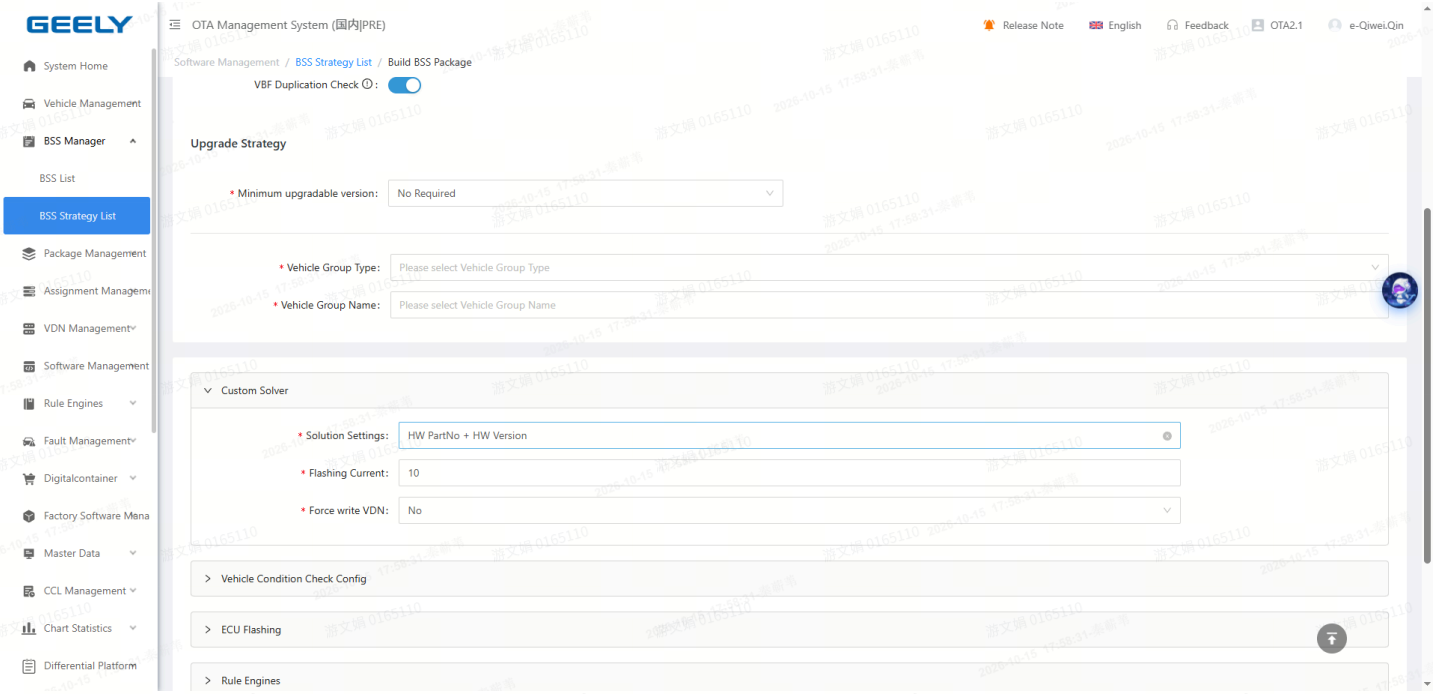
- Function Entry: Navigation Bar > BSS Management > BSS Policy List
- precondition: The BSS policy version status is **Test Release/MRD Release,** and the BSS policy status is released
- BSS Policy List Action Bar >> Select to create a new shortcut package (after the shortcut package is created, the system will automatically generate a quick OTA task)



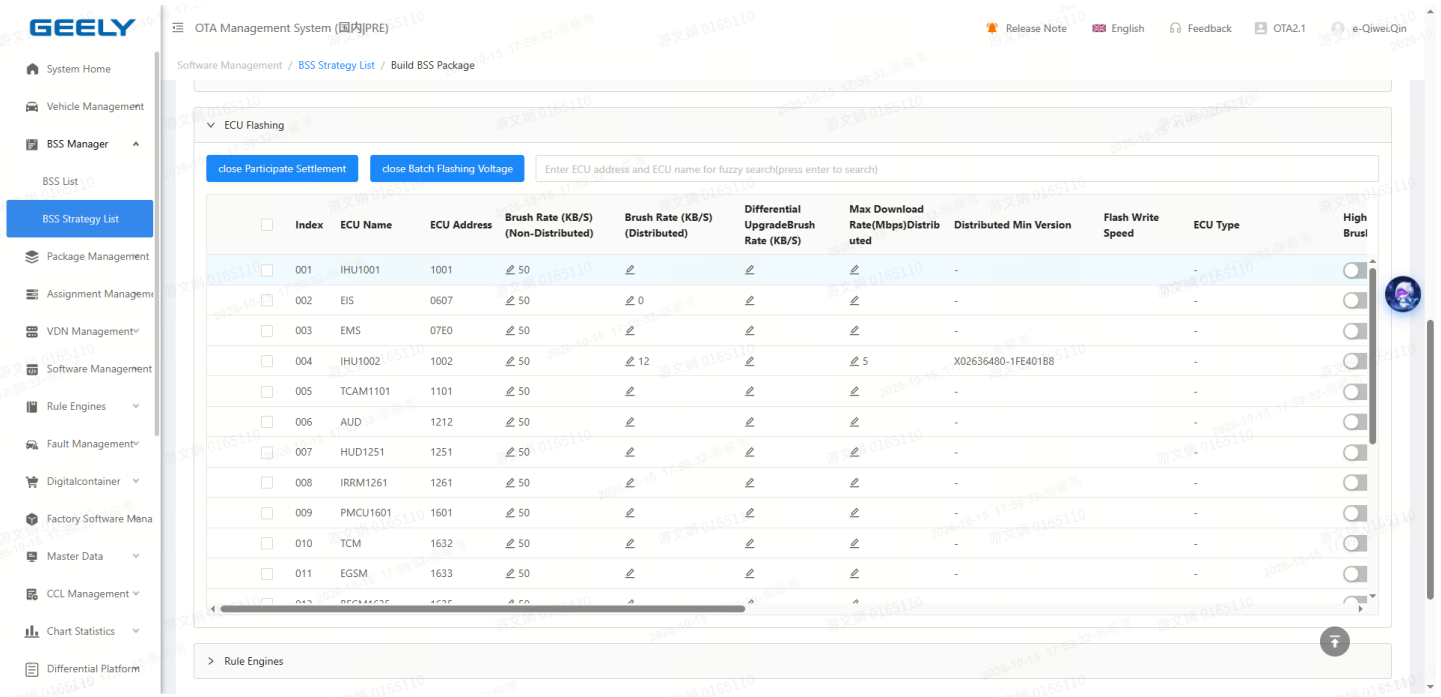
- Fill in the parameter settings for the new shortcut package page as shown in the figure below:



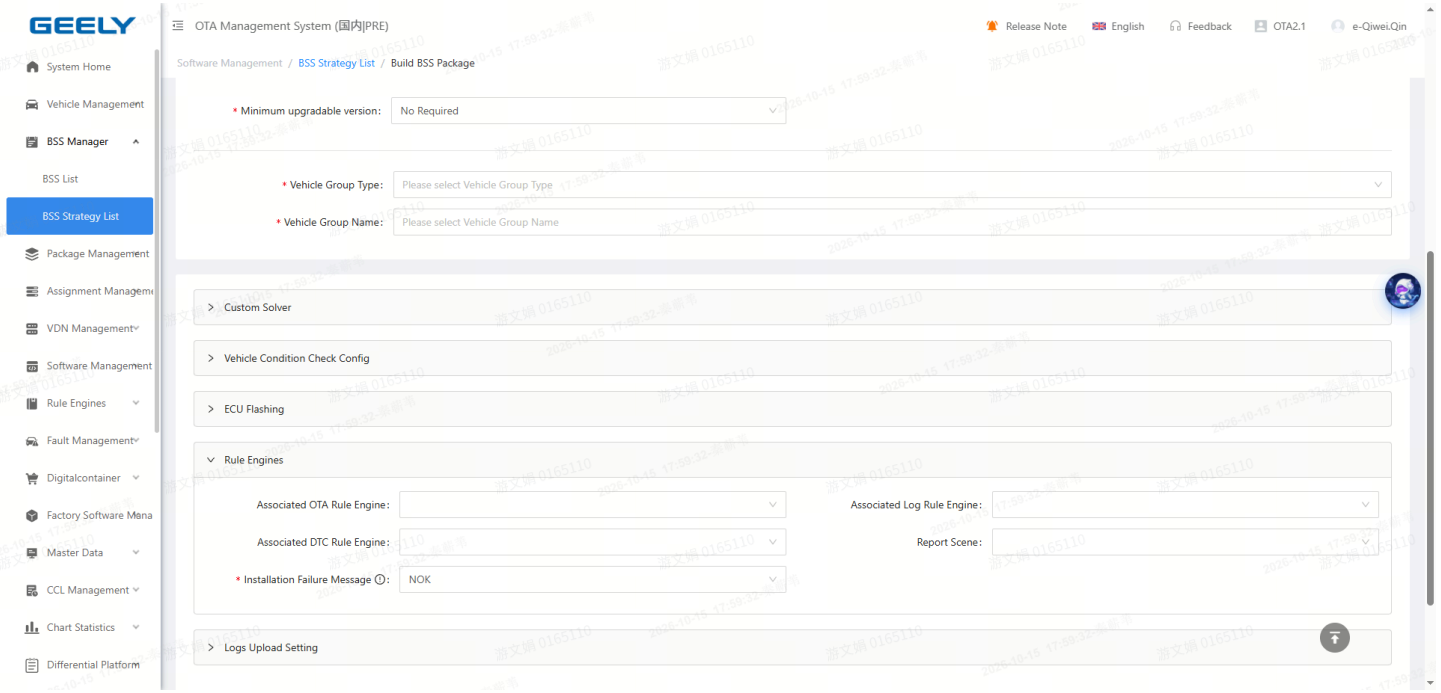
- To restrict the minimum upgradable version, select the corresponding BSSID for the minimum upgradable version. If the drop-down menu for vehicle pool type is empty after selecting the vehicle model during addition (the logged-in user has not created a self-built vehicle pool under this vehicle model, nor is they an administrator of other self-built vehicle pools), the solutions are as follows: 1. Navigate to Vehicle Management >> Vehicle Pool Management >> Add Vehicle Pool in the navigation bar; 2. Navigate to Vehicle Management >> Vehicle Pool Management >> Add Vehicle in the operation bar (self-built vehicle pool for the corresponding vehicle model) >> Enter your own domain account in the Vehicle Pool Administrator field and make a selection >> Submit.
- Solver settings are configured in the custom solver section as shown in the figure below:



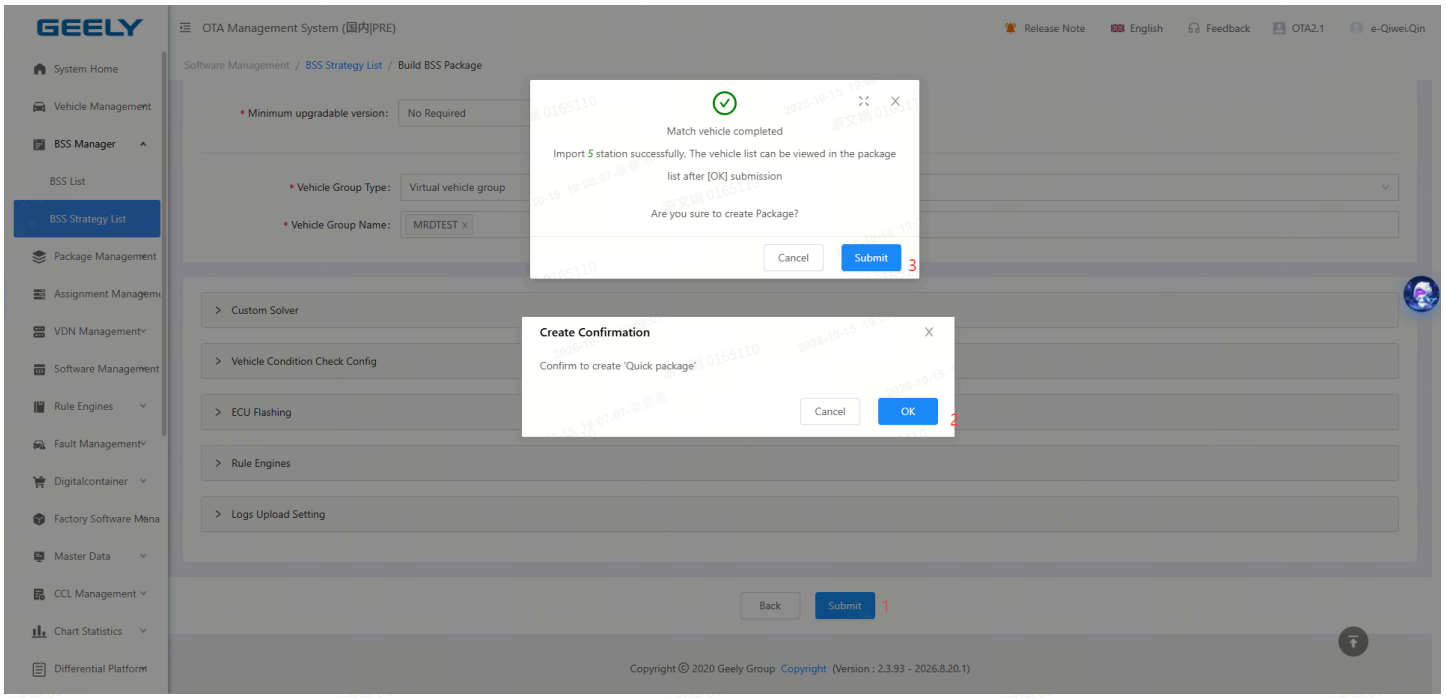
- The flashing settings are configured in the ECU flashing section shown in the figure below (by default, they are consistent with the ECU of the vehicle model):

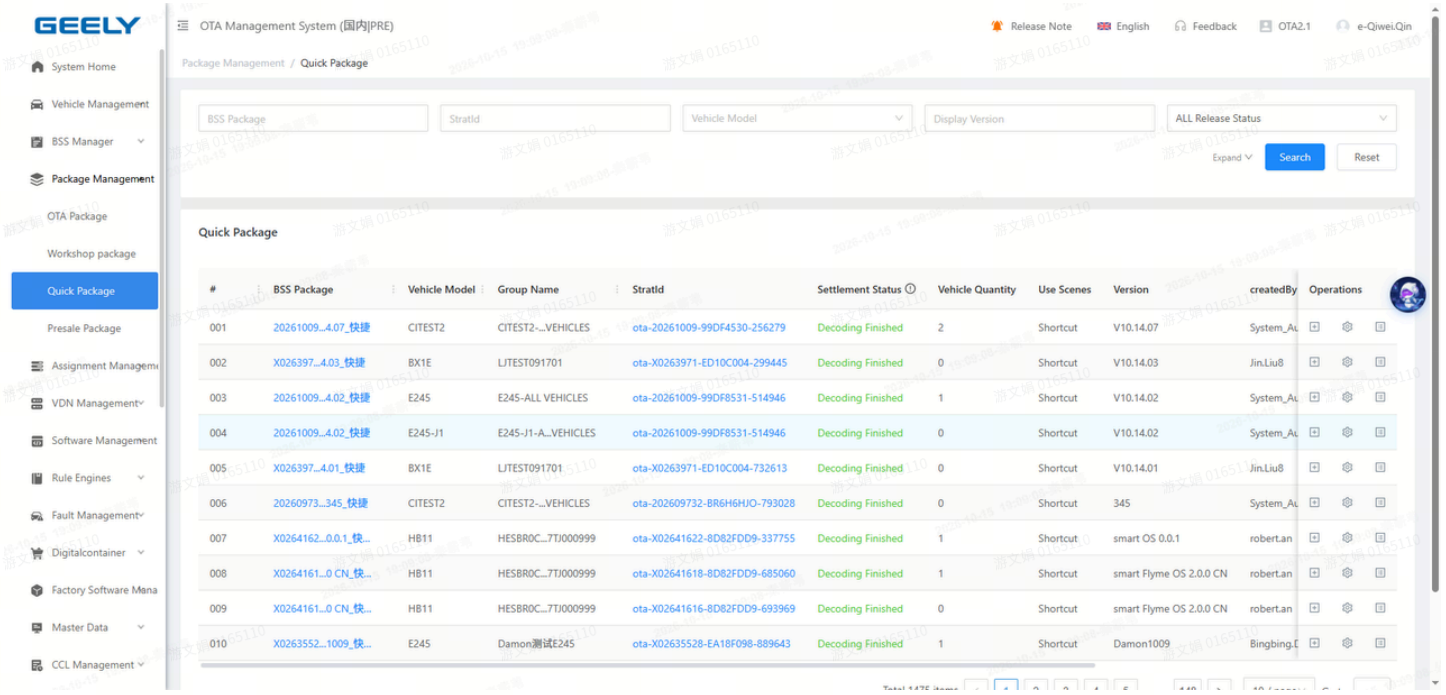


- The rule engine (empty by default) and log upload settings are as follows:



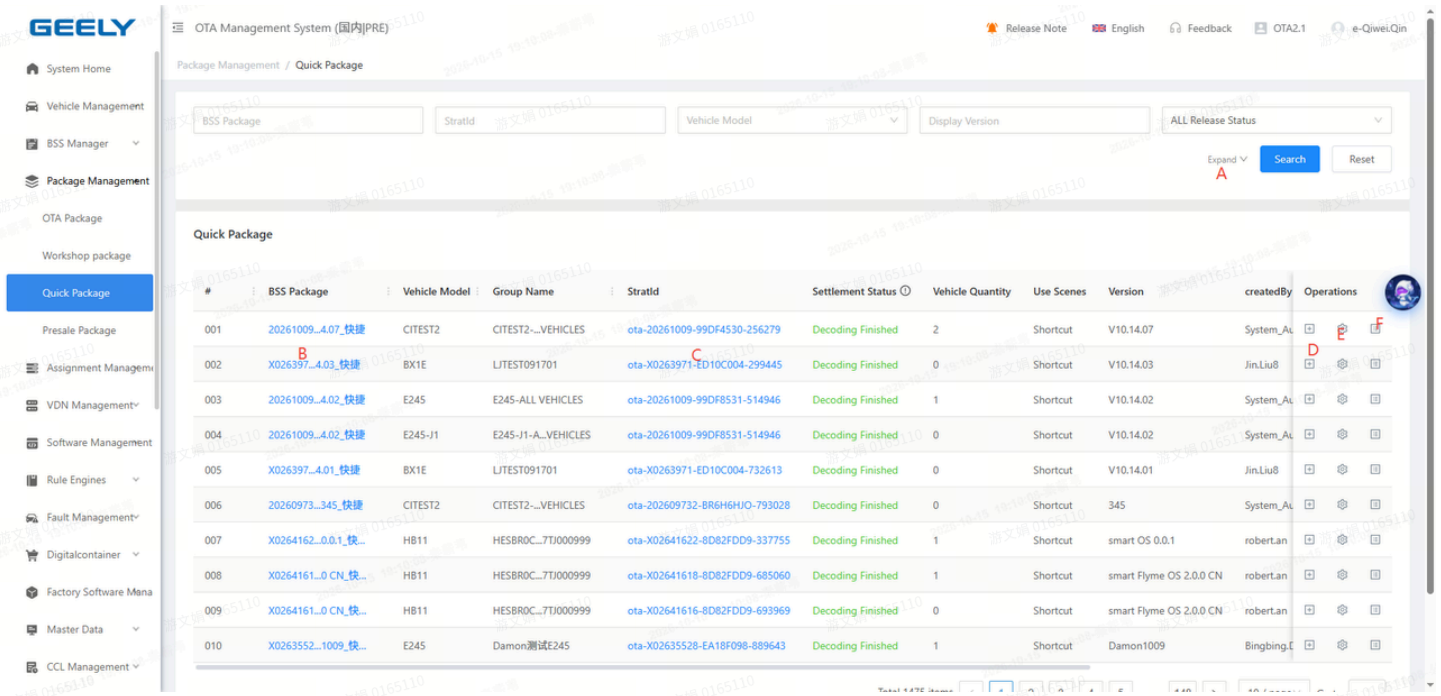
- When associating with the OTA rule engine, enter the rule engine name in the input box (fuzzy match) and select the corresponding rule engine
- Manually create a shortcut package and submit it as shown in the following figure:





3.2.4 Quick Package List

- Function Entry: Navigation Bar > Package Management > Quick Package



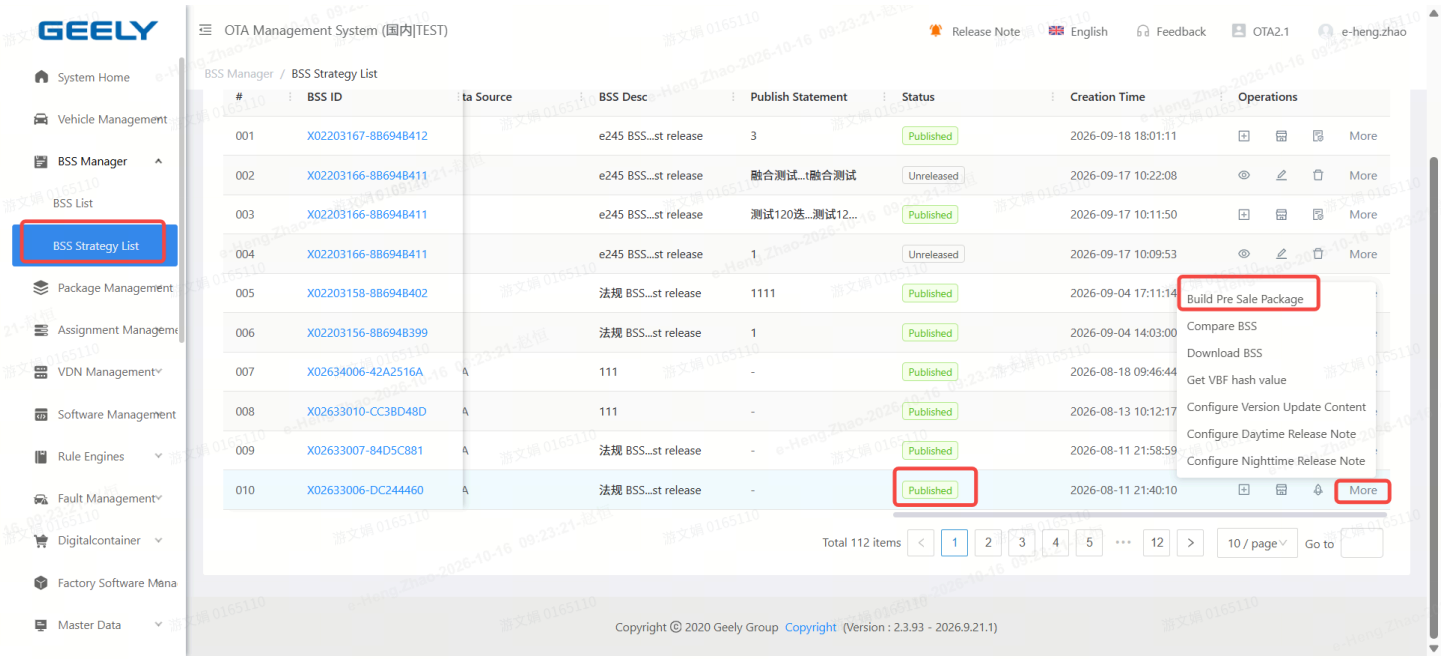
- A: Toggle to expand/collapse the search bar
- B: View Package
- C: View BSS policies
- D: Create OTA Task
- E: Configure the rule engine
- F: Set Software Repair Instructions

3.3 Pre-sales Package

3.3.1 Function Overview

3.3.2 Function Entry

- Navigation Bar: BSS Management >> BSS Policy List
- Prerequisite: BSS policy status is **Published**
- BSS Policy List >> Select to Create Pre-sales Package **(GEEA2.0 does not allow creating pre-sales tasks)**

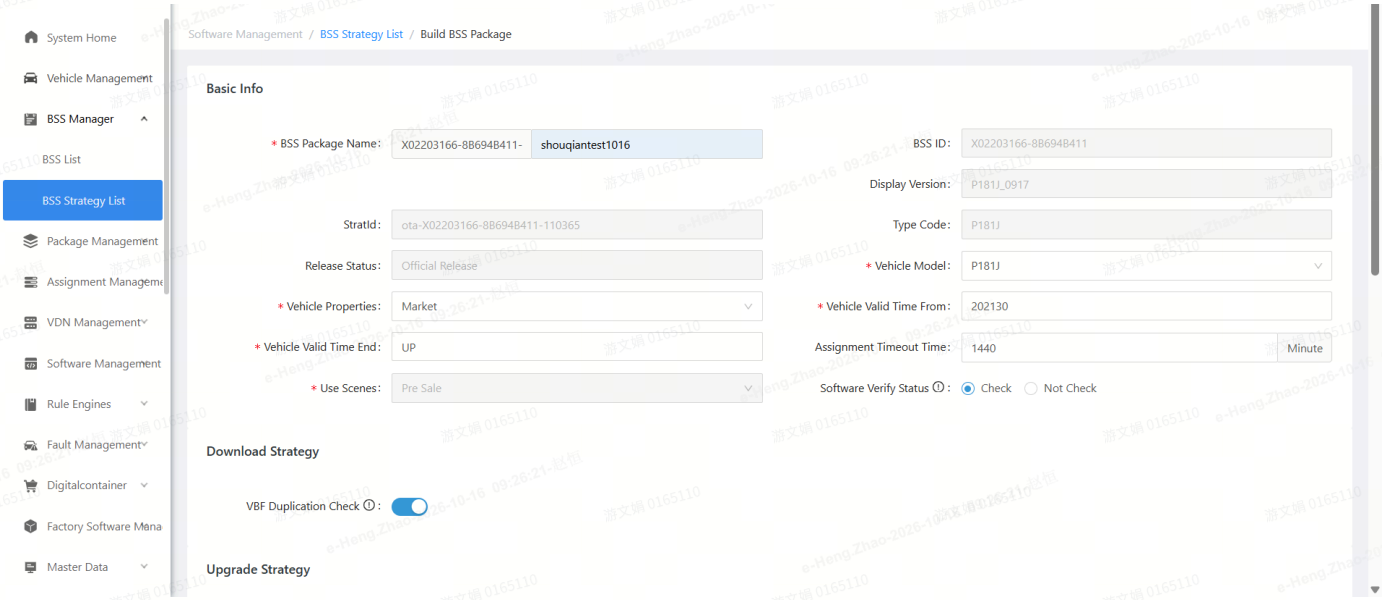


3.3.3 Create Pre-sales Package

3.3.3.1 Basic Information

- BSS Package Name: Enter the BSS Package name, which is limited to 50 alphanumeric characters
- BSS ID: Populated with the selected BSSID and not editable
- StratId: The policy ID corresponding to the selected BSSID, which is not modifiable
- Type code: The type code value corresponding to BSSID, which cannot be modified
- External display version: the external display version corresponding to the BSSID, which cannot be modified
- Version Status: The version status corresponding to the BSSID, which cannot be modified
- Model: All models matching the type code corresponding to BSS, single selection via dropdown menu
- Vehicle Attribute: Dropdown single-select: Test, Marketing, All
- Vehicle batch start time: Match the start time of the structure week corresponding to the vehicle, which is a 6-digit or 8-digit number (example: 202330)
- Vehicle batch end time: Matches the structure week end time of the corresponding vehicle, presented as 6-digit or 8-digit numbers; if there is no such time, it shall be marked as UP (example: UP)
- Task Timeout: Configure the timeout duration for task download (example: 1440)
- Usage scenario: When creating a pre-sales package, the field defaults to pre-sales and cannot be modified

- Software verify check: Whether it is necessary to verify that the software status is verify when the Package contains a vehicle pool with market attributes (this field is displayed when the version status is Official Release, applicable to Geely 2.0/3.0+ 4.0 architecture)

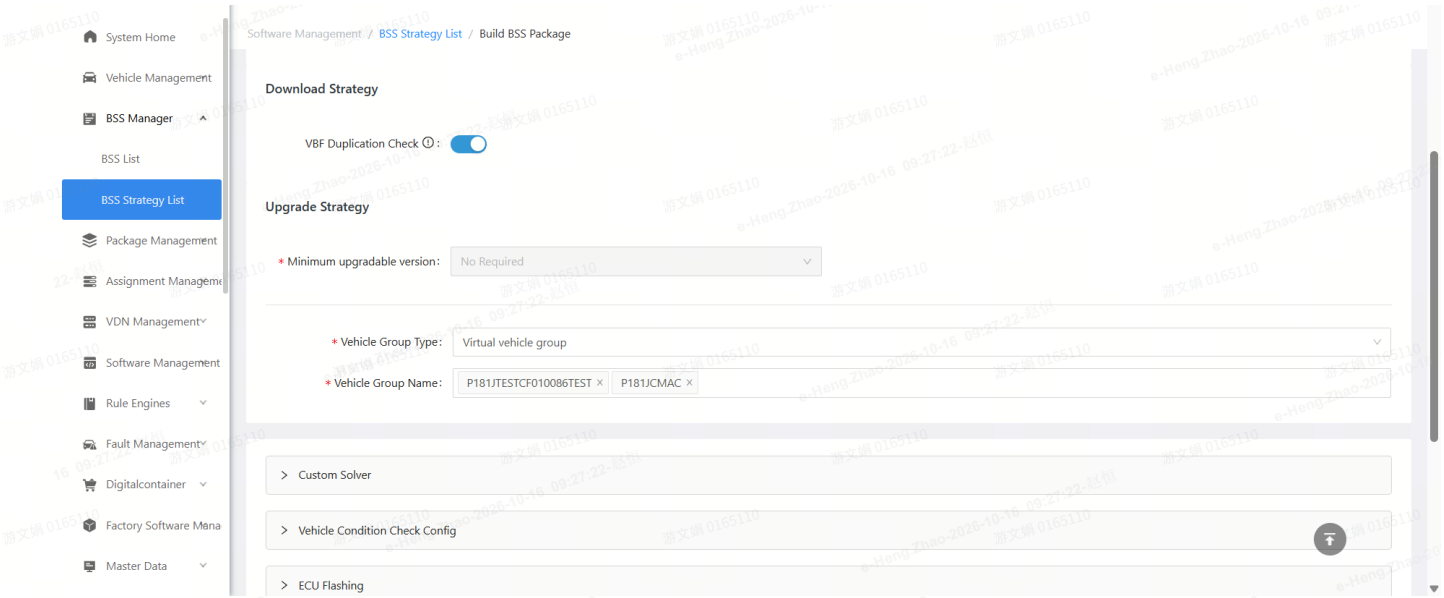


3.3.3.2 Download Strategy

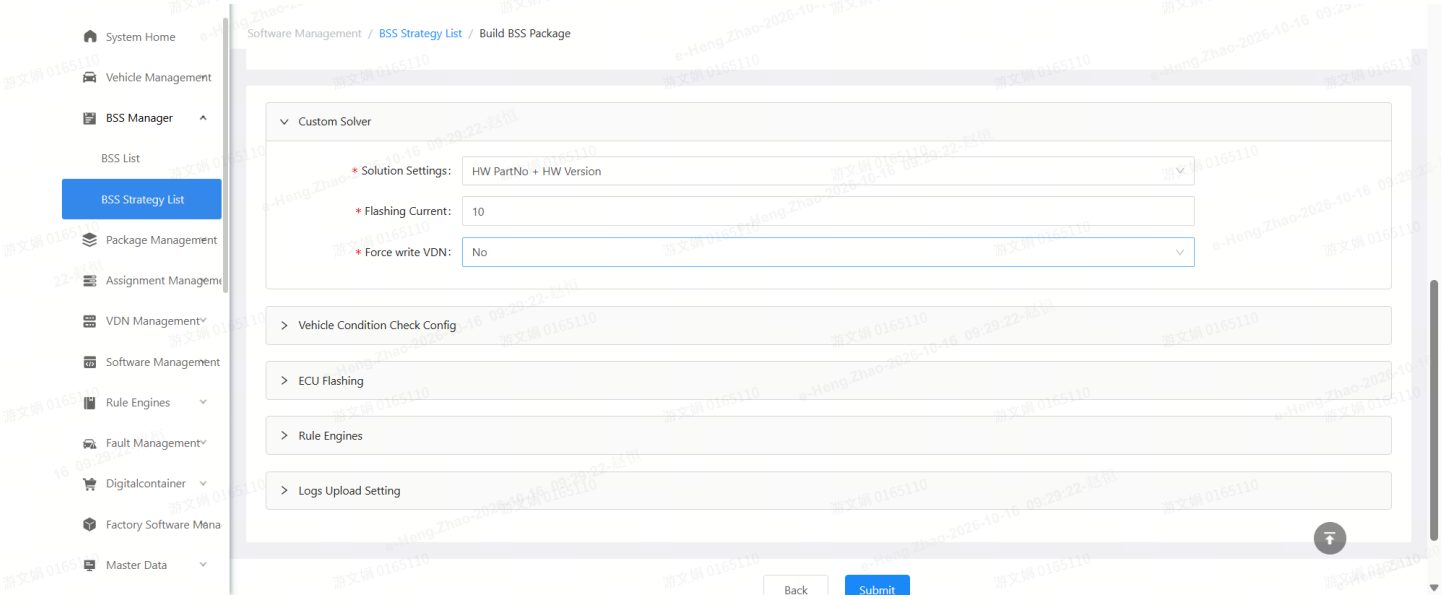
- VBF Repeatability Check: When this function is **enabled**, before OTA downloads a VBF file, it will check whether the same file to be downloaded already exists locally; if so, redundant download is unnecessary. When this function is **disabled**, all files specified in the installation manual need to be downloaded (applicable to ZEEA3.0 and EEA4.0 architectures)

3.3.3.3 Upgrade Strategy

- Minimum upgradable version: Default value is No Required. Currently, the delivered data source only includes: No Required/No Require (non-modifiable, inherited from GDDC policy)
- Vehicle pool type: Dropdown single-select, covering all vehicle pool types under the user's permissions; there is a corresponding relationship between BSS type and vehicle pool.
- Select vehicle pools: Multiple options are supported via the dropdown menu, **and you may choose vehicle pools created by this account or those for which you have access (vehicle pool administrators)**

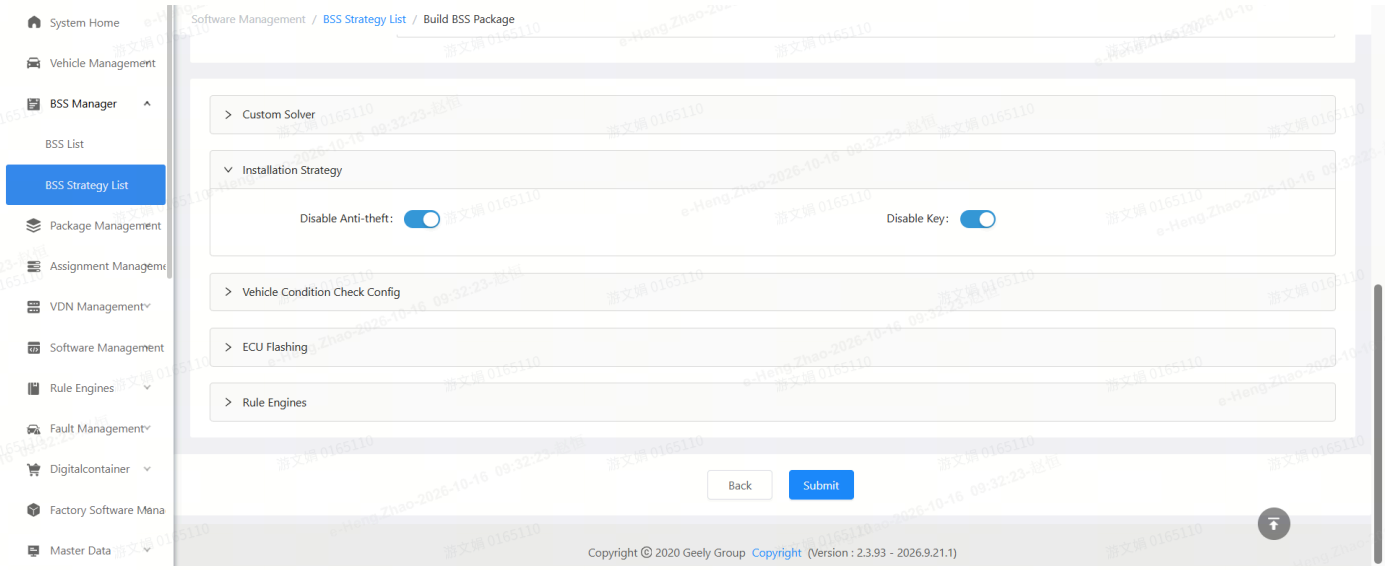


- Custom Solver
  - Solution Settings: Dropdown single-select options: hardware part number, hardware part number + hardware version number, **the default value is derived from the vehicle model's solution settings**
  - Flashing Current: The default value is derived from the flashing current of the vehicle model (applicable to Geely 2.0/3.0+ 4.0 architectures)
  - Force VDN flashing: Defaults to No, options: Yes, No (applicable to Geely 2.0/3.0+ 4.0 architecture; VCD software is delivered via 2.0, VCP is delivered via 3.0+ 4.0)

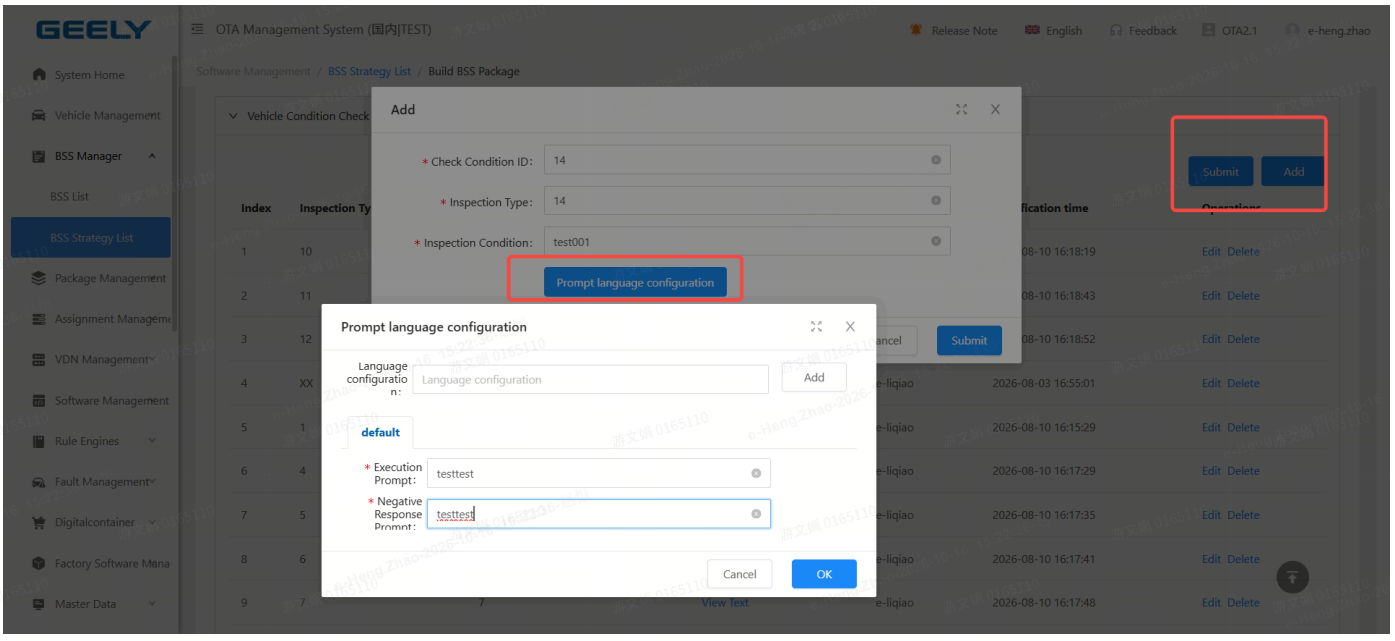


- Installation Policy (ZEEA 3.0 Architecture Only)
  - Anti-theft Disable: When this function is **enabled**, the vehicle's anti-theft function will be automatically turned off during the pre-OTA installation phase, and will be automatically restored after the upgrade is completed. When this function is **disabled**, no operation will be performed on the vehicle's anti-theft function.

- Disable Key: When this function is **enabled**, the vehicle keys will be automatically disabled during the pre-OTA installation phase, and will be automatically restored after the upgrade is completed. When this function is **disabled**, no operation will be performed on the vehicle key function.

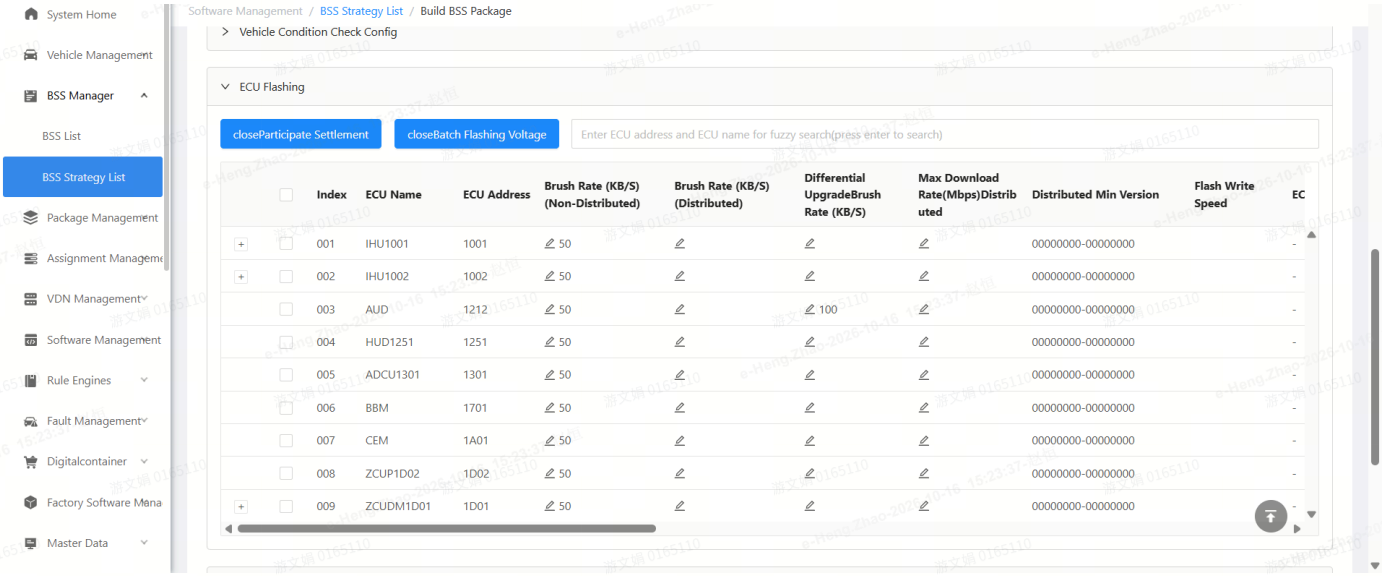


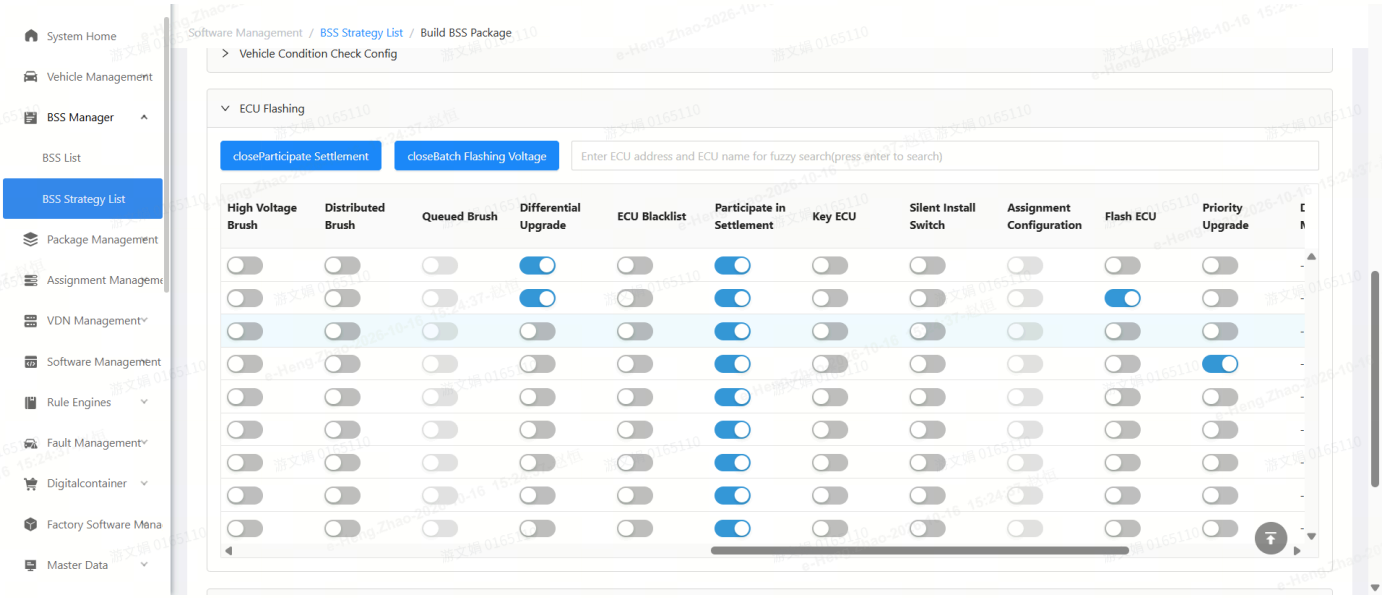
- Vehicle Condition Check Configuration (ZEEA 3.0 and EEA 4.0 Architectures Only)
  - Vehicle Condition Check Instructions: Before an OTA upgrade, prompts for each check item will be displayed via the HMI. If the conditions are not met, not only the prompt for the corresponding check item but also the non-compliance prompt will be displayed. Please configure according to the actual situation.
  - Check Type: Addition and deletion are allowed, but modification of content is not permitted



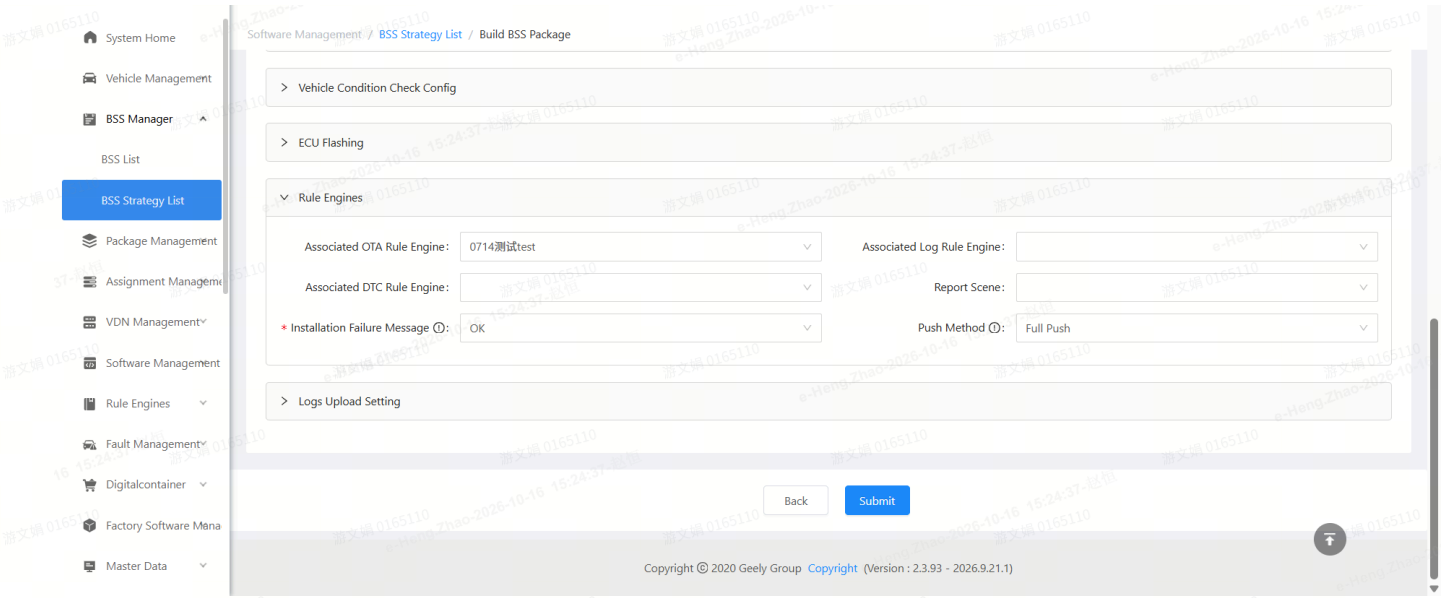
- ECU flashing
  - ECU parameters: The default value is consistent with the configuration of OTA-compatible vehicle models

- The BSS Package ECU inherits the vehicle model ECU of the corresponding vehicle model in BSS by default, and the ECU range within the BSS Package can be manually selected.

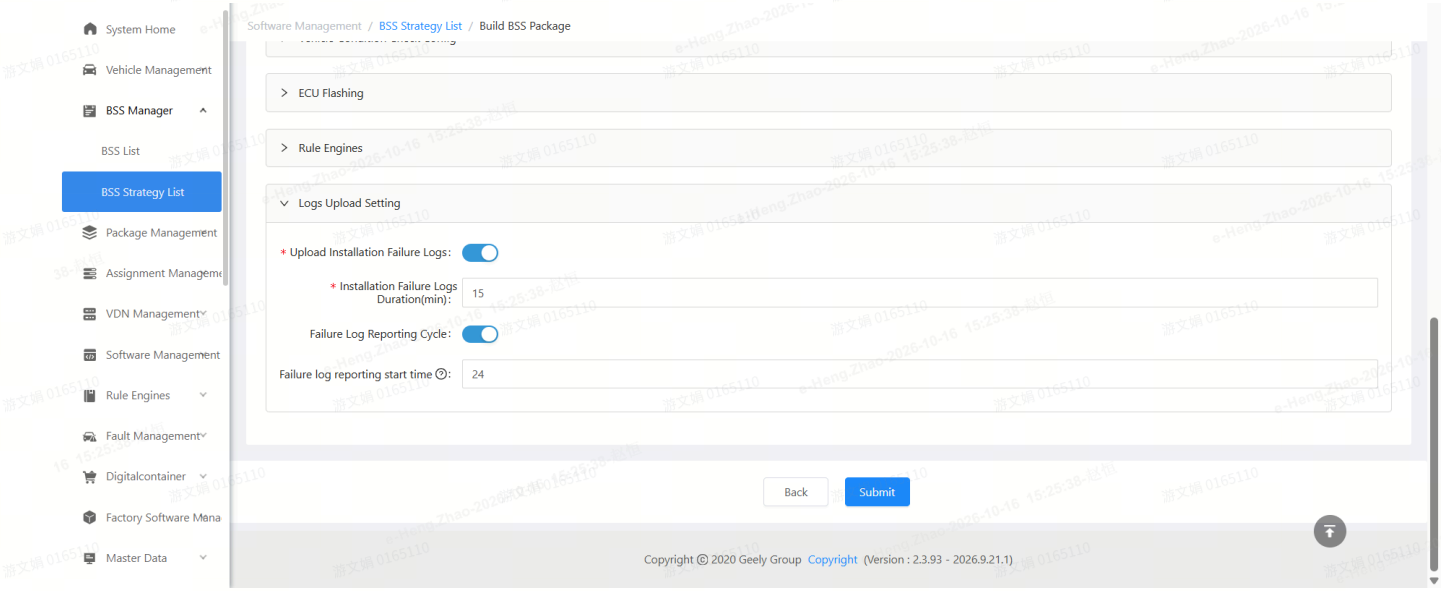




- Rule Engine
  - Associated OTA Rule Engine: Enter the rule name to search for settings
  - Correlation Log Rule Engine: Optional configuration (limited to ZEEA3.0 and EEA4.0 architectures)
  - Associated DTC Rule Engine: Optional configuration available (limited to ZEEA3.0 and EEA4.0 architectures)
  - Installation failure message: No by default; when Yes is selected, the push method must be set to full push or in-store flashing
  - Notification scenario: optional setting (not available in ZEEA3.0)



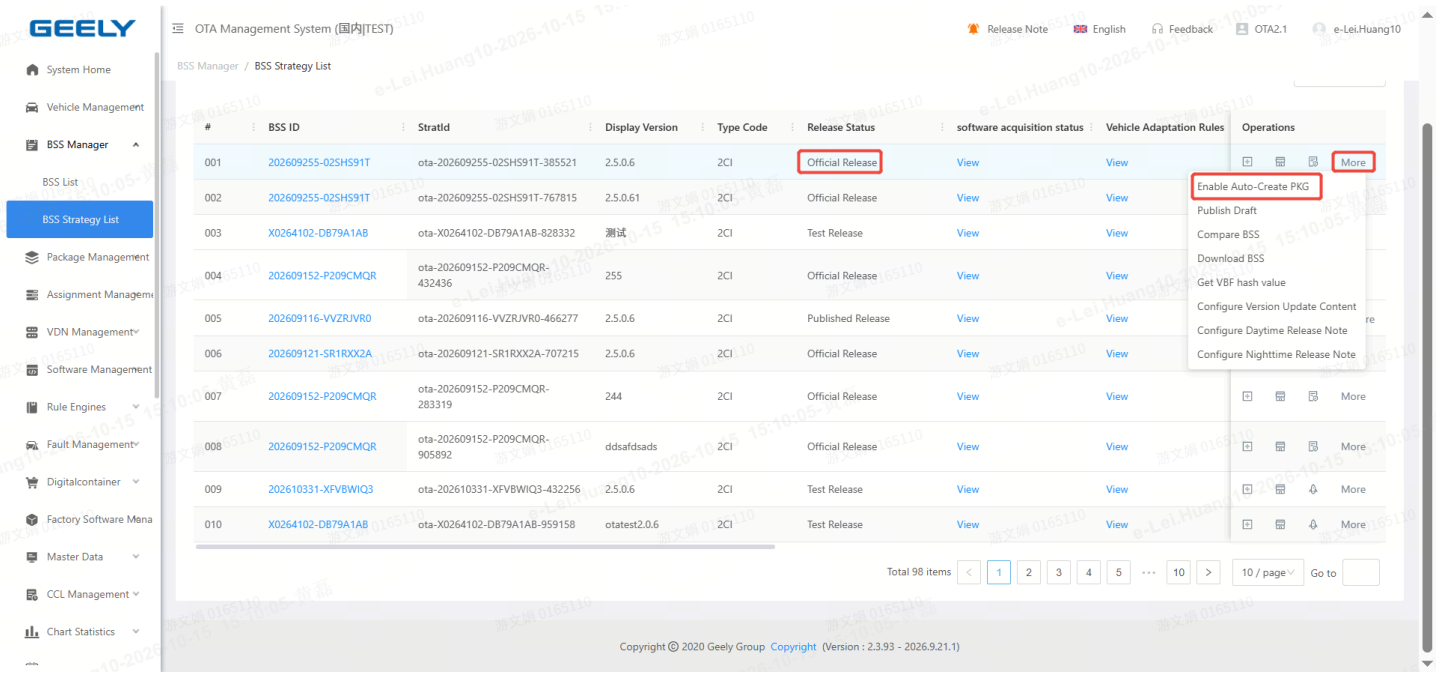
- Log Upload Settings
  - Upload installation failure log: On/Off, enabled by default, can be manually disabled
  - Installation failure log time limit (minutes): The settable time can be entered here
  - Failure log reporting cycle: Disabled by default; after enabling, the following needs to be set: Failure log reporting start time (hour):(format: number, 0-100)



3.4 Workshop Package

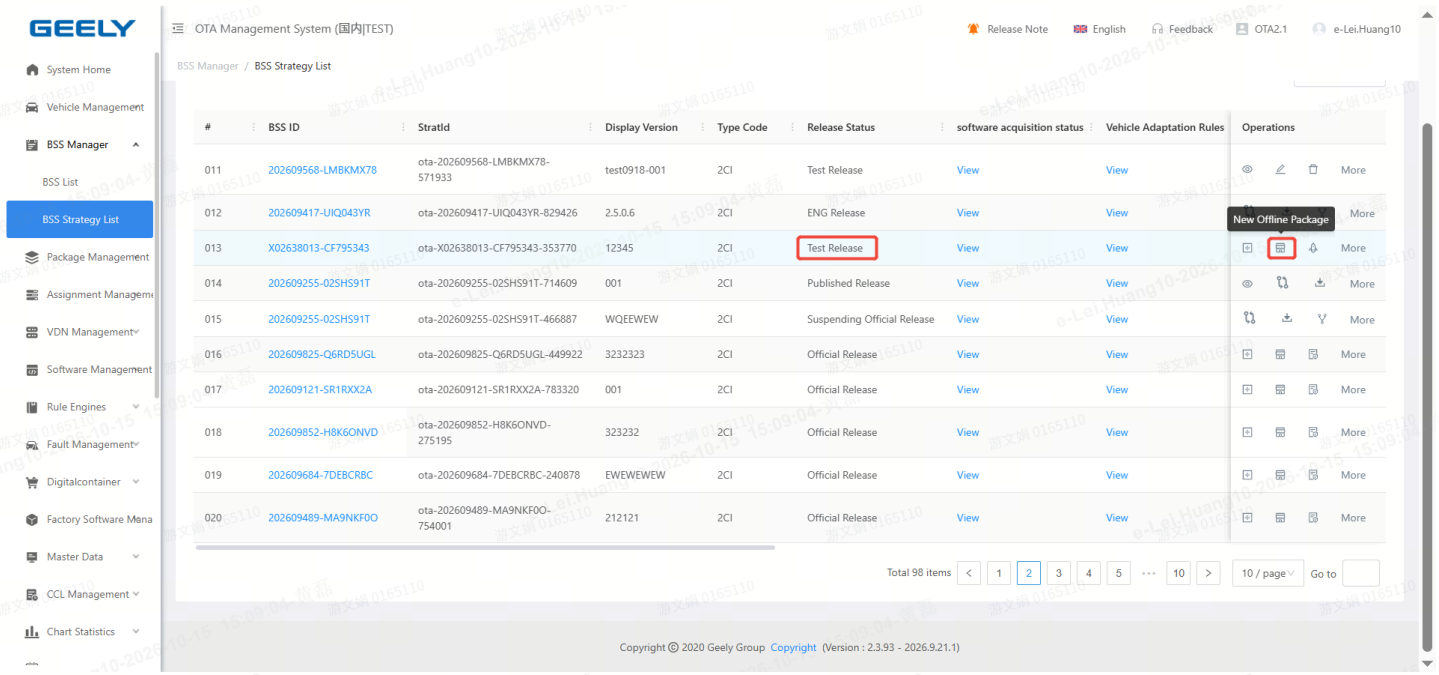
3.4.1 Automatically create Workshop Package

1. Function Entry: Navigation Bar > BSS Management > BSS Policy List
2. Precondition: The data source of the BSS policy is **TC** and the version status is **Official Release,** and the BSS policy status is released
3. BSS Policy List Action Bar > Select Enable Auto Package Creation

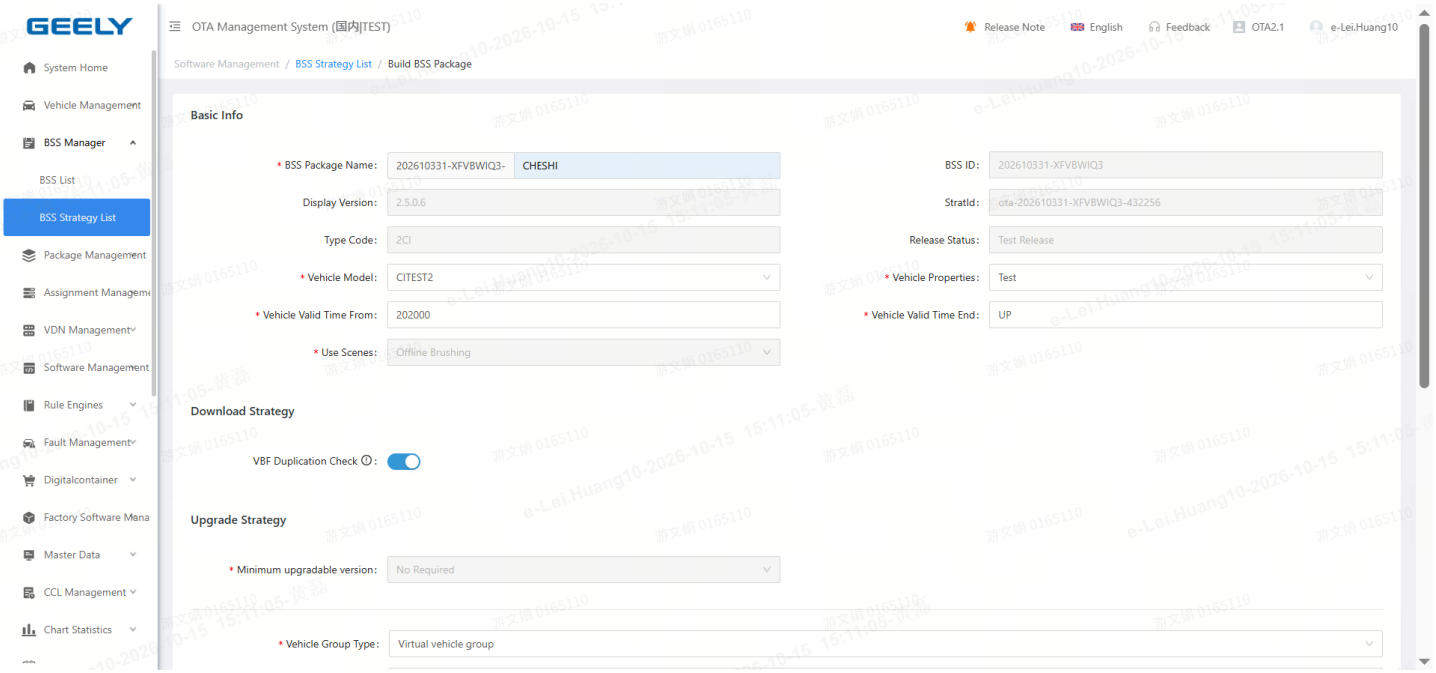


3.4.2 Manually create a Workshop package

- Function Entry: Navigation Bar > BSS Management > BSS Policy List
- precondition: The BSS policy version status is **Test Release/MRD Release,** and the BSS policy status is released
- BSS Policy List Action Bar >> Select to create an offline package (the system will automatically create a quick OTA task after the shortcut package is created)



- The parameter settings for creating a new offline package page are filled in as shown in the following figure:



- To restrict the minimum upgradable version, select the corresponding BSSID for the minimum upgradable version. If the drop-down menu for vehicle pool type is empty after selecting the vehicle model during addition (the logged-in user has not created a self-built vehicle pool under this vehicle model, nor is they an administrator of other self-built vehicle pools), the solutions are as follows: 1. Navigate to Vehicle Management >> Vehicle Pool Management >> Add Vehicle Pool in the navigation bar; 2. Navigate to Vehicle Management >> Vehicle Pool Management >> Add Vehicle in the operation bar (self-built vehicle pool for the corresponding vehicle model) >> Enter your own domain account in the Vehicle Pool Administrator field and make a selection >> Submit.
- Custom Solver Settings:

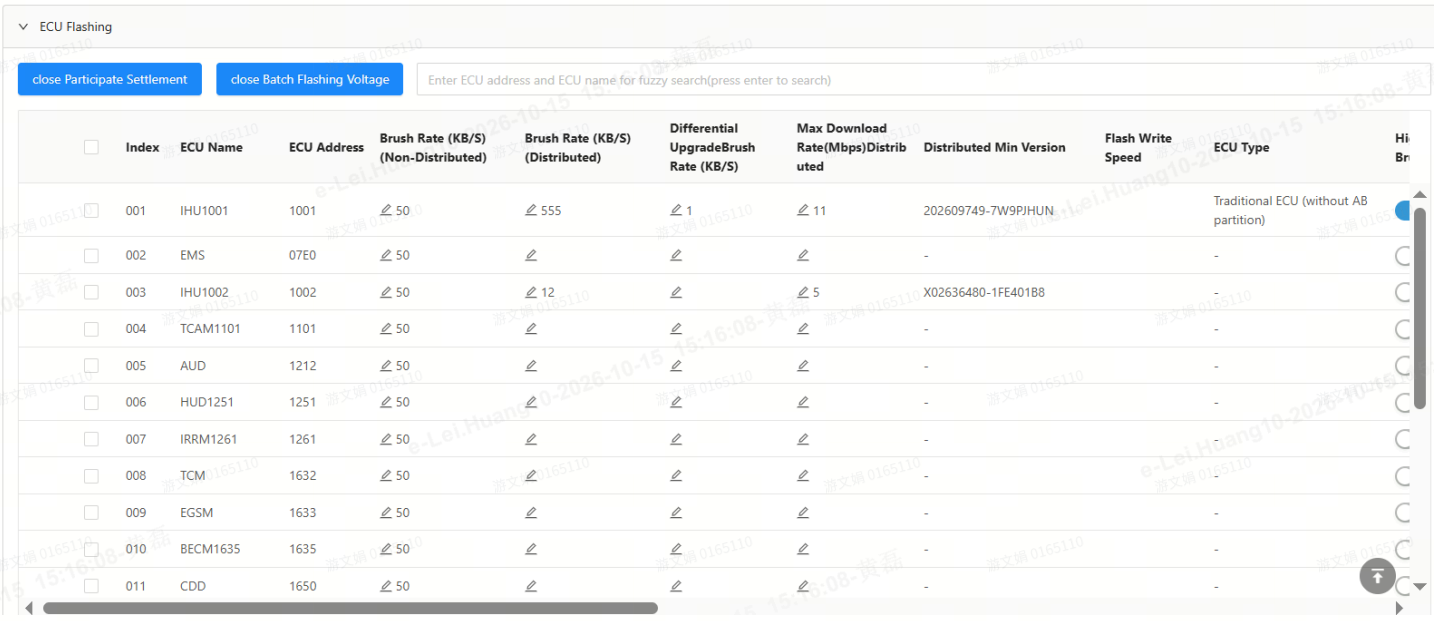


- Vehicle Condition Check Configuration:

Data is synchronized with the vehicle model, and data can be deleted and added.



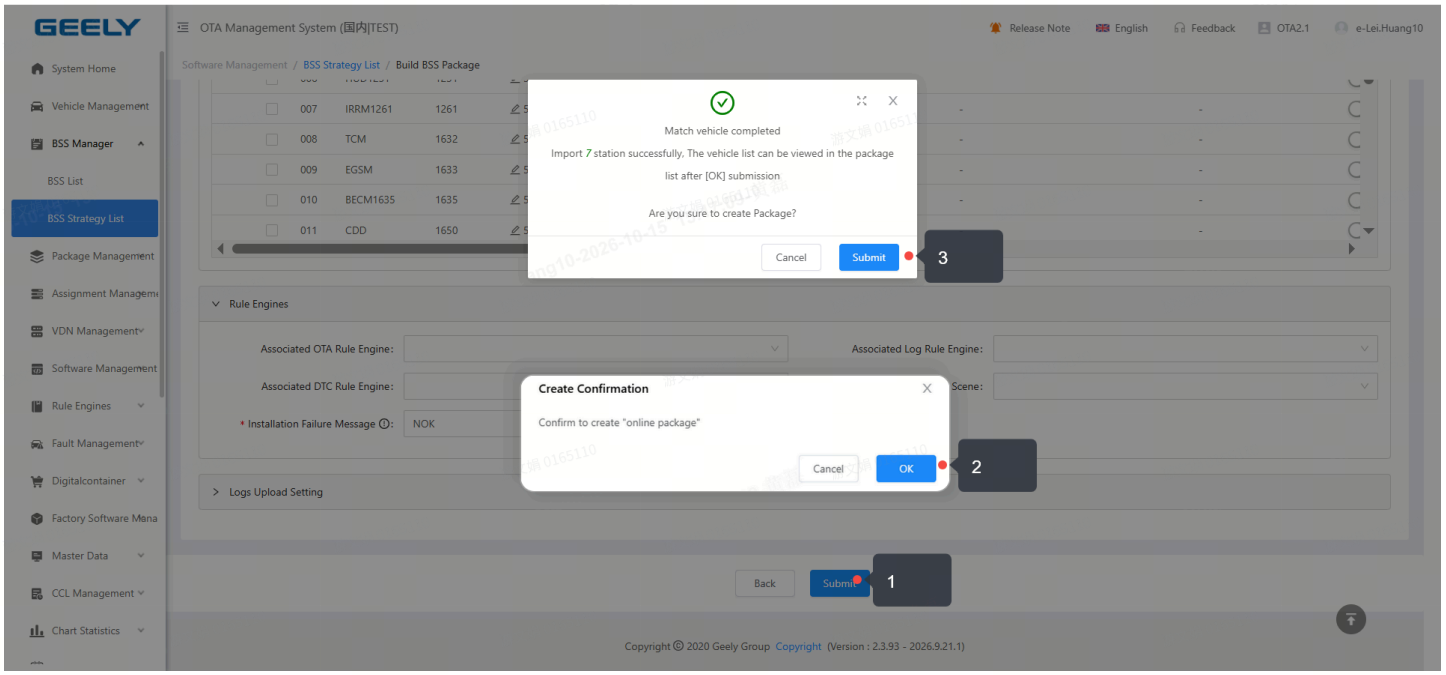
- The flashing settings are configured in the ECU flashing section shown in the figure below (by default, they are consistent with the ECU of the vehicle model):



- The rule engine (empty by default) and log upload settings are as follows:

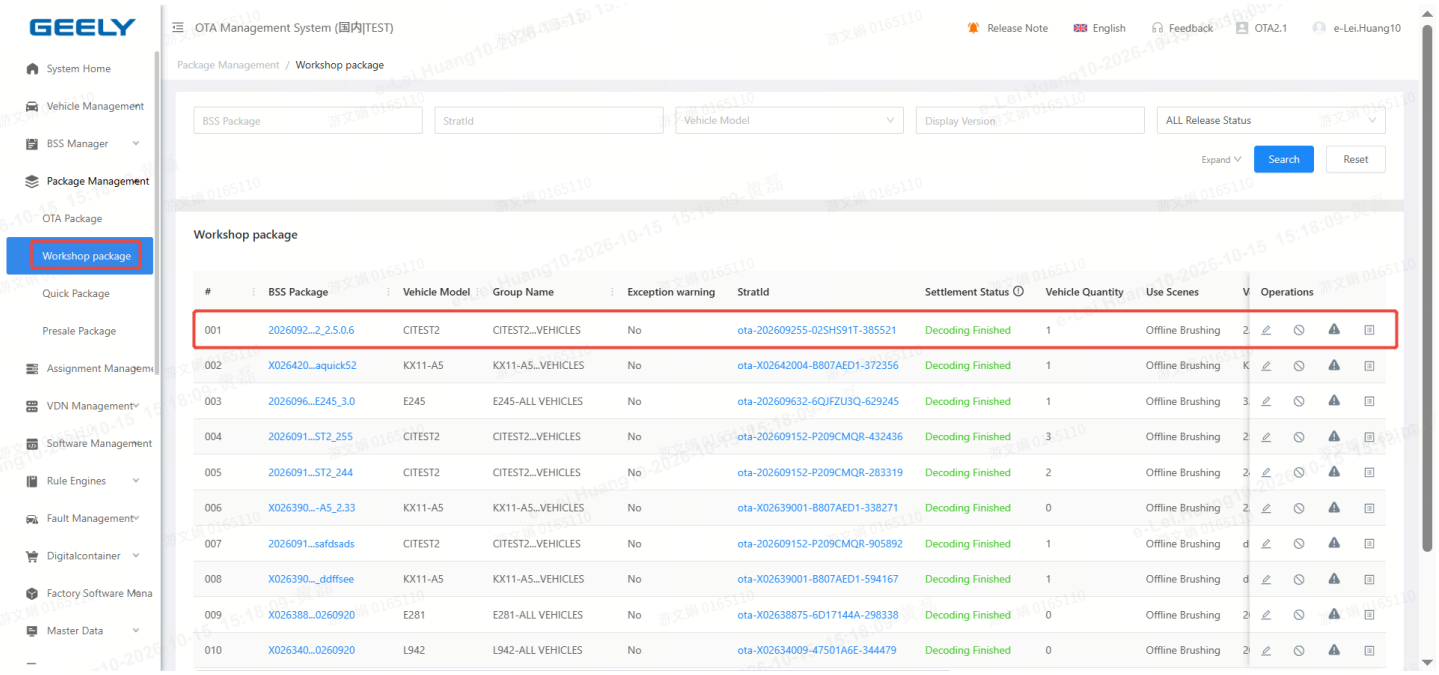


- When associating with the OTA rule engine, enter the rule engine name in the input box (fuzzy match) and select the corresponding rule engine
- Manually create an offline package and submit it as shown in the following figure:



3.4.3 Workshop Package List

- Function entry: Navbar > Package Management > Workshop Package



4. OTA Task Management

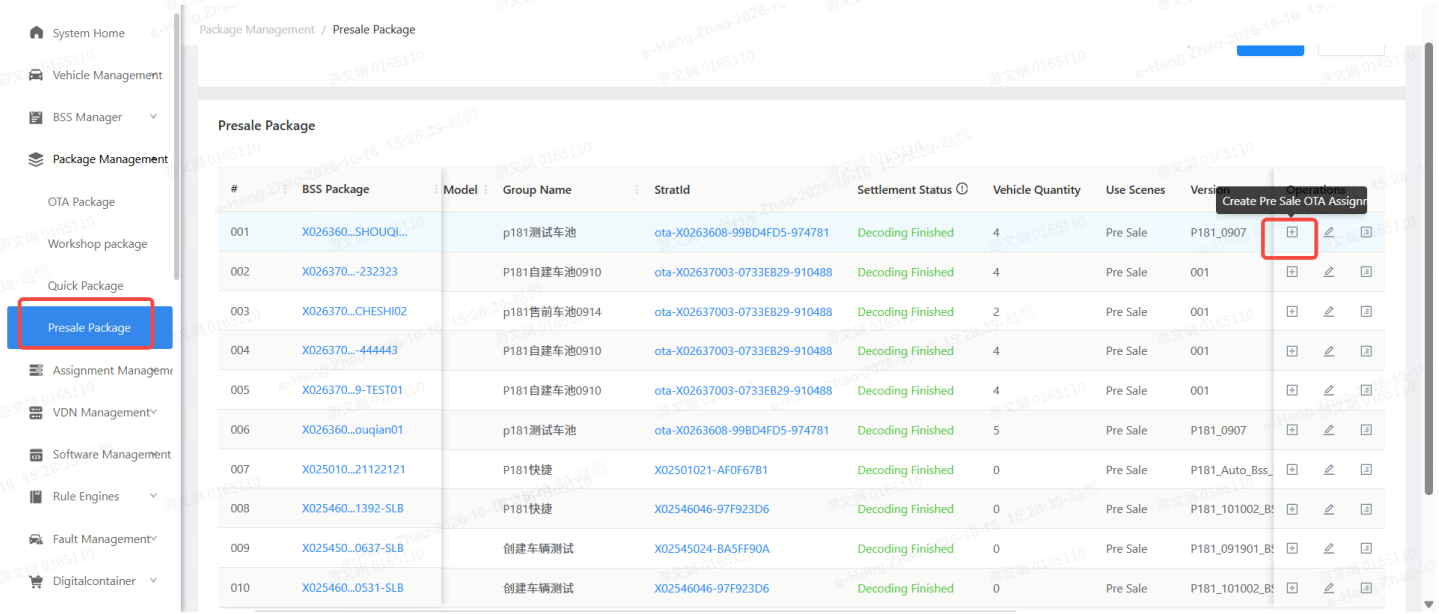
4.1 Regular Task List

4.2 Pre-sales Task List

4.2.1 Function Overview

4.2.2 Function Entry

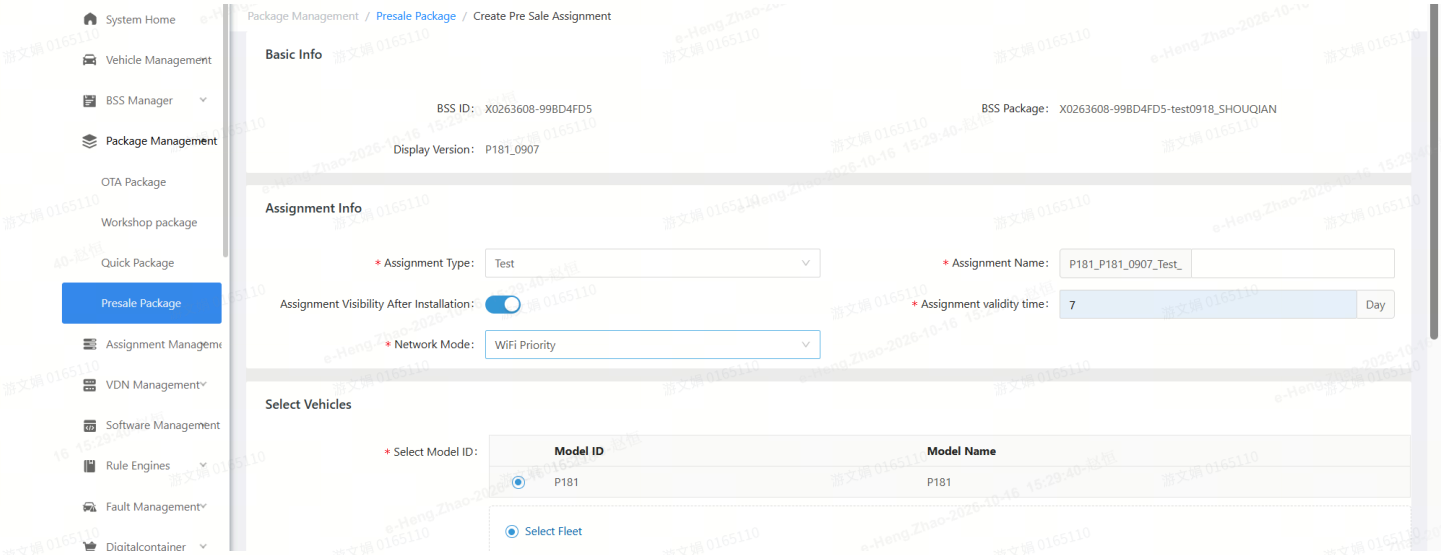
- Navigation Bar: Package Management >> Pre-sales Package
- Prerequisite: Pre-sales Package has been created
- Pre-sales Package >> Create Pre-sales OTA Task



4.2.3 Create a pre-sales task list

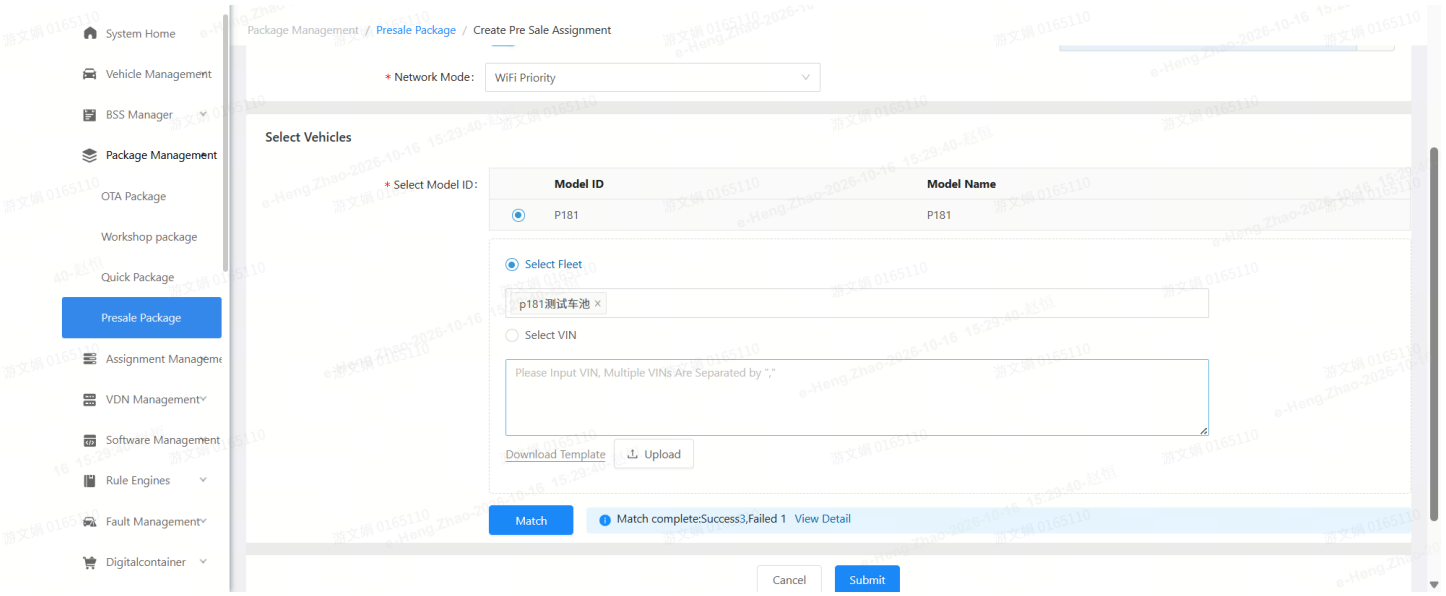
4.2.3.1 Task Information

- BSS Package Name: The selected BSS Package name, which cannot be modified
- BSS ID: Populated by the selected BSS Package and cannot be modified
- External Display Version: the external display version corresponding to the selected BSS Package, which is not modifiable
- Task types: optional testing, phased rollout, marketing, pioneering
- Task Name: Entered task name, limited to 50 alphanumeric characters
- Post-installation Task Visibility: Enabled by default (for Geely 2.0/3.0+ 4.0 architecture)
- Network Mode: Default WIFI Priority (for Geely 3.0+ 4.0 architecture)
- Task valid period: After the expiration time is reached, withdraw the tasks that have not been fully upgraded (applicable to Geely 2.0/3.0+ 4.0 architecture)



4.2.3.2 Vehicle Selection

- Select Vehicle Model: Select the vehicle model associated with the BSS Package
- Vehicle Selection
  - Select Vehicle Pool: Select the vehicle pool under the corresponding BSS Package
  - Select VIN: Enter or upload the vehicle VIN contained in the BSS Package (vehicle quantity limit: 50,000)
- Match: View details of matched vehicles



4.3 Quick Task List

4.3.1 Function Overview

GDDC Platform Operation Manual

GDDC (Global Data Distributed Center), as a subsystem of the OTA platform, is an intelligent data management platform based on cloud native and edge computing architectures, whose design needs to take into account massive data, low latency, high availability, security and scalability. It is not only the "black box" and "command center" of vehicle data, but also the "digital brain" that enables end-to-end software upgrade and user experience throughout the entire vehicle lifecycle covering R&D, production, operation and after-sales services.

1. Master Data Management

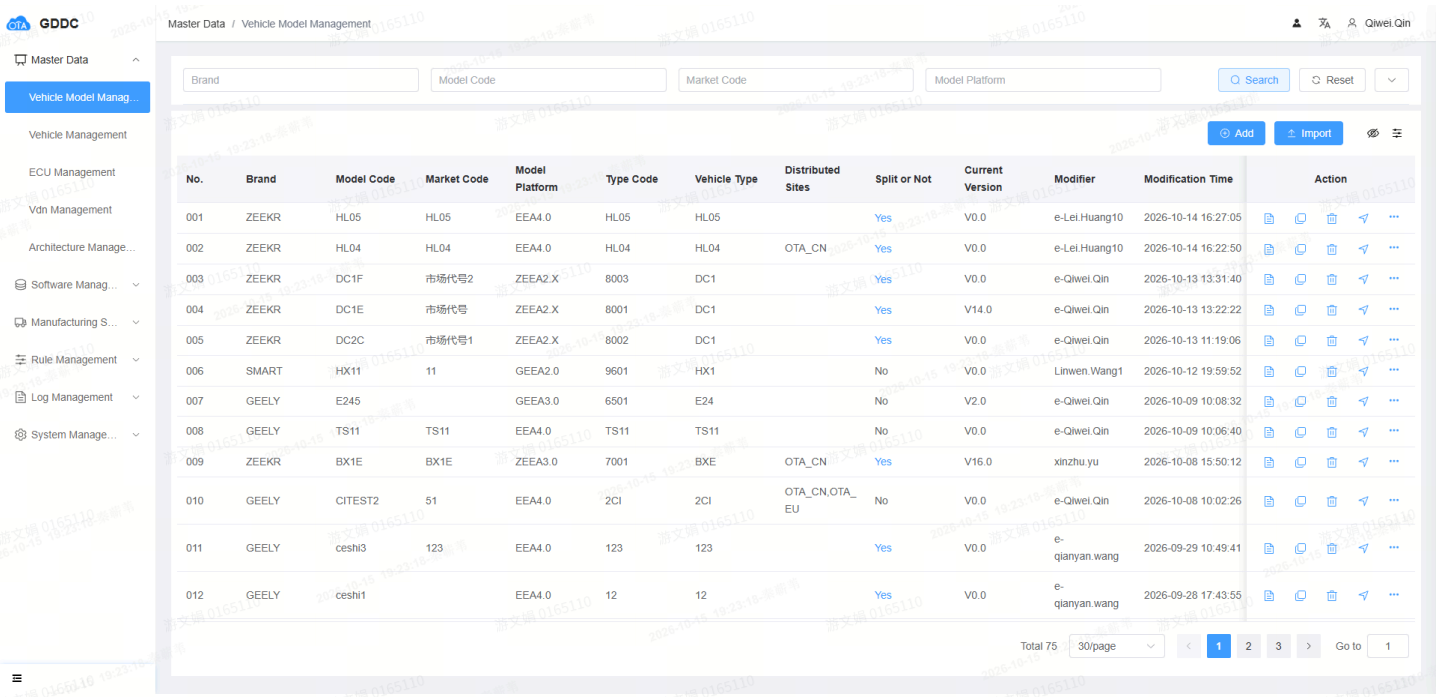
1.1 Vehicle Model Management

1.1.1 Function Overview

Manage and maintain information on all vehicle model projects worldwide, covering contents including basic vehicle model information, vehicle OTA flashing information, vehicle manufacturing plant information, etc.

1.1.2 Function Entry

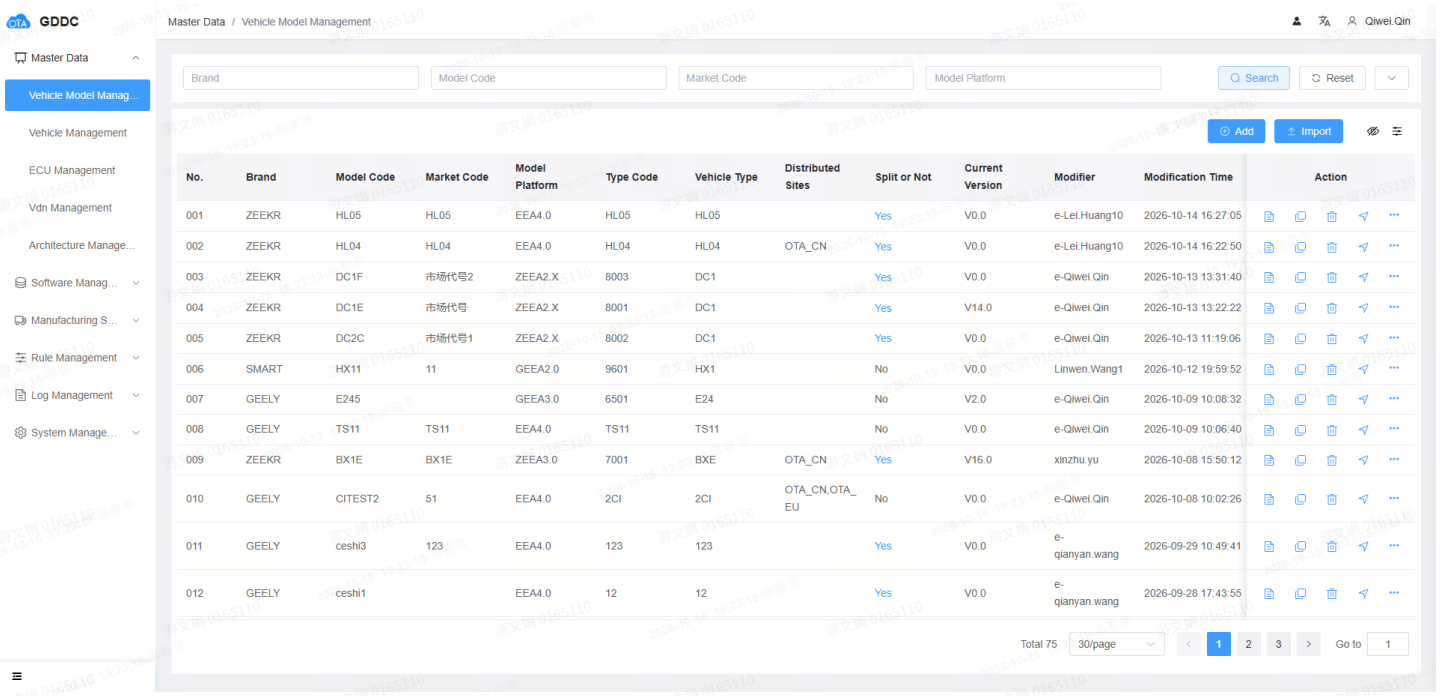
- Navigation Bar: Master Data Management >> Vehicle Model Management
- Precondition: The logged-in user has the corresponding platform permissions



1.1.3 Vehicle Model List

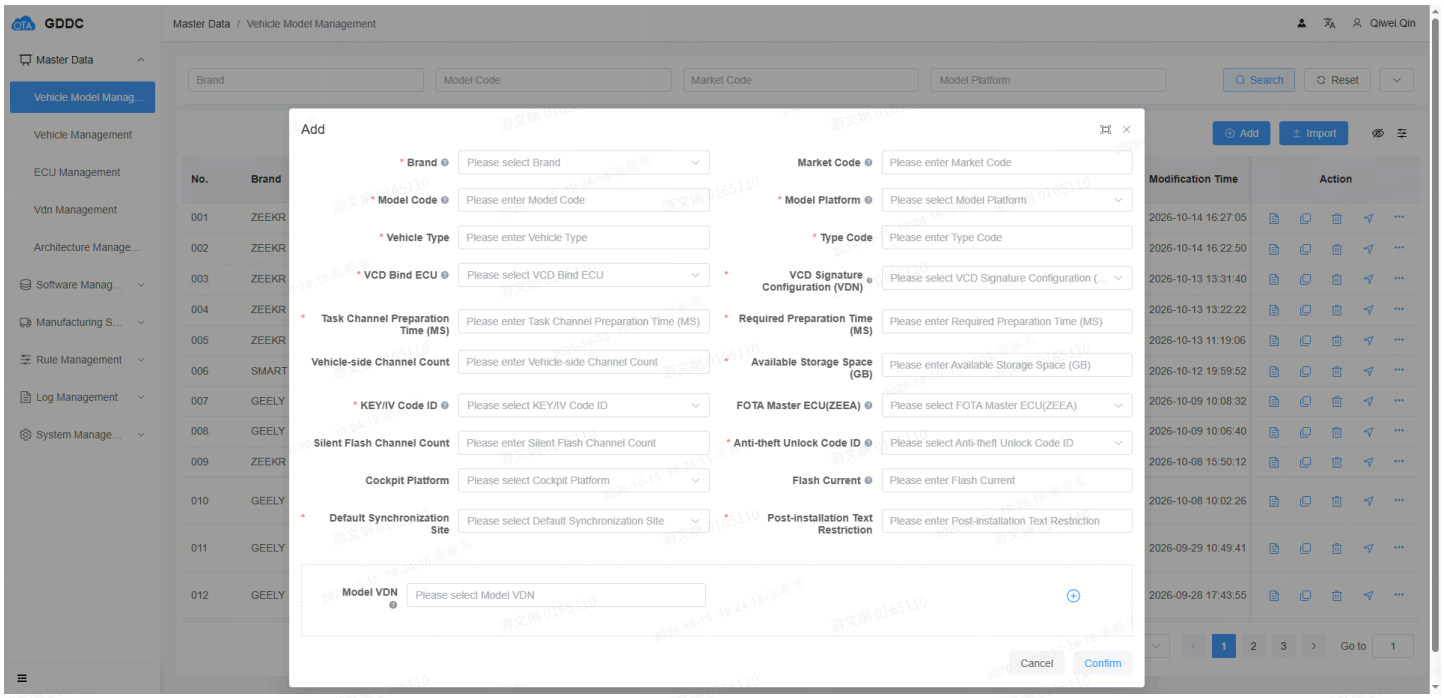
1.1.3.1 Vehicle Model Inquiry

Vehicle models support query by brand (fuzzy match), model code, market code, vehicle platform, Type Code, Vehicle Type, and change time. The query bar supports expansion and collapse; when collapsed, only four query criteria are displayed: brand (fuzzy match), model code, market code, and vehicle model. Reset: Reset the query bar to the default state (empty by default)



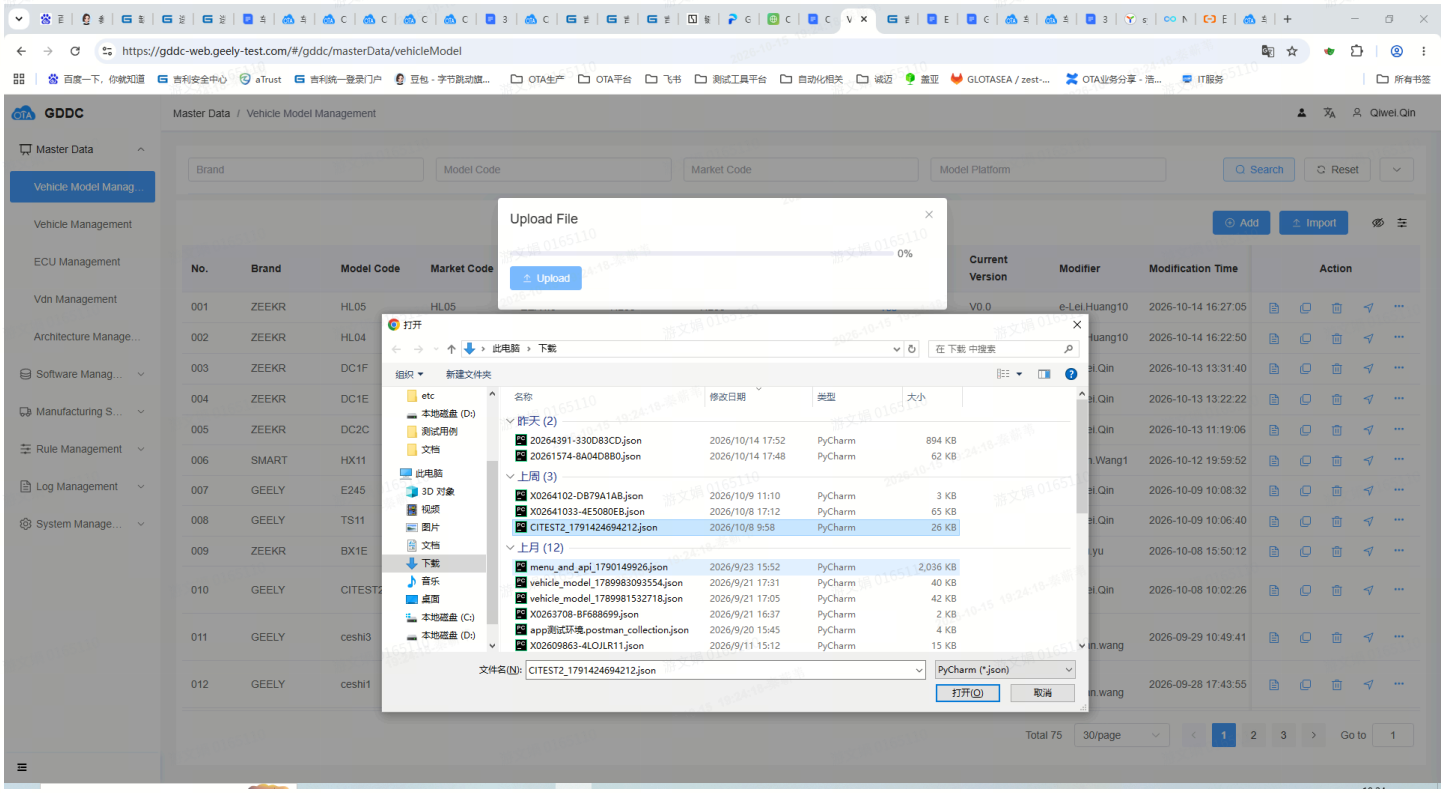
1.1.3.2 New Vehicle Models

- New vehicle model: Please fill in as the following example



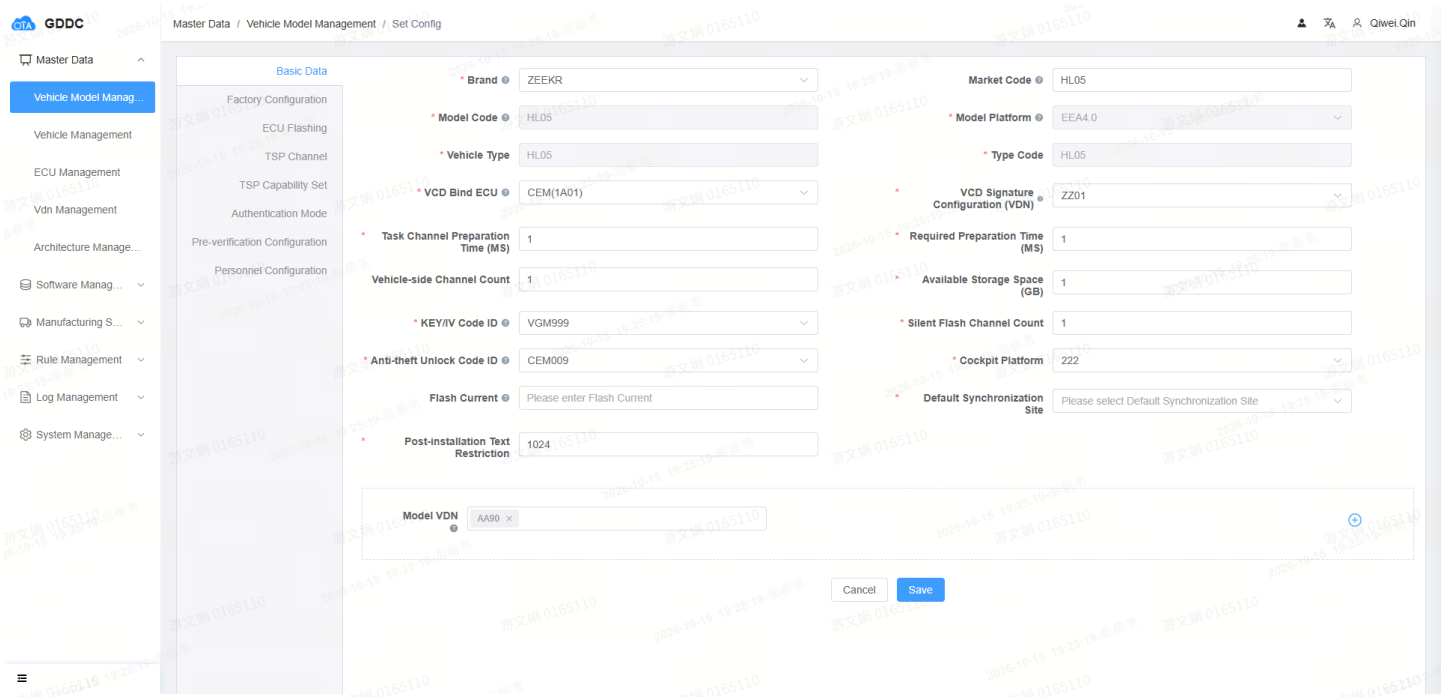
1.1.3.3 Import Vehicle Models

- Import: Support for importing JSON files

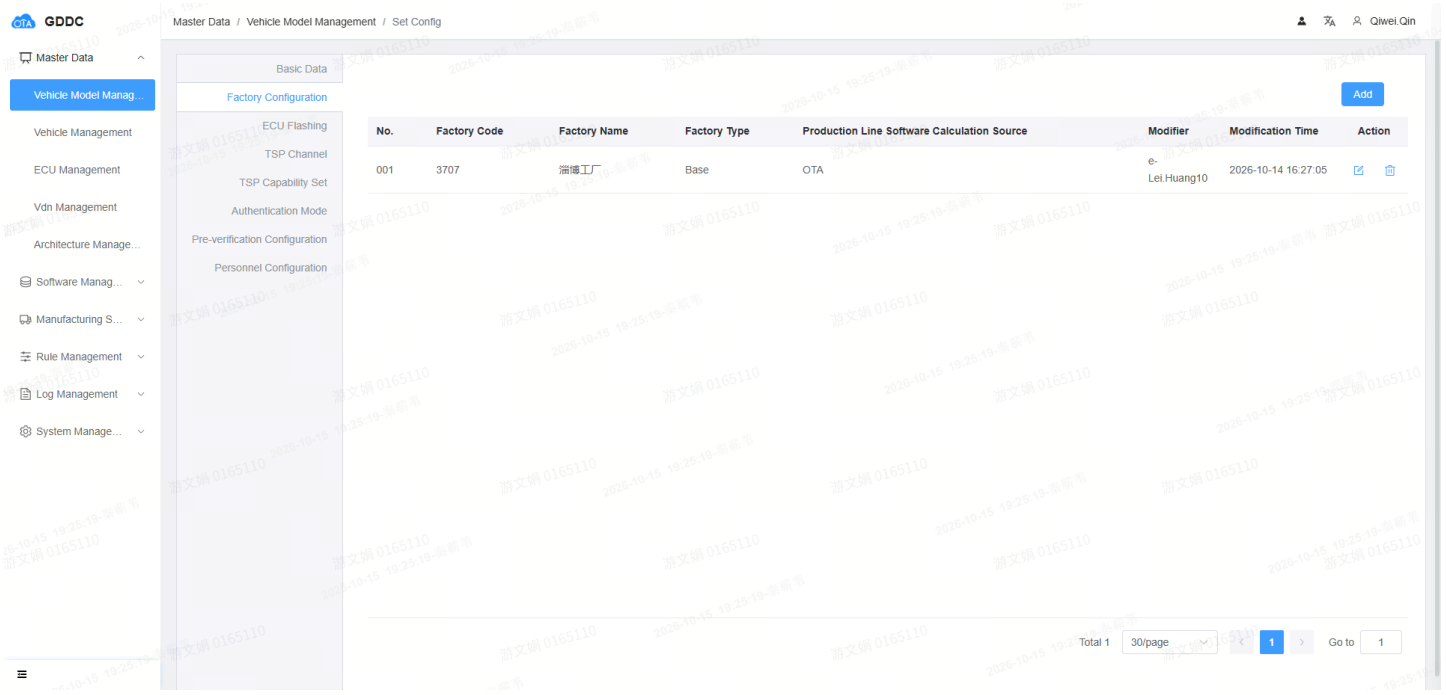


1.1.3.4 Vehicle Configuration

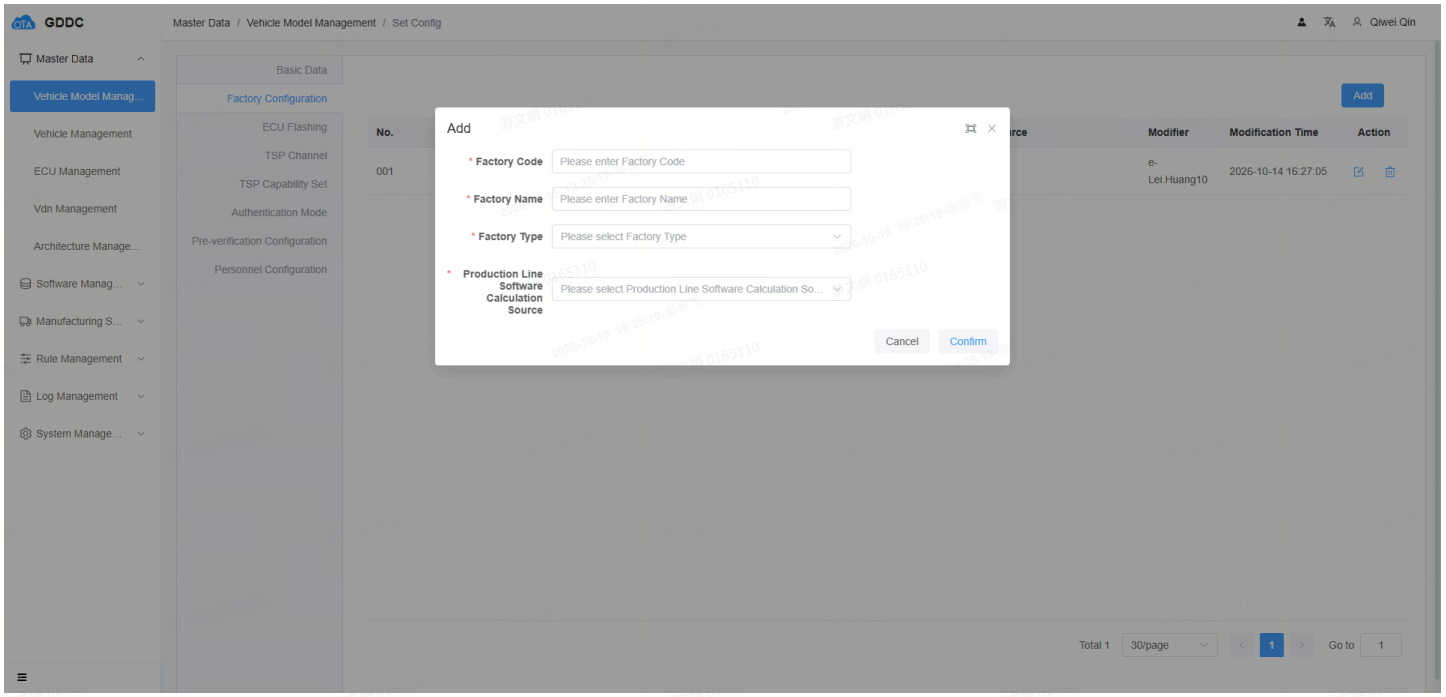
- Function entry: configured in the operation bar of the vehicle model list
- Configuration: For basic data editing, the grayed-out items as shown in the figure below are not editable, while all other fields can be edited and modified



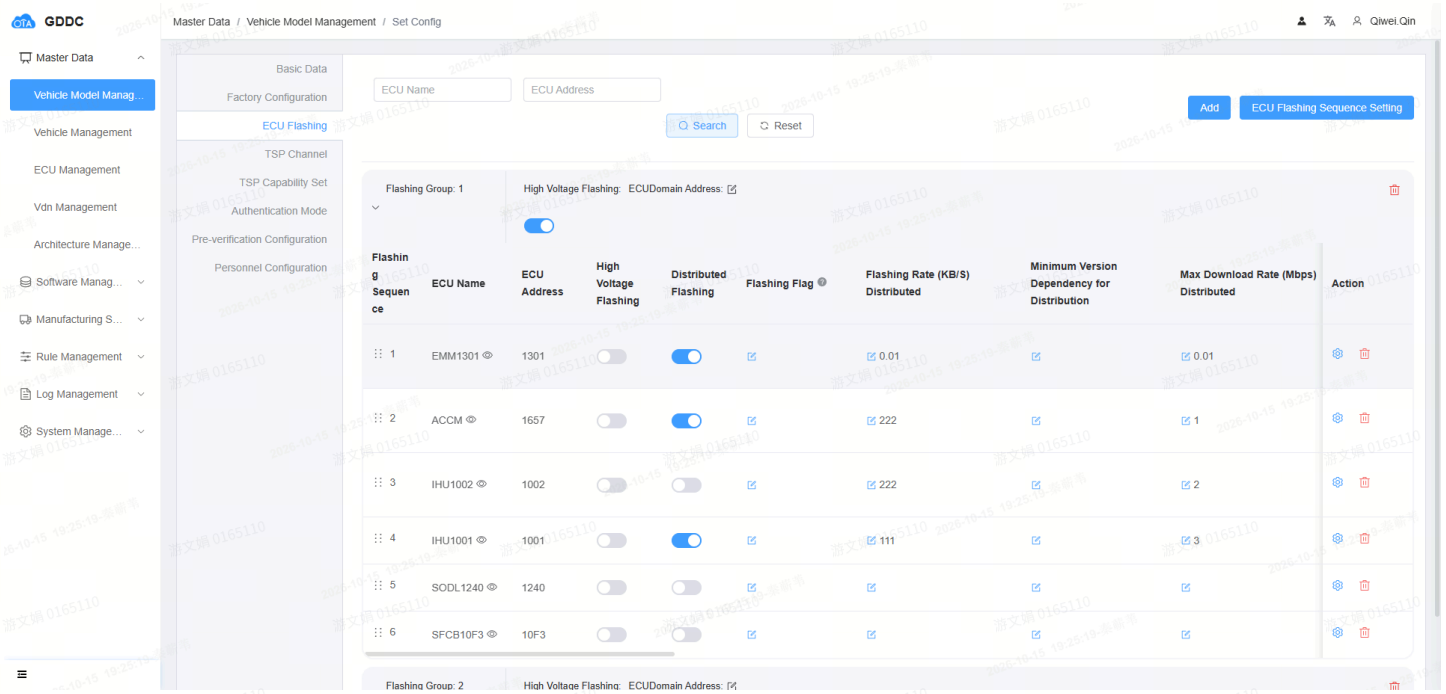
- Configuration: The factory configuration supports adding, editing, and deleting operations.



Refer to the figure below for filling in the new factory configuration information

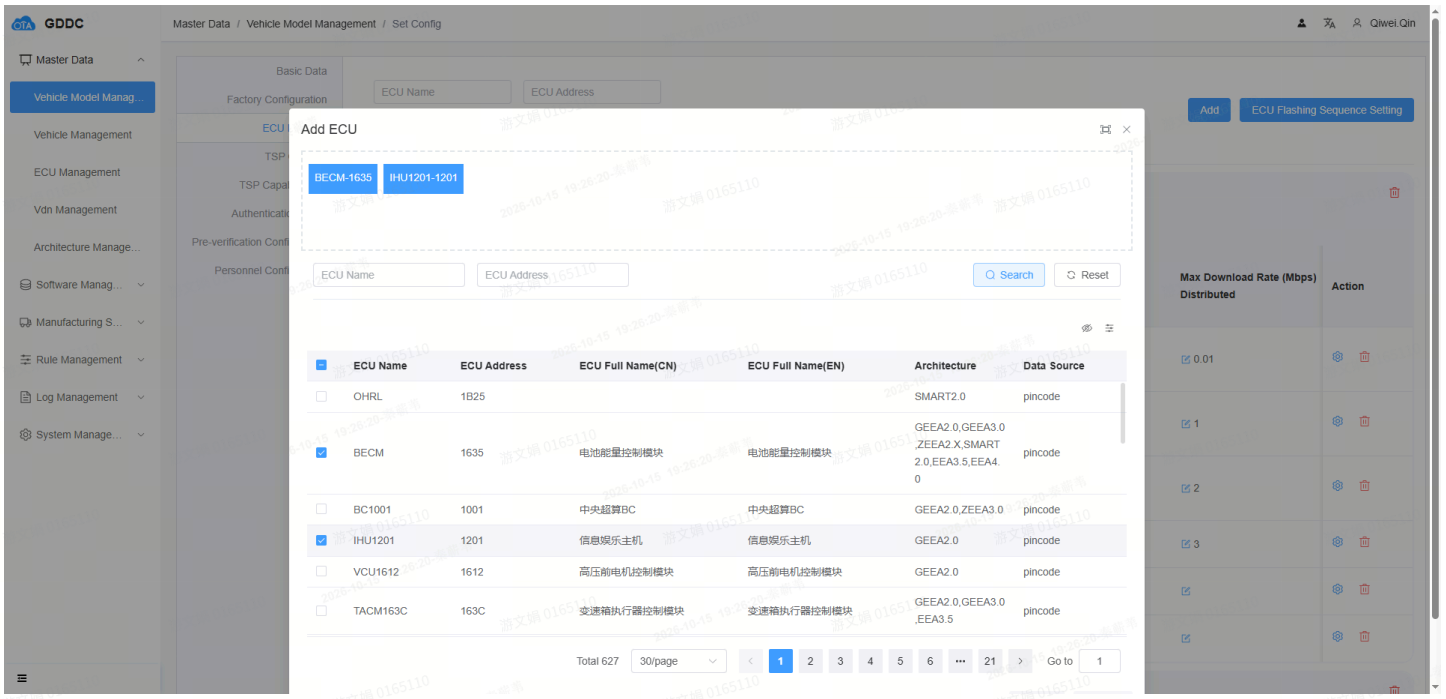


- Configuration: ECU flashing (GEEA2.0, GEEA3.0, GEEA4.0, ZEEA2.X and ZEEA3.0, integrated editing of ECU parameter settings, which ensures the ECU settings are comprehensive and complete; inapplicable architectures will be ignored during site application.)

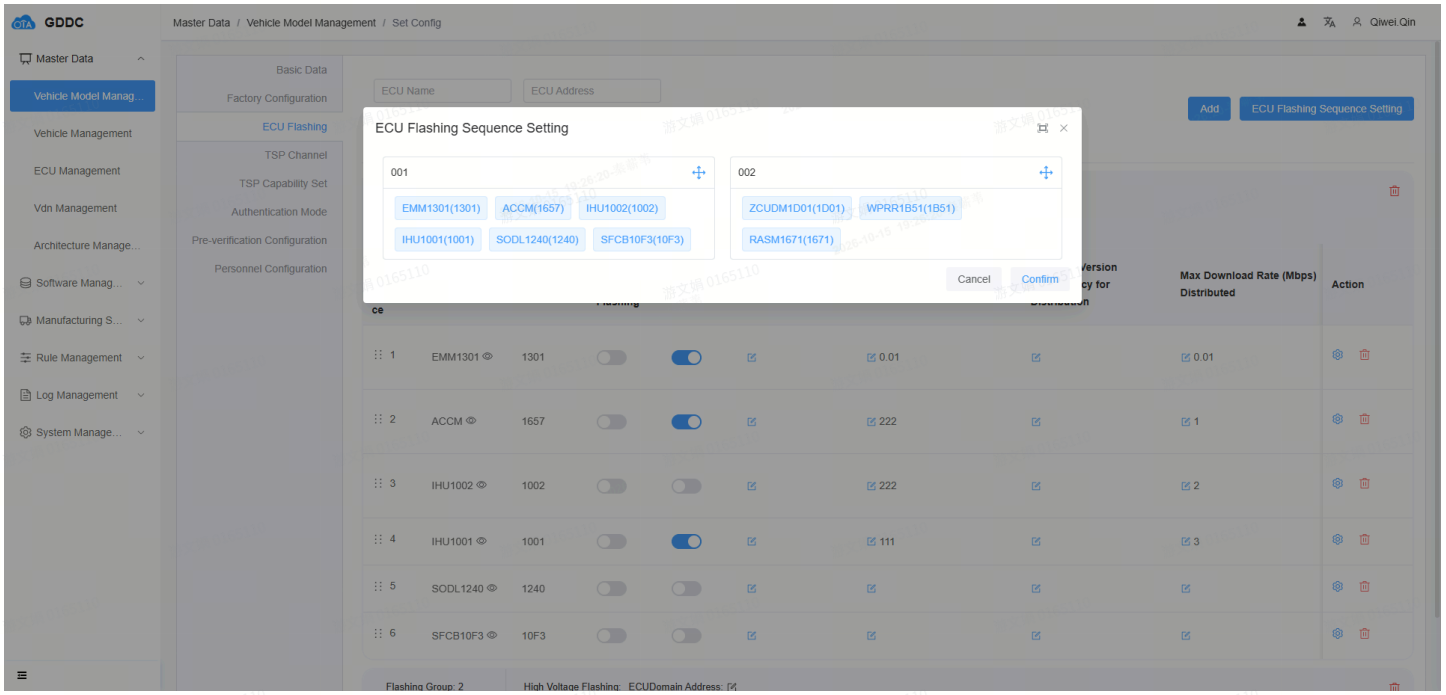


Interaction Instructions:

1. Added support for multi-select for ECUs (see "Add New ECUs"): you can query and tick ECUs multiple times, submit multiple ECUs at once after completing the selection, and the initially added ECUs will be assigned to Group 1 by default.



2. ECU flashing sequence setting: it supports setting ECU groups, the sequence among groups, and the sequence of ECUs within a group. After the settings are saved and take effect, the list page needs to be refreshed directly to display the latest effect;

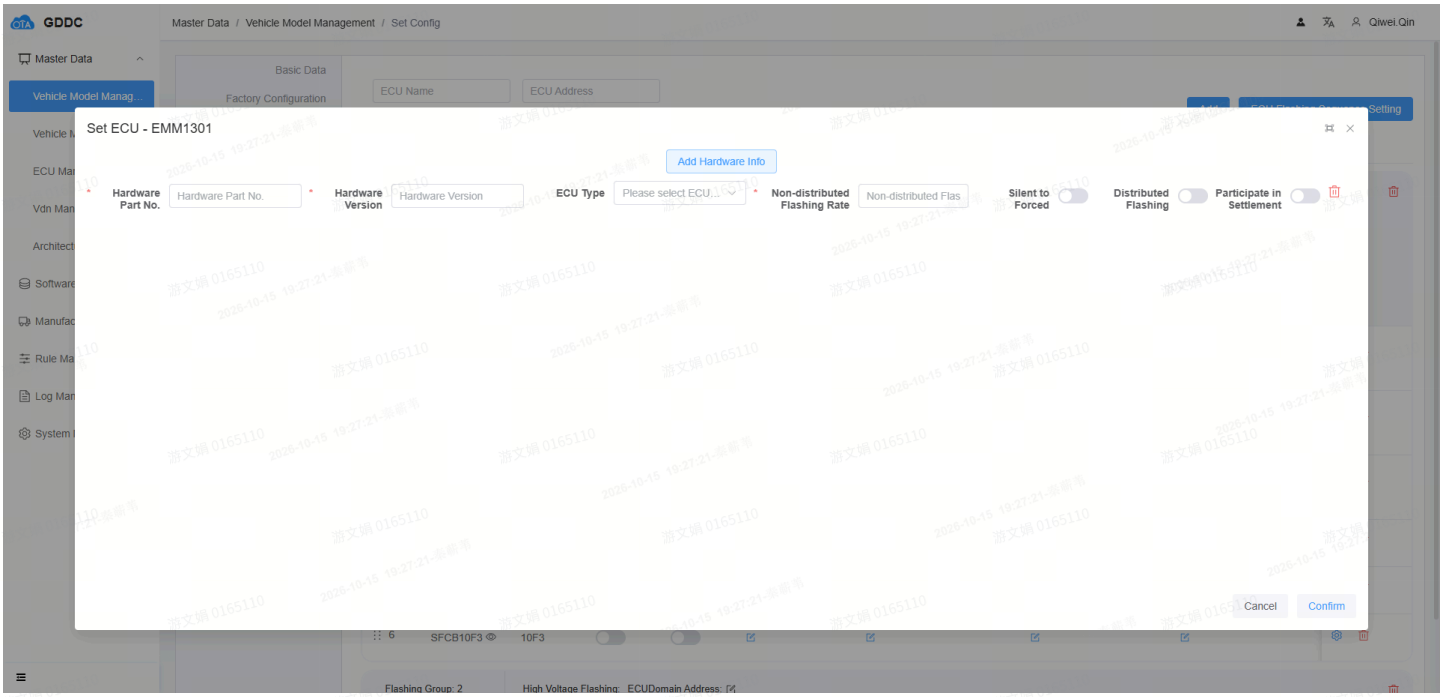


3. Supports dragging the list within a group to adjust the sequence of ECUs in the group;

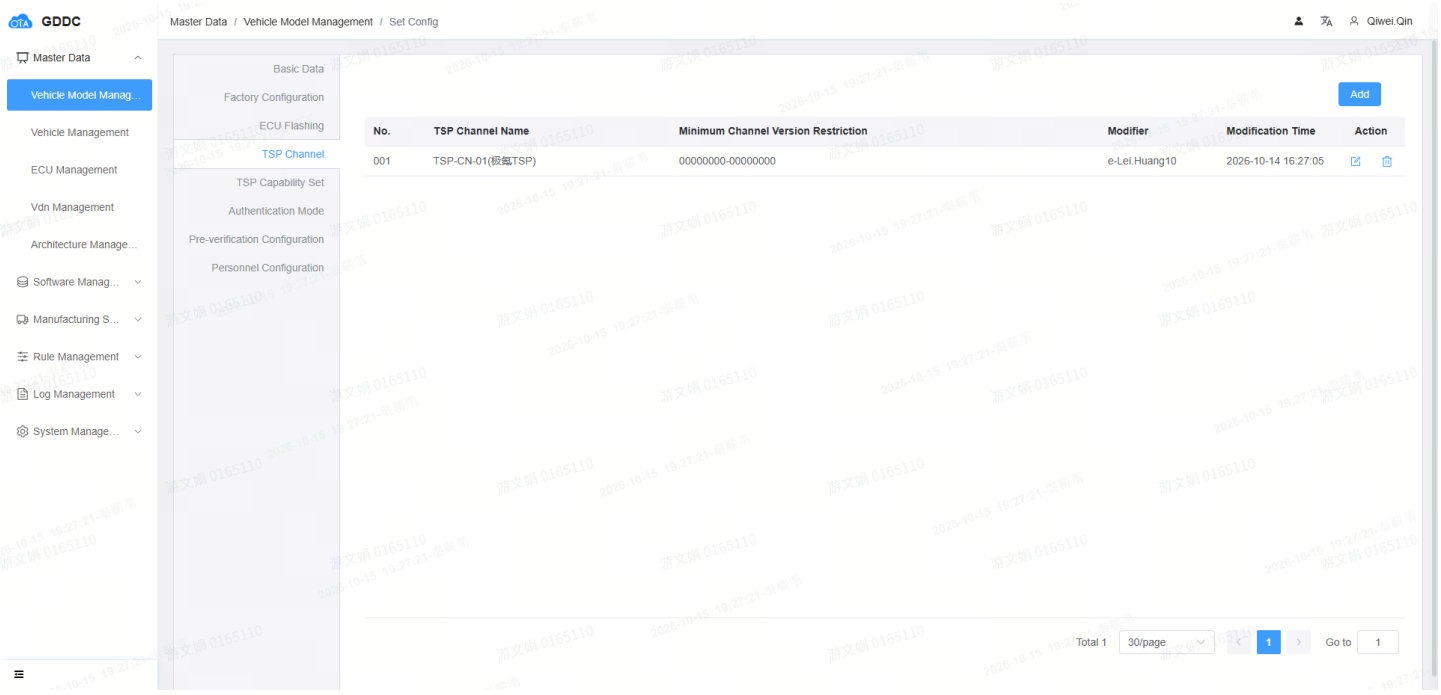
4. Supports setting parameters on groups (high-voltage flashing is disabled by default);

5. The ECU content supports setting and saving of individual fields; the saved changes will not take effect immediately, and will only come into force after the submission is approved through review.

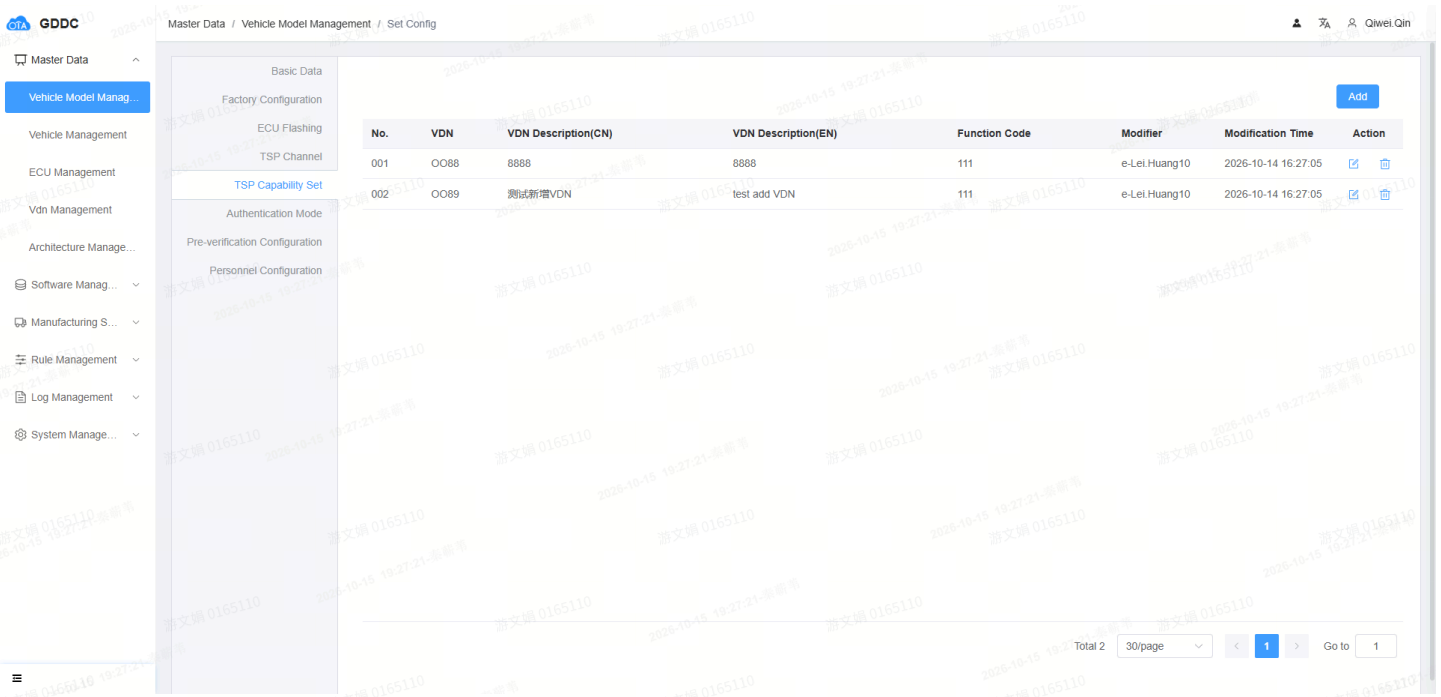
6. Set hardware to support editing and deletion;



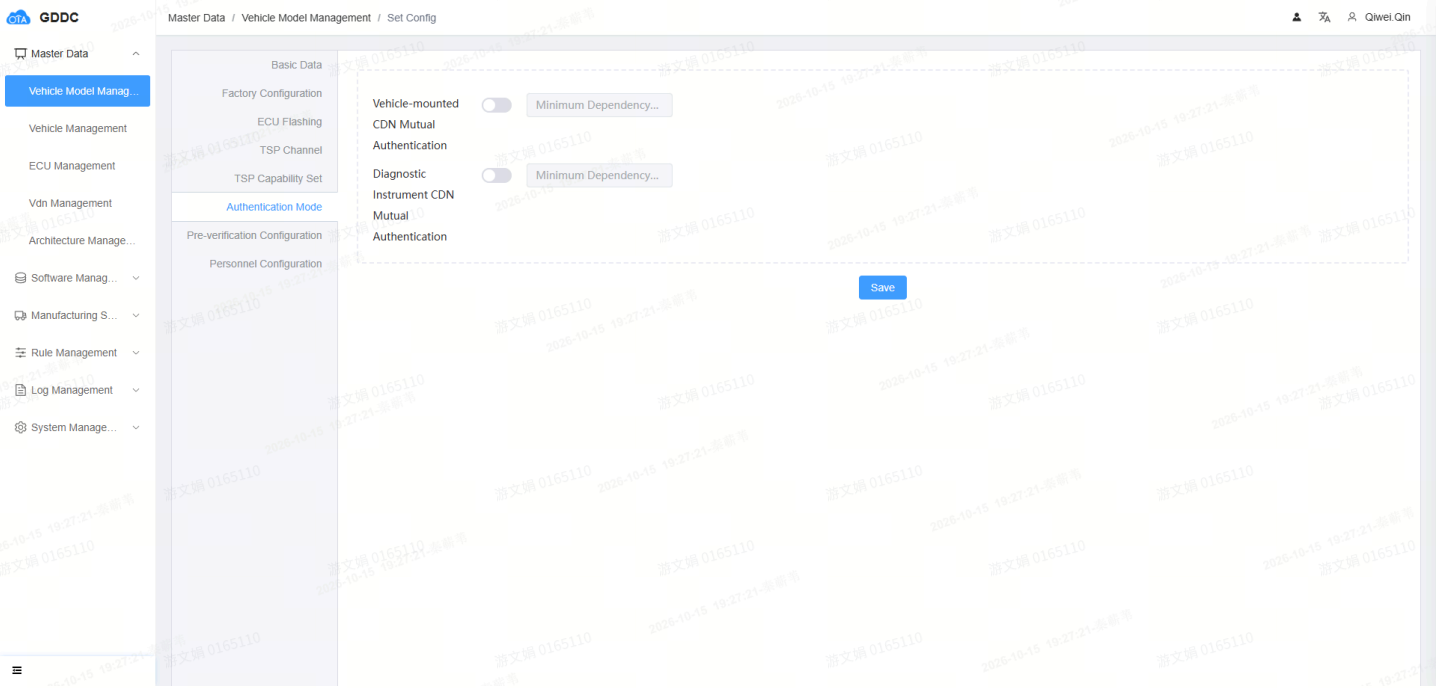
- Configuration: The TSP channel supports adding, editing and deleting operations



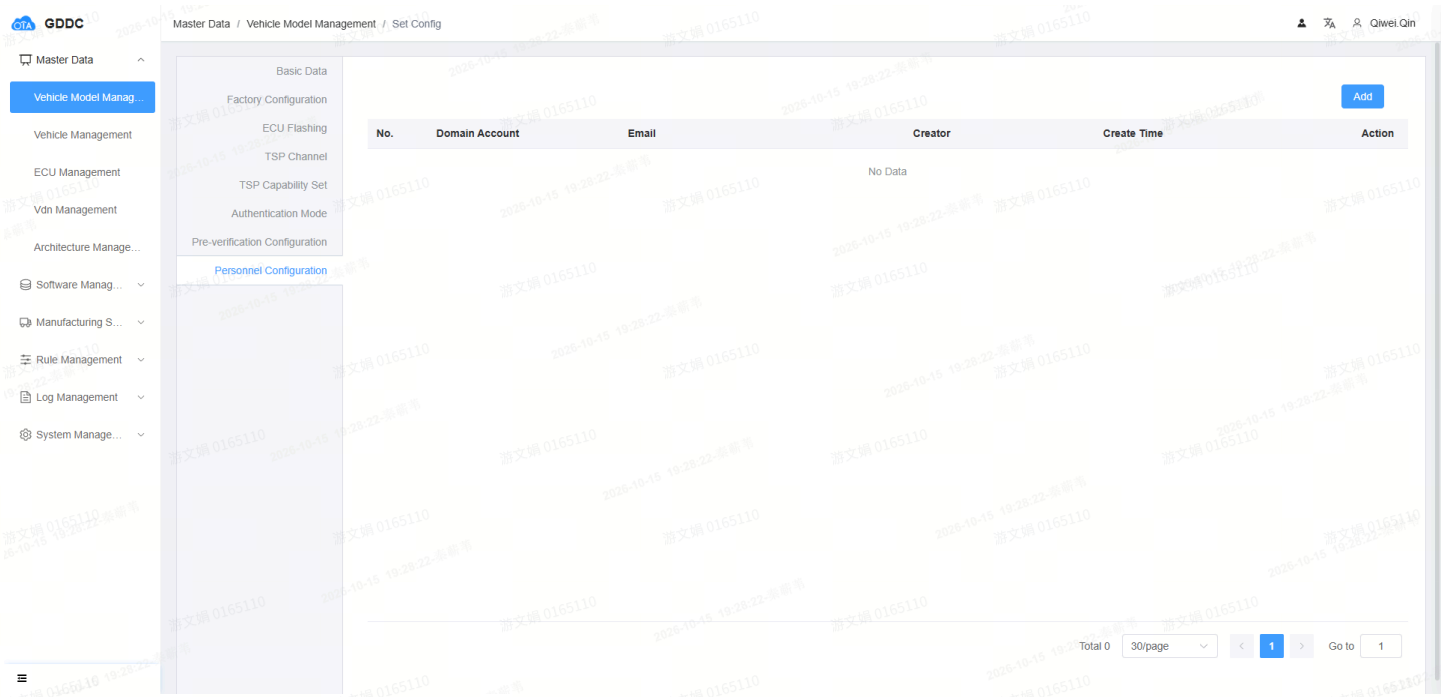
- Configuration: The TSP capability set supports addition, editing and deletion



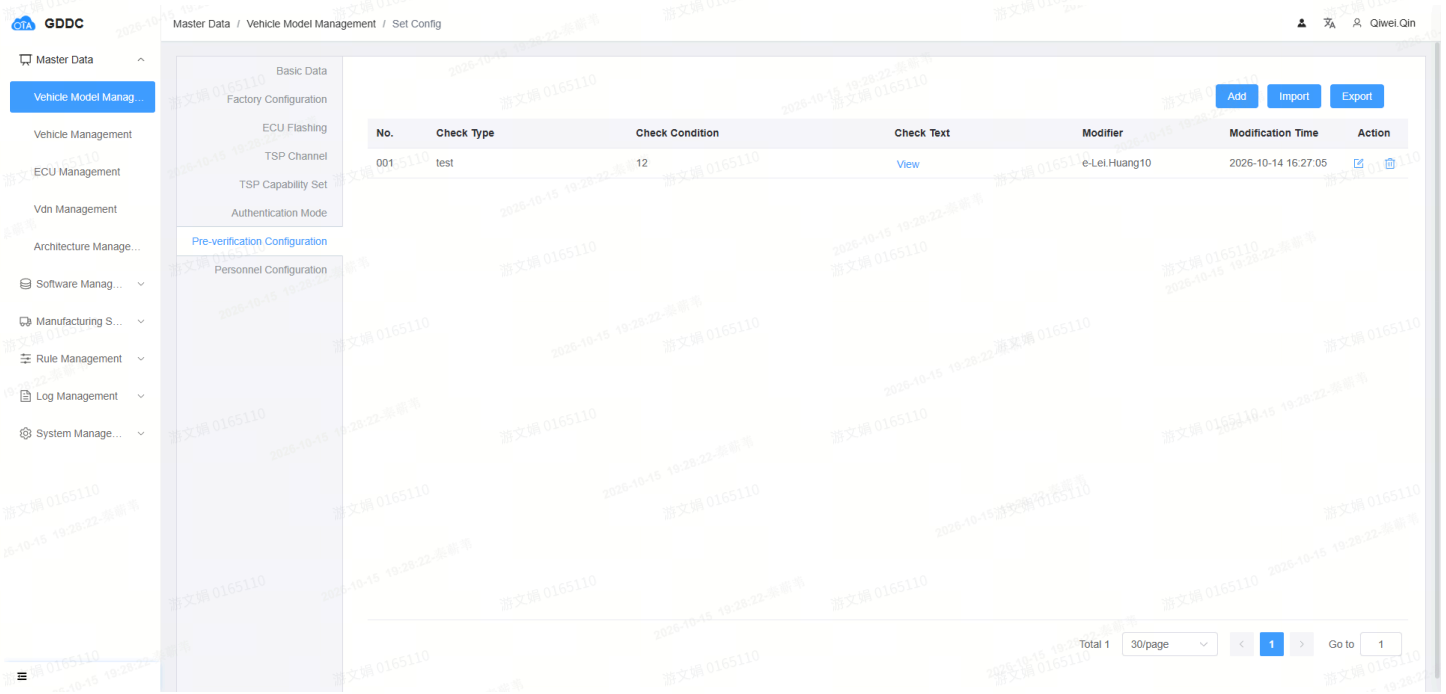
- Configuration: The switches for the authentication modes [Vehicle-end CDN mutual authentication] and [Diagnostic tool CDN mutual authentication] are disabled by default. The minimum required dependency is baseline, which is a mandatory field when the switch is enabled, defaults to empty, and supports selecting no dependency.



- Configuration: Staff configuration supports adding and deleting

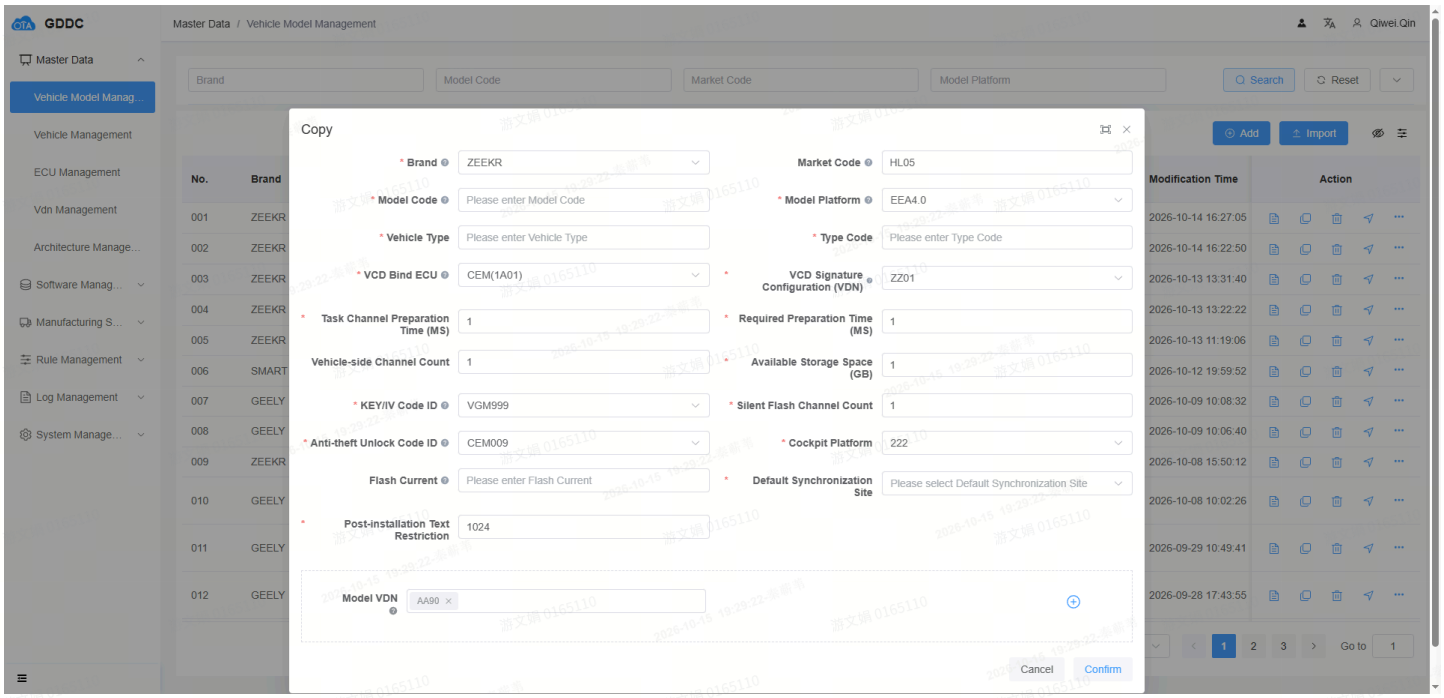


Configuration: Pre-check Configuration (This function menu is available for EEA 4.0 architecture and ZEEA 3.0 architecture, and is not available for other architectures for the time being)



1.1.3.5 Vehicle Model Replication

- Click Copy in the vehicle model copy operation bar. Essentially, the copy function is a shortcut entry for the "create" function, and its final submission logic is completely consistent with that of the create function.

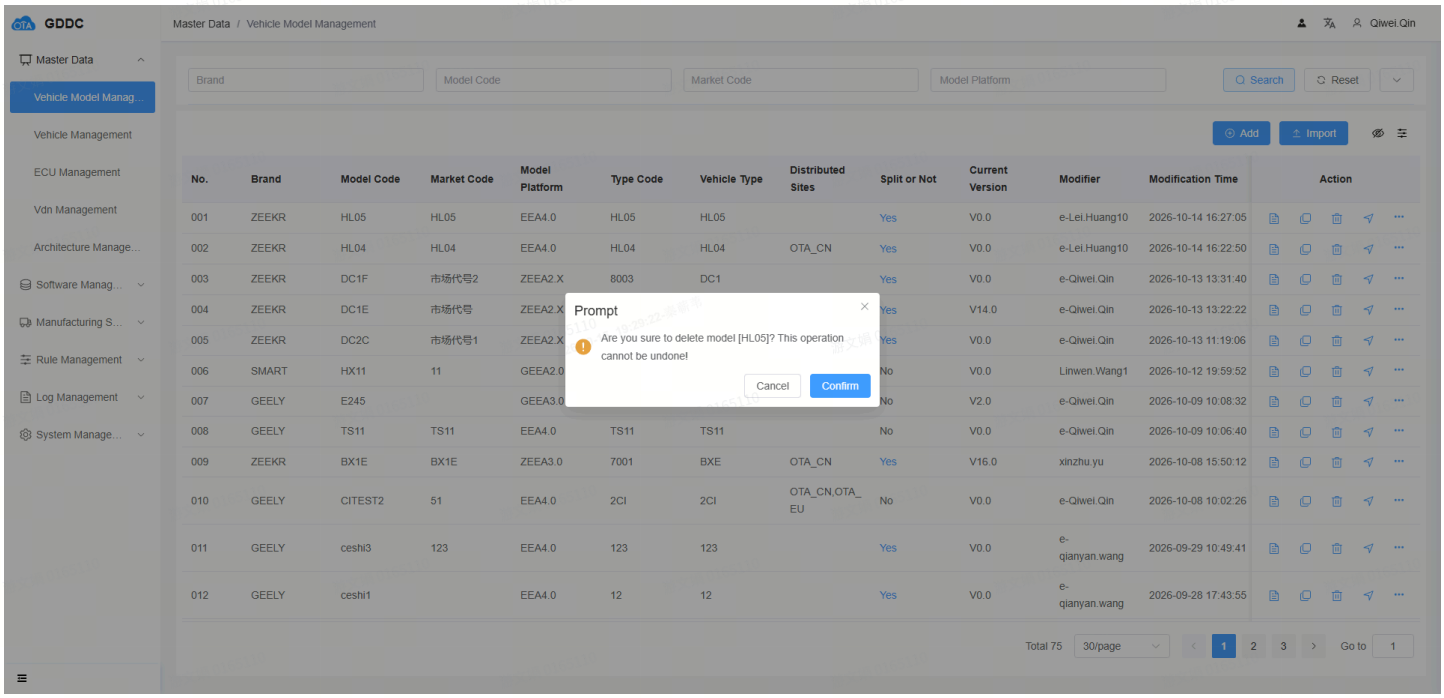


1.1.3.6 Model Deletion

- Vehicle models can be deleted, and deletion is irreversible
- Before performing the deletion action, check whether the vehicle model has been synchronized to any site worldwide, which shall be determined via the field [Distributed Sites]

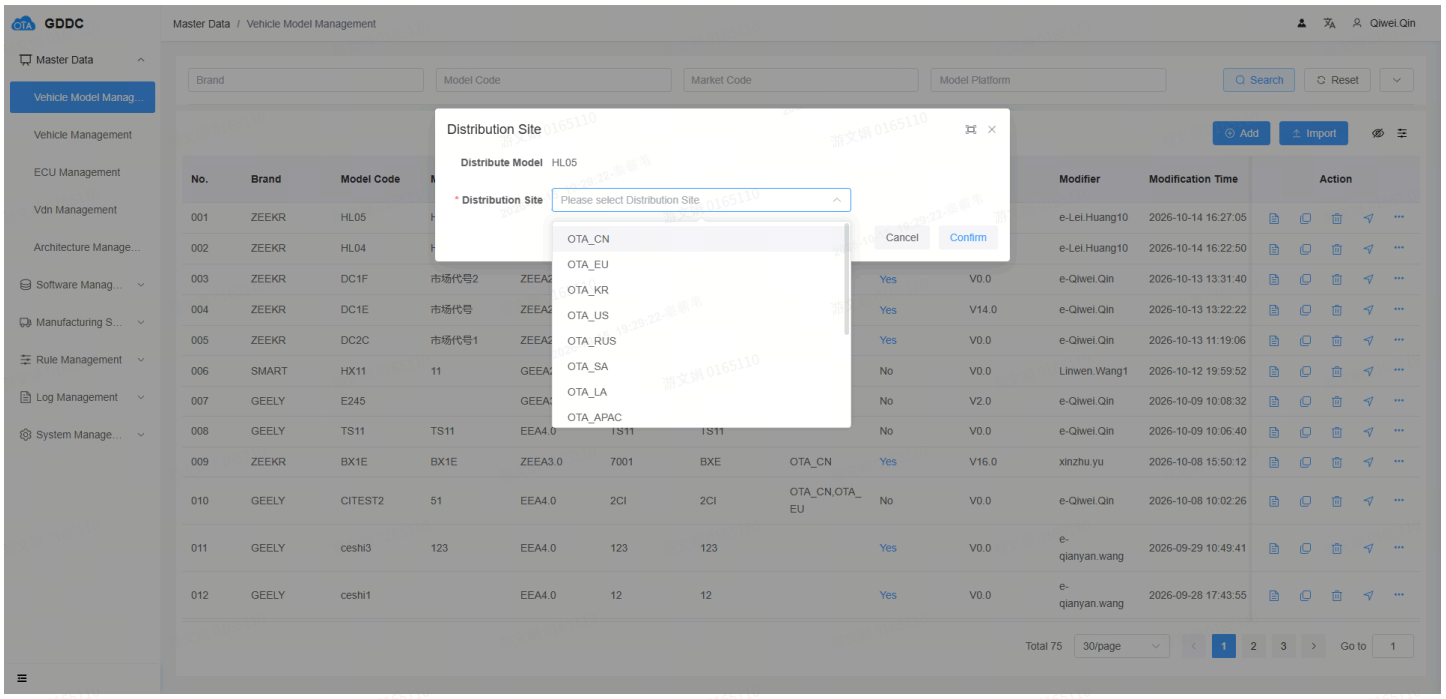
1. If not, submit a deletion application, and a vehicle model deletion approval task will be generated synchronously (which requires the approval of all vehicle model responsible persons)

2. Yes - Deletion is not allowed, and a prompt will be displayed: "Deletion failed. This vehicle model has been referenced and cannot be deleted. ";



1.1.3.7 Model Distribution

- GDDC maintains vehicle model Master Data, and real-time synchronization to all OTA server sites is required after addition, deletion or modification operations;
- Default Synchronization Site: You can configure and select a default distribution site when creating a vehicle model, and once the vehicle model data is approved, it will be automatically distributed to the corresponding site;
- Manual distribution selection: Users can manually select a vehicle model on the vehicle model management interface of the GDDC system, and distribute it to one or more OTA service systems;
- Distribution button display: The distribution button is only displayed when the release status is "Approved", and will not be displayed for statuses of "Pending Approval", "Under Approval" and "Rejected";
- Distribution Verification: Before the user clicks the "Distribute" button, the system shall verify the release status of this verification, as well as the completeness (e. g., whether required fields are filled) and validity (e. g., whether the data complies with the rules) of the vehicle model data. If the data is invalid, the system shall prompt the user and terminate the operation.
- During synchronization, mandatory field validation is required. Synchronization is not allowed if any mandatory field is empty, and a prompt will be displayed: "There are unconfigured mandatory fields for the vehicle model. Please complete the configuration before synchronizing. "



1.1.3.8 Factory Administrator

- Plant administrators are supported to add and delete

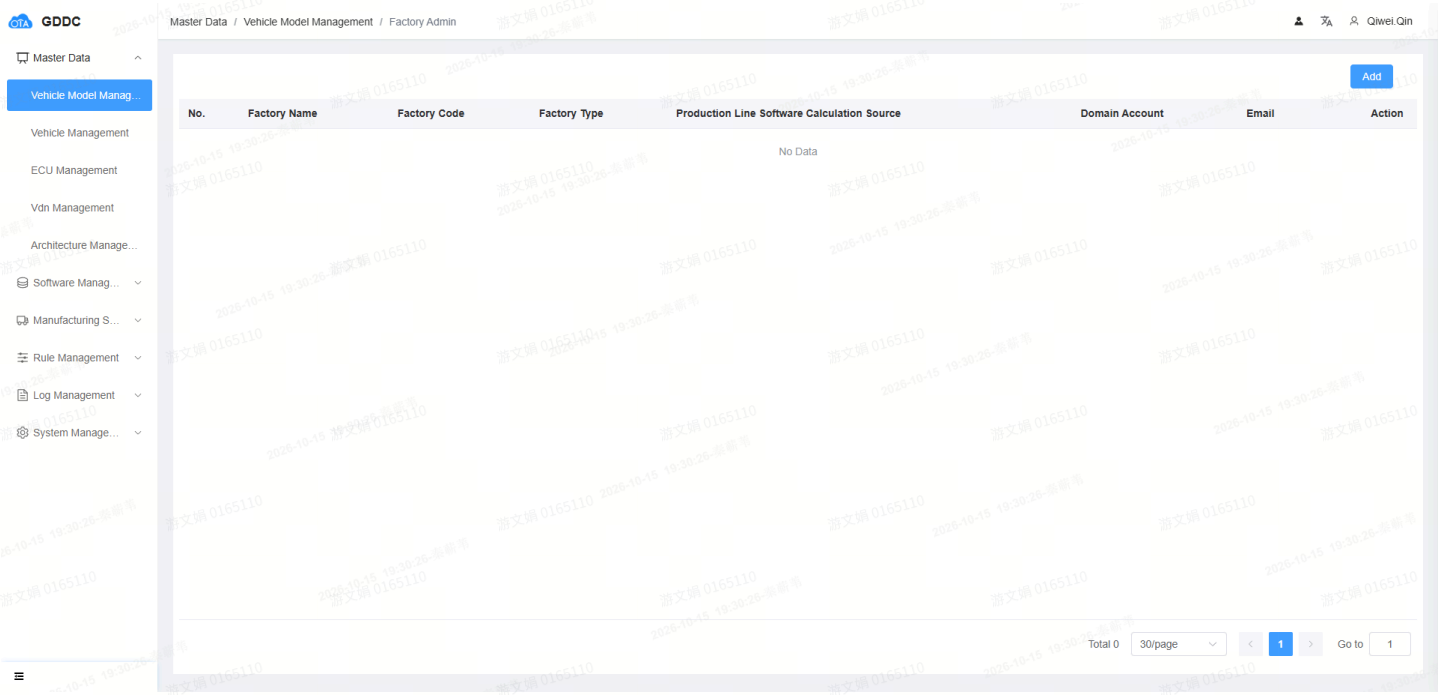
1. The function list is displayed by administrator, i. e., one entry per administrator;

2. The plant name, plant code, plant type and the settlement source of production line software are all synchronized from the plant configuration in the vehicle model configuration, and will be updated and synchronized in real time once there is an update.

3. Add - Factory Code: Required field, no default value, factory codes can be selected, the selectable factory codes must fall within the scope configured in the vehicle model factory settings, and multiple factory codes are supported, meaning one user can be set as the administrator for multiple factories at a time;

4. Add - Domain Account/Email: Required field, no default value, only one entry allowed at a time, and validation will be performed to check whether the domain account is valid;

5. Delete: A secondary confirmation is required for this operation.



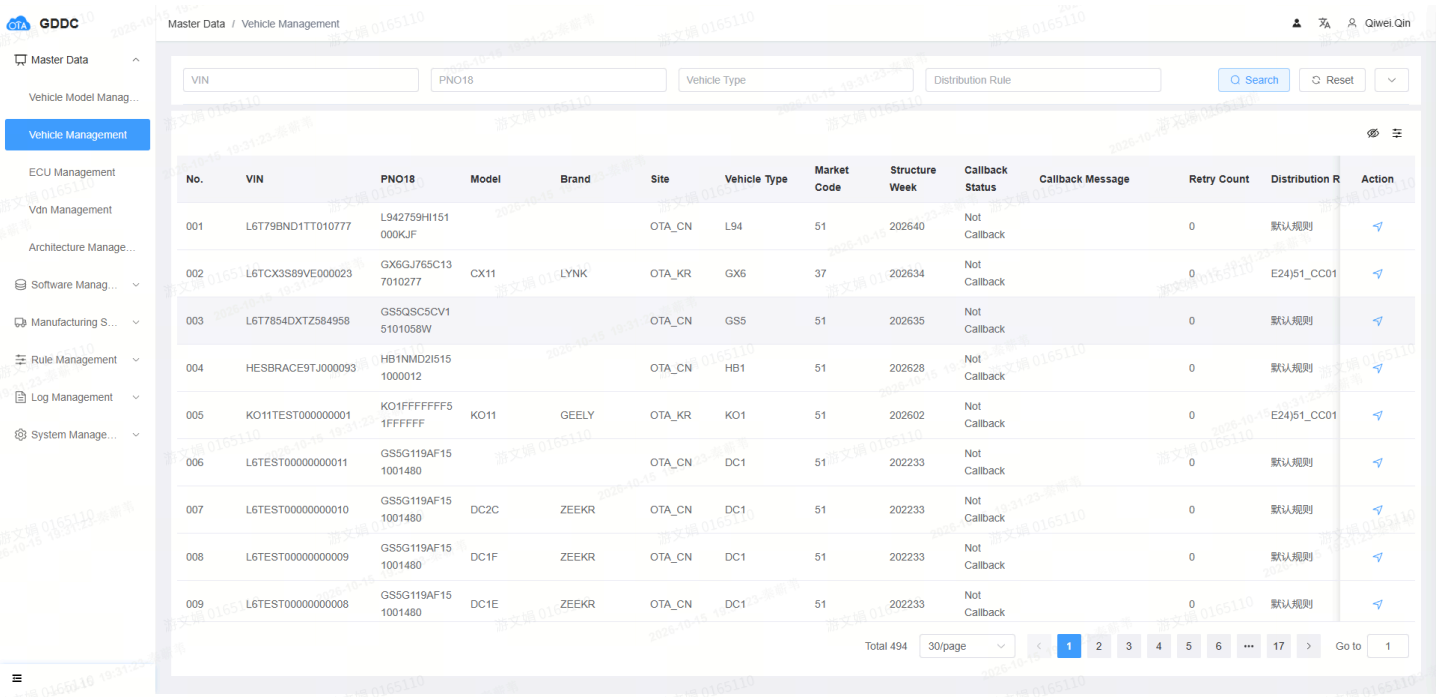
1.2 Vehicle Management

1.2.1 Function Overview

After a vehicle leaves the factory via the MES system, FC data is automatically pushed to the GDDC system. Upon receiving the MES data, the GDDC system parses the basic information, matches it against the [Vehicle Distribution Rules], and then distributes the data to OTA sites as well as stores it in the GDDC repository.

1.2.2 Function Entry

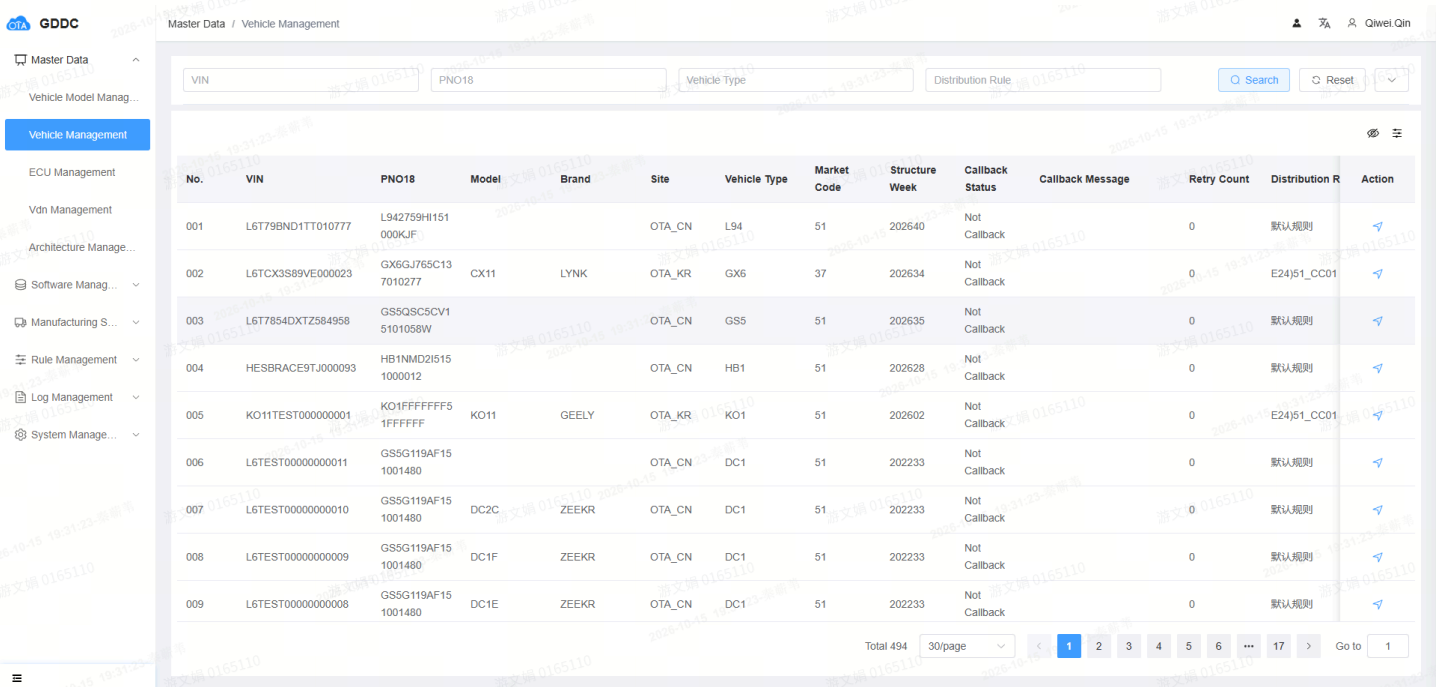
- Navigation Bar: Master Data Management >> Vehicle Management



1.2.3 Vehicle List

1.2.3.1 Vehicle Inquiry

The vehicle query function supports filtering by VIN, PNO18, Vehicle Type, site, market code, callback status, data source and change time. The query bar can be expanded and collapsed; when collapsed, only four query criteria including VIN, PNO18, Vehicle Type and site are displayed. Reset: Restore the query bar to its default state (empty by default).



1.2.3.2 Vehicle Distribution

- The system automatically triggers a distribution decision based on the preset priority of the [Vehicle Distribution Rules], ensuring that vehicles are correctly routed to the target OTA site.

- Manual distribution triggers distribution decisions based on the preset priority of [Vehicle Distribution Rules]. For details of [Vehicle Distribution Rules], refer to Section 4.1.

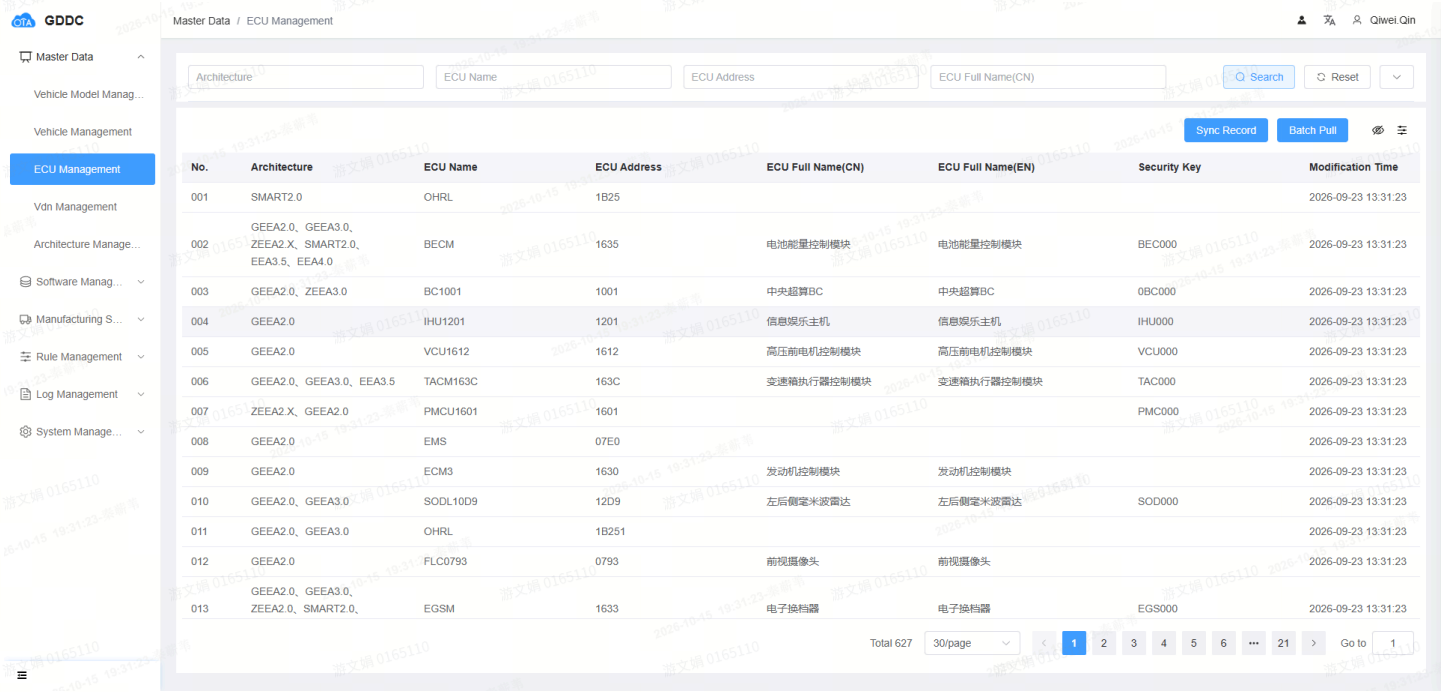
1.3 ECU Management

1.3.1 Function Overview

This function is designed to synchronize the global ECU basic data (ECU NAME, ECU ID) from the TC system to the GDDC system, so as to form an ECU database that is both globally unified and locally applicable, and provide data support for other businesses (such as vehicle model configuration).

1.3.2 Function Entry

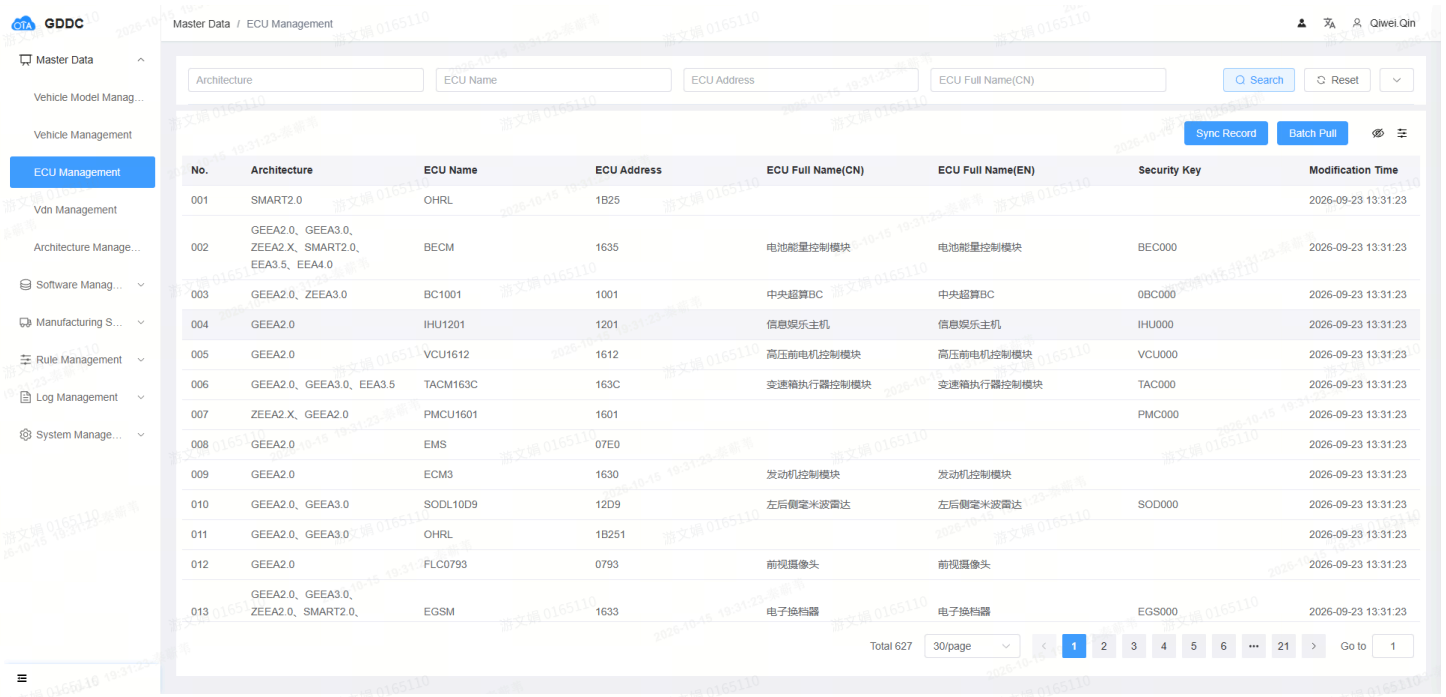
- Navigation Bar: Master Data Management >> ECU Management



1.3.3 ECU List

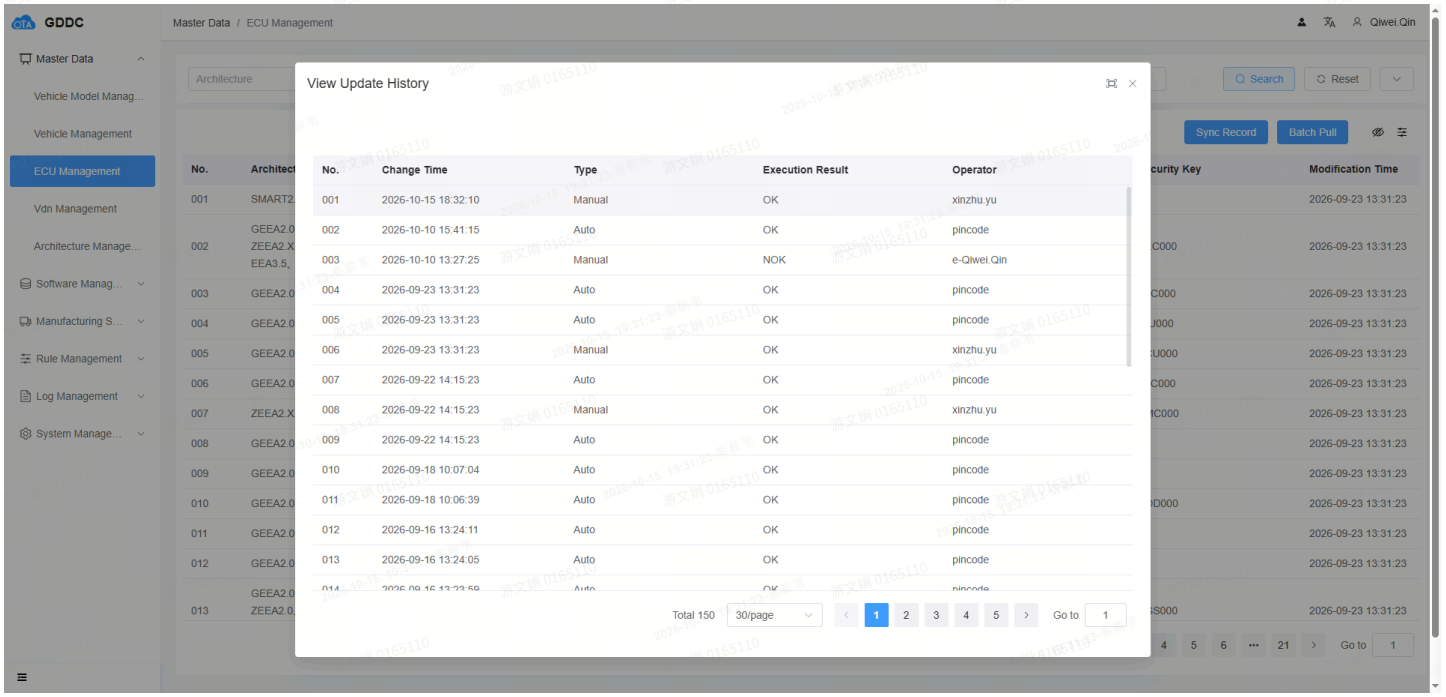
1.3.3.1 ECU Query

The ECU supports query by architecture (fuzzy match), ECU name, ECU address, full Chinese name of ECU, full English name of ECU, and change time. The query bar can be expanded and collapsed; when collapsed, only four query criteria are displayed: architecture (fuzzy match), ECU name, ECU address, and full Chinese name of ECU. Reset: Reset the query bar to the default state (empty by default).



1.3.3.2 Synchronous Recording

- Show all sync records



1.3.3.3 Batch fetch

- Manual synchronization: The GDDC system provides a [Batch Pull] button for administrators to trigger manually when needed.
- Automatic synchronization: The pincode center will identify ECU changes, and when the ECU information changes, it will call the push interface to synchronize the ECU data to GDDC;

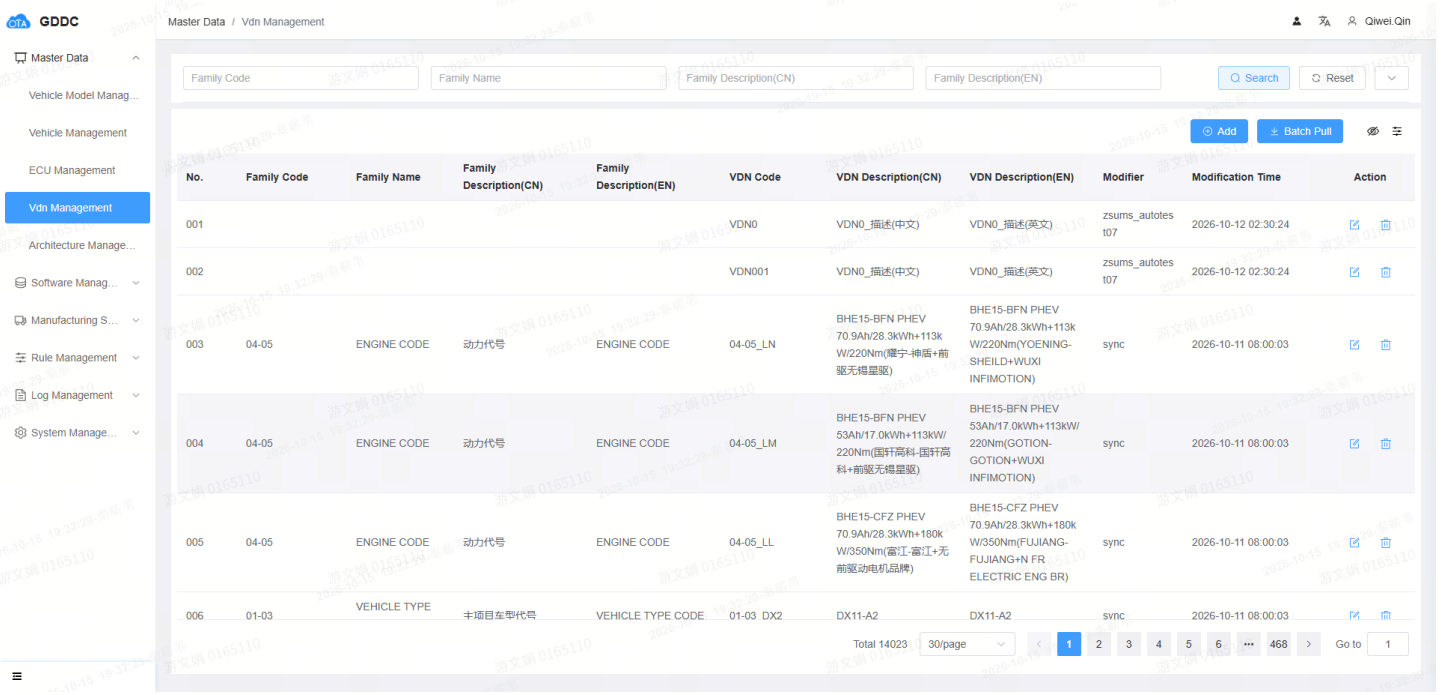
1.4 VDN Management

1.4.1 Function Overview

Integrate with the GBOM system to obtain global EBOC data: the new data in this table is mainly sourced from the regular push of the GBOM system;

1.4.2 Function Entry

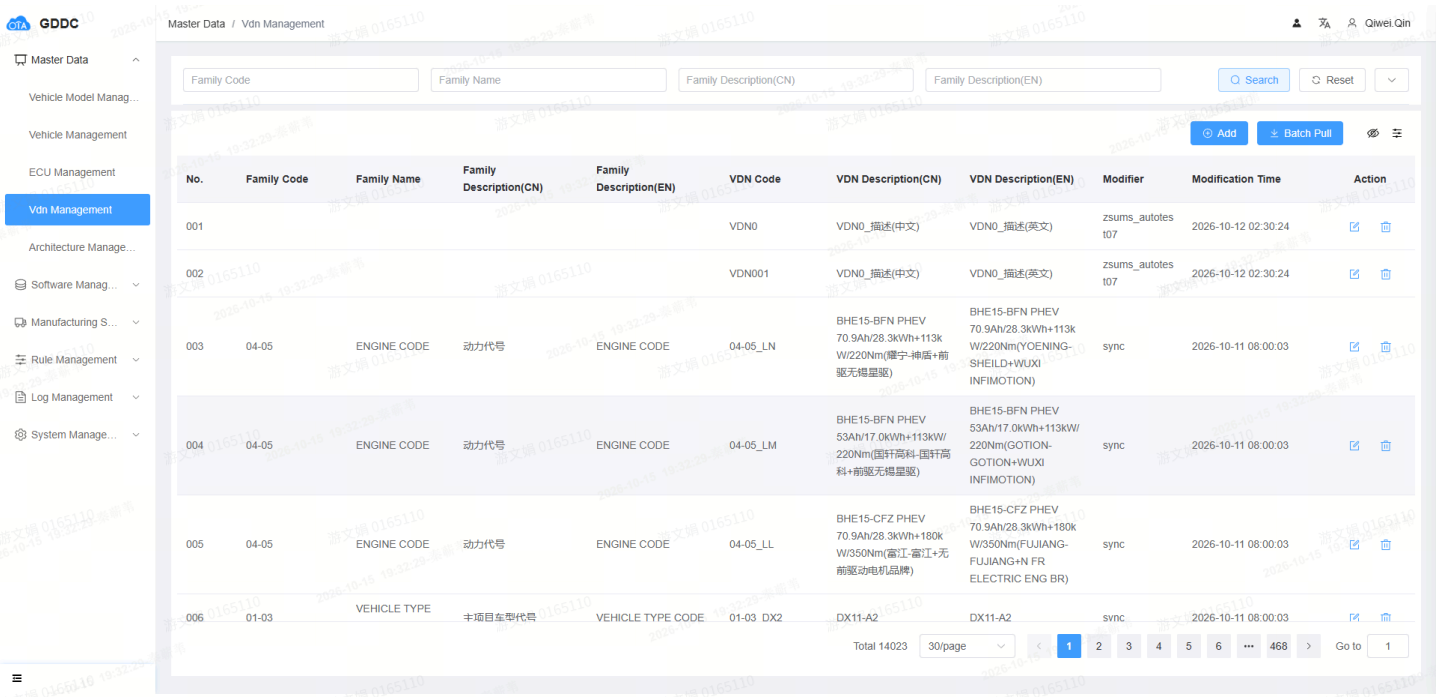
- Navigation Bar: Master Data Management >> VDN Management



1.4.3 VDN List

Operating Instructions:

- Newly added VDN: Manual VDN addition is supported, and the added VDN can be pulled and used by downstream parties; if the family code of the newly added VDN already exists, only the existing family code can be used.
- Batch Pull: Supports managers to manually pull VDN in batches;
- Edit: Supports editing of VDN codes and both Chinese and English descriptions
- Deletion: Deleting VDN is now supported, and a secondary confirmation is required



Synchronization Strategy:

- Full automatic synchronization: Automatically executed once at 00:00 every day to ensure the consistency of basic data.
- Conflict Handling: The GBOM system serves as the authoritative data source. In case of data conflicts detected during synchronization, the data from the GBOM system shall be taken as the sole standard.

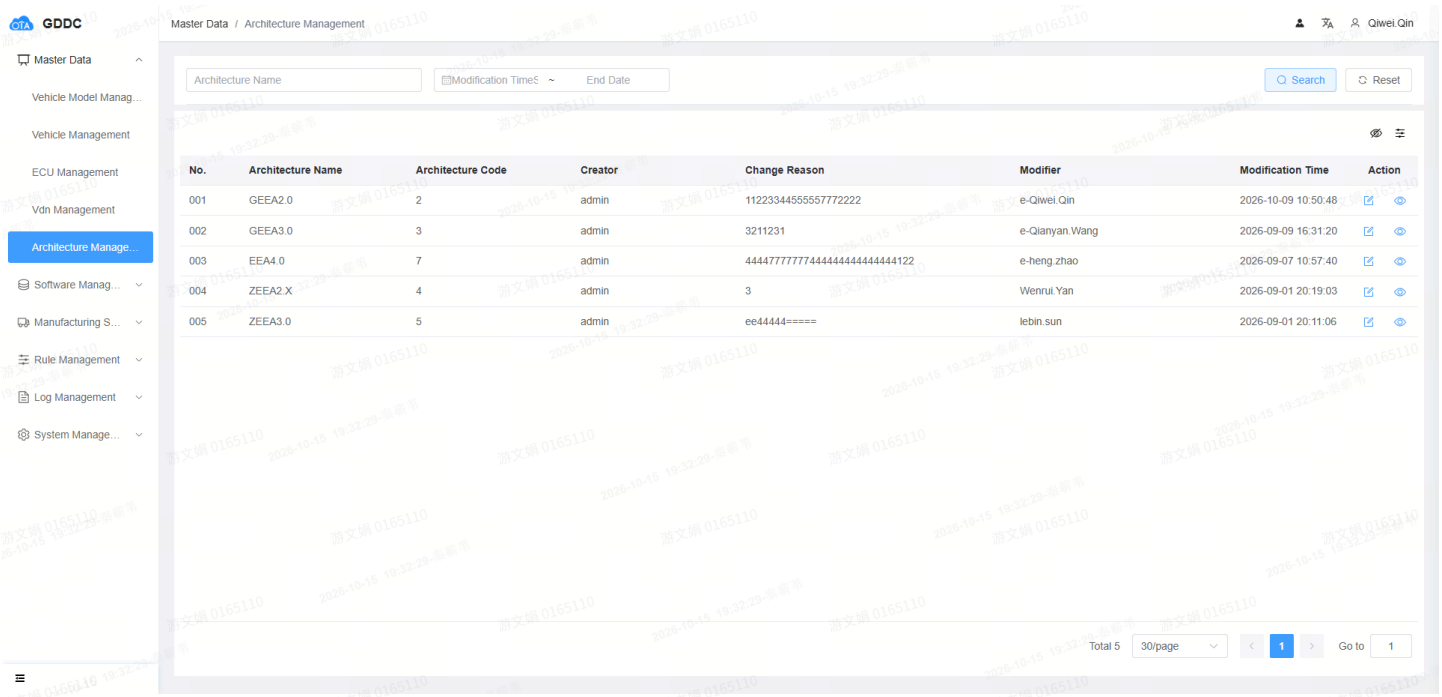
1.5 Architecture Management

1.5.1 Function Overview

The GDDC platform defines the architecture for all vehicle models, supports the configuration of APP push status, and supports the configuration of F18C verification.

1.5.2 Function Entry

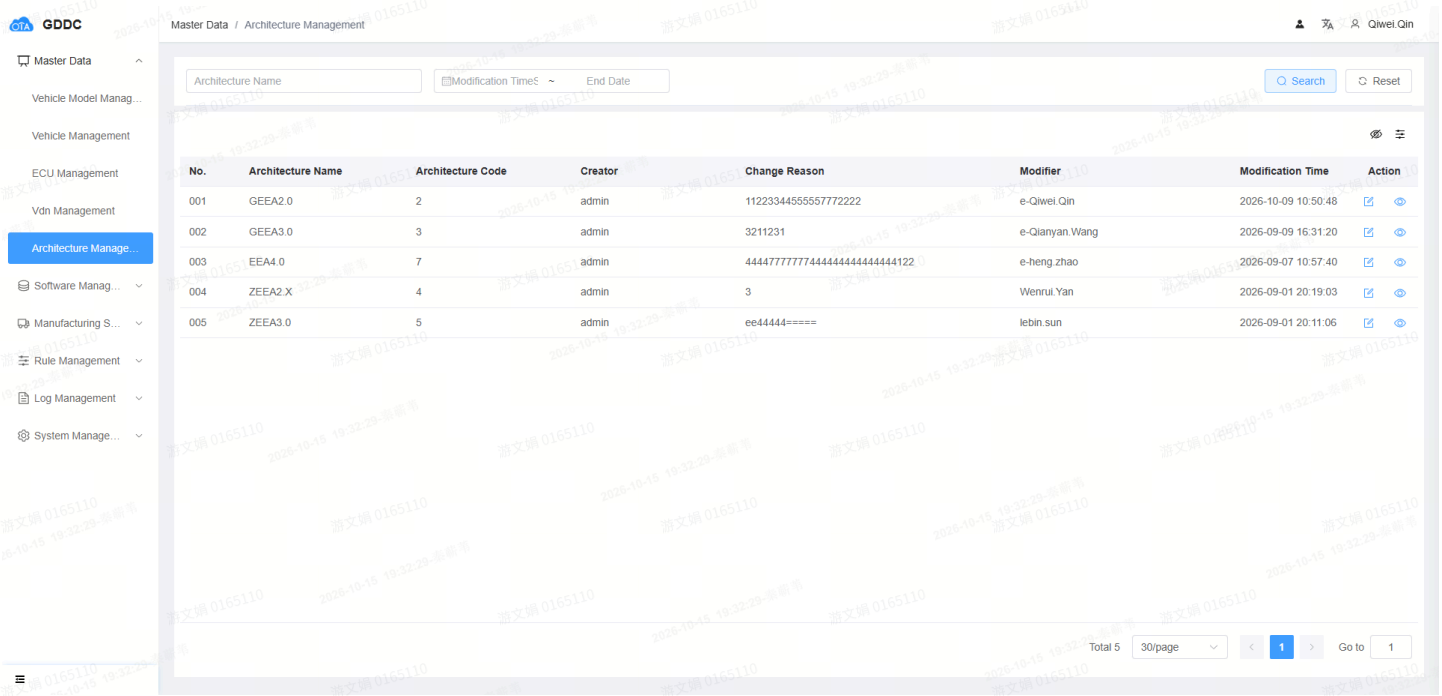
- Navigation Bar: Master Data Management >> Architecture Management



1.5.3 Architecture List

Operating Instructions:

- Query Architecture: Supports querying by architecture name and change time
- Edit: Support for configuring APP push status, and support for configuring F18C verification;



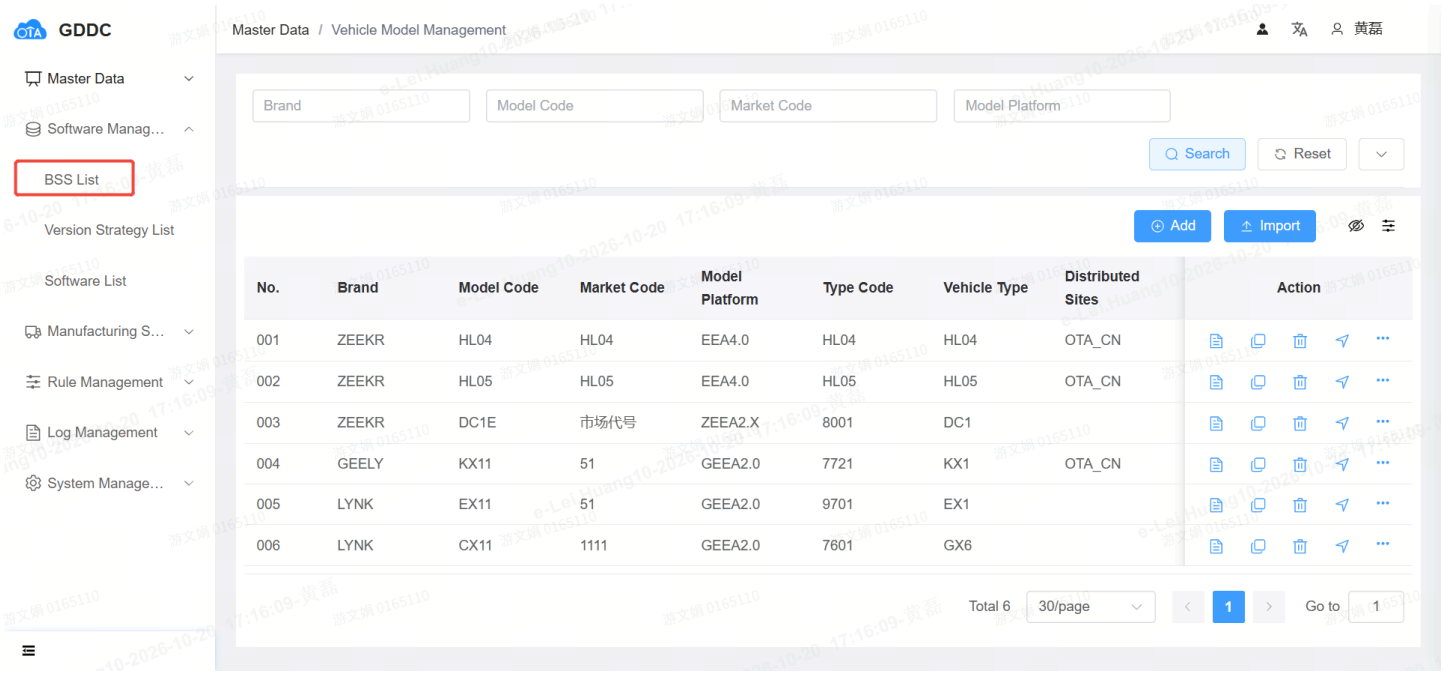
2. Software Management

2.1 BSS List

2.1.1 Function Introduction

2.1.2 Entrance

Menu > Software Management > BSS List



2.1.3 precondition

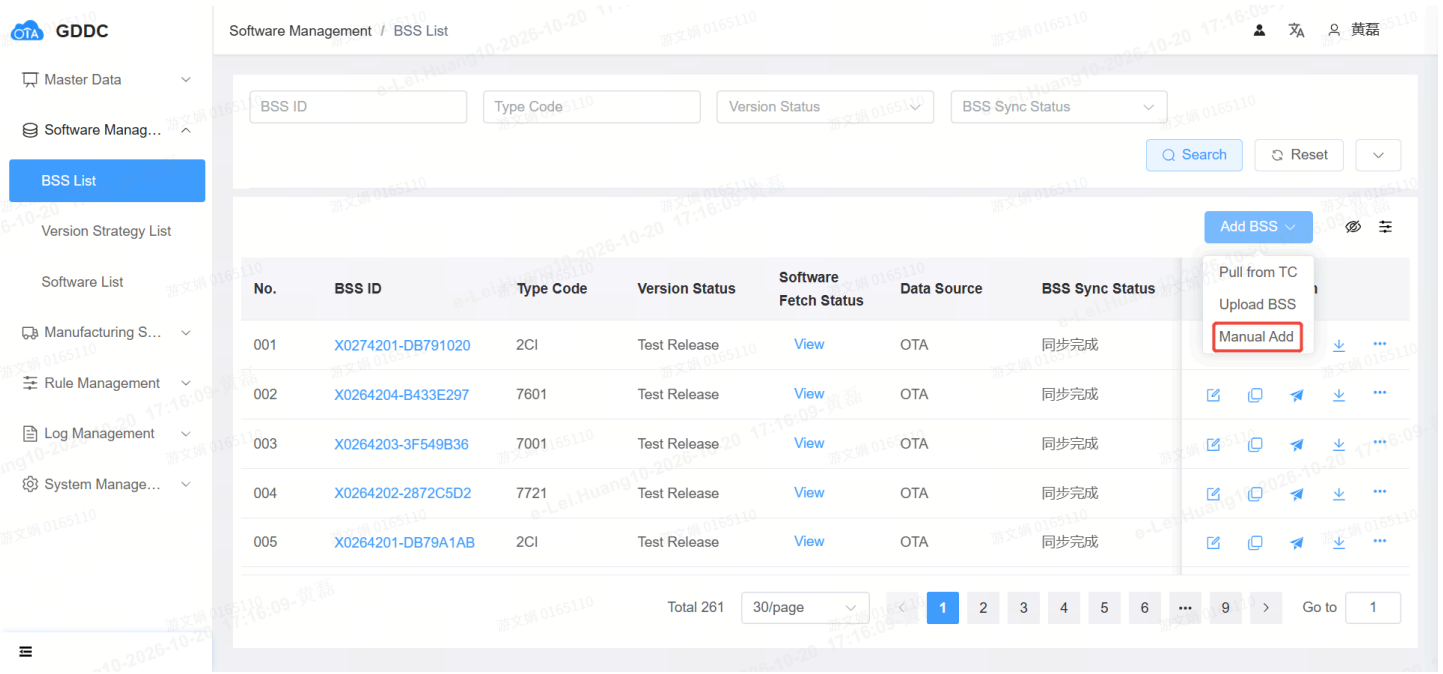
2.1.4 Operation

2.1.4.1 BSS List - Query

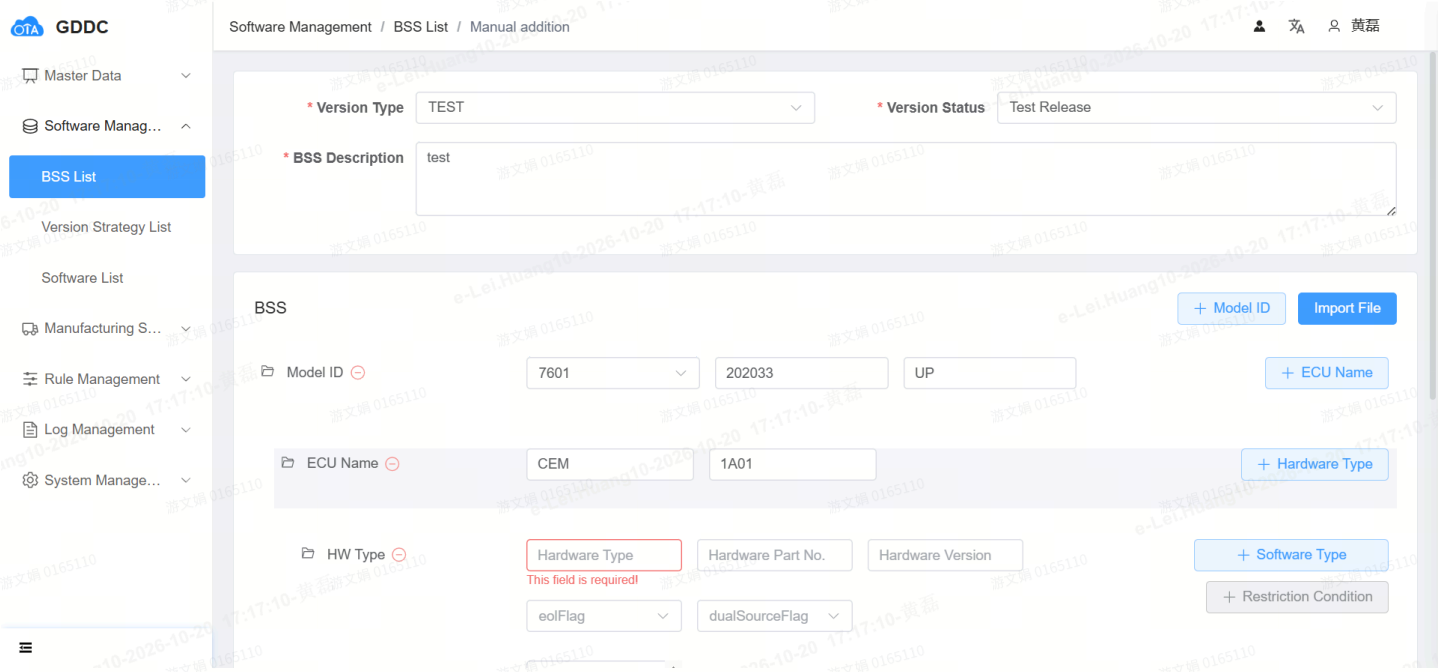
BSS supports query by BSSID, Type Code, version status, BSS synchronization status, and start and end time. The query bar supports expansion and collapse; when collapsed, only four query criteria including BSSID, Type Code, version status, and BSS synchronization status are displayed. Reset: Reset the query bar to the default state (empty by default).

2.1.4.2 BSS List - Manual Addition:

Function Entry: Menu >> Software Management >> BSS List >> Add New BSS >> Addition Method: Manual Addition



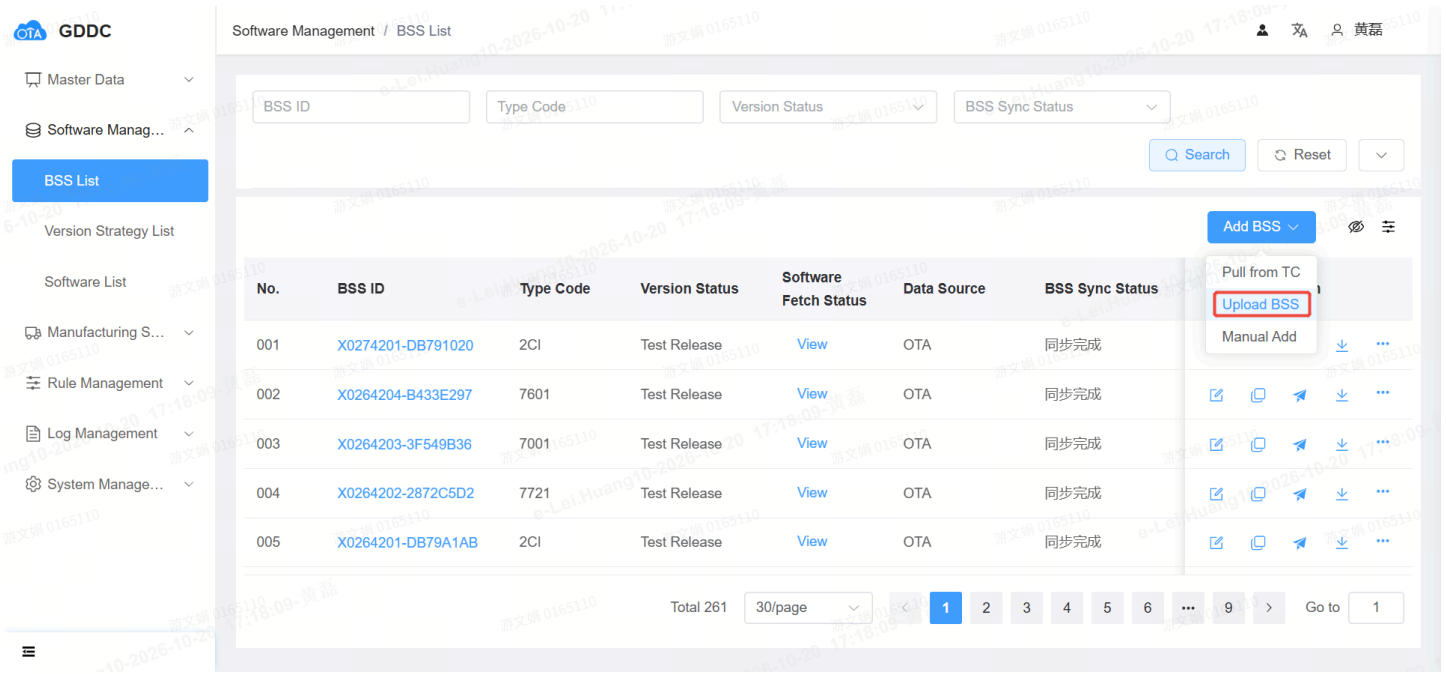
Manually create the BSS page:



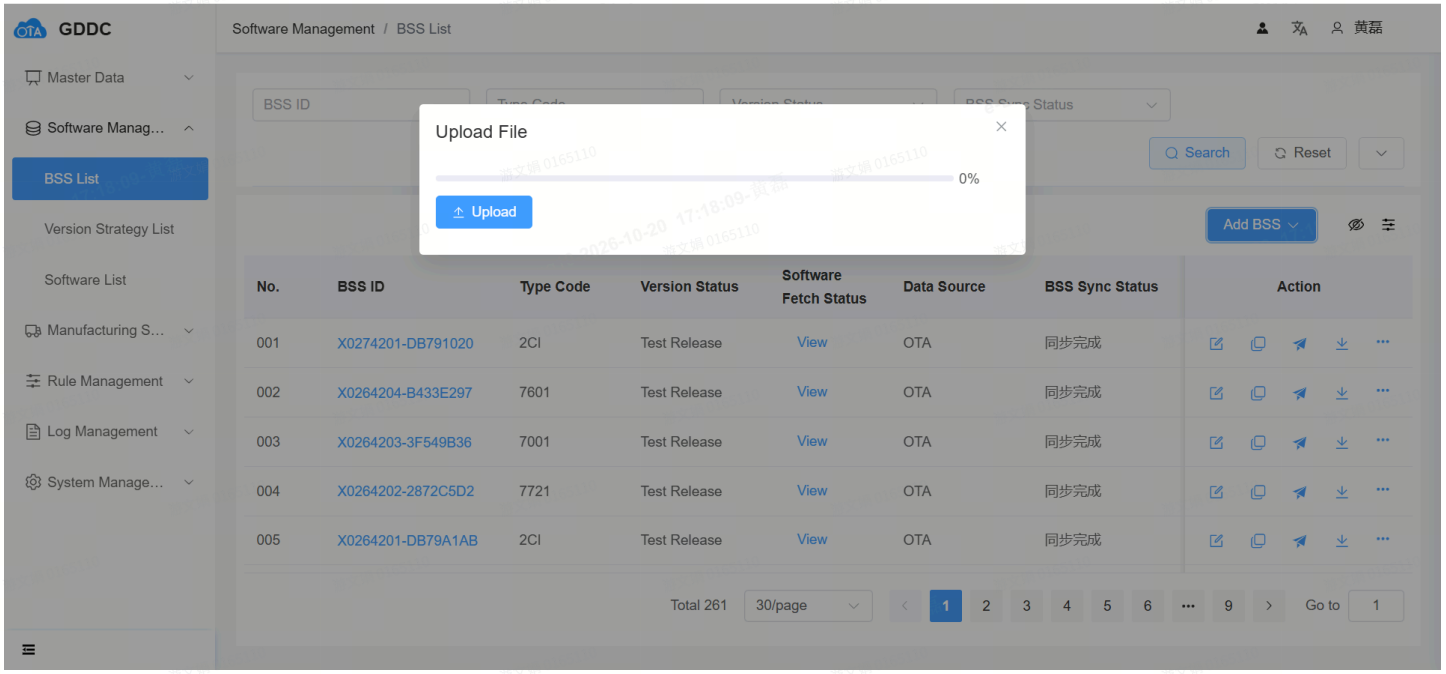
Just fill in the data on this page with reference to relevant information

2.1.4.3 BSS List - Upload BSS:

Function entry: Menu >> Software Management >> BSS List >> Add New BSS >> Addition Method: Upload BSS



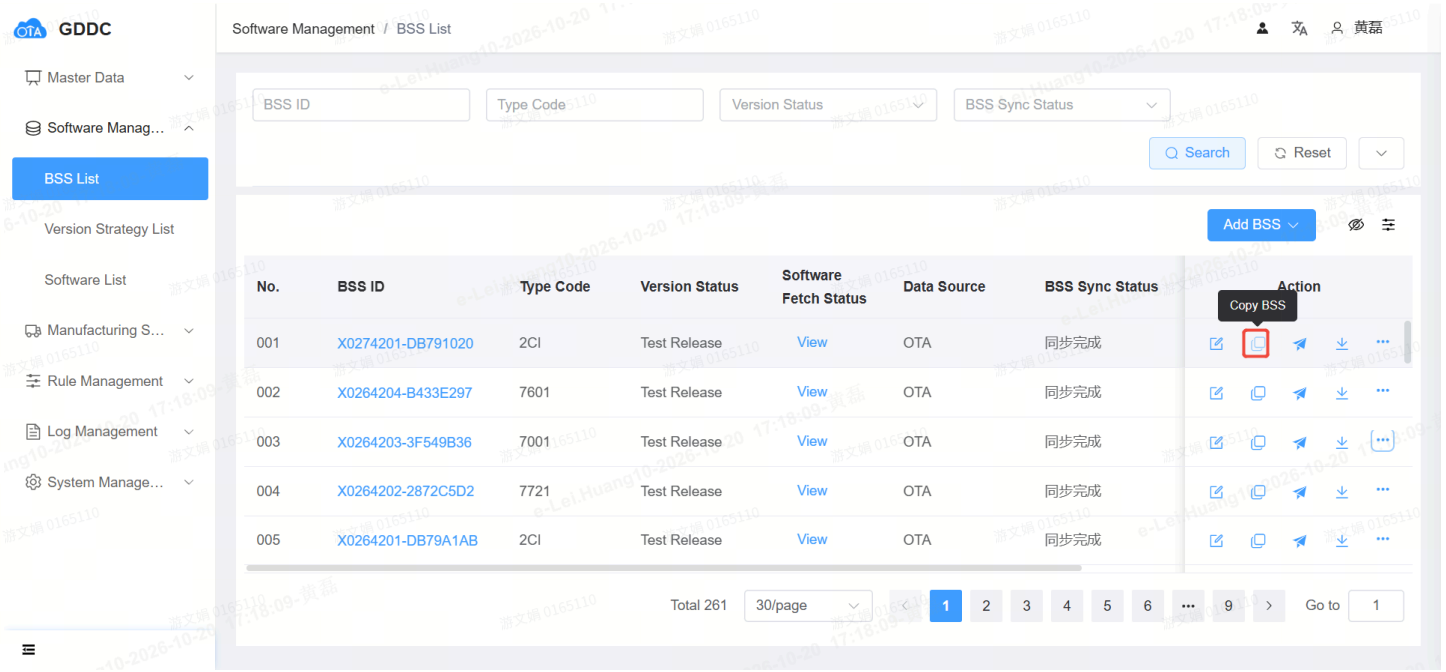
Upload BSS file:



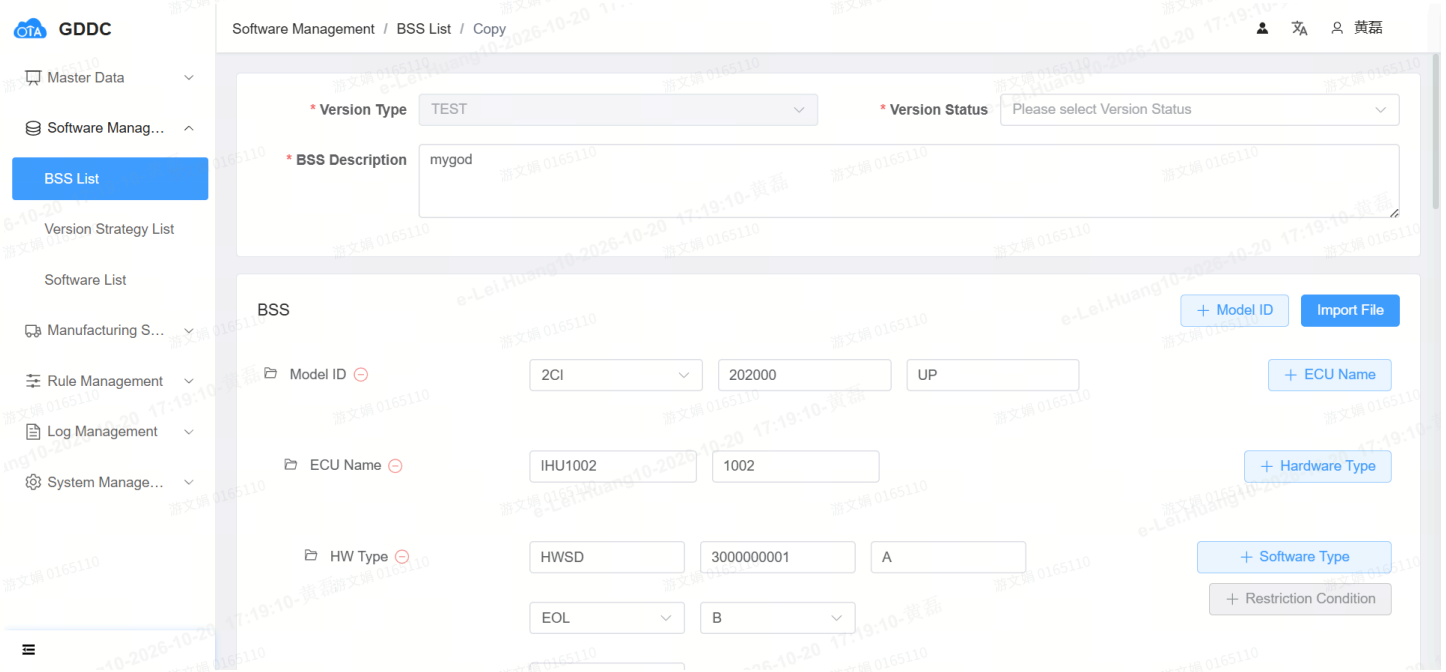
2.1.4.4 BSS List - Copy BSS:

Function Overview:

Function Entry: Menu >> Software Management >> BSS List >> Operation Column >> Copy BSS

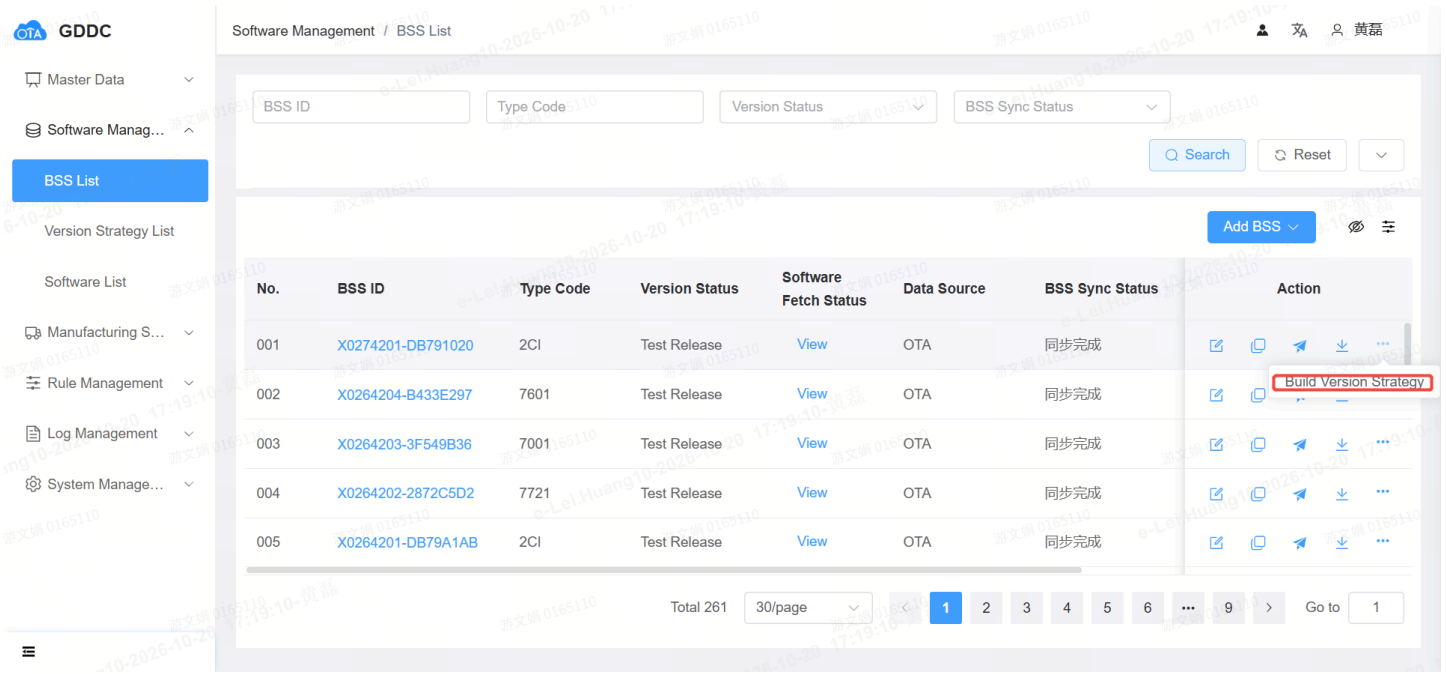


Copy the BSS edit page:



2.1.4.5 BSS List - Build Version Strategy:

- Function Entry: Menu >> Software Management >> BSS List >> Operation Column >> Build Version Policy



- Build Version Policy Edit Page:
- Basic Information:



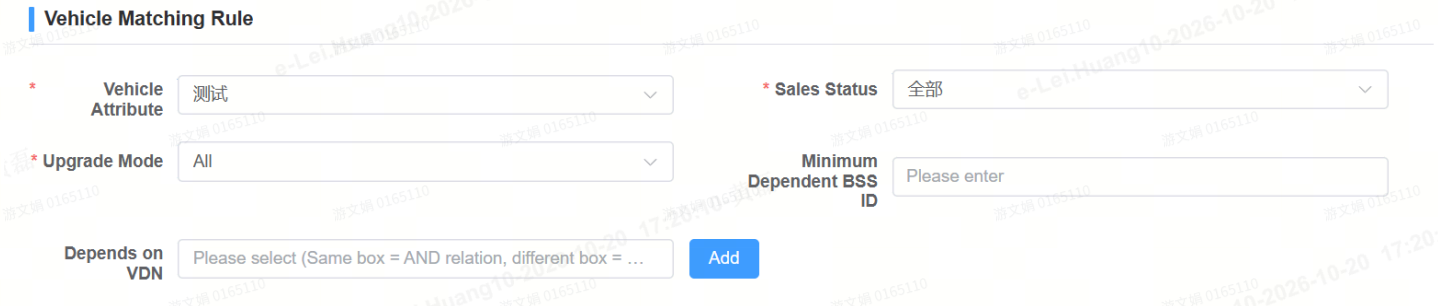
BSSID: The BSSID corresponding to this BSS, which is grayed out and cannot be modified

Version Type: Data source is BSS, grayed out and unmodifiable

Version Status: Data source BSS, grayed out and uneditable

Public Display Version: Text Box, Required

- Vehicle Adaptation Rules:



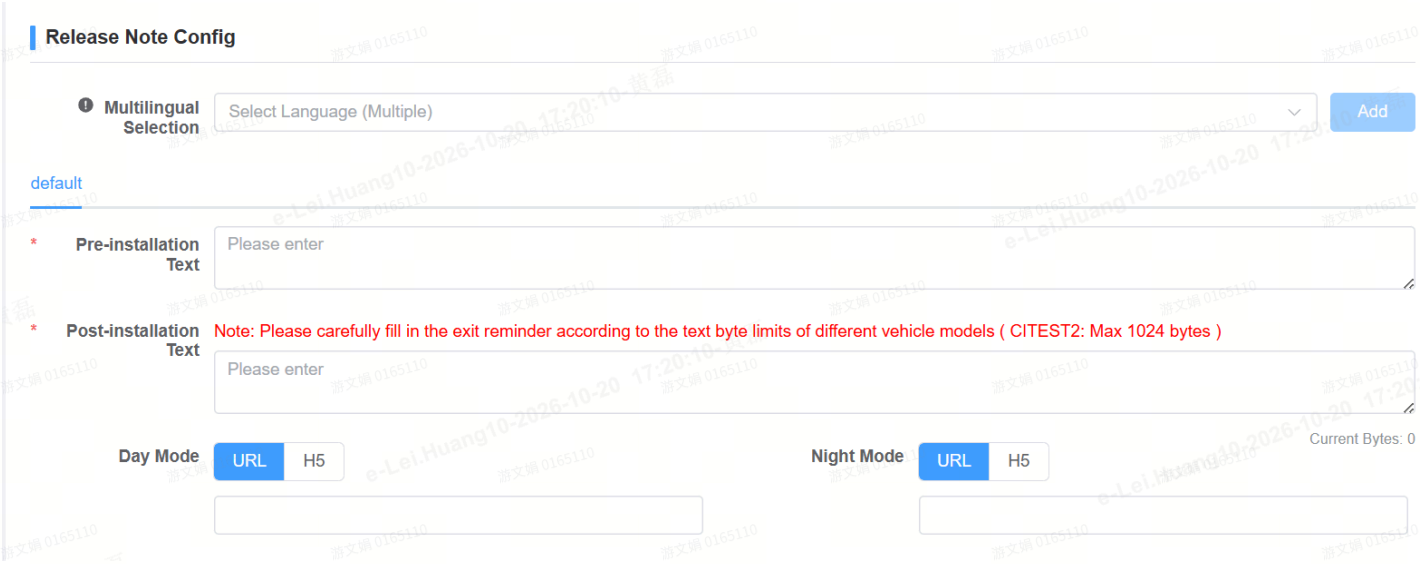
Vehicle Attribute: Dropdown menu, restricted based on the version status of BSS: Official := Market, MRD and test := Test, ENG: all options are selectable

Sales status: Dropdown menu, options include Sold, Unsold, All

Minimum dependent BSS ID: Text box, data source covers all BSS IDs with the same typecode; after selection, vehicles whose current BSS ID release time is earlier than that of the set minimum dependent BSS ID are not allowed to be set as vehicle targets and upgraded; if left blank, no restriction applies by default

VDN dependency: Drop-down menu, with GDDC as the data source. After selection and configuration, the vehicle's target VDN must meet this VDN dependency requirement to use this BSS for OTA upgrade.

- ReleaseNote Configuration (Geely):



Multilingual selection: Default by default, a required configuration, and can be added independently as needed

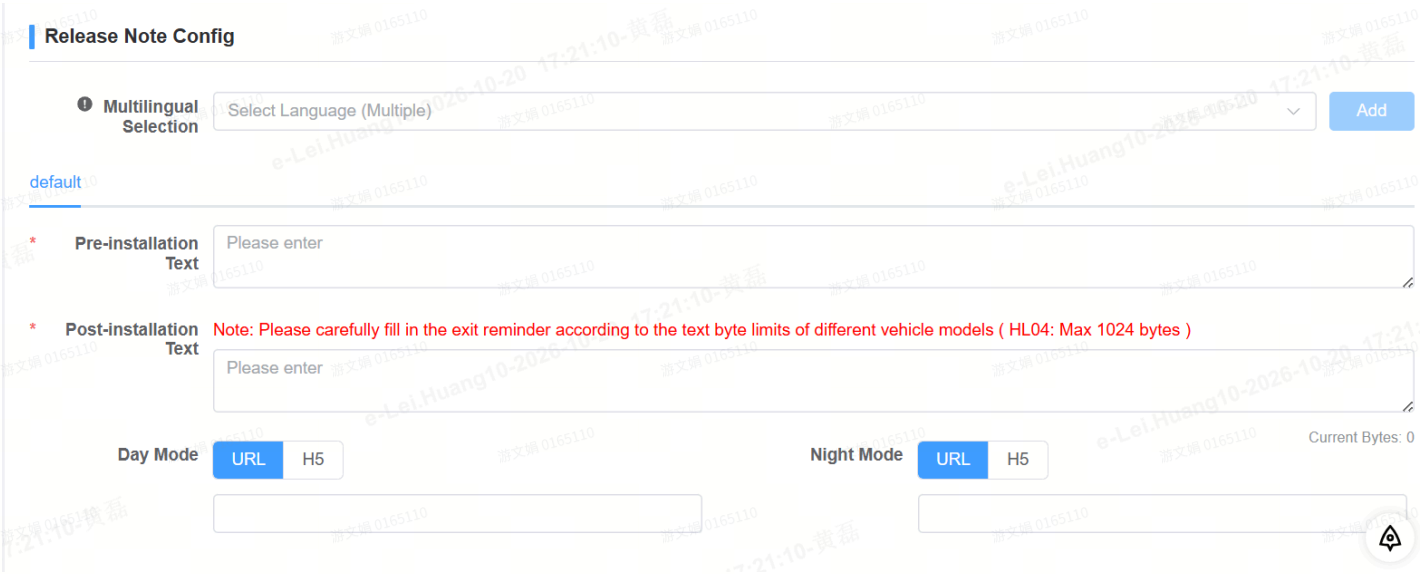
Pre-installation Note: Empty by default, mandatory configuration

Post-installation description: Empty by default, mandatory configuration

Day Mode: Empty by default, optional URL or H5

Dark Mode: Empty by default, optional URL or H5

- ReleaseNote Configuration (ZEEKR):



Multilingual selection: Default by default, a required configuration, and can be added independently as needed

Description: Required, empty by default

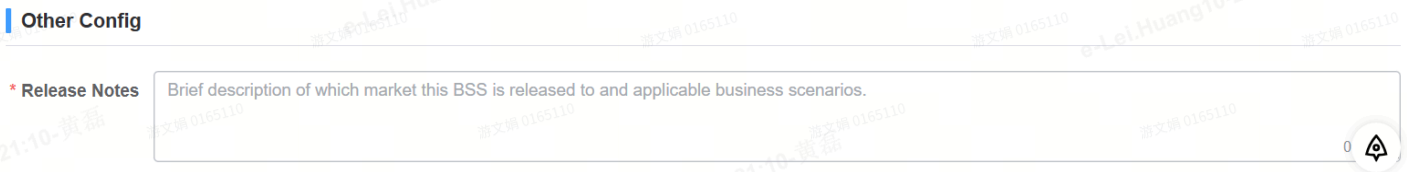
Upgrade failure prompt: Required, ZEEKR's BSS needs maintenance

Upgrade severe failure prompt: Required, Zeekr's BSS requires maintenance

New Feature Introduction: Empty by default, optional URL or H5 is supported

Upgrade feature introduction: default is empty, optional URL or H5

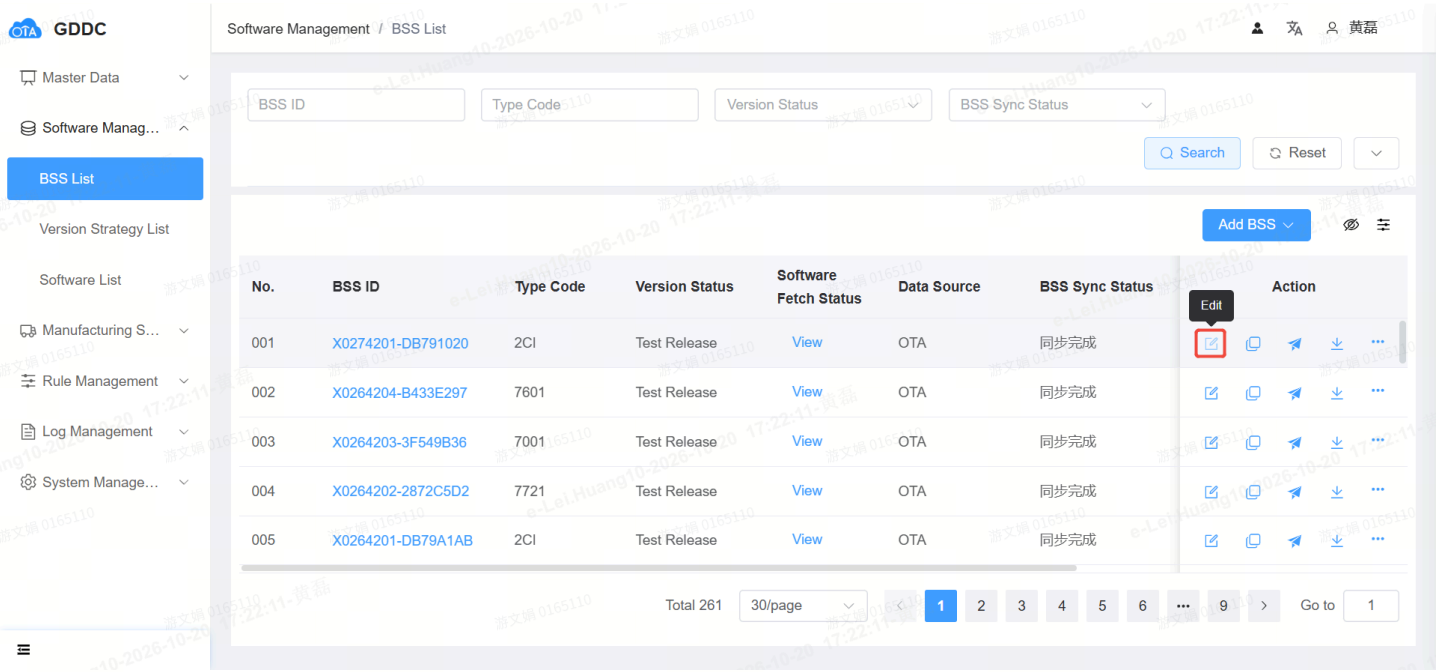
- Other configurations:



Release Notes: Text box, required

2.1.4.6 BSS List - Edit

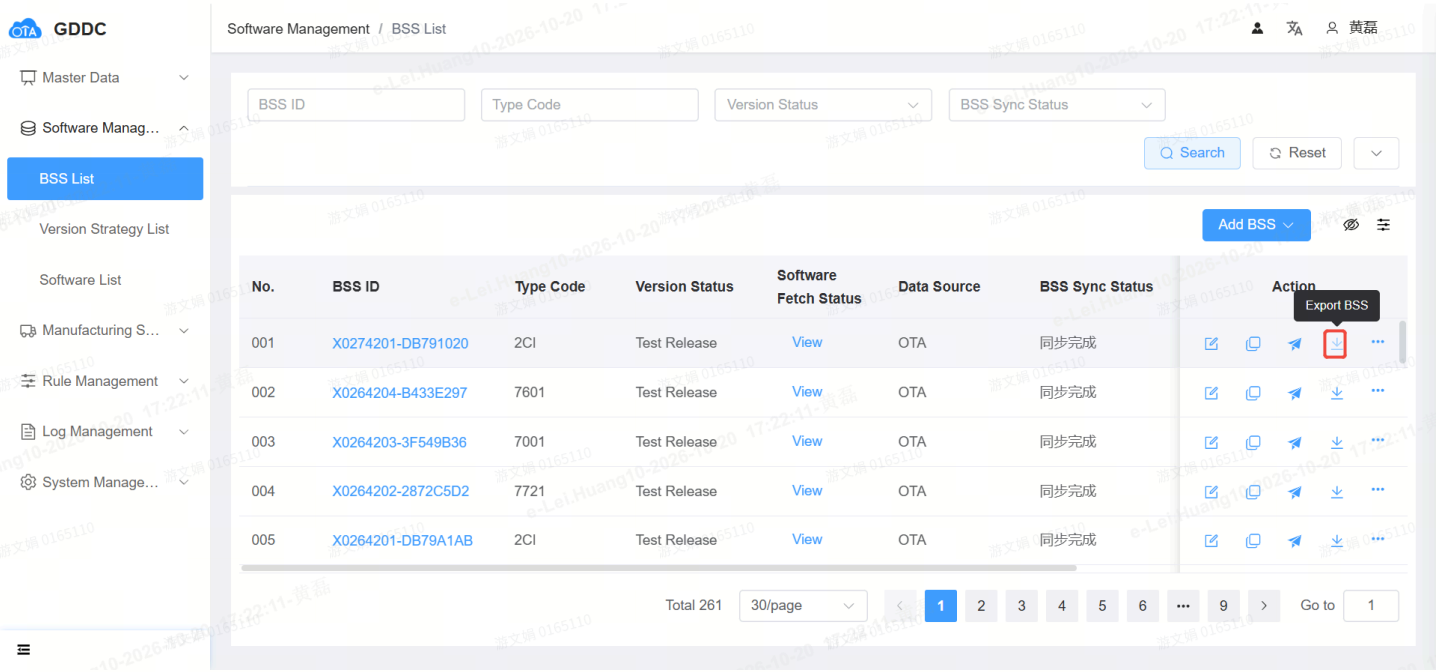
- Function entry: Menu >> Software Management >> BSS List >> Operation Column >> Edit



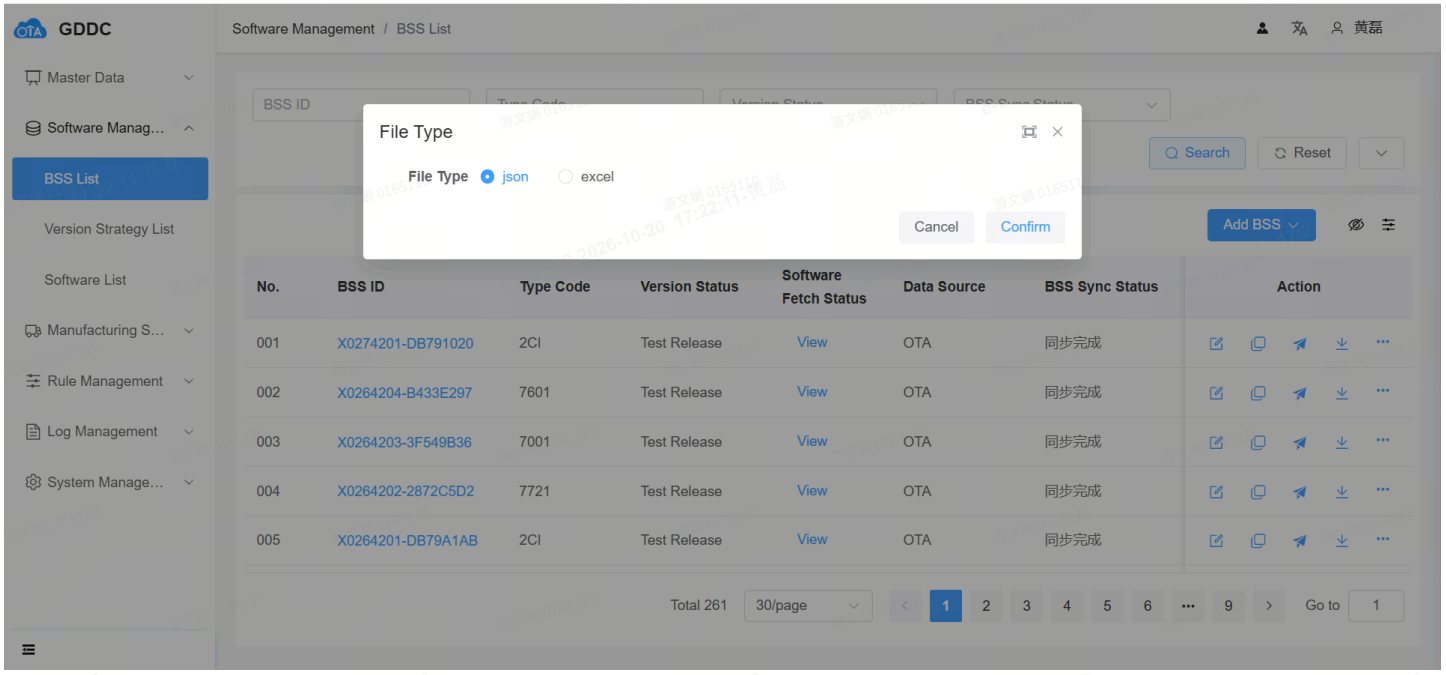
- Editing Instructions: Editing is only allowed when the content is pending release; all information is editable, and the StratId generated from multiple edits will not be regenerated.

2.1.4.7 BSS List - Export

- Function entry: Menu >> Software Management >> BSS List >> Operation Column >> Export



- Export File Type Selection:

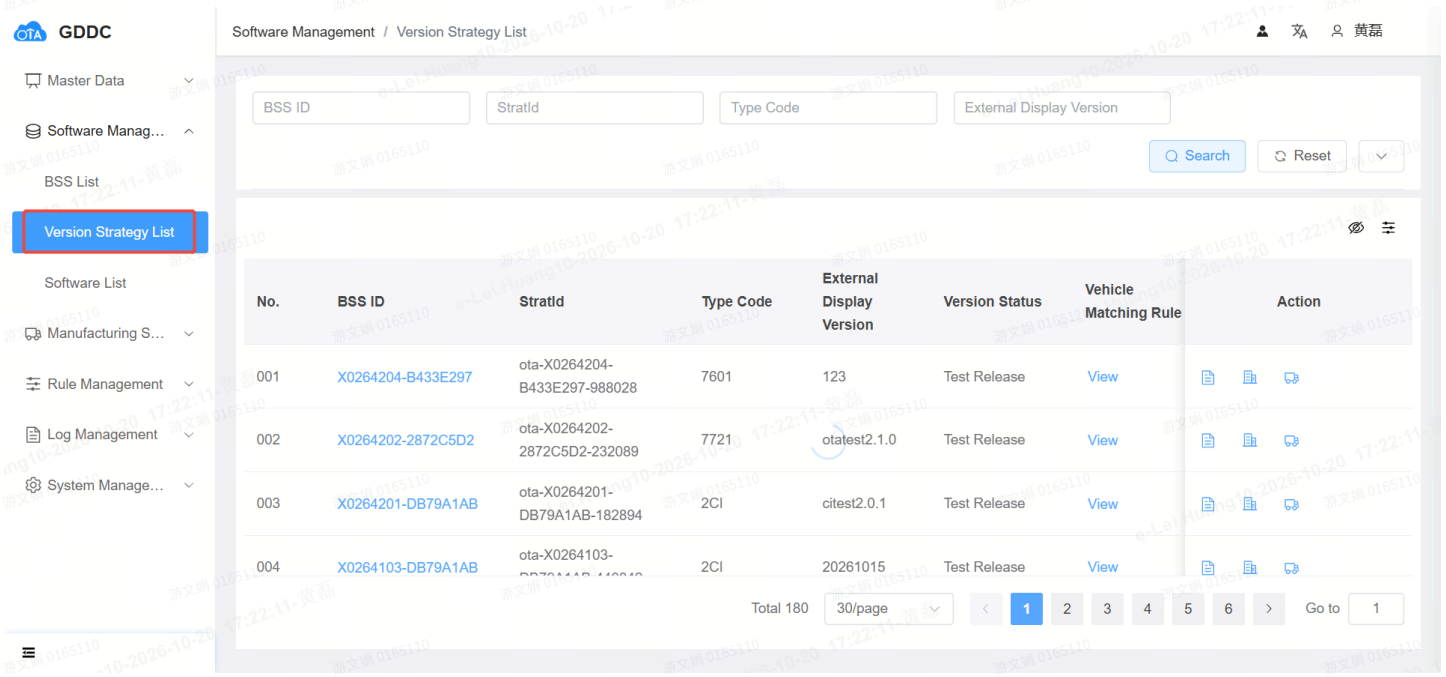


2.2 Version Policy List

2.2.1 Function Introduction

2.2.2 Entrance

Menu > Software Management > Version Policy List



2.2.3 precondition

The data in the list can only be generated for BSSs that have already had version policies configured from the BSS list.

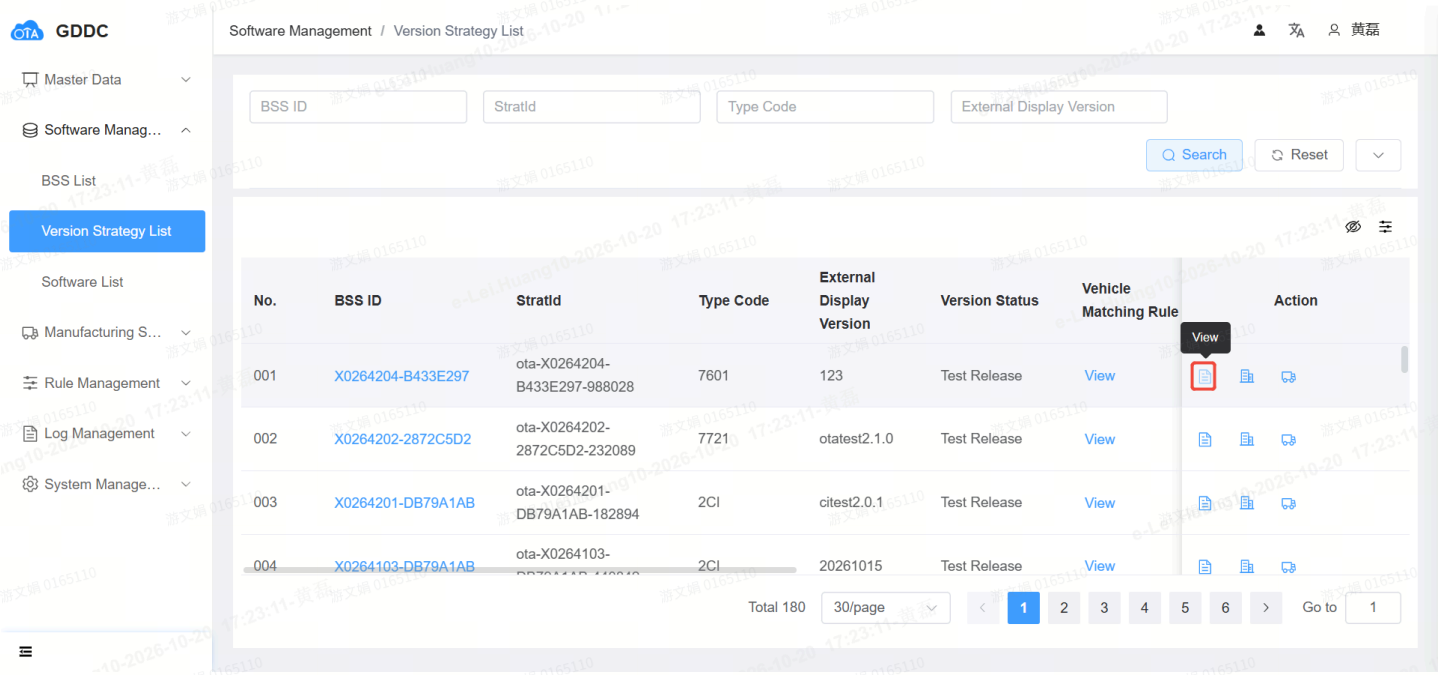
2.2.4 Operation

2.2.4.1 Version Policy List - Query

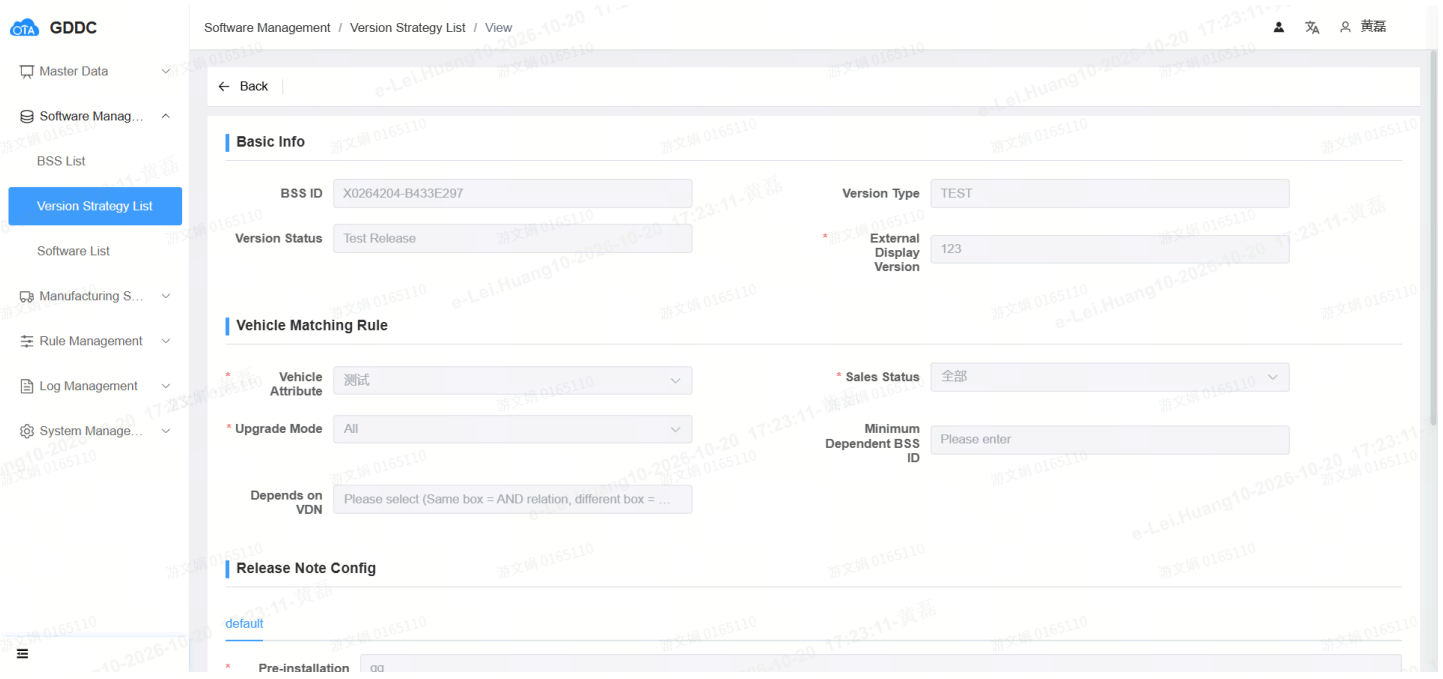
The version policy list supports query by BSSID, StratId, Type Code, externally displayed version, version status, status, policy release time and end time. The query bar supports expansion and collapse; when collapsed, only four query conditions including BSSID, StratId, Type Code and externally displayed version are displayed. Reset: Reset the query bar to the default state (empty by default)

2.2.4.2 Version Policy List - View

Function Entry: Menu >> Software Management >> BSS List >> View



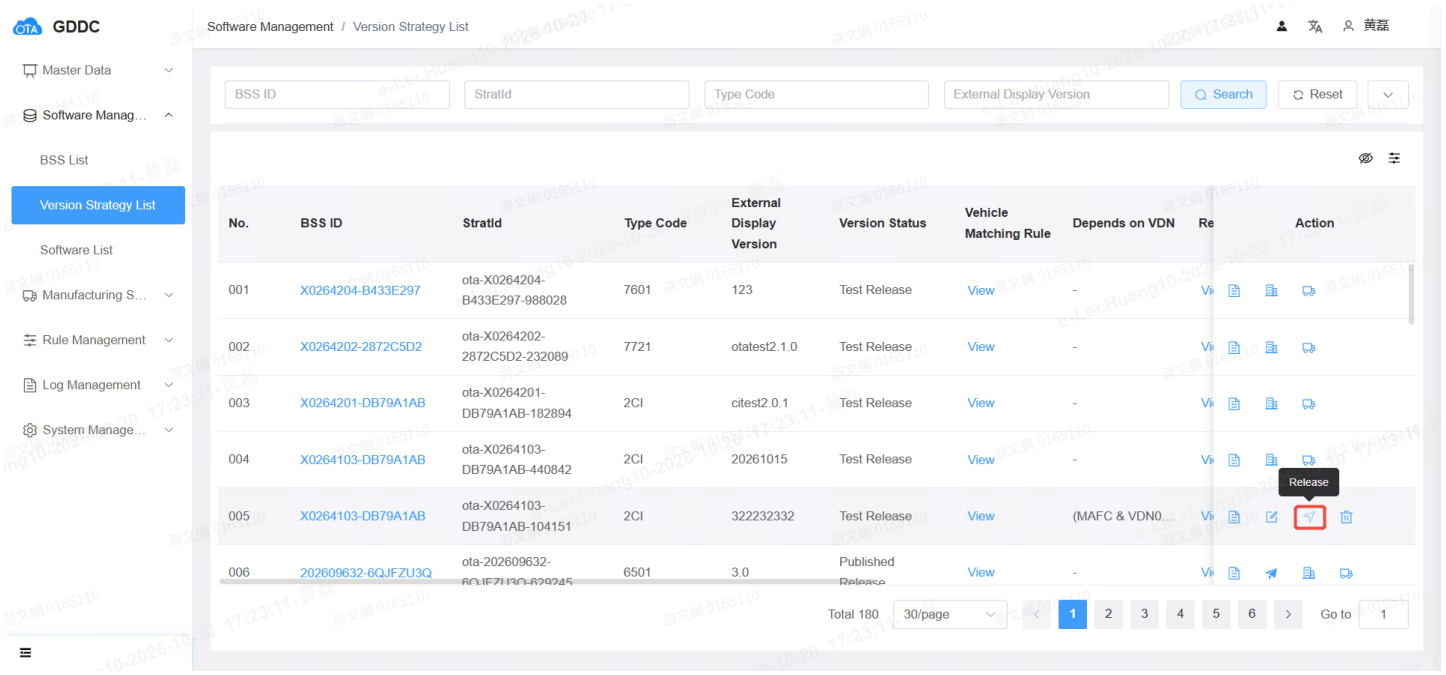
View page:



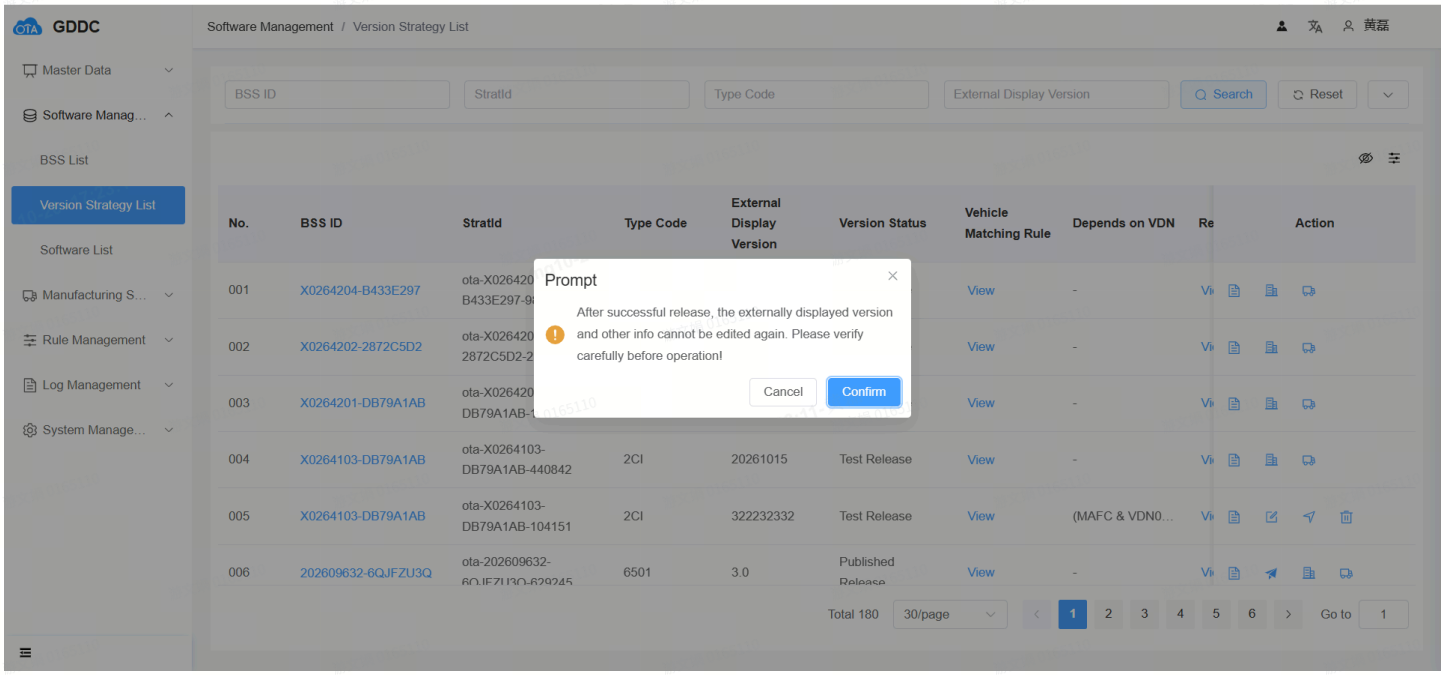
2.2.4.3 Version Policy List - Release

Function entry: Menu >> Software Management >> BSS List >> Release

Precondition: Only unreleased versions can be released



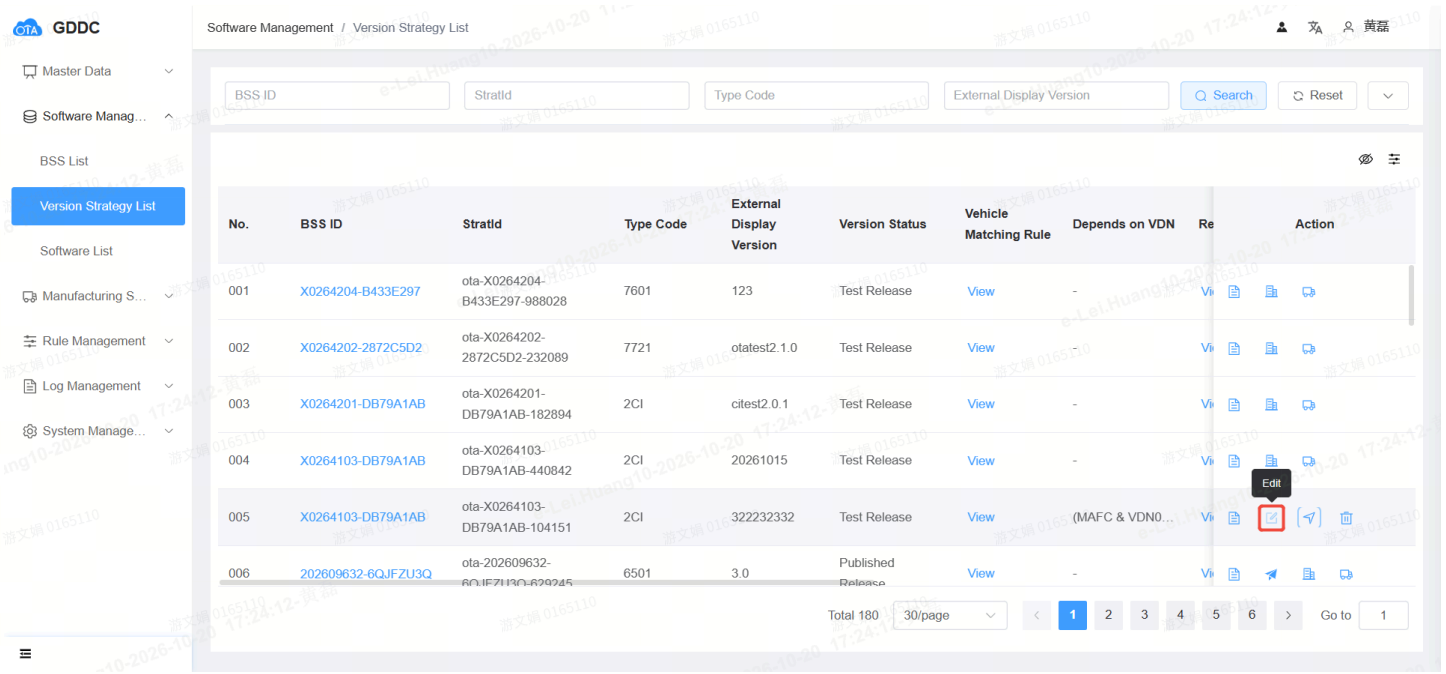
Post Confirmation Pop-up:



2.2.4.4 Version Policy List - Edit

Function entry: Menu >> Software Management >> BSS List >> Edit

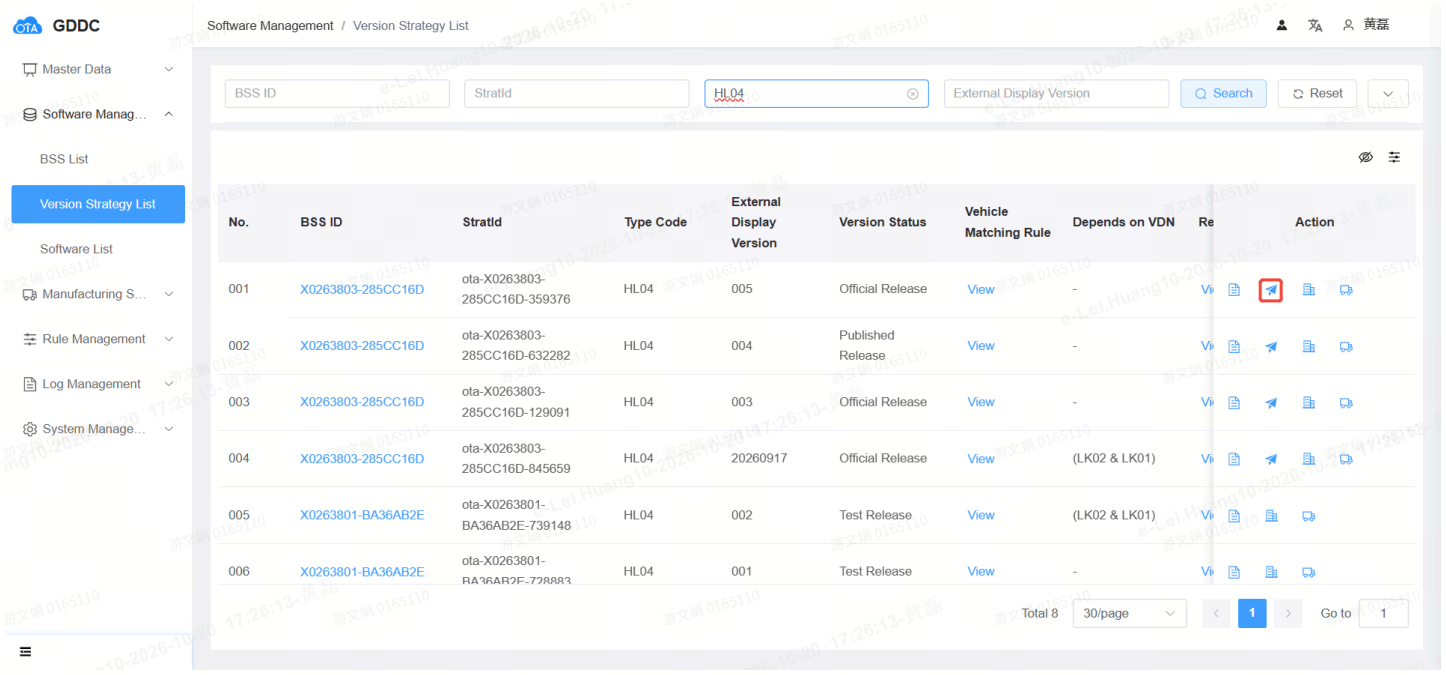
Precondition: Editing is only allowed when the Test Release version has not been released



2.2.4.5 Version Policy List - Release Publish

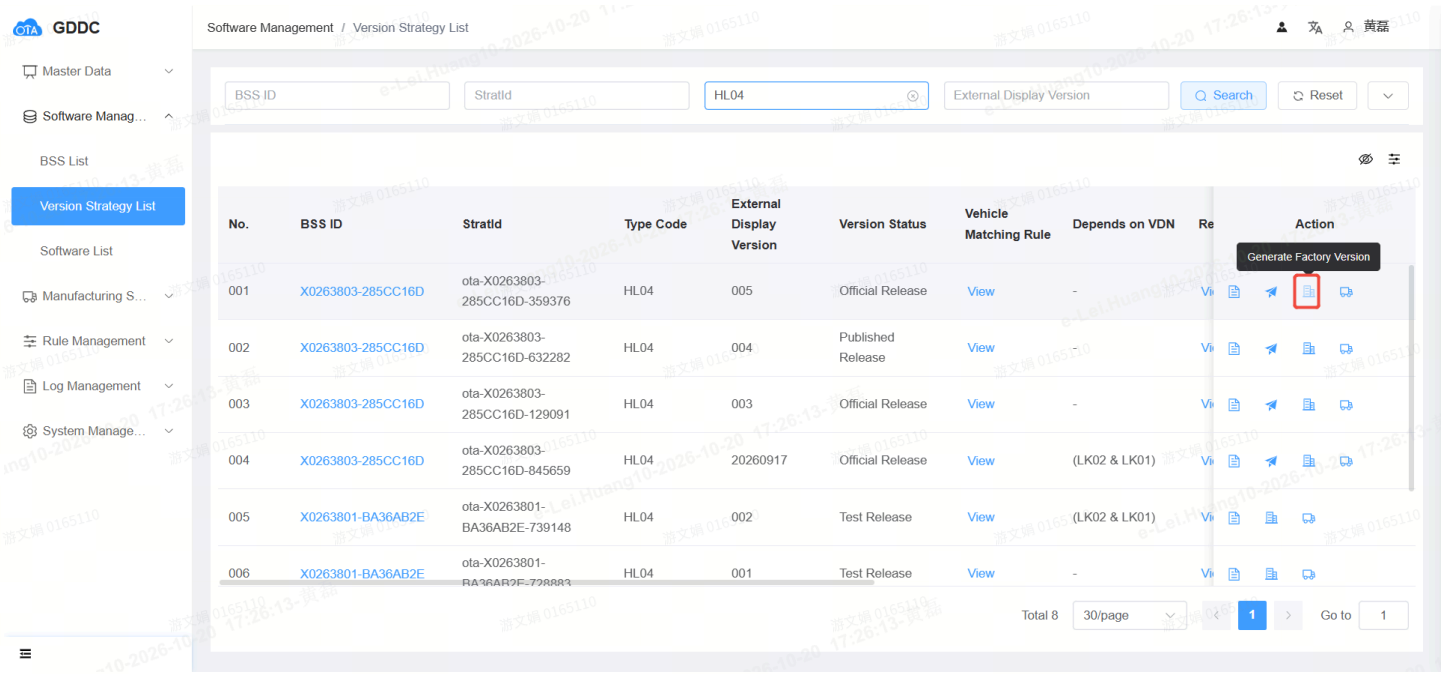
Function Entry: Menu >> Software Management >> BSS List >> Publish

Precondition: Only available in the Official Release version

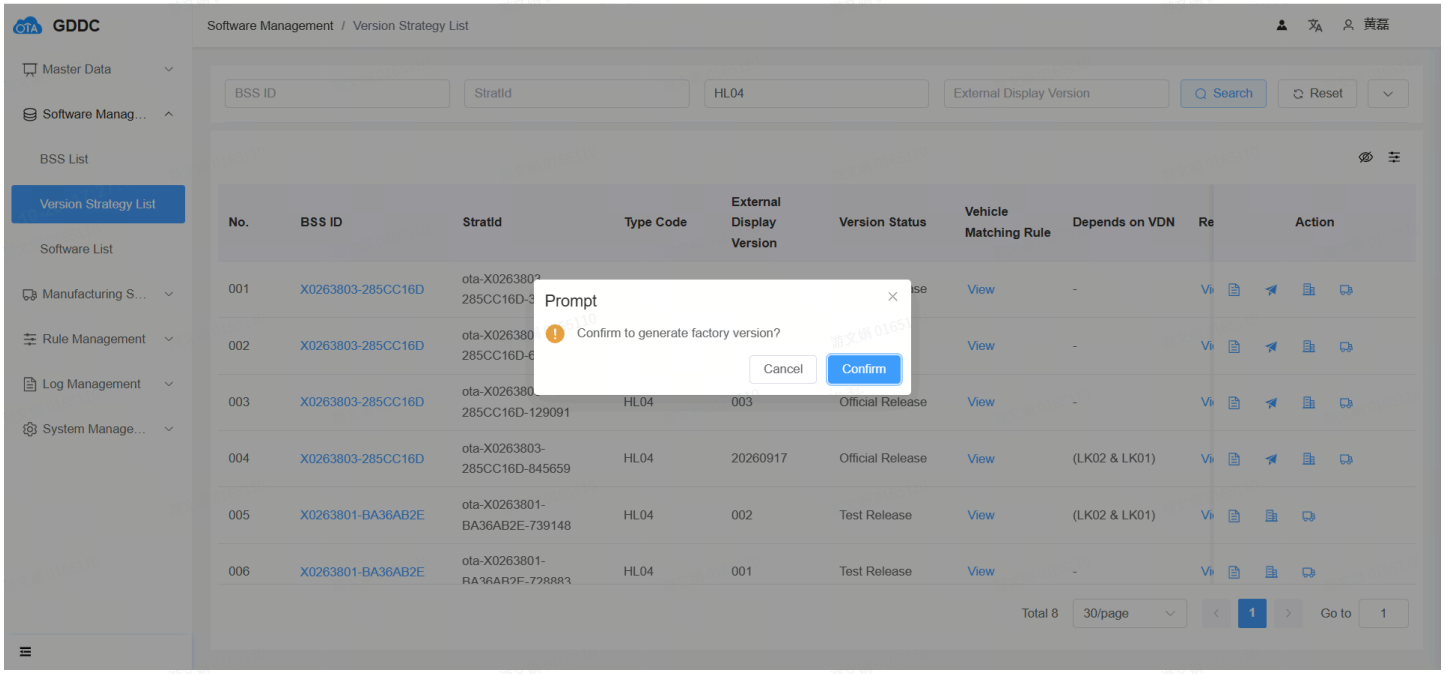


2.2.4.6 List of Version Policies - Factory Version Generation

Function entry: Menu >> Software Management >> BSS List >> Generate Factory Version

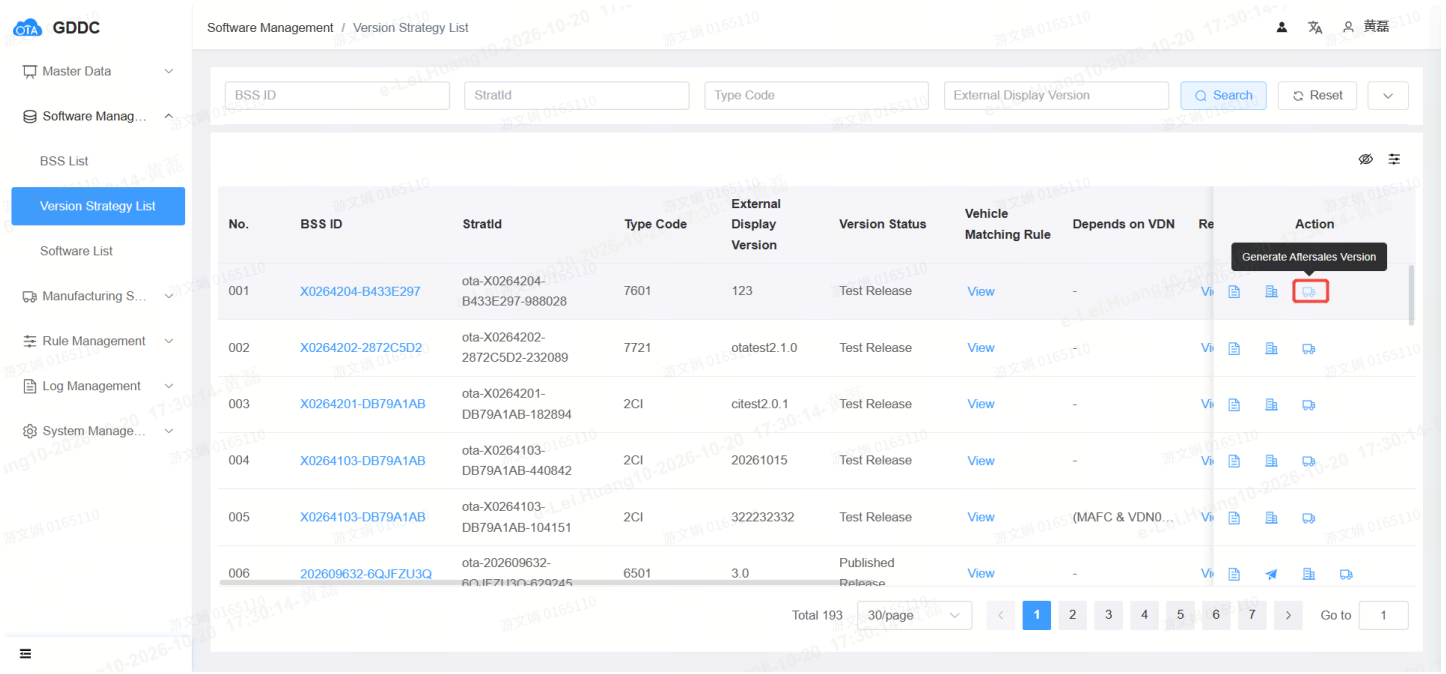


Generate factory version confirmation popup:



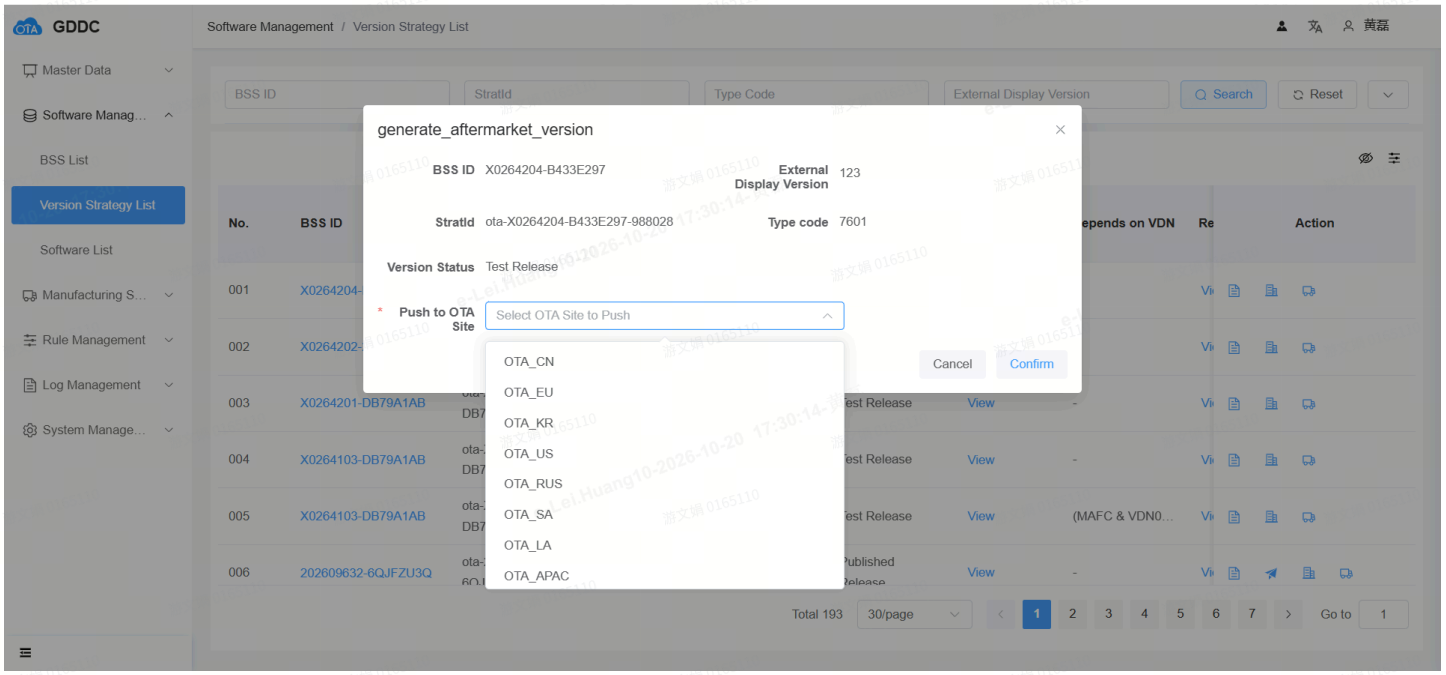
2.2.4.7 Version Policy List - After-sales Version Generation

Function entry: Menu >> Software Management >> BSS List >> Generate After-sales Version



Generate after-sales version confirmation push popup:

Push OTA site: a drop-down box for selecting the site to be pushed



2.3 Software List

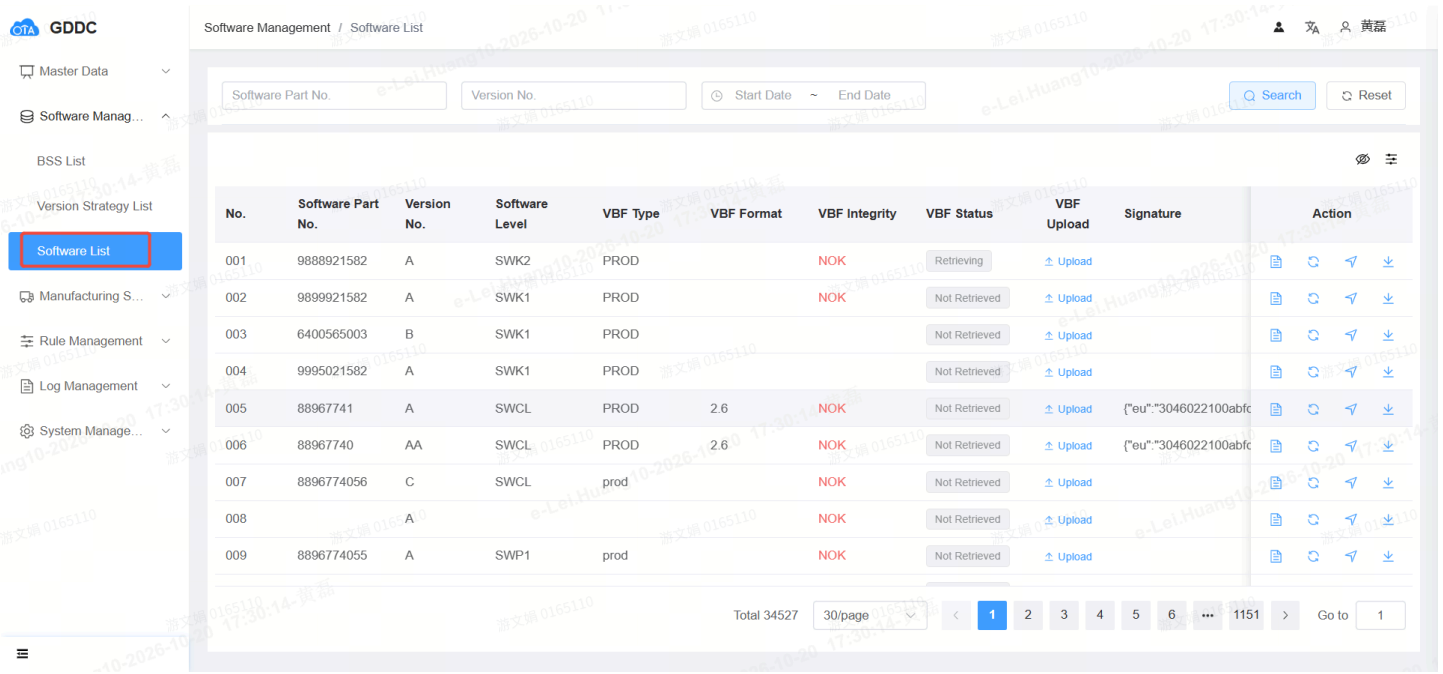
2.3.1 Function Introduction

Software management is primarily responsible for managing global data such as software information, software versions, and VBF files, and interfaces with the TC system to obtain all required software component information and VBF files.

data such as documents

2.3.2 Entrance

Menu > Software Management > Software List



2.3.3 precondition

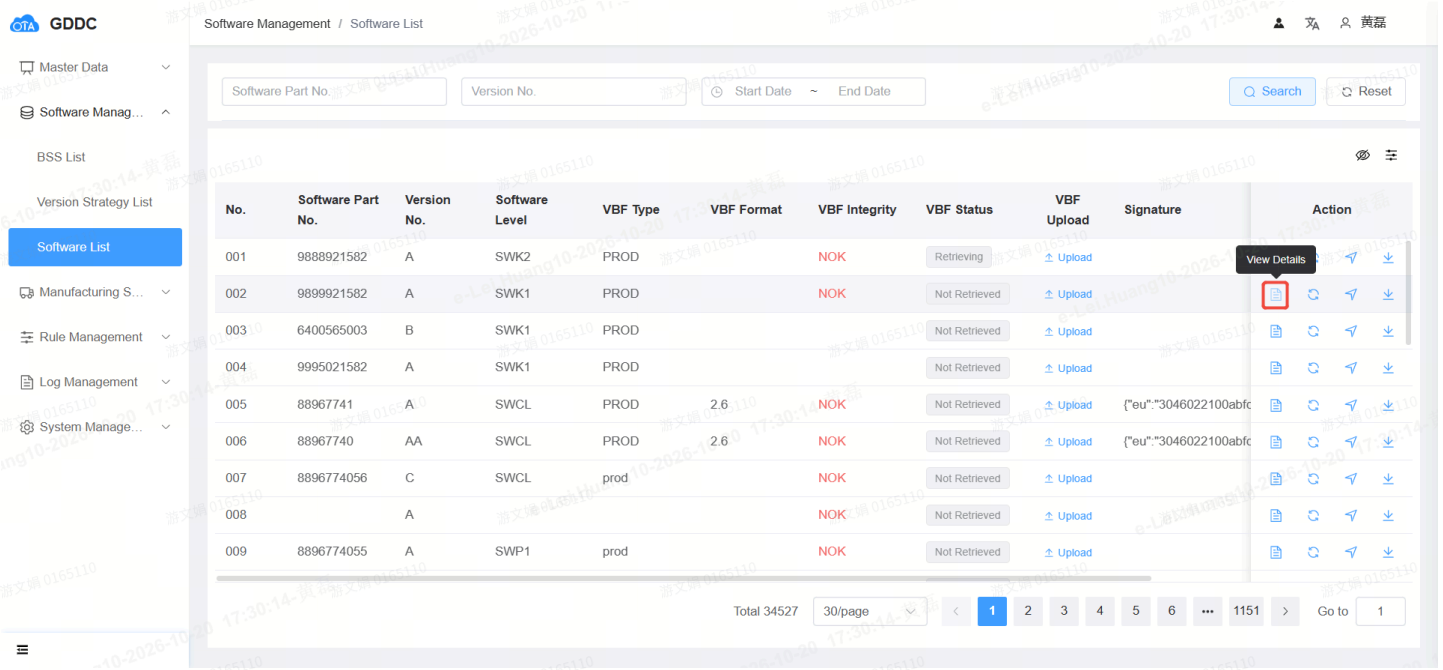
2.3.4 Operation

2.3.4.1 Software List - Query

The software list supports query by software part number, version number, start date and end date, and the query bar supports reset: reset the query bar to the default state (which is empty by default)

2.3.4.2 Software List - View Details

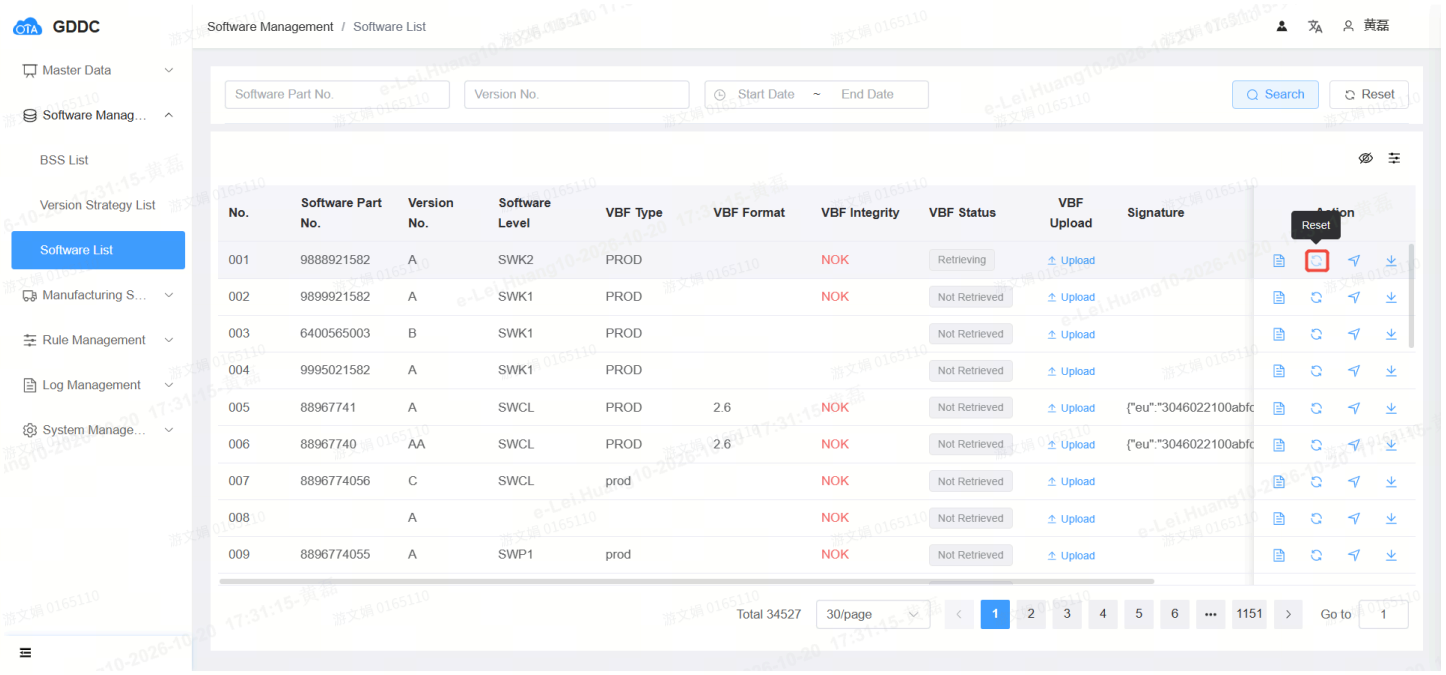
Function Entry: Software Management >> Software List >> Operation Column >> View Details



2.3.4.3 Software List - Reset

Function Entry: Software Management >> Software List >> Operation Column >> Reset

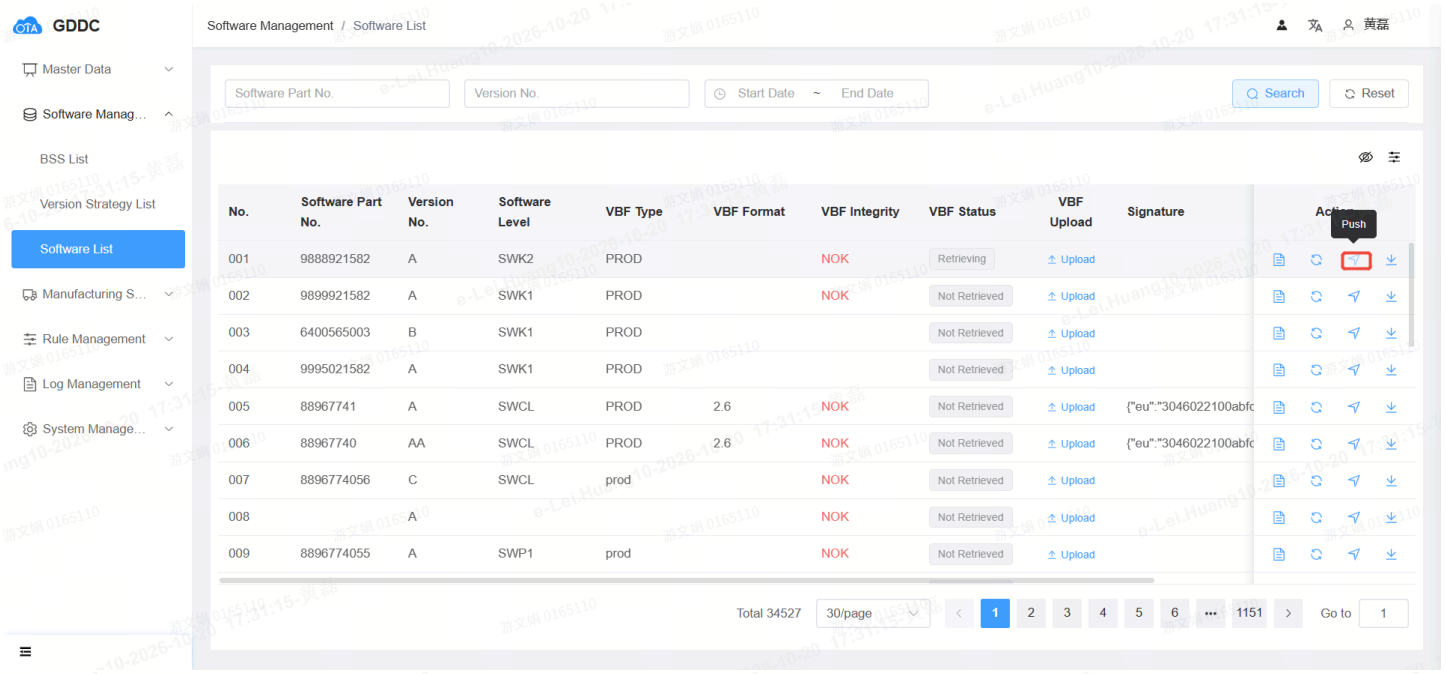
Reset software status



2.3.4.4 Software List - Resend

Function entry: Software Management >> Software List >> Operation Column >> Repush

Re-push the software to the site



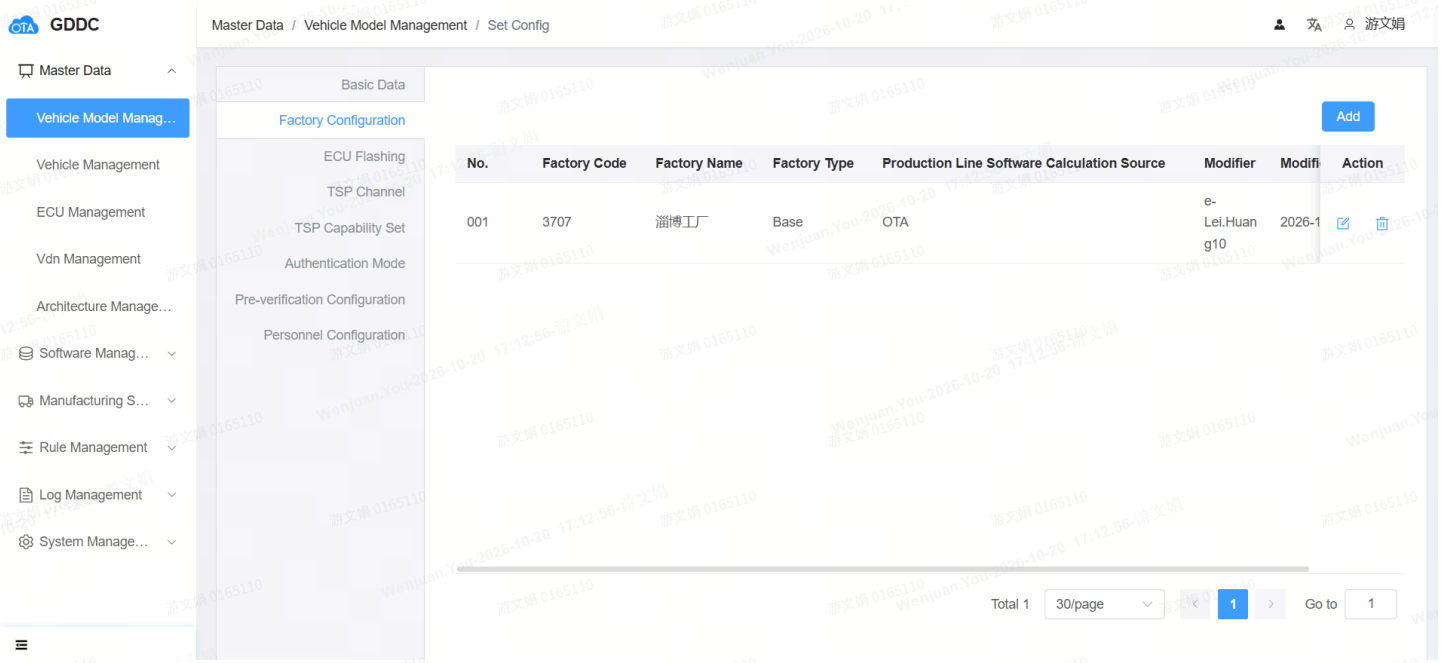
3. Manufacturing Software Management

Function Introduction: To address the issue of homogeneous factory software

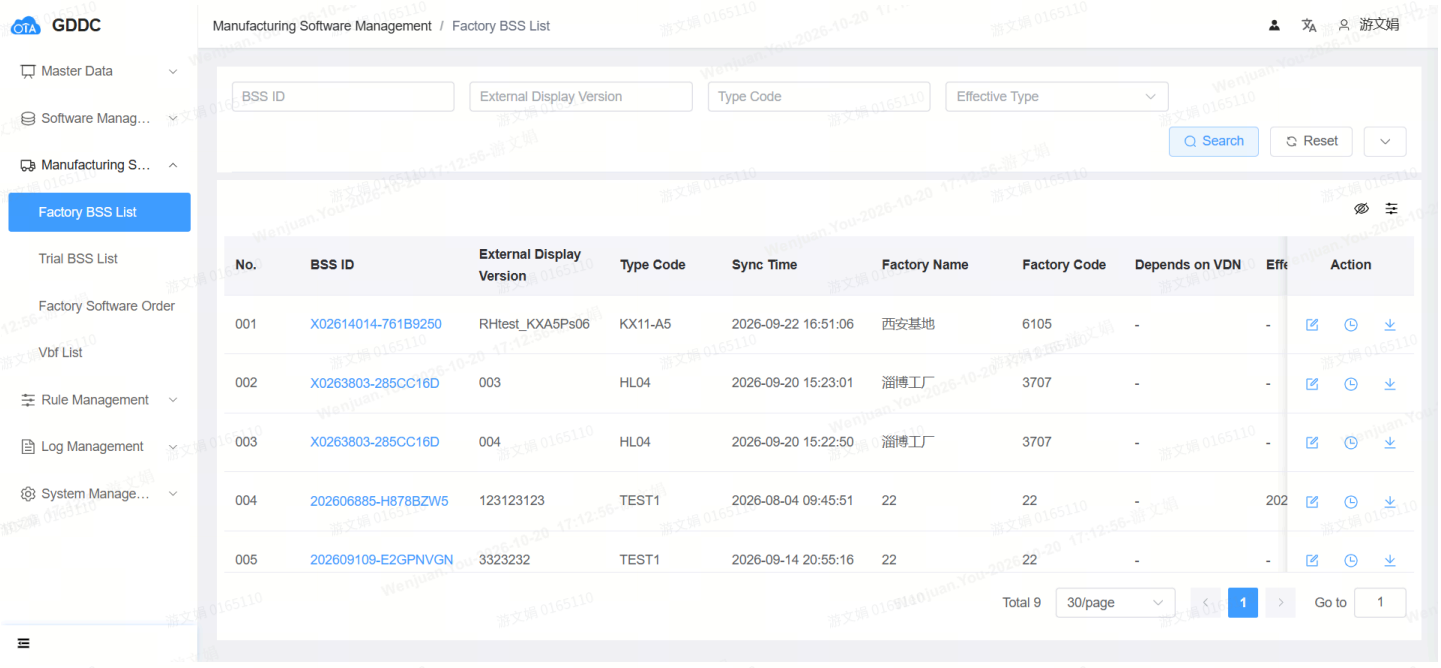
3.1 Factory BSS List

3.1.1 Function Entry: Manufacturing Software Management >> Factory BSS List

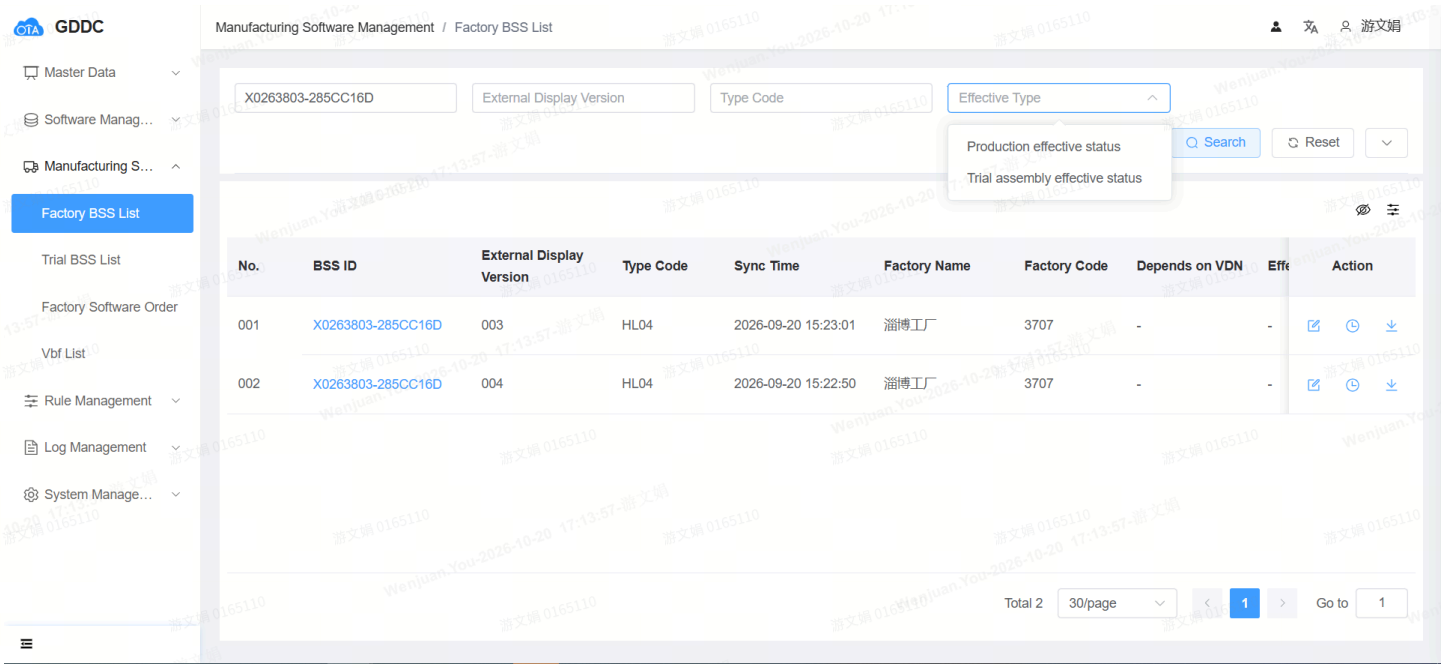
3.1.2 Precondition: BSS of the OfficialReleased type synchronized by TC, and the vehicle model has been configured with a factory code whose factory type is base



BSS supports one-to-many association with plant codes. If multiple base plant codes are configured, the plant BSS list will be displayed in multiple lines.



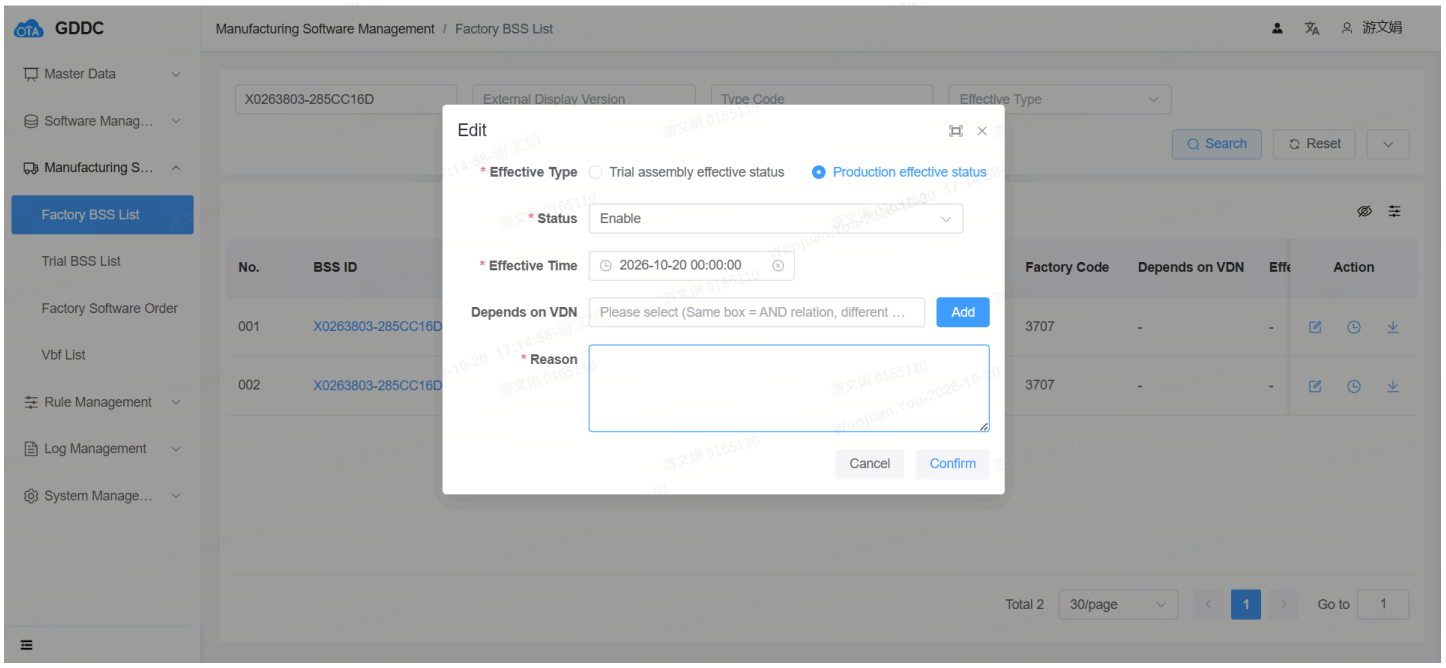
- Query criteria: BSSID, externally displayed version, Type code, effective status (production effective status, trial installation effective status), status (disabled, enabled)
- Actions: Edit, History, Download



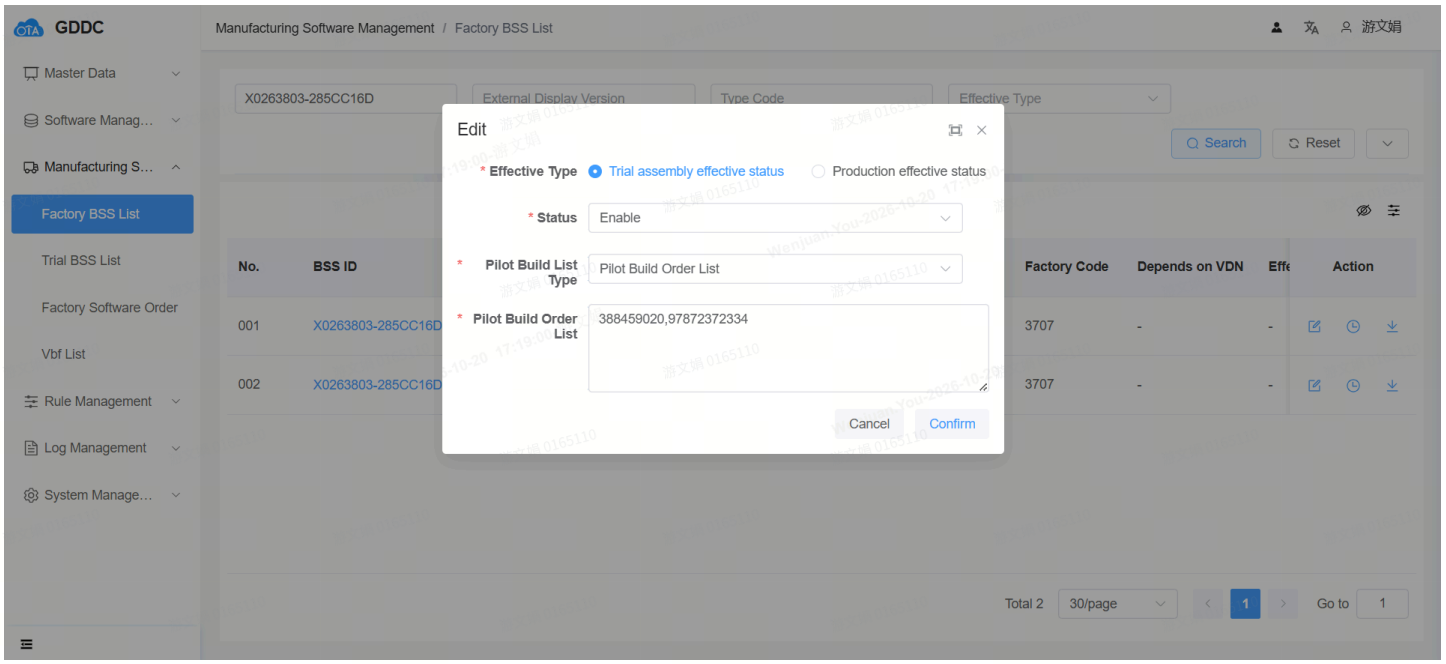
3.1.3 Edit the factory BSS list

Production effective status enabled, set effective time (required), reason (required), dependent VDN (optional)

When a vehicle requests a software homologous interface, if there are multiple enabled BSS entries in the factory BSS list of the same factory and type code, the vehicle will preferentially match the factory BSS list with a later enable time.



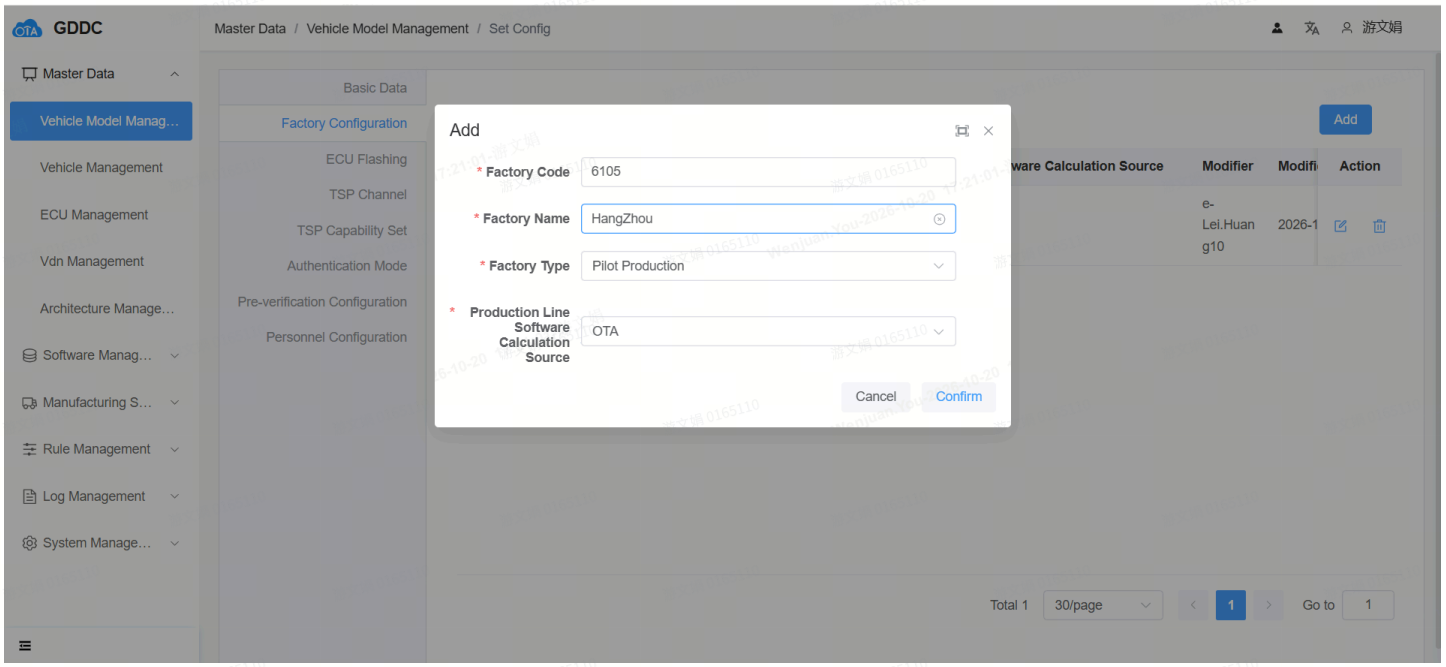
The trial assembly active status is intended for use with trial assembly orders, and multiple trial assembly order numbers can be configured. Once any one of the configured order numbers is successfully used to generate a factory software order, that corresponding trial assembly order number will be cleared; when all configured trial assembly order numbers have been successfully used to generate factory software orders, the status of this entry in the factory BSS list will be automatically disabled.



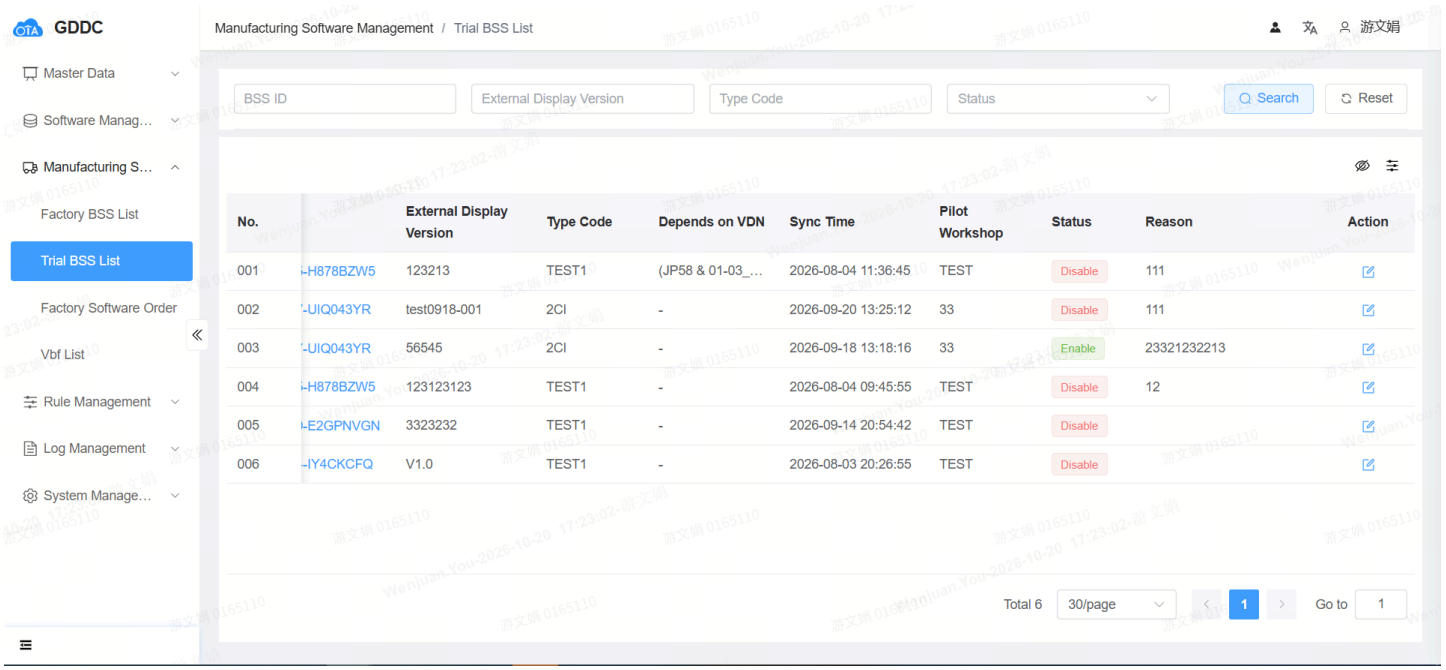
3.2 Pilot Trial BSS List

3.2.1 Function Entry: Manufacturing Software Management >> Trial Production BSS List

3.2.2 Precondition: For the BSS of ENG Released type synchronized by TC, the vehicle model has been configured with a plant code whose plant type is trial-manufacture



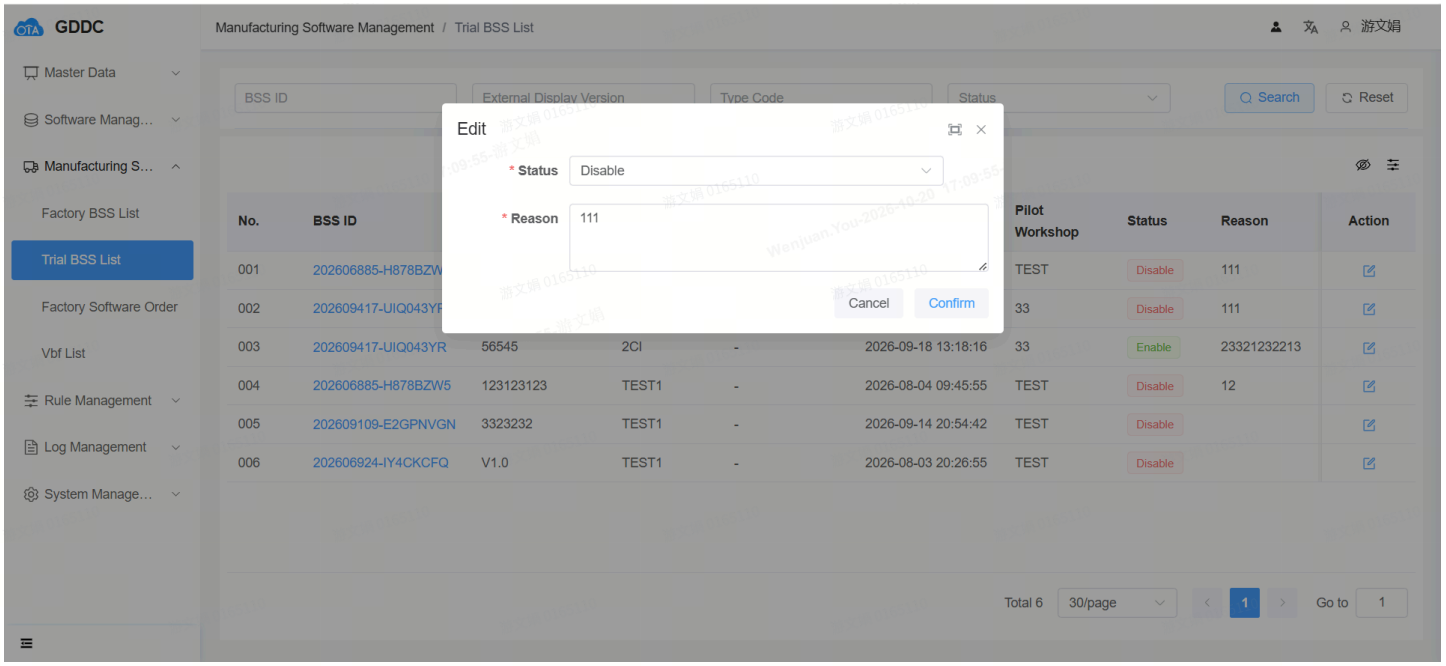
BSS one-to-many associated plant codes: if multiple pilot plant codes are configured, the pilot BSS list will be displayed in multiple rows



- Query Criteria: BSSID, externally displayed version, Type code, status (disabled, enabled)
- Operation: Edit

3.2.3 Edit the trial production BSS list

Status (mandatory), Reason (mandatory), Dependent VDN (optional)



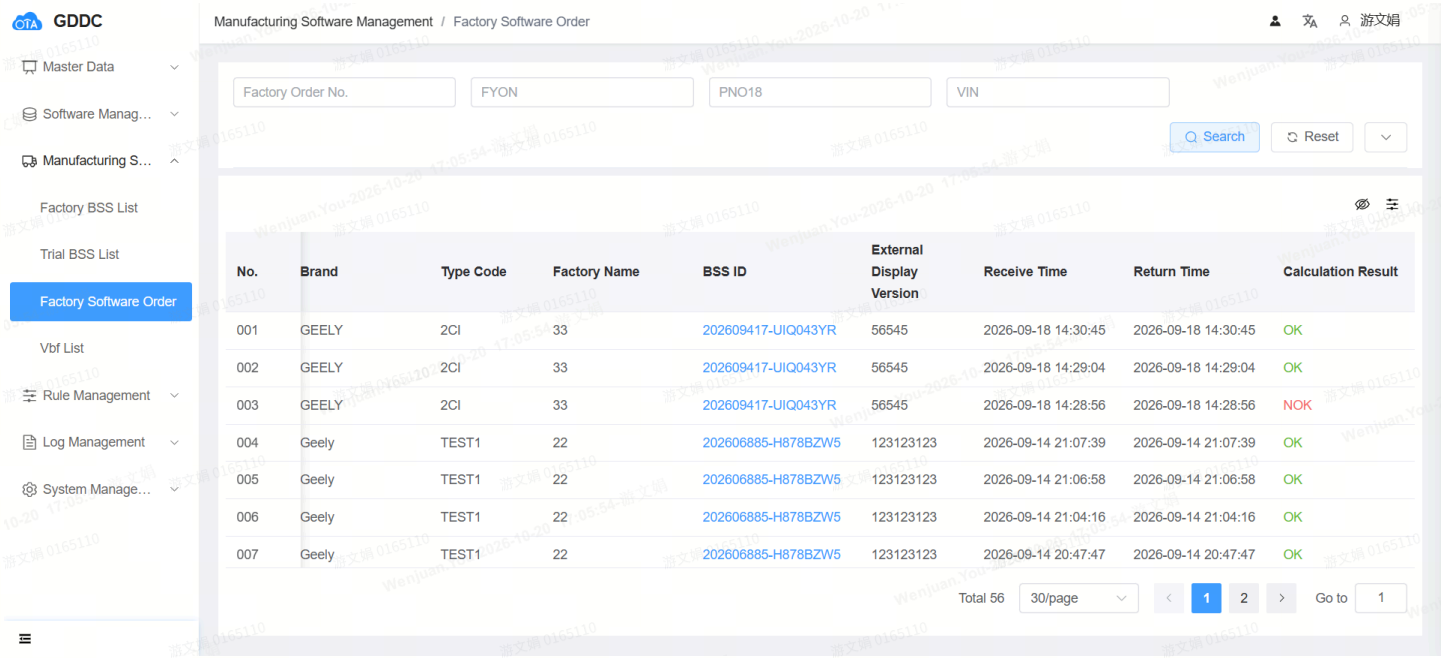
Vehicles request the software homologous interface, and the trial-production orders that are successfully solved will be displayed on the factory software order list page.

3.3 Factory Software Order

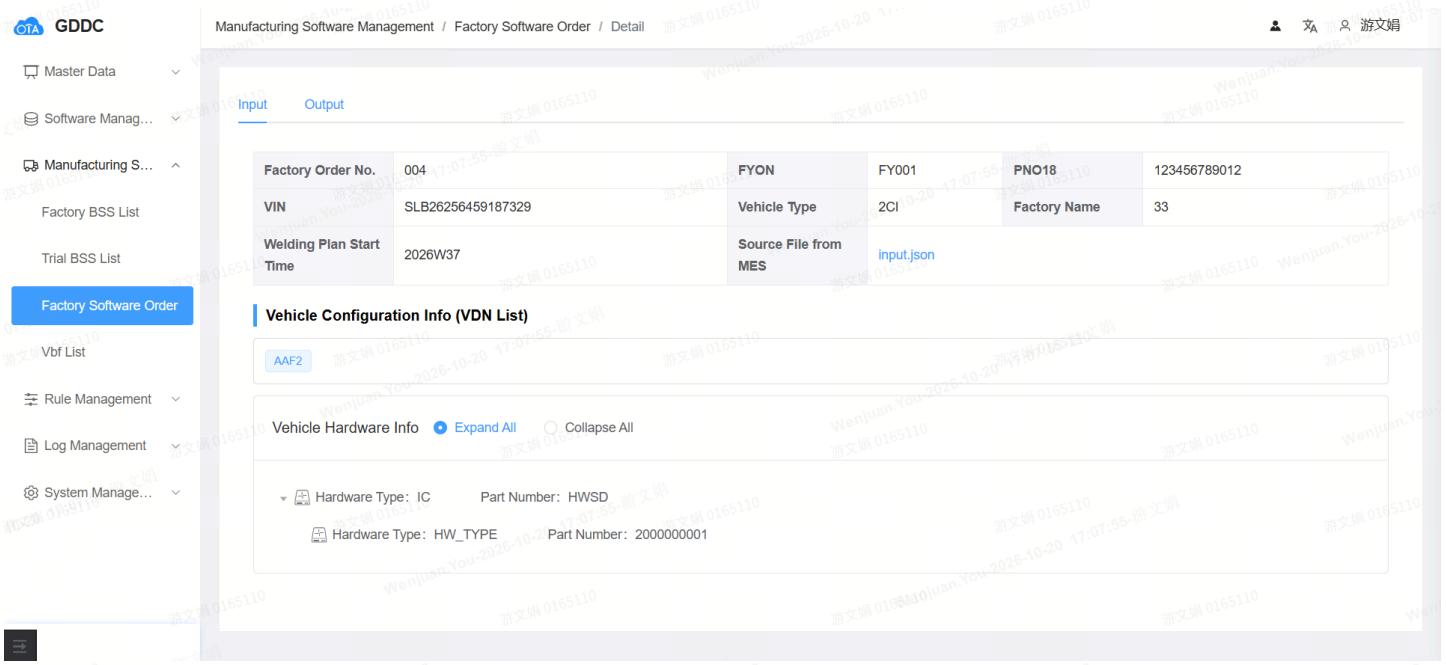
3.3.1 Function Entry: Manufacturing Software Management >> Factory Software Order

3.3.2 Precondition: There are enabled BSSIDs in the factory BSS list

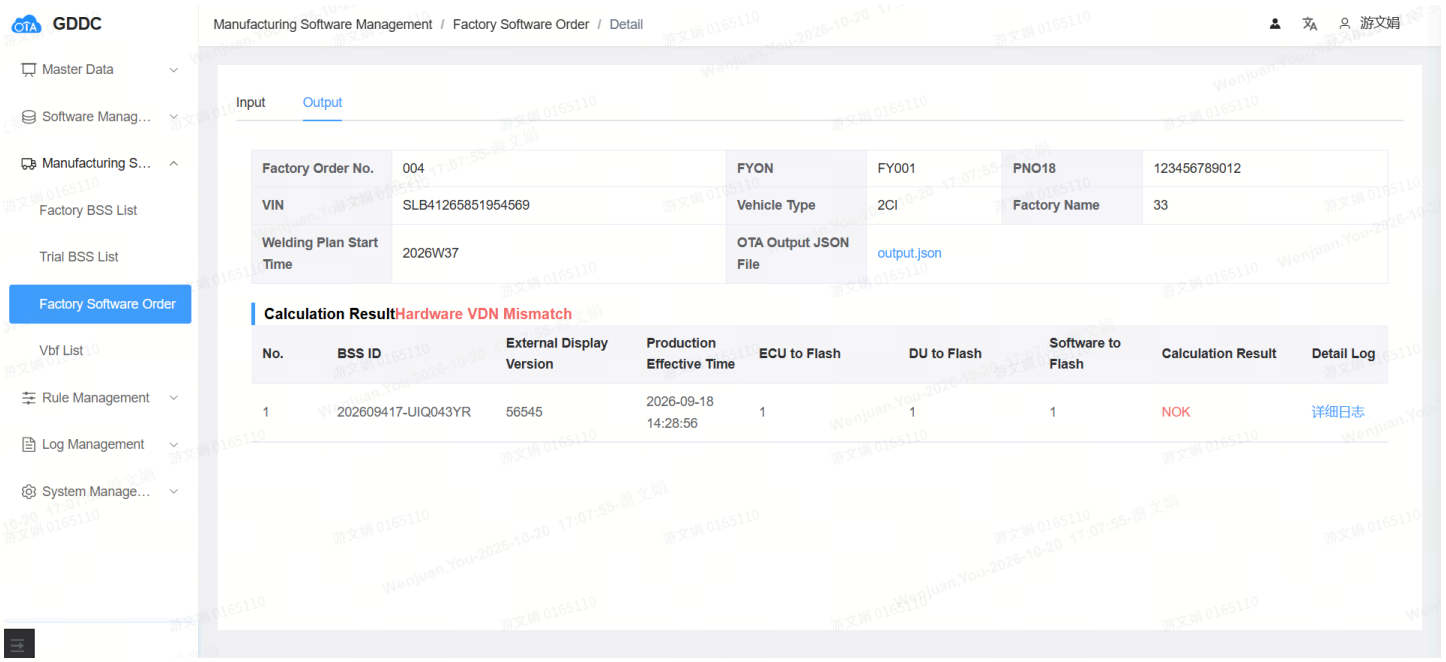
When a vehicle requests the software homologous interface, a record will be generated in the factory software order regardless of whether the request returns success or failure: if the calculation succeeds, the calculation result is marked as OK; if the calculation fails, the calculation result is marked as NOK.



Click the order number to access the order details page, which includes input and output sections. The input section contains all interface request parameters, and you can click input.json to download the complete request parameters.



The output page displays the calculation results, and the reason for the failure will be provided in case of a failure.



3.4 vbf download records

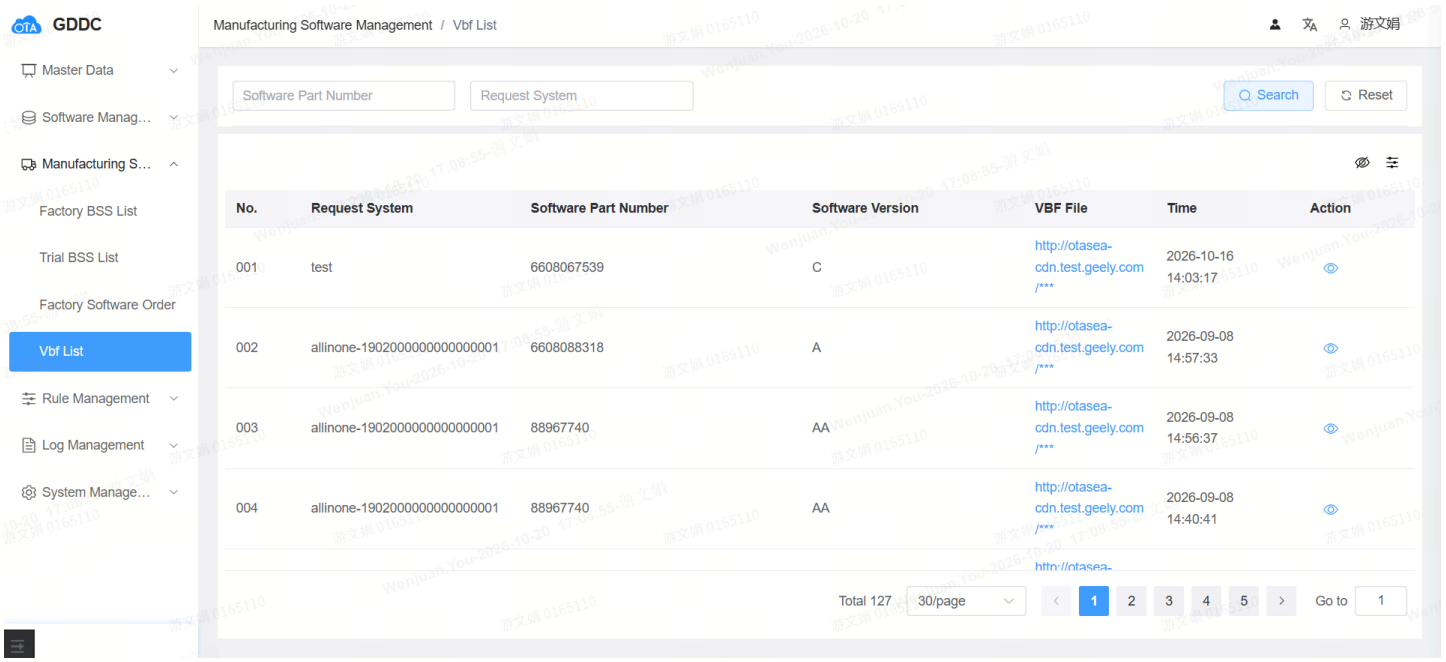
3.4.1 Function Introduction: When the factory electrical inspection equipment VCATS performs electrical inspection, it supports obtaining VBF files from the OTA platform.

3.4.2 Function Entry: Manufacturing Software Management >> Factory Software List

3.4.3 Precondition: VCATS electrical inspection equipment triggered

3.4.4 VBF Matching Rules

- If VCATS only carries softwarePartNum, the platform will return the VBF file with the same softwarePartNum and vbfStatus as prod; if multiple PROD VBF files are matched or no PROD file exists, an error message will be returned.
- If VCATS carries softwarePartNum and softwarePartVersion, the platform will return the matching prod status VBF file
- If VCATS carries softwarePartNum, softwarePartVersion and vbfStatus, the platform will return the VBF file that matches exactly.



4. Rule Management

4.1 Vehicle Distribution Rules

The vehicle distribution rule is the core configuration basis for the GDDC system to automatically/manually route and distribute vehicle data to the corresponding OTA sites.

4.1.1 Function Entry: Rule Management >> Vehicle Distribution Rules

There are three types of vehicle distribution rules, namely vehicleType_marketCode, VDN and Default. The priority of the three types can be changed in the rule configuration sequence as shown in the figure below: the rule listed earlier has a higher priority, while the DEFAULT has the lowest priority which cannot be modified.In the figure below, vehicleType_marketCode has the highest priority, followed by VDN, and Default has the lowest priority. After MES vehicle synchronization is completed, the system will first match the rules of vehicleType_marketCode: if a match is found, the vehicle will be sent to the corresponding site; if no match is found, it will then match the VDN rules, and if there is still no match, the DEFAULT rule will be applied.



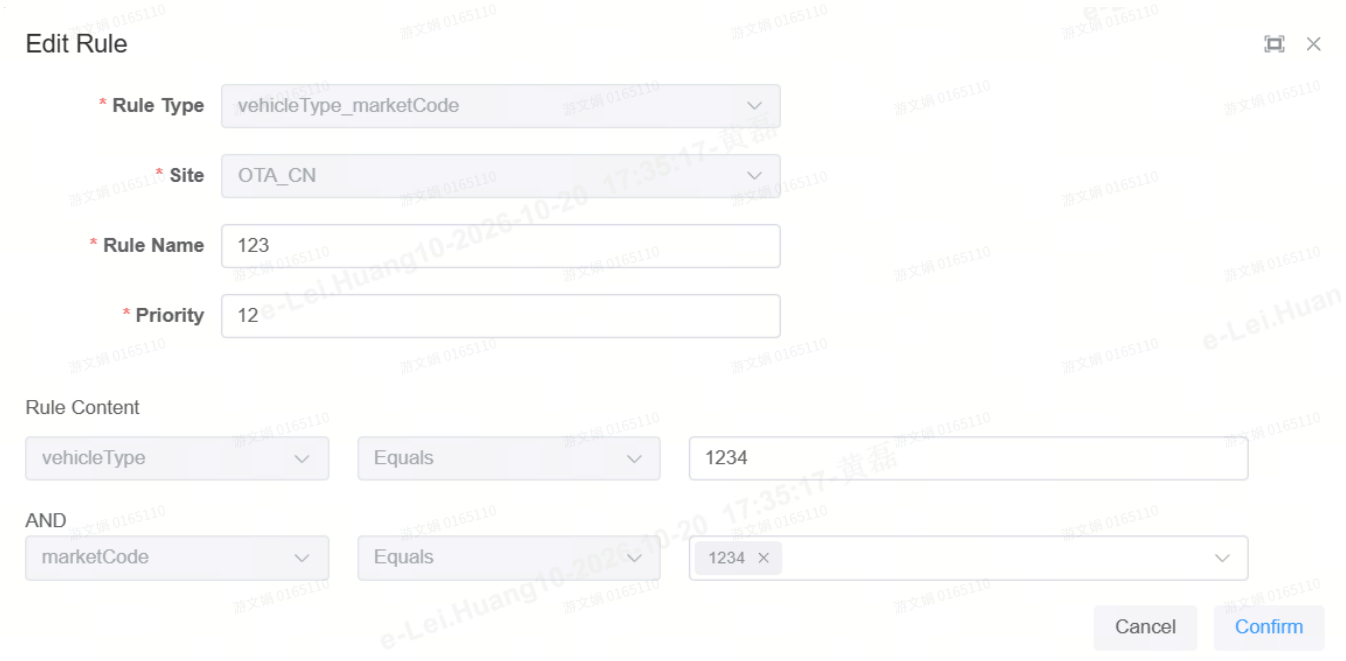
4.1.2 Function Operation

- Rule Type: vehicleType_marketCode



The figure above shows the page for adding a new vehicleType_marketCode type rule (the rule type cannot be modified during editing), where a higher priority number indicates a higher priority; vehicleType corresponds to the Vehicle Type in Master Data Management - Vehicle Model Management, while marketCode refers to the corresponding market code, which supports multiple selections, and the multiple selected items follow an OR relationship.The configuration in the above figure means that for vehicles synchronized from MES, if the vehicleType is 7601 and the marketCode is 51, matching will start from the vehicleType_marketCode rule with the highest priority; if a match is found in the previous steps, the rule in the above figure will be matched and the relevant information will be pushed to the OTA_CN site. The vehicleType_marketCode rules cannot have identical priorities.

- Rule Type: VDN



The figure above shows the page for adding a new VDN type rule (the rule type cannot be modified during editing), where a higher priority number indicates a higher priority.

The VDN for verification requires the VDN value to be verified. The specific VDN to be verified is configured below. VDNs in the same row have an AND relationship, while those in different rows have an OR relationship. A maximum of 10 rows can be added. In the above figure, if the VDN of the upstream synchronized vehicle contains AA01 and AB01, or contains CC01, it will match this rule and be sent to the OTA_EU site.

- Rule Type: DEFAULT

  The DEFAULT type does not support addition. If the upstream synchronizes vehicles and neither the vehicleType_marketCode nor the VDN rule is matched, the DEFAULT rule will be matched. By default, the DEFAULT rule sends to the OTA_CN site and this setting cannot be changed.

4.2 OTA Site Management

Site management refers to the process of converting the external market codes of overseas vehicle models into internal market codes.

4.2.1 Function Entry: Rule Management >> OTA Site Management

4.2.2 Function Operation



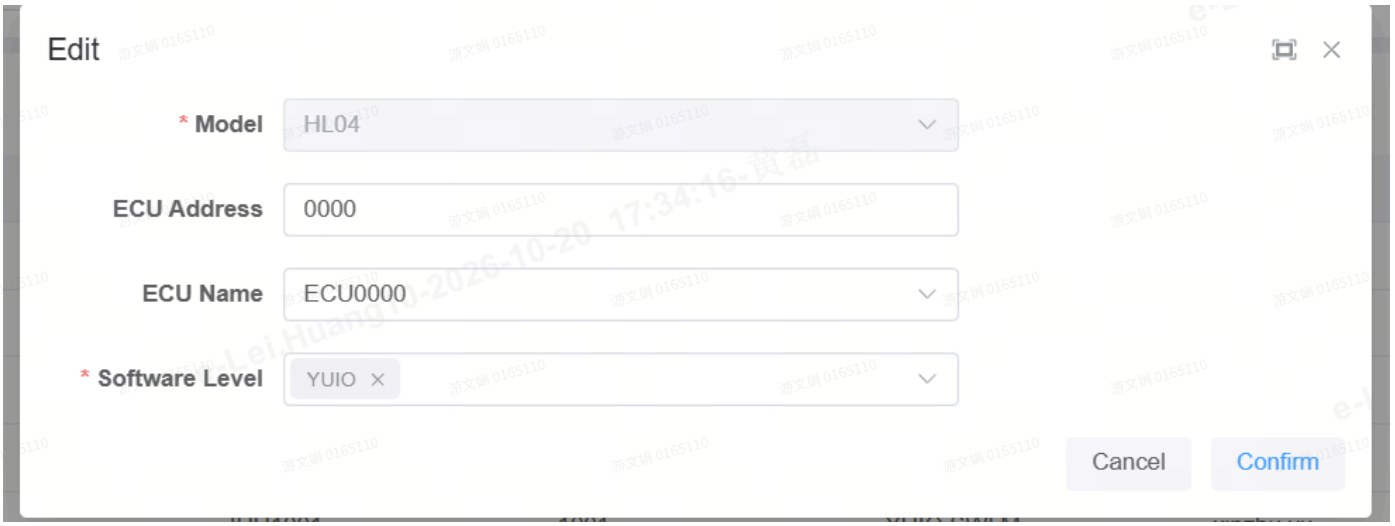
As shown in the figure above, the site name is selected via a drop-down box, which is actually only for display purposes and involves no logic processing; the associated vehicle model corresponds to the mapped model of the type field in the g02 synchronization XML. For example, currently 416 corresponds to V21 and 417 corresponds to V2C. The market code corresponds to the internal market code of our system, while the external market code corresponds to the plant field in the g02 synchronization XML.Taking the above figure as an example, for the vehicle synchronized via the g02 interface, if the type field in the xml is 416 and the plant field is 21, then the vehicleType will be converted to V21, the market code will

be converted to 45, and the vehicle forwarding rule will be matched according to the vehicleType_marketCode type specified in Section 4.1.

4.3 Manufacturing Software Blocklist

4.3.1 Function Entry: Rule Management >> Manufacturing Software Blocklist

4.3.2 Function Operation



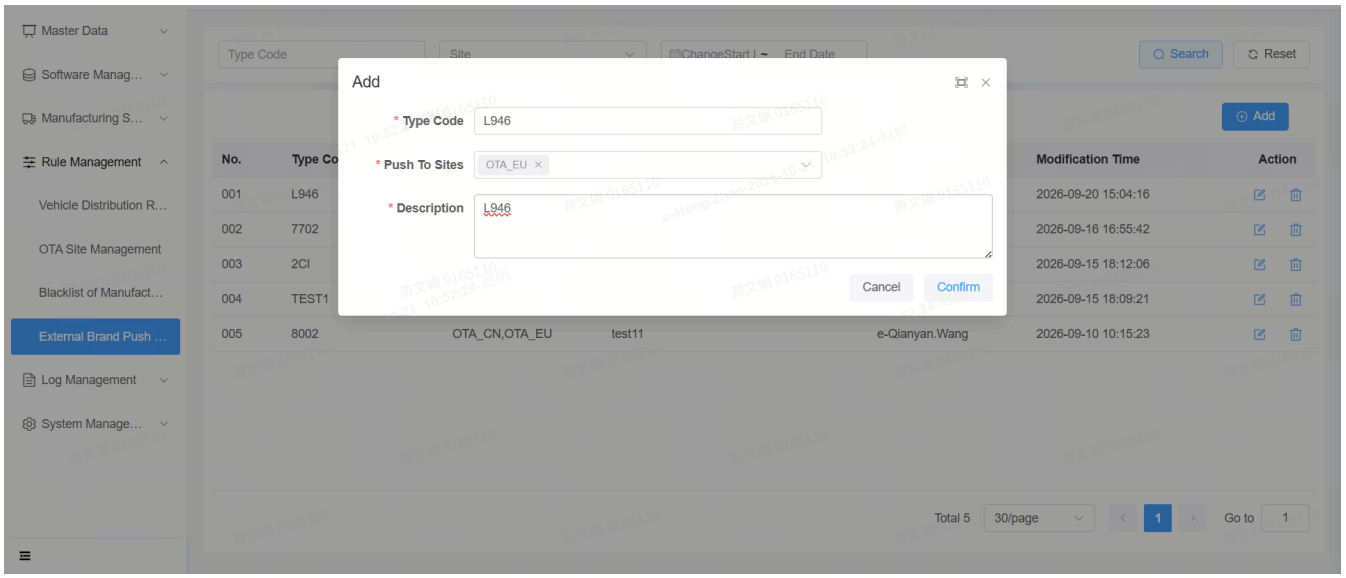
- The function of creating a software blocklist is to filter out specific software layers when the factory requests software homologation. As shown in the figure above, when the factory requests software homologation, if the calculation result contains an ECU address of 1001 and an ECU name of IHU1001, the software at the SWLM layer under IHU1001 will be filtered out and not returned in the interface. The ECU address and ECU name can be empty (they can only be both empty or both non-empty); when they are both empty, the software at the SWLM layer under all ECUs will be filtered out and not returned.

4.4 Foreign Brand Push Management (not yet launched)

4.4.1 Function Entry: Rule Management >> External Brand Push Management

4.4.2 Function Operation

- Menu Function Description: After TC synchronizes with BSS, it reads the configurations under this menu; if the requirements are met, it directly pushes to BSS without the need to construct policies.



5. Log Management

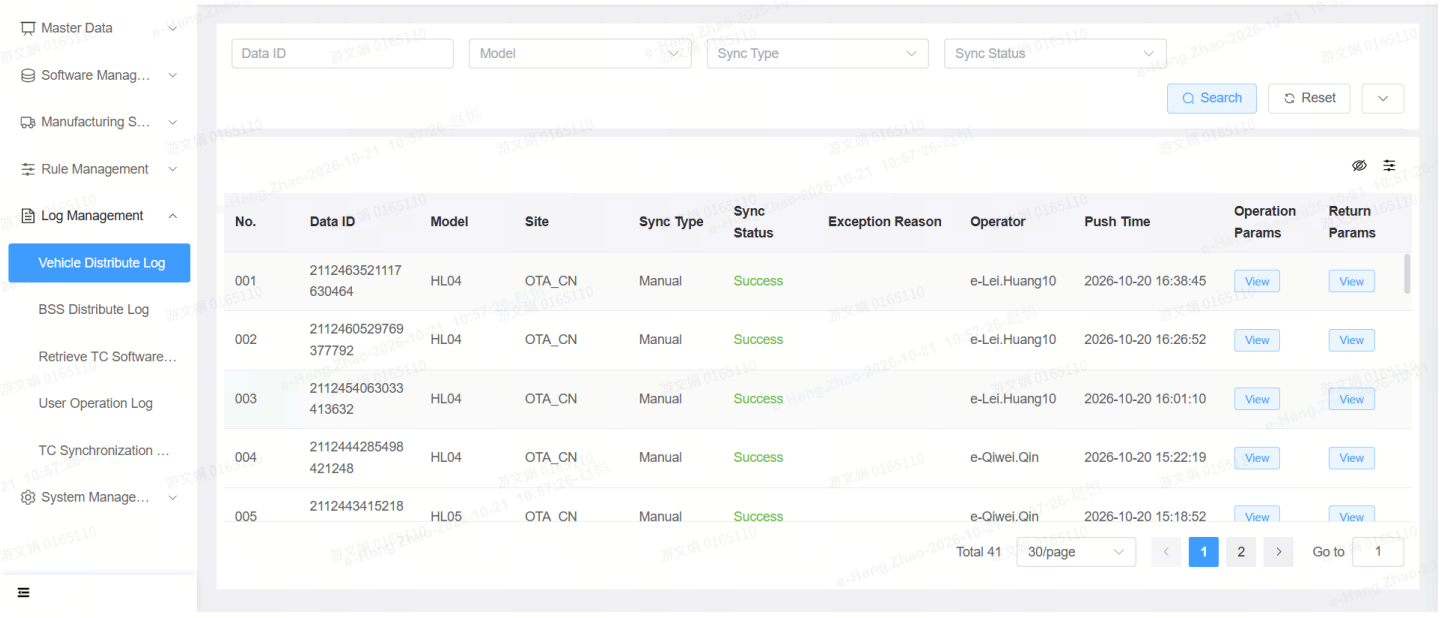
Function Introduction: The log management function is designed to record all operations performed by users via the management backend, as well as the invocation logs of all third-party interfaces and data synchronization records.

Core Value: By recording various events on the platform, it enables users to query the data source and warehousing process, facilitating troubleshooting.

5.1 Vehicle Distribution Log (Not Implemented)

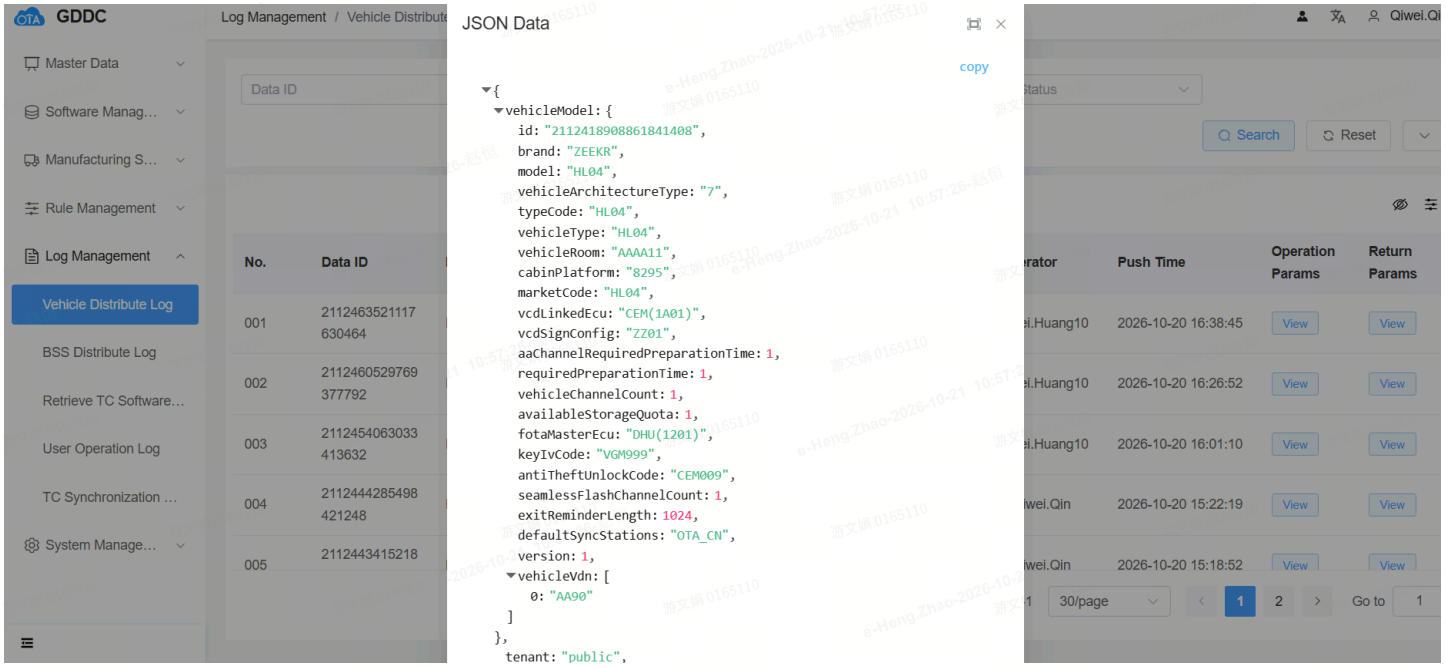
MES vehicle synchronization, or clicking Master Data Management - Vehicle List - Resend after vehicle synchronization will generate logs, which is not yet implemented at present.

5.2 Vehicle Model Distribution Log

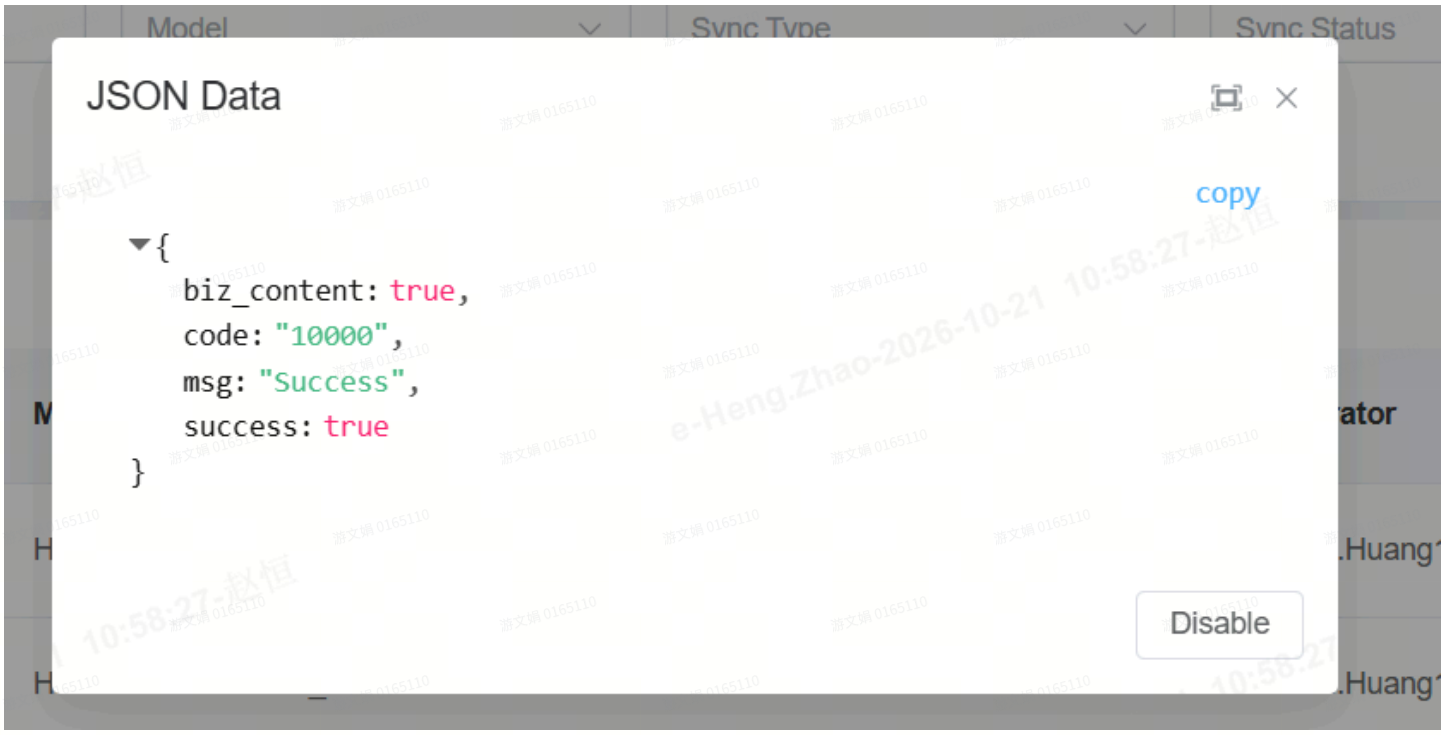


After clicking "Distribute" and confirming in the vehicle model list of Master Data Management - Vehicle Model Management, a new log will be generated in the vehicle model distribution log. In

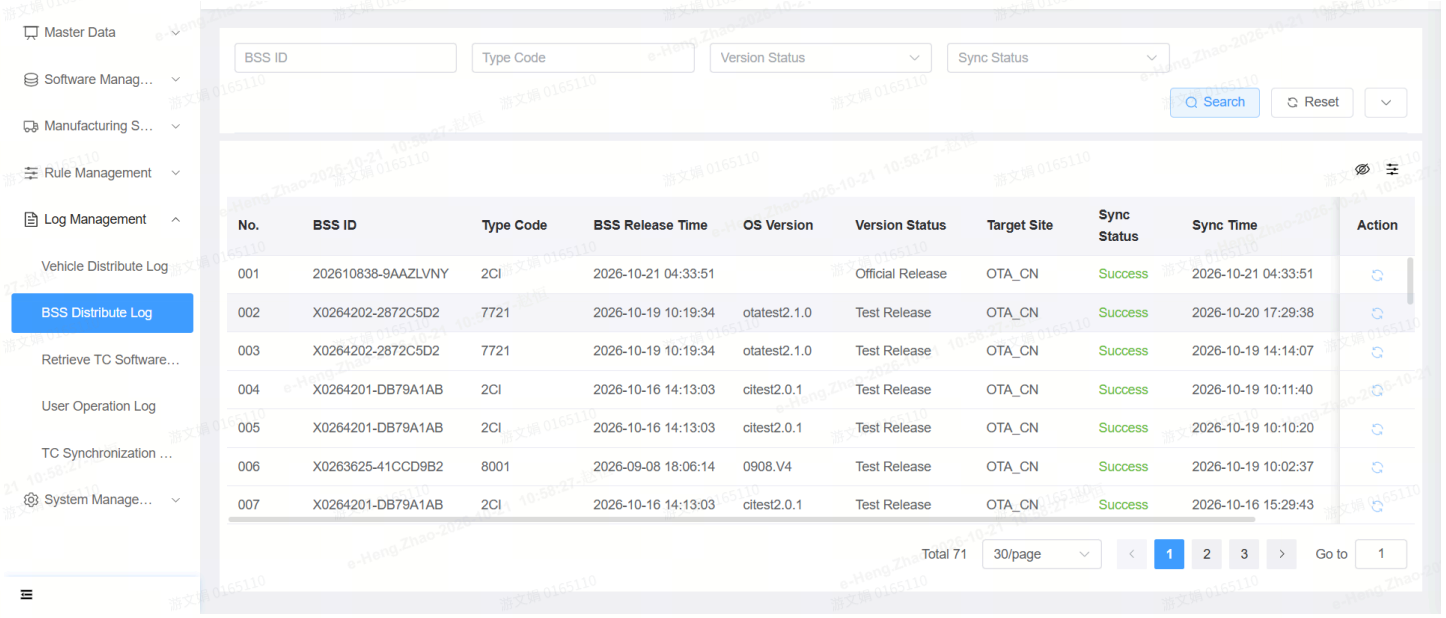
this log, the "Data ID" field is the auto-increment ID of the database, the vehicle model corresponds to the vehicle model code in Master Data Management - Vehicle Model Management, the site refers to the OTA site selected during distribution, and multiple sites are supported. The synchronization type includes automatic and manual; currently GDDC only supports manual push, and the synchronization status includes success and failure.When the operation fails, the specific cause of the exception will be displayed. (Since the synchronization of vehicle models to the site is an asynchronous operation, the "success" status here only indicates that the OTA site has received the synchronization request, rather than the final result of the actual synchronization.) The operator refers to the user who clicked the distribute button and confirmed the operation; the push time refers to the exact time when the user performed the push operation. The operation parameters are the request parameters used when pushing the vehicle model to the site at that time. Click "View" to see the specific content, as shown in the figure below. Click "Copy" to copy the request parameters to the clipboard, the square button to enter full-screen mode, and the X button to close the window.



Click "View Return Parameters" to check the specific content of the return parameters, as shown in the figure below. Click "copy" to copy the request parameters to the clipboard, the square button enables full-screen mode, and click the X to close the window.



5.3 BSS Distribution Log



The above is the BSS distribution log page. Pushing the baseline to the OTA site and generating a log will be triggered in the following scenarios:

1. When receiving the TC push baseline (with backend configuration for whether to enable automatic push), no log will be generated currently.

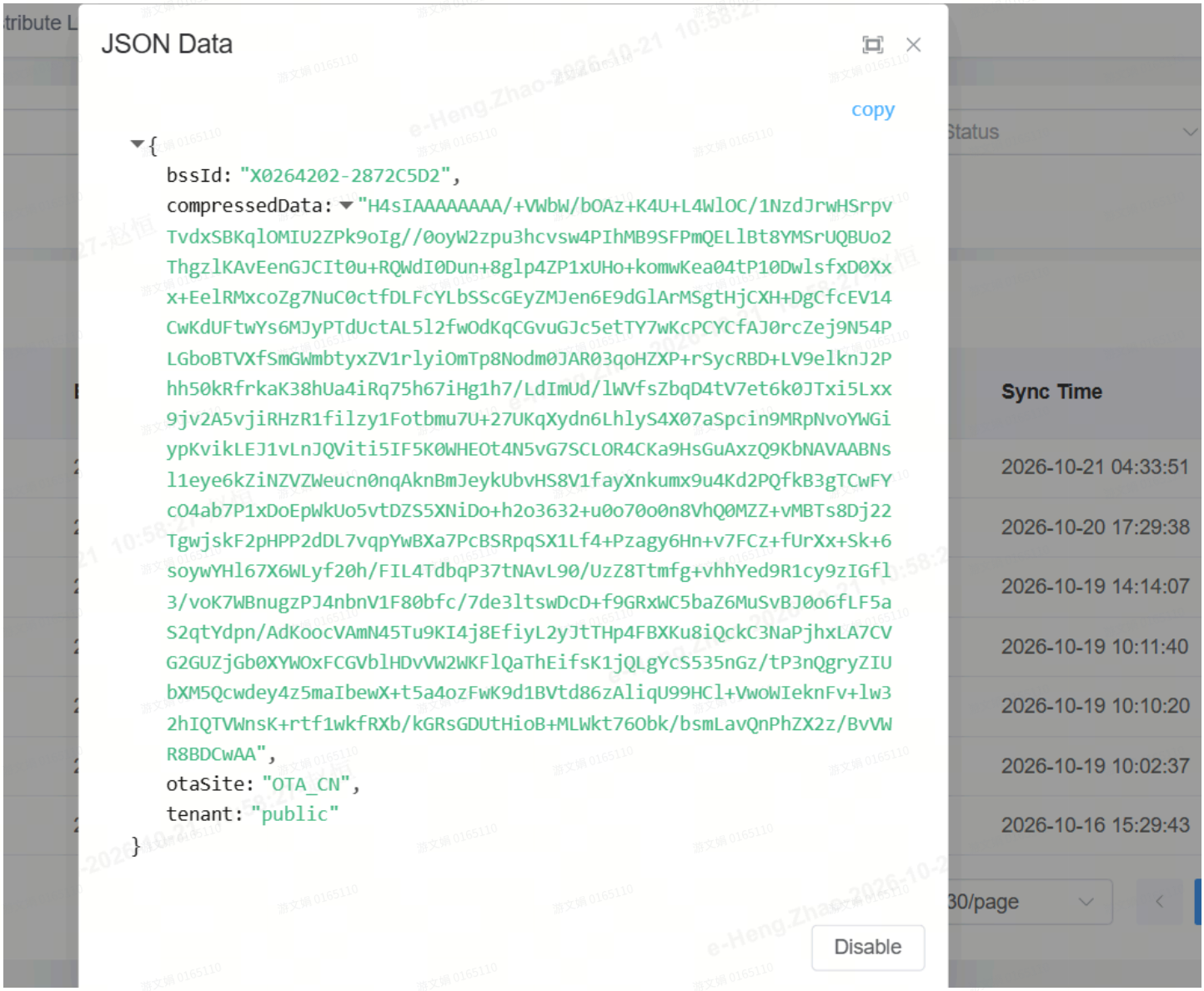
2. Click [Publish] in the Software Management - BSS list (the Publish button is only clickable for items configured in the external brand push management).

3. In BSS, go to Software Management - Version Policy list, then click either [Generate After-sales Version] or [Publish].

Among them, the field BSS ID corresponds to the BSS-ID in the Software Management - BSS List and Software Management - Version Policy List; the Type-code corresponds to the Type-code in

Master Data Management - Vehicle Model Management; the BSS release time corresponds to the BSS creation time in the Software Management - BSS List; the OS version corresponds to the externally displayed version in the Software Management - Version Policy List (ENG Release empty when manually pushed); the version status corresponds to the version status in the Software Management - Version Policy List; the target site corresponds to the site selected when pushing the BSS;The synchronization status includes success and failure (OTA sites execute asynchronously, and "success" only means that the OTA site has received the request, not that the actual execution has been successfully completed); the synchronization time corresponds to the time of manual push.

The operation parameters refer to the request parameters used when the vehicle model was pushed to the site at that time. Click "View" to see the specific content, as shown in the figure below; the content has been compressed via GZIP. Click "Copy" to copy the request parameters to the clipboard, the square button enables full-screen display, and click the X icon to close the window.



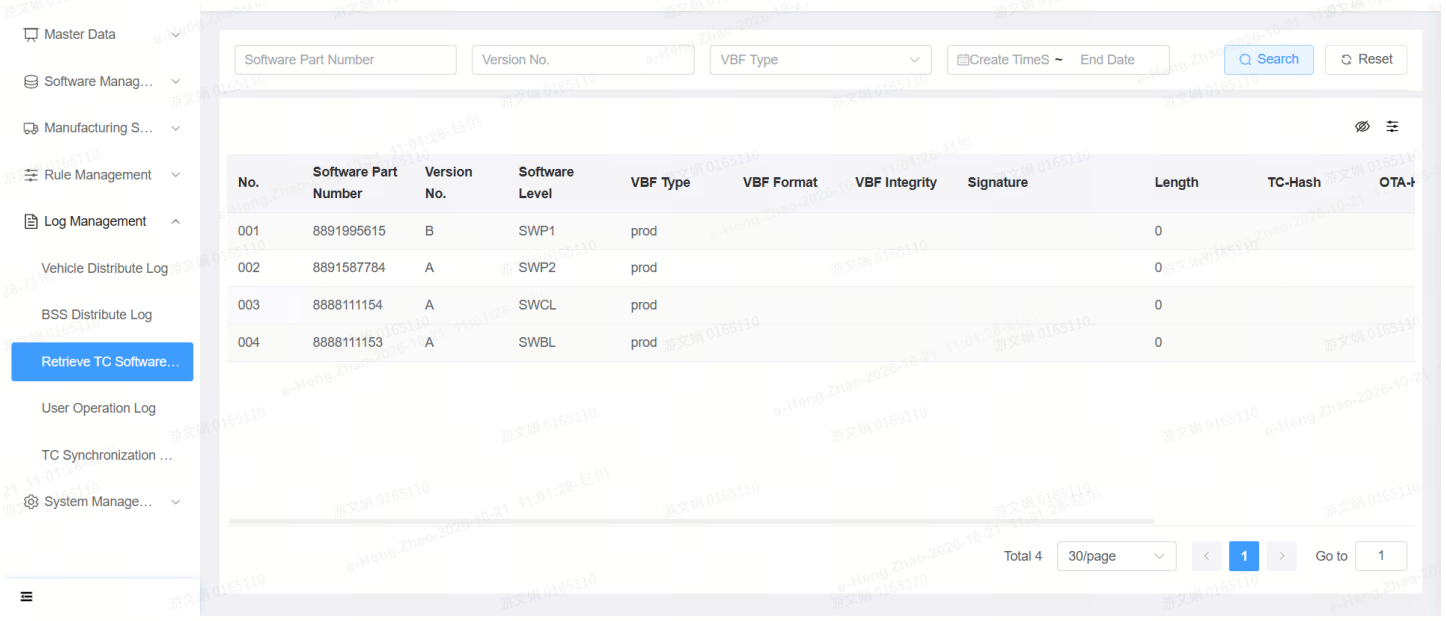
Click "View Return Parameters" to check the specific content of the return parameters, as shown in the figure below. Click "copy" to copy the request parameters to the clipboard, the square

button enables full-screen mode, and click the X to close the window.



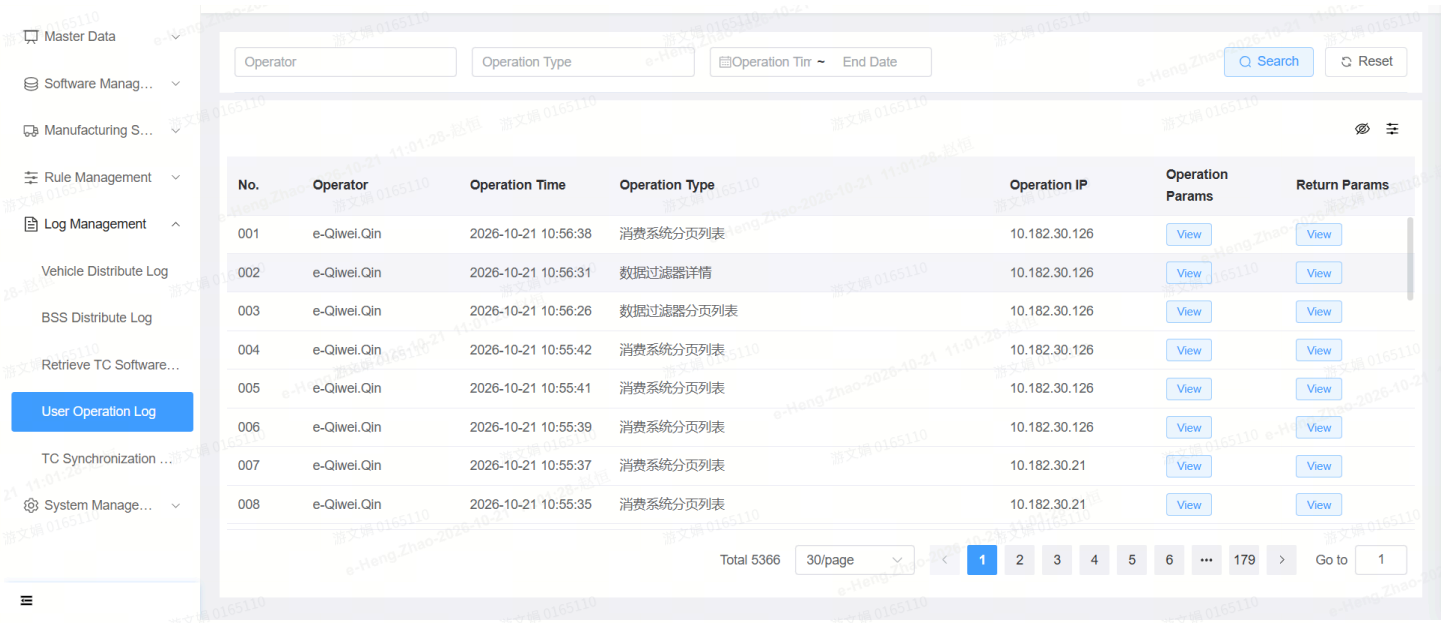
If synchronization fails, you can click [Retry] in the [Actions] column

5.4 Obtain TC software logs



Record the interface logs of the GDDC system calling the TC system to obtain the VBF address

5.5 User Operation Log



Record user operation logs of users operating in the GDDC system; mainly record addition, deletion, modification, as well as important data query records

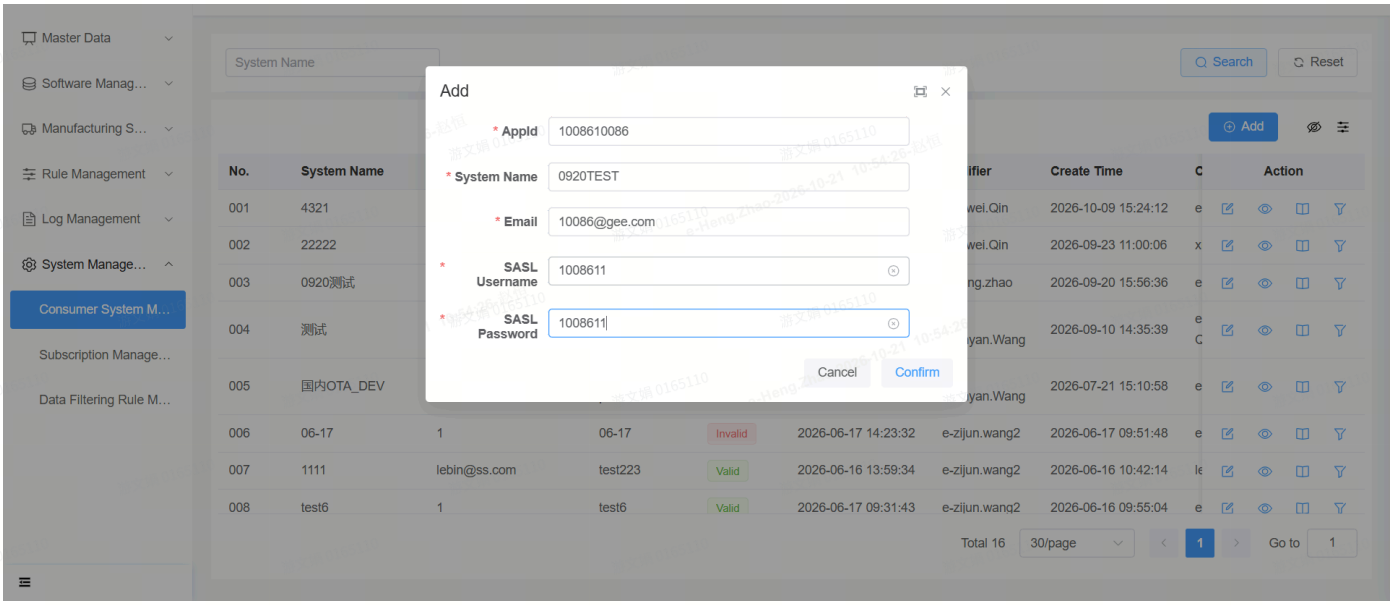
Click on the operation parameters to view the specific content, as shown in the figure below. Click "copy" to copy the request parameters to the clipboard, click the square button to enter full-screen mode, and click the "X" to close the window.



6. System Administration

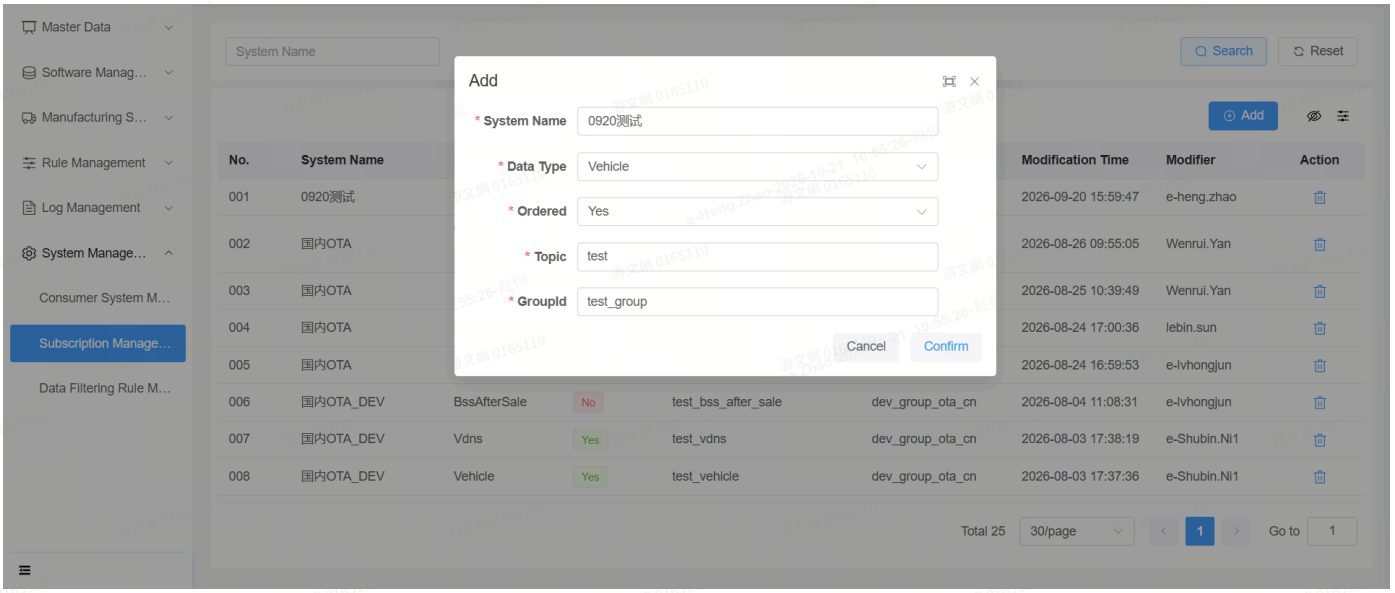
6.1 Consumer System Management

- This function serves as the configuration page for the global vehicle distribution queue, where the content of the data consumption queue is configured on a site-by-site basis.



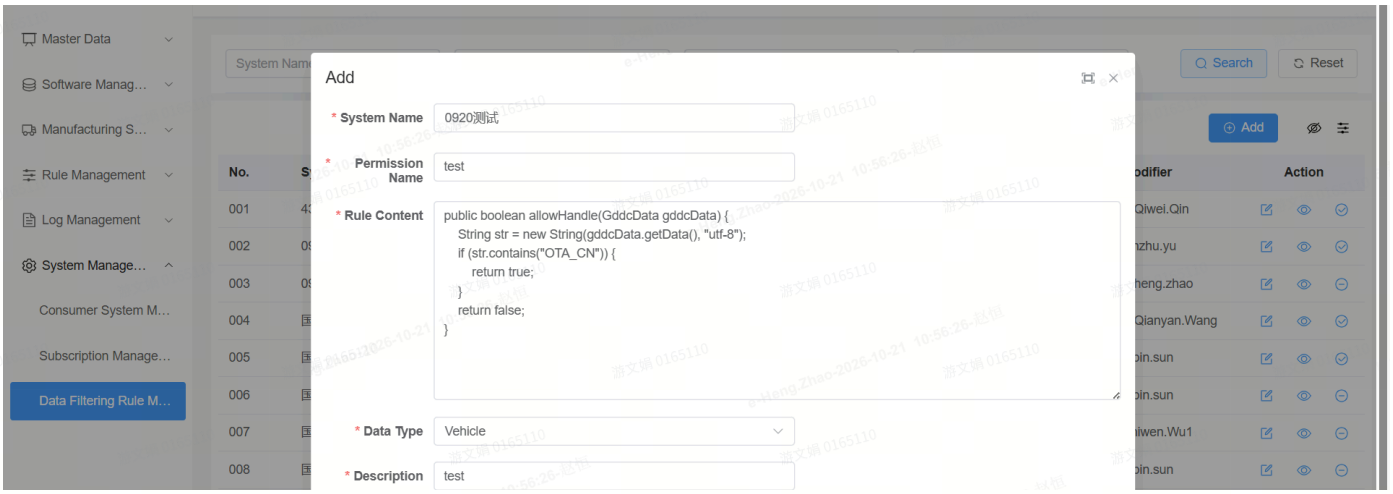
6.2 Subscription Management

- Configuration of Subscription Consumption Queues for Each Site



6.3 Data Filtering Rule Management

- Prior to sending vehicle data to the message queue, this filtering rule needs to be applied to ensure that the vehicles and data sent to the queue are correct


Master Data
Software Manag…
Manufacturing S…
Rule Management
Log Management
System Manage…
Consumer System M…
Subscription Manage…
Data Filtering Rule M…
Search
Reset
Add
No.
Action
Add
* System Name
0920测试
* Permission Name
test
* Rule Content
public boolean allowHandle(GddcData gddcData) {
String str = new String(gddcData.getData(), "utf-8");
if (str.contains("OTA_CN")) {
return true;
}
return false;
}
* Data Type
Vehicle
* Description
test